Aptitude for GATE

The GATE syllabus for General Aptitude is as follows:

Verbal Ability: English grammar, sentence completion, verbal analogies, word groups, instructions, critical reasoning and verbal deduction.

Numerical Ability: Numerical computation, numerical estimation, numerical reasoning and data interpretation.

However, GATE questions in the past have not necessarily followed this syllabus definition. To prepare for GATE general aptitude, it would require you to prepare all topics under aptitude. However, it would be foolhardy to do so as it is illogical to spend the amount of time required to prepare 'everything' vis-à-vis marks in GATE for general aptitude. So we suggest the better option.

We have closely analyzed the GATE question types in General Aptitude. In this book, we will work at two levels. A) Basics that you must know, and b) concepts/tricks that you acquire from solving questions.

Hence, the first part of the book is all the important theory with some practice questions. The second part of the book is a database of questions that cover some question types asked in GATE. After using this book, start solving the online tests.

First go through the concepts and make sure that you know your theory well. Once you have done that, start answering the questions. Wherever you come across a new concept / trick, mark that question as important and come back to it during revision process.

All the best

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Faculty, Career Avenues

Mathematics and General Aptitude

Chapter 1 Even and Odd Numbers

Suggested Time : 30 minutes

The numbers, which leave no remainder when divided by 2 are called even numbers and the numbers, which are not divisible by 2, are called odd numbers. The general form of writing an even number is 2n, and that of an odd number is $2n \pm 1$
Examples of even numbers are 0, 2, 4, 6, – 8, – 12, etc.
Examples of odd numbers are 1, 3, 5, – 7, – 15, etc.

Note:

1. The sum, difference and product of two even numbers are even.
   Example: 6 + 8 = 14, is even.

2. The sum and difference of two odd numbers is an even number.
   Example: 5 + 11 = 16, is even.

3. The product of two odd numbers is an odd number.
   Example: 3 × 7 = 21, is odd.

4. Zero is an even number.

5. (i) Even ^{Even} = Even. Example: 2^4 is even
   (ii) Even ^{Odd} = Even. Example: 2^3 is even
   (iii) Odd ^{Even} = Odd. Example: 3^2 is odd
   (iv) Odd ^{Odd} = Odd. Example: 3^3 is odd

Practice Questions A (Important)

1. If x and y are natural numbers, which of the following is definitely odd?
   1] $x^2y^3(x + y)$ 2] $x^3y^2 (x + y)$ 3] $(2x + 1) (2y + 1)$ 4] $(x + 1)(2y + 1)$

2. How many even integers lie between the two values of x (including the extremes), if $x^2 = 100$?
   1] 5 2] 10 3] 11 4] 4

3. If $7x + 3y = 240$, which of these statements is true?
   1] x and y are both odd. 2] Sum of x and y is even.
   3] If x is odd, y is even. 4] None of these

4. If x is a positive even integer and y a positive odd integer, then which of the following statements are true?
   i. $[(-1)^x]^y$ is always positive ii. $[(-1)^x]^{-y}$ is always negative
   iii. $[(-1)^x]^{y/2}$ is always negative iv. $(-1)^{x+y}$ is always negative

   1] i and ii only 2] i and iv only 3] ii and iv only 4] i and iii only

5. State whether the following statements are true or false?
   (i) Sum of two odd numbers is never odd.
   (ii) Product of two odd numbers is always odd.
   (iii) Sum of product of two odd numbers and product of two even numbers is sometimes even.
   (iv) Any even power of an odd number is even.
   (v) Product of two prime numbers is never prime.
   (vi) Sum of two prime numbers is never prime.

6. $2x + 3y = 240$, where x & y are natural numbers, what can you say about y?
   1] y is odd 2] y is even 3] y is a multiple of 3 4] none of these

Answer Key

1-3 2-3 3-2 4-2 5- True, True, False, False, True, False 6-1

Explanatory Answers

1. (2x + 1) and (2y + 1) are odd.
Odd × Odd is Odd.

2. x = – 10 or + 10. Between these integers in the number line, there are 11 even numbers.
(– 10, – 8, …0, 2, 4,... 10)

3. 7x + 3y = 240
240 is even. Either odd + odd = even, or even + even = even.
If x is odd, 7x is odd.
∴ 3y has to be odd to make 7x + 3y even.
∴ y has to be odd.
Similarly, when x is even, y has to be even.
∴ x and y are either both odd or both even.
∴ x + y is even in both scenarios. Hence [2]

4. i. As x is an even positive integer, $(-1)^x$ is 1 and $[(-1)^x]^y$ is 1. Hence, true.
ii. $(-1)^x$ is 1 and $((-1)^x)^{-y} = \frac{1}{y}$, which is positive. Hence, false.
iii. $(-1)^x$ is 1 and $((-1)^x)^{\frac{y}{2}}$ is positive. Hence, false.
iv. $(-1)^{x+y} = (-1)^{odd} = -1$. Hence, true. Hence [2]

5. (i) is true. (ii) is true. (iii) is false (always odd). (iv) is false (any power of an odd number is odd).
(v) is true. (vi) is false as 2 + 3 = 5, which is again prime.

6. 2x + 3y = 240
Whatever be x, 2x is even.
∴ 3y has to be even to make 2x+ 3y even (i.e. 240).
∴ y has to be even.

Practice Questions B (Optional)

1. Which of the following is not an even number?
   1] 2 2] 0
   3] – 4 4] All are even numbers

2. The number x ! (x factorial) is definitely
   1] Even 2] Odd
   3] Even if x > 1 4] Even if x is even and off if x is odd

3. 3x + 4y is an even number. Then what can you say about x and y?
   1] x is even, y is odd 2] x and y are both even
   3] x is even, y may be even or odd 4] x is odd, y is even

4. a, x and y are positive integers. $a = x^y$. Then we can say that
   1] a is even 2] a is odd
   3] a is even, if x is even 4] a is odd, if x is even

5. Let x, y and z be distinct integers, x and y are odd and positive, and z is even and positive. Which one of the following statements cannot be true?
   1] $(x-z)^2y$ is even 2] $(x-z)y^2$ is odd
   3] (x-z)y is odd 4] $(x-y)^2z$ is even

6. If n = 1 + x, where x is the product of four consecutive positive integers, then which of the following is/are true?
   A. n is odd B. n is prime C. n is a perfect square
   1] A and C only 2] A and B only
   3] A only 4] None of these

Answer Key

1-4 2-3 3-3 4-3 5-1 6-1

Chapter 2 Tests of Divisibility

Suggested Time : 45 minutes

Divisibility by 2	A number is divisible by 2 when its unit's digit is even. Example: 34, 900, 8616, etc. are divisible by 2 since the last digit is even.
Divisibility by 3	A number is divisible by 3 if the sum of its digits is a multiple of 3. Example: In 5412, the sum of the digits is 5 + 4 + 1 + 2 = 12, which is a multiple of 3. Hence, 5412 is divisible by 3.
Divisibility by 4	A number is divisible by 4 if the number formed by its last two digits is divisible by 4. Example: In 43552, the number formed by its last two digits is 52, which is divisible by 4. Hence, the number is divisible by 4.
Divisibility by 6	A number is divisible by 6 if it is divisible by both 2 and 3. Hence, apply the divisibility test of both 2 and 3. Example: 600, 36, 78, etc. are divisible by 6.
Divisibility by 7	A number is divisible by 7, if the sum of the number of tens in the number and five times the unit digit is divisible by 7. This test may be performed successively to get a smaller multiple of 7. Note that number of tens in 562 is 56 (and not 6) as 562 = 500 + 60 + 2 and 500 has 50 tens while 60 has another 6. Example: 735 is divisible by 7 as 73 + 5(5) = 73 + 25 = 98; 9 + 5(8) = 9 + 40 = 49, which is divisible by 7.
Divisibility by 8	A number is divisible by 8 if the number formed by its last three digits is divisible by 8. Example: In 569048, the last 3 digits of the number is 048, which is divisible by 8. Hence, the number is divisible by 8.
Divisibility by 9	A number is divisible by 9, when the sum of its digits is a multiple of 9. Example: 7281 is divisible by 9 since 7 + 2 + 8 + 1 = 18, which is divisible by 9.
Divisibility by 11	A number is divisible by 11, if the difference of the sum of the digits in odd places and the sum of the digits in even places (starting from units' place) is either 0 or a multiple of 11. For example, 8050314052, is divisible by 11 since the difference between the sum of digits in even places (5 + 4 + 3 + 5 + 8 = 25) and sum of digits in odd places (2 + 0 + 1 + 0 + 0 = 3) is 22 (i.e. 25– 3 = 22), which is divisible by 11.
Divisibility by 12	A number is divisible by 12, when it is divisible by 3 and 4. Example: 180 is divisible by 12, since it is divisible by 3 and 4.
Divisibility by 15	A number is divisible by 15, when it is divisible by both 3 and 5. Example: 1125 is divisible by 15, since it is divisible by 3 and 5.
Divisibility by 25	A number is divisible by 25, when the number formed by the last two digits of the number is divisible by 25. Example: 475350 is divisible by 25 since "50" is divisible by 25.

Note: 0 is divisible by all the numbers.

Practice Questions A (Important)

1. How many of the following number/s is / are divisible by 8 and 11 simultaneously?
(i) 12496 (ii) 414206 (iii) 999000 (iv) 48400
1] 1 2] 2 3] 3 4] 0

2. Which of the following numbers is a prime number?
1] 2181 2] 3585 3] 4587 4] 3931

3. What should you divide 11979 by, to get a perfect cube?
1] 3 2] 9 3] 11 4] 11^3

4. 109521ab is divisible by 22. Which of these is a possible value of a and b?
1] 06 2] 60 3] 54 4] 39

5. How many numbers are there between 100 and 200 (both inclusive), and divisible by 2 or 3?
1] 67 2]68 3] 84 4] 100

6. A 4-digit number is of the form "36ab" where a and b are positive integers, not necessarily unequal. If 36ab is divisible by 4, and also by 6, then how many possible values of "36ab" exist?
1] 6 2] 9 3] 8 4] 7

7. Fill in the missing digit in "53 _ 3559575" so that the number is a multiple of 11.
1] 3 2] 1 3] 7 4] 6

Answer Key

1-2 2-4 3-2 4-4 5-2 6-4 7-4

Explanatory Answers

1. (a) and (d)

2. 2181 – divisible by 3
3585 – divisible by 5
4587 – divisible by 11
3931 – prime number (check up to divisibility by prime numbers upto square root of 3931)

3. $11979 = 11^3 \times 9$. So it needs to be divided by 9.

4. ab has to be even and 109521ab divisible by 11. Using divisibility by 11, 6+b = 12 + a or b = 6 + a
Answer is (1). 39 cannot fit the bill, as it is odd.

5. There are 101 numbers between 100 and 200, both inclusive (starting with an even number).
$\therefore$ 51 of these numbers are even, i.e. divisible by 2.
Again, the numbers divisible by 3 start from 102 (34×3) and end at 198 (66×3).
From 34 to 66, there are 33 numbers.
Similarly, as $16 \times 6 = 96$, there are sixteen numbers among those 101 that are divisible by both 2 and 3.
$\therefore$ The required number = 51 + 33 – 16 = 68.

6. For a number N to be divisible by 4 and 6, N must be divisible by LCM of (4, 6). Now 3600 is divisible by 12.
The possible numbers are 3612, 3624, 3636,... 3696. (i.e. $3600 + 12 \times 1$, $3600 + 12 \times 2$, …$3600 + 12 \times 8$).
Total = 7 numbers.
Note: 3600 is divisible by 4 and 6, but 0 is neither positive nor negative.
The number 3660 is not possible, because in "36ab", 'a' and 'b' are positive integers

7. For a number to be divisible by 11, the difference of the sum of digits in even position and sum of the digits in odd position must be 0 or a multiple of 11.
In the number 53 x 3559575
Sum of digits in odd position = 5 + 5 + 5 + 3 + 3 = 21
Sum of digits in even position = 7 + 9 + 5 + x + 5 = 26 + x
$\therefore 26 + x - 21 = 11 \Rightarrow 5 + x = 11 \Rightarrow x = 6$
$\therefore$ The number is 536 355 9575. Hence [4]

Practice Questions B (Optional)

1. Which of the following is divisible by 11?
1] 123456789 2] 123454321 3] 112233221 4] None of the above

2. The number x ! (x factorial) is definitely
1] divisible by 6 2] divisible by 3
3] divisible by 6 if x > 2 4] not divisible by 6 if x < 5

3. How many 3-digit numbers of the form 36X are divisible by 3?
1] 2 2] 3 3] 4 4] 1

4. How many 4-digit numbers of the form 36AB are divisible by 15?
1] 6 2] 8 3] 7 4] 9

5. If 8a765 + 8765a (where 'a' is an integer between 0... 9)is divisible by 9 then what is 'a'?
1] 1 2] 3 3] 2 4] 4

6. Let N = Product of any three consecutive natural numbers. Then N is certainly divisible by
1] 6 2] 4 3] 5 4] 7

7. What are the digits a,b so that the number a759b is divisible by 72?
1] 4,2 2] 2,4 3] 3,6 4] 6,3

8. Let $N = 2^{2n} - 1$. Then which of the following statements are true?

i. N is always divisible by 3
ii. N is always divisible by 3 and 5
iii. N is divisible by 5 if n is odd
iv. N is divisible by 5 if n is even

1] i and iii 2] ii only 3] i and iv 4] iii and iv

Answer Key

1-4 2-3 3-3 4-3 5-1 6-1 7-1 8-3

Chapter 3 Power Cycles

Suggested Time : 30 minutes

Look very closely at the following values

2^1	2^2	2^3	2^4	2^5	2^6	2^7	2^8	2^9	2^{10}	2^{11}	2^{12}
2	4	8	16	32	64	128	256	512	1024	2048	4096

The unit's place of the numbers show a trend. They are 2, 4, 8, 6, 2, 4, 8, 6, …. In other words, the units place repeats in cycles of 4. Now try the same for other numbers like 3, 4, 5, 6, 7, …

Power	x^1	x^2	x^3	x^4	x^5	x^6	x^7	x^8
Unit's Digit of powers when x = 3	3	9	7	1	3	9	7	1
Unit's Digit of powers when x = 4	4	6	4	6	4	6	4	6
Unit's Digit of powers when x = 5	5	5	5	5	5	5	5	5
Unit's Digit of powers when x = 6	6	6	6	6	6	6	6	6
Unit's Digit of powers when x = 7	7	9	3	1	7	9	3	1
Unit's Digit of powers when x = 8	8	4	2	6	8	4	2	6
Unit's Digit of powers when x = 9	9	1	9	1	9	1	9	1
Unit's Digit of powers when x = 12 (same as 2)	2	4	8	6	2	4	8	6
Unit's Digit of powers when x = 13 (same as 3)	3	9	7	1	3	9	7	1

We can say that the unit's digit in powers of the number 3 shows the trend 3, 9, 7, 1, …, i.e. a cycle of 4. In other words every 4th power is 1. Or alternately, 3^{4n} ends in 1. Hence, all numbers of the form 3^{4n} end in 1, 3^{4n+1} end in 3, 3^{4n+2} end in 9 and 3^{4n+3} end in 7.

We can say that the unit's digit in powers of the number 4 shows the trend 4, 6, 4, 6, …i.e. a cycle of 2. However, a cycle of 2 also means a cycle of 4. In other words every 4th power is 6. Or alternately, 4^{4n} ends in 6. Hence, all numbers of the form 4^{4n} end in 6, 4^{4n+1} end in 4, 4^{4n+2} end in 6 and 4^{4n+3} end in 4.

This is true for all numbers, i.e. **For all numbers, the unit's digit repeats in a cycle of 4**.

Hence, $13^{13} = 13^{12+1} = 13^{4n+1}$. Now 3 shows a cycle of 3, 9, 7, 1…. So 3^{4n} ends in 1 and 3^{4n+1} end in 3. So 13^{13} ends in 3, or in other words it has 3 in the unit's place.

Practice Questions A (Optional)

1. What is the digit in the unit's place in the expansion of 3^{83}?
   1] 3 2] 9 3] 7 4] 1

2. What digit comes in the unit's place if 12^{12} is multiplied by 13^{13}?
   1] 6 2] 8 3] 2 4] 4

3. What is the last (unit's place) digit in the expansion of 17^{286}?
   1] 9 2] 3 3] 1 4] 7

4. What is the last (unit's place) digit in the expansion of $17^{28} \times 19^{86}$?
   1] 9 2] 3 3] 1 4] 7

Answer Key

1-3 2-2 3-1 4-4

Explanatory Answers

1. $3^{83} = 3^{80+3} = 3^{4 \times 20 + 3} = 3^{4k+3}$.
   Power cycle of 3 is 3, 9, 7, 1, …, i.e. 3^{4k} ends in 1
   Hence, 3^{4k+3} ends is 7.

Hence [3]

2. $12^{12} = 12^{4k}$ and $13^{13} = 13^{4k+1}$
   Power cycle of 2 is 2, 4, 8, 6, … and power cycle of 3 is 3, 9, 7, 1, …
   Hence, 12^{12} ends in 6 and 13^{13} ends in 3.

$\therefore 12^{12} \times 13^{13}$ ends in $6 \times 3 = 18$, i.e. the number ends in 8.

Hence [2]

3. 17 will have the same power cycle as 7.

$7^1 = 7$

$7^2 = 7 \times 7 = 49$

$7^3 = 49 \times 7 = 343$

$7^4 = 343 \times 7$ ... this has 1 at the unit's place

$\therefore$ Obviously, 7^5 will have $1 \times 7 = 7$ in the unit's place.

Thus, powers of 7 have 7, 9, 3, and 1 occurring simultaneously at the unit's place.

$286 = 71 \times 4 + 2$

$\therefore 17^{285}$ will have 7 at the unit's place.

$\therefore 17^{286}$ will have 9 at the unit's place. Hence, (1).

4. 17 will have the same power cycle as 7. Thus, powers of 7 have 7, 9, 3, and 1 occurring simultaneously at the unit's place.

$17^{28} = 17^{4k}$. Hence 17^{28} will end with 1.

19 will have the same power cycle as 9. Thus, powers of 9 have 9, 1, 9, and 1 occurring simultaneously at the unit's place.

$19^{86} = 19^{4k+2}$. Hence 19^{86} will end with 1.

Hence $17^{28} \times 19^{86}$ will be written as ___1 × ___1, i.e. It will be _____1. i.e. end with 1.

Chapter 4 Highest Power

Suggested Time : 45 minutes

A less popular but an easy type of question: Let us take an example. 5! is 120.

Note : 5! means 5 factorial. A factorial of any number 'n' is the value of the expression $n \times (n-1) \times (n-2) \times (n-3) \times \ldots \times 1$. Hence, $5! = 5 \times 4 \times 3 \times 2 \times 1 = 120$

Is 5! divisible by 2? Yes.
Is 5! divisible by 4? Yes.
Is 5! divisible by 8? Yes.
Is 5! divisible by 16? No.

5! is divisible by 2^1, 2^2 and 2^3 but not 2^4. So the highest power of 2 that divides 5! is 3.

Alternately, we can try and divide all the numbers in the expansion of 5! by 2.
1 is not divisible by 2.
2 is divisible by 2.
3 is not divisible by 2.
4 is divisible by 2, and the quotient is 2, which is further divisible by 2.
5 is not divisible by 2.
So 2 and 4 are the only numbers that are divisible by 2, and the highest power of 2 that divides (2×4) is 2^3.



Illustration

What is the largest possible value of 'x' in the following case, given that the numerator is perfectly divisible by the denominator?

4. $\frac{10!}{2^x}$

1] 5 2] 6 3] 7 4] 8

10! is $10 \times 9 \times \ldots \times 2 \times 1$
Of these, only 2, 4, 6, 8 and 10 are divisible by 2.
When 2 is divided by 2, the quotient is 1.
When 4 is divided by 2, the quotient is 2, which is further divisible by 2.
When 6 is divided by 2, the quotient is 3.
When 8 is divided by 2, the quotient is 4. This is again divided by 2, and the quotient is 2, which is further divisible by 2.
When 10 is divided by 2, the quotient is 5.

In other words, 2, 6 and 10 are divisible once, 4 is divisible twice and 8 is divisible thrice by 2. So 10! is divisible 1 + 1 + 1 + 2 + 3 = 8 times by 2. So the highest power of 2 that divides 10! is 8.

Googly Question

What is the highest power of 6 that divides 10!?
The answer is not 1. It is 4.

Note that 6 is 2 × 3. Hence, we should follow the following steps
1. What is the highest power of 2 that divides 10!? $\Rightarrow$ 8
2. What is the highest power of 3 that divides 10!? $\Rightarrow$ 4
3. What is the highest power of 6 that divides 10!?

From steps 1 and 2 we know that 10! is divisible by $2^8 \times 3^4$. Alternately, 10! is divisible by $2^4 \times (2^4 \times 3^4)$. Hence, 10! is divisible by $2^4 \times 6^4$. So the highest power of 6 that divides 10! is 4.

Shortcut 1

What is the largest possible value of 'x' in the following case, given that the numerator is perfectly divisible by the denominator?

$$\frac{10!}{3^x}$$

Upto 10, the number of numbers divisible by 3^1 are 3. (only 3, 6, 9)
Upto 10, the number of numbers divisible by 3^2 are 1. (only 9)
Hence, answer is 3 + 1 = 4

Shortcut 2

What is the largest possible value of 'x' in the following case, given that the numerator is perfectly divisible by the denominator?

$$\frac{10!}{6^x}$$

Note : Dividing by 6^x is dividing by $2^x \times 3^x$. Every 2nd number is divisible by 2, but every 3rd number is divisible by 3. Hence, there are lesser numbers that are divisible by 3.

If $6^x = 2^a \times 3^b$, then b < a. This means that we need to only find out the highest power of 3 that divides 6!, as the number of numbers in 6! that are divisible by 2 are more.

i.e. The question is the same as $\frac{10!}{3^x}$

Now, divide 10 by 3 and write the quotient, and then repeat this exercise by dividing the quotient by 3, and so on, i.e. $\frac{10}{3} = \frac{3}{3} = 1$.

Finally, add all the quotients you get in this process. In the above example it is 3 + 1 = 4.

Extension of shortcut 2

Another way of expressing the above is

$$\begin{array}{r|l} 3 & 10 \\ 3 & 3 \quad \leftarrow \\ & 1 \quad \leftarrow \end{array}$$

Taking the sum of the quotients we get 3 + 1 = 4.

Practice Questions A (Optional)

1. How many of the following numbers divide 5! (i.e. 5 factorial)?
(a) 3 (b) 5 (c) 10 (d) 15 (e) 16

1] 2 2] 3 3] 4 4] 5

Directions for questions 4-7

What is the largest possible value of 'x' in each of the following cases, given that in each case the numerator is perfectly divisible by the denominator?

2. $\frac{10!}{2^x}$

1] 5 2] 6 3] 7 4] 8

3. $\frac{10!}{3^x}$

1] 2 2] 3 3] 4 4] 5

4. $\frac{10!}{6^x}$

1] 1 2] 2 3] 3 4] 4

5. $\frac{10!}{10^x}$

1] 1 2] 2 3] 4 4] 5

6. What is the highest power of 4 that can divide 40!?
1] 19 2] 40 3] 12 4] 38

Answer Key

1-3 2-4 3-3 4-4 5-2 6-1

Explanatory Answers

1. $5! = 5 \times 4 \times 3 \times 2 \times 1 = 5 \times 2 \times 2 \times 3 \times 2 \times 1 = 5 \times 2^3 \times 3$.
Hence 5! is divisible by 3, 5, 10 (i.e. 2×5), 15 (i.e. 3×5), but not 16 (i.e. 2^4). Hence [3]

2. Using extension of shortcut 2,

$2 \overline{)10}$

$2 \overline{)5}$ Adding the quotients, we get 5 + 2 + 1 = 8. Hence [4]

$2 \overline{)2}$

1

3. Using extension of shortcut 2,

$3\,|\,\underline{10}$

$3\,|\,\underline{3}$ Adding the quotients, we get 3 + 1 = 4. Hence [3]

1

4. For 6, we divide by 3 as $6 = 2 \times 3$, and the constraining factor is 3.

We get $\frac{10}{3} = \frac{3}{3} = 1$. Hence answer = 3 + 1 = 4. Hence [4]

5. For 10, we divide by 5 as $10 = 2 \times 5$, and the constraining factor is 5.

We get $\frac{10}{5} = 2$. Hence answer = 2. Hence [2]

6. $\frac{40!}{4^x} = \frac{40!}{(2\times 2)^x}$

$2\,|\,\underline{40}$

$2\,|\,\underline{20}$

$2\,|\,\underline{10}$

$2\,|\,\underline{5}$

$2\,|\,\underline{2}$

1

Highest power of 2 that divides 40 ! is 20 + 10 + 5 + 2 + 1 = 38, i.e. $\frac{40!}{2^{38}}$

$\therefore$ Highest power of 2×2 that divides 40! is $\frac{38}{2}$ = 19, i.e. $\frac{40!}{\left(2^2\right)^{19}}$. Hence [1]



Chapter 5 Factors

Suggested Time : 45 minutes

Factor: A factor is a number which divides another number. For example, the factors of 10 are 1, 2, 5 and 10.

NUMBER OF FACTORS OF A GIVEN NUMBER

If N is a composite number such that $N = a^x b^y c^z \ldots$ where a, b, c are prime factors of N and x, y, z are positive integers, then

The number of factors of N = (1 + x) (1 + y) (1 + z)…

SUM OF FACTORS

If N is a composite number such that $N = a^x b^y c^z \ldots$ where a, b, c are prime factors and x, y, z are positive integers, then

The sum of the factors of N = $\left(\frac{a^{x+1}-1}{a-1}\right)\left(\frac{b^{y+1}-1}{b-1}\right)\left(\frac{c^{z+1}-1}{c-1}\right)\ldots$

Example:
Find out the sum of the factors of 200

$N = 200 = 2^3 \times 5^2$

The sum of the factors = $\left(\frac{2^{3+1}-1}{2-1}\right)\left(\frac{5^{2+1}-1}{5-1}\right) = 15 \times 31 = 465$

SOME IMPORTANT RESULTS ON FACTORS

If N is a composite number such that $N = a^x b^y c^z \ldots$ where a, b, c are prime factors and x, y, z are positive integers, then

1. $\left.\begin{matrix}\textbf{The number of ways of expressing N} \\ \textbf{as product of two different factors}\end{matrix}\right\} = \frac{(1+x)(1+y)(1+z)\ldots}{2} = \frac{\textbf{Number of factors}}{2}$

2. If N is a perfect square, then

$$\left.\begin{matrix}\textbf{The number of ways of expressing N} \\ \textbf{as product of two different factors}\end{matrix}\right\} = \frac{\{[(1+x)(1+y)(1+z)\ldots]-1\}}{2}$$

Note: All perfect squares (and only perfect squares) have odd number of factors.
Example: Number of factors of 36 is 9.
$36 = 4 \times 9$
$36 = 2^2 \times 3^2$
Number of factors = $(2 + 1) \times (2 + 1) = 9$

Practice Questions A (Optional)

1. How many rectangles with integral sides and with an area 224 m^2 are possible?
   1] 10 2] 12 3] 6 4] 8

2. What is the number of positive integers that can divide 200 without remainder?
   1] 10 2] 6 3] 12 4] 5

3. The number N has 144 factors including 1 and itself. What is the maximum and minimum possible number of prime factors that N can have?
   1] 144 & 1 2] 6 & 1 3] 8 & 2 4] 4 & 2

4. Find the number of positive integers, which divide 10^{999} but not 10^{998}.
   1] 1999 2] 999 3] 2999 4] 1799

Answer Key

1-3 2-3 3-2 4-1

Explanatory Answer

1. To find out the number of rectangles with integral sides, we need to find out the number of ways of expressing 224 as a product of two factors.

$24 = 8 \times 28 = 8 \times 4 \times 7 = 2^5 \times 7^1$

Number of ways of expressing 224 as a product of two factors (OR)

Number of rectangles possible = $\frac{(1+5)(1+1)}{2} = 6$

2. If $N = a^x b^y c^z \ldots$ where a, b, c are prime factors of N and x, y, z are positive integers, then
The number of factors of N = (1 + x) (1 + y) (1 + z)…
$200 = 2^3 \times 5^2$
The number of factors = (3 + 1) (2 + 1) = $4 \times 3 = 12$

3. If $N = a^x b^y c^z \ldots$ where a, b, c are prime factors of N and x, y, z are positive integers, then
The number of factors of N = (1 + x) (1 + y) (1 + z)…
Given that the number of factors is 144 $= 2 \times 2 \times 2 \times 2 \times 3 \times 3$
If the number N should be expressed with maximum prime factors then $N = a \times b \times c \times d \times e^2 \times f^2$ whose number of factors = $2 \times 2 \times 2 \times 2 \times 3 \times 3 = 144$.
$\therefore$ The maximum possible prime factors of N = 6
If the number N should be expressed with minimum prime factors then $N = a^{143}$ whose number of factors = 143 + 1 = 144.
$\therefore$ The minimum possible prime factor of N = 1

4. If $N = a^x b^y c^z \ldots$ where a, b, c are prime factors of N and x, y, z are positive integers, then
The number of factors of N = (1 + x)(1 + y) (1 + z)…

Let $N_1 = 10^{999} = 5^{999} \times 2^{999}$

$\therefore$ Number of divisors of N = (999 + 1) (999 + 1) = 1000^2

Let $N_2 = 10^{998} = 2^{998} \times 5^{998}$

$\therefore$ Number of divisors = (998 + 1) (998 + 1) = 999^2

$\therefore$ The number of positive integers, which divide 10^{999} but not 10^{998} is $1000^2 - 999^2$
= (1000 + 999) (1000 – 999) = 1999. Hence [1]

Chapter 6 HCF, LCM

Suggested Time : 120 minutes

HIGHEST COMMON FACTOR (HCF)

The Highest Common factor (HCF) of two or more numbers is the greatest number which divides each of the given numbers completely, leaving no remainder.

Example: Find out the Highest Common Factor of 12 and 18?
The Factors of 12: 1, 2, 3, 4, 6 and 12
The Factors of 18: 1, 2, 3, 6, 9 and 18
The common factors of 12 and 18: 1, 2, 3, and 6
Among them the highest common factor is 6. Hence the H. C. F of 12 and 18 is 6.

The terms Greatest Common Divisor (GCD), highest common divisor (HCD), greatest common factor (GCF) are also used to denote HCF.

HCF BY FACTORISATION METHOD

The numbers are resolved to its prime factors and then the highest of the common factors can be written down at a glance.

Example: Find the HCF of 105, 135 and 180

Step 1: Express the given number in terms of its prime factors.

$$\left.\begin{aligned} 105 &= 3\times5\times7 \\ 135 &= 3\times3\times3\times5 \\ 180 &= 2\times2\times3\times3\times5 \end{aligned}\right\} \Rightarrow \left.\begin{aligned} 105 &= 3\times5\times7 \\ 135 &= 3^3\times5 \\ 180 &= 2^2\times3^2\times5 \end{aligned}\right\}$$

Step 2: Now, take the lowest power of each of the common prime factors. The highest of the common factors, HCF = $3 \times 5 = 15$.

HCF BY LONG DIVISION METHOD

To find out the HCF of large numbers we use the long division method. In this method, divide the larger number by the smaller number. Then divide the divisor by the remainder; then divide the divisor of the latter division by the next remainder and continue this process till the remainder becomes zero. The last divisor is the HCF of the two numbers.

Example: Find the HCF of 851 and 943

```
851)943(1
  –851
   92 )851(9
     –828
       23)92(4
         –92
          0          ∴ HCF =23
```

Step 1: Divide 943 by 851. The remainder of the division = 92
Step 2: Divide 851 by 92. The remainder of the division = 23
Step 3: Divide 92 by 23. The remainder of the division = 0
The HCF of (851, 943) = 23.

LEAST COMMON MULTIPLE (LCM)

The least common multiple (LCM) of two or more numbers is the smallest number, which is divisible by each of the given numbers completely leaving no remainder.

LCM BY FACTORIZATION METHOD

The numbers are resolved to its prime factors and the least of the common multiple can be written at a glance.

Example: Find the LCM of 30, 32, 36, 40 and 48

$30 = 2 \times 3 \times 5$; $32 = 2^5$; $36 = 2^2 \times 3^2$; $40 = 2^3 \times 5$; $48 = 2^4 \times 3$

Now take the highest power of each of the prime factors.

LCM of (30, 32, 36, 40 and 48) $= 2^5 \times 3^2 \times 5 = 1440$.

LCM BY DIVISION METHOD

This method is explained by the following example.

Example: Find out the LCM of 30, 32, 36, 40 and 48. 2 | 30, 32, 36, 40, 48 2 | 15, 16, 18, 20, 24 2 | 15, 8, 9, 10, 12 2 | 15, 4, 9, 5, 6 3 | 15, 2, 9, 5, 3 5 | 5, 2, 3, 5, 1 1, 2, 3, 1, 1 LCM of (30, 32, 36, 40, 48) $= 2 \times 2 \times 2 \times 2 \times 3 \times 5 \times 2 \times 3 = 1440$	**Step 1:** Write the given numbers in a line and select one prime factor common to at least two of the given numbers and divide all the numbers with the prime factor. **Step 2:** Write down the quotient for every number under the number itself. If any number is not divisible by the prime factor selected in step 1, write the number unaltered in the next line of the quotient. Repeat this process until you get a line of prime numbers or 1as the quotients. **Step 3:** The product of all the divisors and the prime numbers in the last line will be the LCM of the given numbers.

SOME IMPORTANT RESULTS ON HCF AND LCM

1. HCF of Fractions $= \dfrac{\text{HCF of numerators}}{\text{LCM of denominators}}$ and LCM of Fractions $= \dfrac{\text{LCM of numerators}}{\text{HCF of denominators}}$

   For example, find LCM and HCF of $\frac{3}{4}$ and $\frac{6}{14}$.

   LCM $= \dfrac{6}{2} = 3$ HCF $= \dfrac{3}{28}$

2. Relationship between LCM and HCF

   Let A, B be the numbers and let H be their HCF. Then,

   A = some factor $\times$ H = a $\times$ H

B = b $\times$ H and a and b have no common factors, if they had, H can not be the HCF of A and B. Clearly LCM of A and B = a $\times$ b $\times$ H

Therefore, LCM $\times$ HCF = a $\times$ b $\times$ H $\times$ H

= (a $\times$ H) $\times$ (b $\times$ H)

= A $\times$ B

Hence, Product of LCM and HCF of two numbers = Product of the numbers.

SOME IMPORTANT RULES

RULE 1
If the number N when divided by divisor D leaves the quotient Q and remainder R, then the number can be written in the form, **N = D× Q + R,** where Q = 0, 1, 2, 3…
Example: 45 when divided by 6 the remainder is 3 and the quotient is 7
The number can be written as = 6× 7 + 3 = 45

RULE 2
If the number N when divided by divisor D leaves the remainder R, then **(N – R)** is divisible by D.
Example: 83 when divided by 7 the remainder is 6
N – R = 83 – 6 = 77.
77 is exactly divisible by 7

RULE 3
If the number N when divided by divisor D leaves the remainder R, then **N + (D – R)** is divisible by D.
Example: 69 when divided by 5 the remainder is 4
N + (D – R) = 69 + (5 – 4) = 70
70 is exactly divisible by 5

Note: The above three rules have been also stated in the note on Remainder Theorem and are used in solving the questions based on LCM and HCF.

QUESTION TYPES BASED ON LCM AND HCF
Based on LCM and HCF, there are five types of questions that we generally come across in the aptitude tests.

LCM - TYPE 1
(REMAINDER IN EACH DIVISION IS SAME)

Example:
What is the smallest number which when divided by 6, 7 and 8 leaves 2 as the remainder in each case?

Here the remainder in each of the division is the same.

Step 1: A. Let the required number and the remainder in each case be N and R respectively.
Using **Rule 2**, N – 2 is divisible by 6, 7 and 8.
B. ∴N – 2 must be a multiple of 6, 7 and 8.
C. The smallest possible value of N – 2 must be the LCM of (6, 7, 8).
Any such value of N – 2 can be written as follows
N – 2 = K × LCM of (6, 7, 8) where K = 1, 2, 3…

Step 2: K × LCM of (6, 7, 8) = K × 2^3 × 3 × 7 = 168K (1)

Step 3: To find out the least possible value of N – 2,
Substituting K = 1in (1), we get
N – 2 = 168
N = 170

Visualization Approach (To understand concept of negative remainder, refer to the next chapter on Remainder Theorem)

Let the number be x. It is given that

$6\overline{)\ x}$ $7\overline{)\ x}$ $8\overline{)\ x}$

$\underline{2 \text{ or } -4}$ $\underline{2 \text{ or } -5}$ $\underline{2 \text{ or } -6}$

i.e. x divided by 6 leaves a remainder of 2 or – 4. In other words, x – 2 or x + 4 is divisible by 6.
Similarly, x divided by 7 leaves a remainder of 2 or – 5. In other words, x – 2 or x + 5 is divisible by 7.
Similarly, x divided by 8 leaves a remainder of 2 or – 6. In other words, x – 2 or x + 6 is divisible by 8.
Hence, we can say that x – 2 is divisible by 6, 7 and 8.

The smallest number that s divisible by 6, 7 and 8 is their LCM, i.e. 168. The other numbers are multiples of the LCM, i.e. LCM × 2, LCM × 2, LCM × K.
Here, x – 2 = 168.
$\therefore x = 168 + 2 = 170$.

Challenger Question

What is the smallest four-digit number which when divided by 6, 7 and 8 leaves 2 as the remainder in each case?

Clue : x – 2 = 168k, where k = 1, 2, 3 …

LCM - TYPE 2
(THE DIFFERENCE BETWEEN DIVISOR AND REMAINDER IN EACH DIVISION IS SAME)

Example:
What is the smallest number which when divided by 6, 7 and 8 leaves 4, 5 and 6 as remainders respectively?

Here, (Divisor – Remainder) in each division is same.

Divisor – Remainder = (6 – 4) = (7 – 5) = (8 – 6) = D – R = 2

Step 1: A. Let the required number be N.
Using **Rule 3**, N + 2 is divisible by 6, 7 and 8.
B. $\therefore$ N + 2 must be a multiple of 6, 7 and 8.
C. The smallest possible value of N + 2 must be the LCM of (6, 7, 8).
Any such value of N + 2 can be written as follows
N + 2 = K × LCM of (6, 7, 8) where K = 1, 2, 3…

Step 2: K × LCM of (6, 7, 8) = $K \times 2^3 \times 3 \times 7 = 168K$ (1)

Step 3: To find out the least possible value of N + 2,
Substituting K = 1in (1), we get
N + 2 = 168
N = 166

Now try solving the above using the visualization approach. The first step has been worked out for you.

Let the number be x. It is given that

$6\overline{)\ x}$ $7\overline{)\ x}$ $8\overline{)\ x}$

$\underline{4 \text{ or } -2}$ $\underline{5 \text{ or } -2}$ $\underline{6 \text{ or } -2}$

i.e. x divided by 6 leaves a remainder of 4 or – 2. In other words, x – 4 or x + 2 is divisible by 6.

Challenger Questions

1. What is the smallest four-digit number which when divided by 6, 7 and 8 leaves 4, 5 and 6 as remainders respectively?

2. What is the largest four-digit number which when divided by 6, 7 and 8 leaves 4, 5 and 6 as remainders respectively?

LCM - TYPE 3
(REMAINDER IN EACH DIVISION IS NOT THE SAME & THE DIFFERENCE BETWEEN THE DIVISOR AND THE REMAINDER IN EACH DIVISION IS NOT THE SAME)

Example:
What is the smallest number which when divided by 7 and 8 leaves 4 and 6 as remainders respectively?

Step 1: Let the required number be N. Using **Rule 1**, N can be expressed as $N = D \times Q + R$.
When N is divided by 8, the remainder is 6.
Hence, $N = 8Q + 6$ Equation (1)

Step 2: When N is divided by 7, the remainder is 4. $\therefore$ By **Rule 2**, (N – 4) must be divisible by 7.
But, $N = 8Q + 6$.
$\therefore N - 4 = 8Q + 6 - 4 = 8Q + 2$

Since N – 4 is multiple of 7, 8Q + 2 can be written as 7K where K is 0, 1, 2, 3…
Hence, $8Q + 2 = 7K$ and

$K = \frac{8Q+2}{7} = Q + \frac{Q+2}{7}$.

$\frac{Q+2}{7}$ has to be an integer ($\because$ K is an integer)

Now Q can take the values 5, 12, 19…

Step 3: $Q \neq 12, 19, \ldots$ ($\because$ The problem asks for smallest possible number)
$\therefore Q = 5$; Substituting the value of Q in equation (1), we get
$N = 8 \times 5 + 6 = 46$.
The required number is 46.

Note 1 : Any such value of N can be written as 46 + (7 × 8) K, where K = 0, 1, 2, 3, …
Note 2 : Alternate approach to find Q
In step 2, we reach the stage $N - 4 = 8Q + 2$
We know that $8 \equiv 1 \bmod 7$
$8Q \equiv Q \bmod 7$
$8Q + 2 \equiv Q + 2 \bmod 7$
But we know that when 8Q + 2 is divided by 7, the remainder is 0, i.e. $8Q + 2 = 0 \bmod 7$.
Hence, Q + 2 should be 7 or a multiple of 7. Hence, Q = 5, 12, 19, …

HCF - TYPE 1
(REMAINDER IN EACH DIVISION IS DIFFERENT)

Example:
What is the largest number which divides 132 and 387 leaving 12 and 27 as remainders respectively?

Step 1: Let the largest divisor be H. Using **Rule 2**,
132 when divided by H leaves 12 as remainder. ∴ 132 – 12 = 120 is divisible by H
387 when divided by H leaves 27 as remainder. ∴ 387 – 27 = 360 is divisible by H
Since H divides 120 and 360 without remainder, H is a common factor of 120 and 360.

Step 2: The largest number that divide 120 and 360 without remainder is HCF of 120 and 360.
HCF of (360, 120) = 120

HCF - TYPE 2
(REMAINDER IN EACH DIVISION IS SAME)

Example:
What is the largest number which divides 150, 210 and 375 leaving the same remainders in each case?

Step 1: Let the largest divisor and the remainder be H and R respectively. Using **Rule 2**,
150 when divided by H the remainder is R. ∴ 150 – R is divisible by H
210 when divided by H the remainder is R. ∴ 210 – R is divisible by H
375 when divided by H the remainder is R. ∴ 375 – R is divisible by H

Step 2: The largest factor possible is HCF of [(150 – R), (210 – R) and (375 – R)]
Since (150 – R) and (210 – R) are divisible by H, their difference will be divisible by H, i.e.
(210 – R) – (150 – R) = 60 is also divisible by H.
Since (210 – R) and (375 – R) are divisible by H, their difference will be divisible by H, i.e.
(375 – R) – (210 – R) = 165 is also divisible by H.

Step 3: H the largest number that divides 60 and 165. ∴H is the HCF of 60 and 165
HCF (60, 165) = 15.
The required divisor is 15.

Practice Questions A (Important)

1. Find the greatest number that will divide 964, 1238, and 1400 leaving remainder of 41, 31, and 51 respectively.
1] 58 2] 64 3] 69 4] 71

2. Find the largest number with which when 250, 490 and 850 are divided, the remainders are same.
1] 140 2] 105 3] 120 4] 136

3. Find the smallest number which when divided by 6, 7, 8, 9 and 10 leaves 4, 5, 6, 7 and 8 respectively as remainders.
1] 2518 2] 2532 3] 2638 4] 2218

4. Four bells begin tolling at the same time and continue to toll at intervals of 20, 25, 28 and 50 seconds respectively. Find the smallest integral number of minutes after which all the bells toll together again.
1] 700 2] 70 3] 1400 4] 35

5. What is the greatest length that can be used to measure exactly the lengths 20 ft, 13 ft 9 inches, 17 ft 6 inches and 21 ft 3 inches? (Take 1ft = 12inches)
1] 1ft. 3inches 2] 1ft. 6inches 3] 1ft. 9inches 4] 2ft. 3inches

6. The LCM of two numbers is 28 times its HCF; the sum of the LCM and HCF is 1740. If one of number is 240, what is the other number?
1] 60 2] 402 3] 420 4] Data inconsistent

7. A heap of pebbles when made up into groups of 32, 40 or 72 leave 10, 18 and 50 as leftover pebbles respectively. Find the least possible number of pebbles in the heap.
1] 1440 2] 1462 3] 458 4] 1418

8. What is the least number that should be subtracted from 1936 so that when the result is divided by 9, 10 and 15, leaves 7 as the remainder in each case?
1] 46 2] 39 3] 29 4] 38

9. The LCM of two numbers is 140 and their HCF is 19. Find the product of the numbers.
1] 2660 2] 266 3] Indeterminate 4] Data inconsistent

10. An army has got 378 swords and 675 spears, which are kept in boxes such that there is equal number of weapons in each box. It should be noted that each box contains either swords or spears. What is the least possible number of boxes the army requires?
1] 28 2] 39 3] 49 4] 52

11. How many pair of numbers will have 36 as their LCM and 1 as their HCF?
1] 2 2] 1 3] 3 4] 4

12. Find the largest 4 digit number which when divided by 20, 25, and 35 will leave a remainder of 2.
1] 7002 2] 9798 3] 9998 4] 9802

13. How many pairs of numbers are possible such that their HCF is 2 and LCM is 72.
1] 4 2] 3 3] 1 4] 0

14. In a party the host orders 3 cakes. There is a $49\frac{1}{2}$ kg vanilla cake, a $33\frac{3}{4}$ kg black forest cake and a $20\frac{1}{4}$ kg strawberry cake. Each of the cake needs to be divided into pieces such that across varieties, each piece is of equal weight. If the pieces are as heavy as possible then what is the maximum number of pieces possible?
1] 46 2] 92 3] 100 4] 112

Answer Key

1-4	2-3	3-1	4-4	5-1	6-3	7-4	8-2	9-4	10-2
11-1	12-4	13-3	14-1						

Explanatory Answers

1. This problem is of HCF type – I. The required number = HCF of [(964 – 41), (1238 – 31), (1400 – 51)]
HCF of (923, 1207, 1349) = 71. Hence [4]

2. This problem is of HCF type – II. The required number = HCF of [(850 – 250), (490 – 250)]
HCF of (600, 240) = 120. Hence [3]

3. This problem is of LCM type – II and D – R = 2. Let the required number be N.
N + 2 = K × LCM of (6, 7, 8, 9, 10). The value of K = 1 (∵ N is the smallest possible number)
LCM of (6, 7, 8, 9, 10) = 2520.
N = 2520 – 2 = 2518. Hence [1]

4. The smallest time after which all the bells toll together again = K × LCM of (20, 25, 28, 50), where K = 1, 2, 3, …
$20 = 2 \times 10 = 2^2 \times 5$; $25 = 5^2$;
$28 = 2 \times 2 \times 7 = 2^2 \times 7$; $50 = 5 \times 10 = 2 \times 5^2$
Time = K × LCM of (20, 25, 28, 50) = $K \times 2^2 \times 5^2 \times 7 = 700K$ sec
When K = 1, Time = 700 sec, which is not an integral minute.
When K = 3, Time = 3 × 700 sec = 2100 sec = 35 minutes.

5. Converting all dimensions in inches we get, 240 inches, 165 inches, 210 inches and 255 inches.
The greatest length that can measure all these lengths = HCF of (240, 165, 210, 255).
HCF of (240, 165, 210, 255) = 15 inches or l ft 3 inches.

6. LCM = 28 × HCF
LCM + HCF = 1740
$28\ HCF + HCF = 1740 \Rightarrow HCF = \frac{1740}{29} = 60$
LCM = 28 × 60.
We know that product of two numbers = LCM × HCF. Hence, the required number = $\frac{28 \times 60 \times 60}{240} = 420$

7. Here we find the problem is of LCM type –II
Let the required number be N.
Divisor – Remainder = D – R = 22
Any such value of N + (D – R) can be written as,
N + (D – R) = K × LCM of (32, 40, 72).
$32 = 2^5$; $40 = 2^3 \times 5$; $72 = 2^3 \times 3^2$
N + (D – R) = K × LCM of (32, 40, 72)
$N + (D - R) = K \times 2^5 \times 5 \times 3^2$
Here the value of K = 1
N + 22 = 1440 (∵ D – R = 22)
N = 1440 – 22 = 1418.

8. Let N be the number which when divided by 9, 10 and 15 leaves 7 as remainder in each case, i.e. N – 7 is exactly divisible by 9, 10 and 15
N – 7 = K × LCM of (9, 10, 15)

$$\begin{array}{r|l} 2 & 9,\ 10,\ 15 \\ 3 & 9,\ 5,\ 15 \\ 5 & 3,\ 5,\ 5 \\ & 3,\ 1,\ 1 \end{array}$$

LCM of (9, 10, 15) = $2\times3\times5\times3 = 90$

N = 90K + 7. So N could be 97, 187, 277…
For K = 21, N = 1897 which is close to 1936.
The number to be subtracted = 1936 – 1897 = 39

9. LCM of two numbers should be a multiple of their HCF. But here 140 is not a multiple of 19.
Hence, the given data is inconsistent

10. The maximum number of swords or spears that can be kept in each box such that they are equal in number = HCF of (378, 675) = 27

The minimum number of boxes required = $\frac{378+675}{27} = 39$.

11. 36 can be factorised as (1, 36) or (2, 18) or (3, 12) or (4, 9). For the pairs (1, 36) and (4, 9) the LCM is 36 and the HCF is 1. Hence [1]

12. This problem is of LCM type - I. Let the required number be N.
N – 2 = K × LCM of (20, 25, 35)
LCM of (20, 25, 35) = 700.
N – 2 = K × 700
When K = 14, we get the largest four digit number possible.
When K = 14, N – 2 = 9800
N = 9800 + 2 = 9802. Hence [4]

13. Only one such pair is possible, i.e. 2 and 72. Hence [3]

14. The largest number that can divide $\frac{99}{2}, \frac{135}{4}$ and $\frac{81}{4}$ is their HCF.

HCF of Fractions = $\frac{\text{HCF of numerator}}{\text{LCM of denominator}} = \frac{\text{HCF of (99,135,81)}}{\text{LCM of (2,4,4)}} = \frac{9}{4}$

Number of pieces = $\frac{\frac{99}{2}}{\frac{9}{4}} + \frac{\frac{135}{4}}{\frac{9}{4}} + \frac{\frac{81}{4}}{\frac{9}{4}} = 46$. Hence [1]



Practice Questions B (Optional)

1. In going up a flight of stairs, at a time if I go up 3, 4, 5, 6 or 7 stairs, then there will be 2, 3, 4, 5 or 6 stair(s) left over respectively. If the number of stairs lies between 400 and 450, then how many stairs are there?
1] 431 2] 419 3] 421 4] Indeterminate

2. What is largest of three numbers such that the first multiplied by the second gives 527; the product of the second and third is 713 and that of the first and third is 391.
1] 17 2] 23 3] 31 4] 37

3. The LCM of 68, 170 and another number A is 2380 and the HCF of the three numbers is 17. What could be the smallest value of A?
1] 17 2] 119 3] 51 4] 68

4. Find the largest number which when divides 34100 and 307100 leaves 63 and 93 as remainders respectively.
1] 337 2] 674 3] 1011 4] 1348

5. Find the least number which when divided by 32, 42 and 52 leaves 22, 32 and 42 as remainders respectively.
1] 8476 2] 8726 3] 8746 4] 8276

6. The LCM of two numbers is 244188 and their HCF is 84. If one of the numbers is 1428, find the other.
1] 12654 2] 14114 3] 171 4] 14364

7. Find the greatest number that divides 13956 and 14565 and leaves 7 as the remainder each case.
1] 29 2] 58 3] 87 4] None of these

8. What is the smallest of two four-digit numbers that have 119 as their HCF and 13923 as their LCM?
1] 119 2] 221 3] 1071 4] 1547

Answer Key

1-2 2-3 3-2 4-1 5-2 6-4 7-1 8-3

Chapter 7 Remainder Theorem

Suggested Time : 60 minutes

Meaning of Remainder: In the calculation below, 19 is the dividend, 5 is the divisor, 3 is the quotient and 4 is the remainder. The remainder is that extra portion of the dividend that is left when the dividend is divided by the divisor. Hence, Dividend = Divisor × Quotient + Remainder

$$5\overline{)19}\ 3,\quad 15,\quad 4$$

If the remainder is subtracted from the dividend, then the new dividend will be totally divisible by the divisor. Hence, if 4 is subtracted from 19, the balance is 15. This is totally divisible by 5.

Negative remainder: In the above calculation, 4 was the excess portion in the dividend. To make 19 divisible by 5, we subtracted 4. Instead, if we add 1 to 19, then the new dividend (19 + 1 = 20) would have again become divisible by the divisor. So we can say that the remainder is – 1. In other words, when the remainder is subtracted from the dividend, then the new dividend will be totally divisible by the divisor. Here, when – 1 is subtracted from the dividend, we get 19 – (– 1) = 20. This is now totally divisible by 5.

Note : Observe the following calculation. Is it correct?

$$5\overline{)19}\ 2,\quad 10,\quad 9$$

While there is apparently nothing wrong in the calculation, it is clear that 9 cannot be the Remainder when any number is divided by 5 as Remainder is always less than Divisor. Hence, one more 5 can be extracted from 9, i.e. 9 reduced by 5, and the Quotient increases by 1.

Mod Notation: In mod notation, Dividend ≡ Remainder Mod Divisor

Examples : a) $19 \equiv 4 \bmod 5$
b) $8 \equiv 1 \bmod 7$
c) $17 \equiv 2 \bmod 3$
d) $18 \equiv 2 \bmod 4$

Using the negative remainder, we can also write the above examples as

a) $19 \equiv -1 \bmod 5$
b) $8 \equiv -6 \bmod 7$
c) $17 \equiv -1 \bmod 3$
d) $18 \equiv -2 \bmod 4$

Or if it is given that $19 \equiv 9 \bmod 5$, we can say that

$19 \equiv 9 \bmod 5, \Rightarrow 19 \equiv 4 \bmod 5.$

Similarly $7 \equiv 4 \bmod 3 \Rightarrow 7 \equiv 1 \bmod 3$, and $8 \equiv -6 \bmod 7 \Rightarrow 8 \equiv 1 \bmod 7$

Remainder Theorem :
(1) If $a \equiv b \bmod c$, then
$a + k \equiv (b + k) \bmod c$
$a - k \equiv (b - k) \bmod c$
$a^k \equiv b^k \bmod c$
$ak \equiv bk \bmod c$

Example, we know that $10 \equiv 3 \bmod 7$
$10 + 1 \equiv (3 + 1) \bmod 7$, i.e. $11 \equiv 4 \bmod 7$
$10 - 2 \equiv (3 - 2) \bmod 7$, i.e. $8 \equiv 1 \bmod 7$
$10^2 \equiv 3^2 \bmod 7$, i.e. $100 \equiv 9 \bmod 7$, or $100 \equiv 2 \bmod 7$
$10 \times 2 \equiv 3 \times 2 \bmod 7$, i.e. $20 \equiv 6 \bmod 7$

(2) If $a \equiv b \bmod c$ and $p \equiv q \bmod c$, then
$a + p \equiv (b + q) \bmod c$
$a - p \equiv (b - q) \bmod c$
$ap \equiv bq \bmod c$

It is important to note that both expressions had the same divisor 'c', i.e. both expressions are of the form 'mod c'.

Example, we know that $10 \equiv 3 \bmod 7$ and $8 \equiv 1 \bmod 7$
$10 + 8 \equiv 3 + 1 \bmod 7$, i.e. $18 \equiv 4 \bmod 7$
$10 - 8 \equiv 3 - 1 \bmod 7$, i.e. $2 \equiv 2 \bmod 7$ (i.e. quotient is 0 and remainder is 2)
$10 \times 8 \equiv 3 \times 1 \bmod 7$, i.e. $80 \equiv 3 \bmod 7$



ILLUSTRATION

What is the remainder when 2^{25} is divided by 3?

1] 1 2] 2 3] 0 4] 3

Step 1 : We know that $2 \equiv 2 \bmod 3$ or $2 \equiv -1 \bmod 3$
Step 2 : Using the negative remainder expression, and the concept $a^k = b^k \bmod c$, we get
$2 \equiv -1 \bmod 3$
$2^{25} \equiv (-1)^{25} \bmod 3$, i.e. $2^{25} \equiv (-1) \bmod 3$, i.e. $2^{25} \equiv 2 \bmod 3$
Hence, the remainder when 2^{25} is divided by 3 is 2.

Practice Questions A (Optional)

1. The remainder when 9^{27} is divided by 5 is
   1] 4 2] 3 3] 1 4] None of these

2. What is the remainder you get when you divide 2^{66} by $2^6 - 1$?
   1] 1 2] 62 3] 6 4] 5

3. What is the remainder when $17^{25} + 21^{25}$ is divided by 19?
   1] 0 2] 2 3] 14 4] 15

4. What is the remainder when $112 \times 115 \times 117$ is divided by 11?
   1] 1 2] 2 3] 4 4] 10

Answer Key

1-1 2-1 3-1 4-3

Explanatory Answers

1. $9 \equiv -1 \bmod 5$
$9^{27} \equiv (-1)^{27} \bmod 5$
$\equiv -1 \bmod 5$
$\equiv 4 \bmod 5$
$\therefore$ 9^{27} when divided by 5 leaves a remainder 4.
Hence [1]

2. $2^6 - 1 = 64 - 1 = 63$
$2^6 = 64 \equiv 1 \bmod 63$
$(2^6)^{11} \equiv 1^{11} \bmod 63$
$\therefore 2^{66} \equiv 1 \bmod 63$
Hence, when 2^{66} is divided by 63, the remainder is 1.
Hence [1]

3. $17 \equiv -2 \bmod 19$
$17^{25} \equiv -2^{25} \bmod 19$
$21 \equiv 2 \bmod 19$
$21^{25} \equiv 2^{25} \bmod 19$
Using the concept, "If $a \equiv b \bmod c$ and $p \equiv q \bmod c$, then $a + p \equiv b + q \bmod c$"
$17^{25} + 21^{25} \equiv -2^{25} + 2^{25} \bmod 19$
$17^{25} + 21^{25} \equiv 0 \bmod 19$, i.e. 0 is the remainder. Hence [1]
Alternately,
Using the concept [$a^n + b^n$ is divisible by $a + b$, when n is odd]
$17^{25} + 21^{25}$ is divisible by 17 + 21 i.e. 38. If a number is divisible by 38, it is definitely by 19. Hence, the remainder = 0.

4. $112 \equiv 2 \bmod 11$
$115 \equiv 5 \bmod 11$
$117 \equiv 7 \bmod 11$
Using the concept, "If $a \equiv b \bmod c$ and $p \equiv q \bmod c$, then $a \times p \equiv b \times q \bmod c$
$112 \times 115 \times 117 \equiv 2 \times 5 \times 7 \bmod 11$
$112 \times 115 \times 117 \equiv 70 \bmod 11$
$112 \times 115 \times 117 \equiv 4 \bmod 11$ (extracting 11 six times), i.e. 4 is the remainder.
Hence [3]

Chapter 8 Percentages, Profit & Loss

Suggested Time : 120 minutes

A PERCENTAGE is a fraction, whose denominator is 100. The numerator of the fraction is called the rate percent. It is usually denoted by "%".

Examples

1. $\frac{30}{100}$ is called 30 percent (or 30%). 30 is called the rate percent

2. 5% means $\frac{5}{100}$ (or 5 out of 100)

Note

1. 10% of Rs. 200 means $\frac{10}{100} \times 200$ = Rs. 20

2. $\frac{1}{4}$ is equivalent to 25% (since 25 %, i.e. $\frac{25}{100} = \frac{1}{4}$), i.e. 1 out of 4 is equal to 25%. In order to convert fractions in to percentages, we multiply the fraction with 100, i.e. $\left(\frac{1}{4} \times 100 = 25\ \%\right)$

PERCENTAGE INCREASE OF A QUANTITY

The increase in a quantity experessed as a percentage of original quantity is called as "percentage increase" of that quantity.

$$\text{Percentage increase} = \frac{\text{Increase}}{\text{Original quantity}} \times 100$$

Note

1. If a number A is increased to 2A, then the percentage increase = $\frac{2A-A}{A} \times 100 = \frac{A}{A} \times 100 = 100\%$.

   i.e. when a number (A) becomes 2 times itself (2A), then the percentage increase = 100 %

2. Similarly, if a number becomes 3 times itself, then the percentage increase is 200%. Hence, if a number becomes n times itself, then the percentage increase = (n – 1) × 100%

PERCENTAGE DECREASE OF A QUANTITY

The decrease in a quantity experessed as a percentage of original quantity is called as "percentage decrease" of that quantity.

$$\text{Percentage decrease} = \frac{\text{Decrease}}{\text{Original quantity}} \times 100$$

Note

If a number 2A is decreased to A, then the percentage decrease = $\frac{2A-A}{2A} \times 100 = \frac{A}{2A} \times 100 = 50\%$.

If a number becomes half of itself, then the percentage decrease is 50%. Similarly, if a number becomes one forth of itself, the decrease is of 75% and if a number becomes 35% of itself, the decrease is of 65%.

SOME IMPORTANT RESULTS

1. Percentage Error = $\frac{\text{Estimated value} - \text{Actual value}}{\text{Actual value}} \times 100$

In questions, if the actual and estimated values are given we can find the percentage error by using the above result.

2. If the value of A is x % more than B, then the value of B is $\left[\frac{x}{100+x} \times 100\right]$ % less than A

3. If the value of A is x % less than that of B, then the value of B is $\left[\frac{x}{100-x}\times 100\right]$ % more than A

This result is similar to the result discussed above.

4. If the price of a commodity increases by x % then percentage reduction in consumption, so as not to increase the expenditure is $\left[\frac{x}{100+x}\times 100\right]$ %

5. If the price of a commodity decreases by x %, then the percentage increase in consumption so as not to decrease the expenditure is $\left[\frac{x}{100-x}\times 100\right]$%

6. If there are successive increases of p %, q % and r % in three stages, the effective percentage increase is $\left[\left(\frac{100+p}{100}\right)\left(\frac{100+q}{100}\right)\left(\frac{100+r}{100}\right)-1\right]\times 100$

PROFIT

When an article is sold for more than what it costs (i.e. SP > CP), we say that there is a profit or gain.

$\therefore$ Profit = Selling Price – Cost Price

Profit Percentage = $\frac{\text{Selling Price} - \text{Cost Price}}{\text{Cost Price}}\times 100$ (or) Profit Percentage = $\frac{\text{Profit}}{\text{CP}}\times 100$

LOSS

When an article is sold for less than what it costs (i.e SP < CP), we say there is a loss.

$\therefore$ Loss = Cost Price – Selling Price

Loss Percentage = $\frac{\text{Cost Price} - \text{Selling Price}}{\text{Cost Price}}\times 100$ (or) Loss Percentage = $\frac{\text{Loss}}{\text{CP}}\times 100$

Examples

1. If Cost Price = Rs. 200, Selling Price = Rs. 250, then profit = 250 – 200 =Rs. 50

$\therefore$ Profit Percentage = $\frac{50}{200}\times 100 = 25\%$

2. If Cost Price = Rs. 200, Selling Price = Rs. 180 then Loss = 200 – 180 = Rs. 20

$\therefore$ Loss Percentage = $\frac{20}{200}\times 100 = 10\%$

IMPORTANT TERMS USED IN THE QUESTIONS

1. **GROSS PROFIT (INCOME)**

It is the total profit without deducting any tax or marketing and administration expenses.
i.e. Gross profit = Sales – Cost of manufacturing.

2. **NET PROFIT (INCOME)**

It is the profit after deducting the taxes and expenses.
i.e. Net profit = Sales – All costs expenses

3. **MARKED PRICE OR LIST PRICE (MP)**

Sometimes the trader puts a tag price on the product. This price is called the marked price. However, he may actually sell it at a lower price, which is called as the 'selling price'.

4. TRADE DISCOUNT

Some shopkeepers offer a certain percentage of the marked price (MP) of an article as discount to dispose off the old or damaged goods. The discount so given is called trade discount.

5. SUCCESSIVE DISCOUNTS

In case of successive discounts, the first discount, is calculated on the marked price and is subtracted from it. The second discount is then calculated on the resulting amount and is again subtracted. The amount left after subtracting the last discount is the selling price of the article.

Example

If there are three successive discounts of p %, q % and r % on a product whose marked price is y, then the effective price after the discounts is given by,

Discounted price = $y \times \frac{(100-p)}{100} \times \frac{(100-q)}{100} \times \frac{(100-r)}{100}$

Example

Which is a better discount for you as a customer?

1. 20 % flat on marked price
2. 10 % on marked price and another 5%
3. 15 % on marked price and another 5%
4. 5% on marked price and another 15%

Let the marked price be Rs. 100

Case 1: When the discount is 20%, the selling price = Rs. 80

Case 2: When the discount is 10 %, the selling price of the article = Rs. 90. Another 10 % is given on Rs. 90. Hence, the final selling price = Rs. 81

Case 3: When the discount is 15 %, the selling price of the article = Rs. 85. Another 5 % is given on Rs. 85. Hence, the final selling price = Rs. (85 – 4.25) = Rs. 80.75

Case 4: When the discount is 5%, the selling price = Rs. 95. Another 15% is given on Rs. 95. Hence, the final selling price = Rs. 80.75

The selling price is minimum in case 3 and case 4. Hence, both the cases are better as a customer.

6. NET PRICE

The price after deducting the trade discount is called the net price.

Net Price = Marked Price – Discount

SOME IMPORTANT RESULTS

1. If the Cost Price of ‘x’ items is equal to the Selling Price of ‘y’ items then we have:

(i) Profit if x > y, then Profit Percentage = $\frac{x-y}{y} \times 100$

(ii) Loss if x < y then Loss Percentage = $\frac{y-x}{y} \times 100$

2. When two articles are sold at the same price such that there is a profit of p% on one article and a loss of p% on the other, then irrespective of what the CP is, the net result is a loss which is given by $\frac{p^2}{100}$ %.



ILLUSTRATION

A man sells two tables at the same price. On one he makes a profit of 10% and on the other he suffers a loss of 10%. Find the loss or gain percentage on the whole transaction.

1] 1 % 2] 0.1 % 3] 0 % 4] Indeterminate

Let the Selling Price of each table be Rs. 100.

If Profit = 10% then

$$CP = SP\left(\frac{100}{100+\text{gain}\%}\right) = \frac{100\times100}{110} = \text{Rs. } \frac{1000}{11}$$

If loss =10% then

$$CP = SP\left(\frac{100}{100-\text{loss}\%}\right) = \frac{100\times100}{90} = \text{Rs. } \frac{1000}{9}$$

$$\therefore \text{Total CP} = \frac{1000}{11}+\frac{1000}{9} = \frac{20000}{99}$$

Total SP = Rs. 200

$$\therefore \text{ Loss} = CP - SP = \frac{20000}{99} - 200 = \frac{200}{99}$$

$$\therefore \text{ Loss Percentage } = \frac{200}{99}\times\frac{99}{20000}\times100 = 1\%$$

Alternately,

$$\text{Loss Percentage } = \frac{p^2}{100} = \frac{10^2}{100} = 1\%$$

SUMMARY

1.	Profit = SP – CP	2.	Loss = CP – SP
3.	Profit Percentage = $\frac{\text{Profit}}{CP}\times100$	4.	Loss Percentage = $\frac{\text{Loss}}{CP}\times100$
5.	SP = CP × $\left(\frac{100 + \text{Gain Percentage}}{100}\right)$	6.	SP = CP × $\left(\frac{100 - \text{Loss percentage}}{100}\right)$
7.	CP= SP × $\left(\frac{100}{100+\text{Gain percentage}}\right)$	8.	CP = SP × $\left(\frac{100}{100-\text{Loss percentage}}\right)$

Practice Questions A (Important)

1. A boy spends 40% of his pocket money, and is left with Rs. 6. What is his pocket money?
1] Rs. 15 2] Rs. 10 3] Rs. 17.5 4] Rs. 20

2. A man receives an annual salary of Rs. 36600 in the year 1985. This is 20% more than his salary in 1984. What was his salary in 1984?
1] Rs. 29280 2] Rs. 43920 3] Rs. 30500 4] Rs. 35000

3. In a certain examination there are three papers and it is necessary for a candidate to get 35% of the total marks to pass in the examination. In one paper the candidate scored 65 out of 120, and in the second paper he scored 35 out of 150. How much must he score in the third paper to pass in the examination, if the maximum marks for the third paper is 180?
1] 115 2] 57.5 3] 55 4] 110

4. If A's income is 20% more than B's income, then by what percentage is B's income less than A's income?
1] 20 % 2] 16.66 % 3] 25 % 4] 16.33 %

5. After 40 kg of water was evaporated from a solution of salt water which had 15% of salt, the remaining solution had 20% salt. Find the weight of the original solution.
1] 160 kg 2] 100 kg 3] 120 kg 4] 150 kg

6. The price of a certain commodity is reduced by 20%. By what percent can its consumption be increased so that the expenditure remains unchanged?
1] 20 % 2] 16.66 % 3] 25 % 4] 16.33 %

7. The cost of manufacturing a machine is divided between material, labour and overheads in the ratio 4:5:1. Later on the cost of material rose by 25%, labour by 40% and overhead by 10%. Find the overall percentage increase in the cost of the machine.
1] 31 % 2] 22 % 3] 32 % 4] 27 %

8. The total population of a town is 294000, of which 150000 are males. Out of every 100 males, 98 can read and write; but only 53% of the total population can do so. Find the percentage of women who can read and write.
1] 61.25 % 2] 6.125 % 3] 56.25 % 4] 5.75 %

9. A shopkeeper marks the selling price of an article at 20% above the cost price, but allows 6% discount on the marked price. What price will a customer have to pay for an article that costs Rs. 1000?
1] Rs. 848 2] Rs. 1128 3] Rs. 1218 4] Rs. 978

10. A florist bought 240 roses at Rs. 9 per dozen. If he sold all of them at 100 paise each, find the profit percentage that the florist makes.
1] 33.33 % 2] 25 % 3] 20 % 4] 40 %

11. A man sells a pen at a profit of 20 %. Had he bought it at 20% less and sold it for Rs. 5 less, he would have gained 25%. What is the cost price of the pen?
1] Rs. 25 2] Rs. 30 3] Rs. 40 4] Rs. 20

12. A man loses 10% by selling a book for Rs. 144. What should be the selling price to gain 15 %?
1] Rs. 149 2] Rs. 184 3] Rs. 186 4] Rs. 190

13. A shopkeeper marks the price of an article at Rs. 80. Find the selling price if he allows a discount of 10%.
1] Rs. 72 2] Rs. 68 3] Rs. 76 4] Rs. 74

14. Find the single discount which is equivalent to successive discounts of 20%, 15% and 10%
1] 38. 8 % 2] 61.2 % 3] 30 % 4] 51.8 %

15. Twenty percentage more can be gained, if a piece of cloth is sold for Rs. 83 instead of Rs. 78. Find the cost price of the piece of cloth.
1] Rs. 62.25 2] Rs. 50 3] Rs. 45 4] Rs. 25

16. If the price of wheat is raised by 30% then by what percentage a householder must reduce the consumption of wheat so that there is no increase in the expenditure?
1] $23\frac{12}{13}\%$ 2] 21 % 3] $23\frac{1}{13}\%$ 4] 24 %

17. The length of a rectangle is decreased by 5% and its breadth is increased by 5%. What is the percentage change in the area of the rectangle?
1] 20 % 2] 25 % 3] 2.5 % 4] 0.25 %

18. In an examination, 45% of the total number of candidates is under 15 years of age. Of these 65% were boys and 441 were girls. Find the total number of candidates.
1] 1540 2] 1260 3] 2800 4] 2500

19. If price of sugar falls by 10%, how many quintals can be bought for the same money which was sufficient to buy 18 quintals at the higher price?
1] 22 2] 21 3] 19 4] 20

20. By selling 33 meters of cloth, the merchant gains the cost price of 11 meters. Find the gain percentage.
1] 50 % 2] 33.33% 3] 66.66 % 4] 55 %

Answer Key

1-2	2-3	3-2	4-2	5-1	6-3	7-1	8-2	9-2	10-1
11-1	12-2	13-1	14-1	15-4	16-3	17-4	18-3	19-4	20-2

Explanatory Answers

1. The boy spends 40% of his pocket money. ∴ He has 60% of the whole left.
60% of the whole = Rs. 6
100% of the whole = $\frac{6}{60} \times 100$ = Rs. 10

2. Let the salary in 1984 be Rs. x.
Salary in 1985 = $x + \frac{20}{100}x = \frac{6x}{5}$
$\frac{6x}{5} = 36600$. Hence $x = \frac{36600 \times 5}{6}$ = Rs.30500

3. Let the marks in the third paper be 'x'
Pass mark is 35% of the total
65 + 35 + x = 35% (120 + 150 + 180)
$100 + x = \frac{35}{100} \times 450$
200 + 2x = 315
$2x = 115 \Rightarrow x = \frac{115}{2} = 57.5$.
He should get 57.5 out of 180 in the third paper

4. B's income is less than A by $\left[\left(\frac{x}{100+x}\right)\times 100\right]$%, i.e. $\frac{20}{120}\times 100 = 16.66$ %

Alternately,

Let B's income be Rs. 100. A's income is 20 % more than B's income.
Hence, A's income = Rs. 120
B's income is Rs. 20 less compared to A's income.

B's income is $\frac{20}{120}\times 100 = 16.66\%$ less compared to that of A's income

5. Let the weight of the original solution be x kg.

Weight of salt in the solution = $\frac{15}{100}\times x = \frac{15}{100}x$

After 40 kg of water is evaporated, weight of remaining solution = (x – 40) kg

Weight of the salt in it = $\frac{20}{100}(x-40)$ kg

Since the total weight of salt in both cases are equal, $\frac{15}{100}x = \frac{20}{100}(x-40)$

$3x = 4x - 160$
$x = 160$ kg.

Weight of the original solution = 160 kg

6. Percentage increase in consumption = $\left[\frac{x}{100-x}\right]\times 100\ \% = \frac{20}{100-20}\times 100 = \frac{20}{80}\times 100 = 25\%$

Alternately,

Let the price of the sugar be Rs. 10 per kg and the consumption of the sugar be 10 kg
Total expenditure = 10 × 10 = Rs. 100
Decreased price of the sugar = Rs. (10 – 2) = Rs. 8

The increased consumption of the sugar = $\frac{100}{8} = 12.5$ kg.

The percentage increase in the consumption of the sugar = $\frac{12.5-10}{10}\times 100 = 25\%$

7. Let the total cost of the machine be Rs. x
Ratio of material, labour and overheads costs is 4:5:1

Material cost = $\frac{4}{10}x$

Labour cost = $\frac{5}{10}x$

Overheads cost = $\frac{1}{10}x$.

Increase in overall cost = $\left[1.25\left(\frac{4}{10}x\right)+1.4\left(\frac{5}{10}x\right)+1.1\left(\frac{1}{10}x\right)\right]$ = 0.5x + 0.7x + 0.11x = 1.31x

Percentage increase = $\frac{\text{Increase}}{\text{Original cost}}\times 100 = \frac{1.31x-x}{x}\times 100 = 31\%$

8. Total Population = 2, 94, 000
Number of Men = 150,000

Number of men who can read and write = $1,50,000 \times \frac{98}{100} = 1,47,000$

Number of people who can read and write = $\frac{53}{100} \times 294000 = 155,820$

$\therefore$ Number of women who can read and write = 155820 – 147000 = 8820

$\therefore$ Percentage of women who can read and write $= \frac{8820}{294000-150000} \times 100 = 6.125\%$

9. Marked price of the article = $1000 \times \frac{100+20}{100} = 1000 \times \frac{120}{100} = \text{Rs. } 1200$

Price after discount = $1200 \times \frac{100-6}{100} = 12 \times 94 = \text{Rs. } 1128$

$\therefore$ The customer will have to pay Rs. 1128.

10. Cost price = $\frac{240}{12} \times 9 = \text{Rs. } 180$

Selling price = 240×1=Rs. 240
Profit = Selling Price – Cost Price =240 – 180 =Rs. 60

$\therefore$ Profit percentage = $\frac{60}{180} \times 100 = 33\frac{1}{3}\%$

Alternately,
Cost of 12 roses = Rs. 9
Cost of 1 rose = 75 paise

$\therefore$ Profit percentage = $\frac{100-75}{75} \times 100 = 33.33\%$

Note: Profit percentage remains the same when selling price and cost price are fixed, irrespective of the number of units sold.

11. Let the Cost Price = Rs x.
Profit = 20 %

$\therefore$ SP = CP $\left(\frac{100+\text{gain}\%}{100}\right) = x\left(\frac{120}{100}\right) = \frac{12x}{10}$

If CP = 0.8x and SP = $\frac{12x}{10} - 5$

Then gain Percentage = 25, i.e $25 = \frac{\frac{12x}{10} - 5 - \frac{8x}{10}}{\frac{8}{10}x} \times 100$

$25 = \left(\frac{\frac{4x}{10} - 5}{\frac{8}{10}x}\right) \times 100$, i.e. $1 = \frac{4x-50}{10} \times \frac{10}{8x} \times 4$, i.e. 2x =4x – 50,

i.e. 2x = 50 => x = Rs. 25

$\therefore$ Cost Price of the pen = Rs. 25

Alternately,

Case 1

Let the cost price = Rs. 100 (our assumption)

If there is a profit of 20 %, then the selling price = Rs. 120.

Case 2

Had he bought it 20 % less, and then the cost price would have been Rs. 80.
If he has to gain 25%, then the selling price at which he has to sell is = 80 + 25 % of 80 = Rs. 100

Compare the selling price in the first and the second case. The selling price in the second case is Rs. 20 less than the selling price in the first case. But in the question it is mentioned that "if he sells Rs. 5 less he would have gained 25 %".
When it is Rs. 20, the cost price = Rs. 100. (our assumption)
When it is Rs. 5 the cost price = $\frac{5\times100}{20}$ = Rs. 25.

12. Selling Price = Rs. 144, loss Percentage = 10%
CP = $\frac{100\times SP}{100-loss}=\frac{100\times144}{90}$ = Rs. 160
$\therefore$ SP = CP $\left(\frac{100+gain\%}{100}\right)$ = 160$\left(\frac{115}{100}\right)$ = Rs. 184
$\therefore$ Selling Price = Rs. 184

13. Marked Price = 80
Discount = $\frac{10}{100}\times80$ = Rs. 8
$\therefore$ Selling Price = 80 – 8 = Rs. 72

14. Let Marked Price = Rs 100
1st discount = 100 × $\frac{20}{100}$ = Rs. 20
2nd discount = (100–20) × $\frac{15}{100}$ = 80 × $\frac{15}{100}$ = Rs. 12
3rd discount = (80 – 12) × $\frac{10}{100}=\frac{68}{10}$ = Rs. 6.8
$\therefore$ Single discount = Sum of all discounts = 20 + 12 + 6.8 = 38.8 %

Alternately,

Discounted price = $100\times\frac{(100-20)}{100}\times\frac{(100-15)}{100}\times\frac{(100-10)}{100}$ = 100 × 0.80 × 0.85 × 0.90 = 61.2
Discount = 100 – 61.2 = 38.8
Percentage discount = 38.8 %

15. Let Cost Price = x , Selling Price = 78 , gain = y %
Cost Price = x , Selling Price = 83 , gain = (y + 20) %
$\left(\frac{78-x}{x}\right)100 = y$ (1)
$\left(\frac{83-x}{x}\right)100 = y+20$ (2)
From, equation (2) – equation (1), we have
20x = 500
x = 500/20 = Rs. 25. Hence [4]

16. Percentage reduction in consumption = $\frac{x}{100+x}\times100=\frac{30}{130}\times100=23\frac{1}{13}$ % . Hence [3]

17. Let the sides of the rectangle be ℓ_1 and b_1
Initial area of the rectangle, $A_1=\ell_1 b_1$
Final area of the rectangle, A_2= (0.95 ℓ_1) (1.05) b_1 = 0.9975 $\ell_1 b_1$
Change in the area = $\ell_1 b_1$ – 0.9975 $\ell_1 b_1$ = 0.0025 $\ell_1 b_1$
Percentage change = $\frac{0.0025\, l_1 b_1}{l_1 b_1}\times100$ = 0.25% . Hence [4]

Alternately,

Percentage change = $\frac{P^2}{100} = \frac{5^2}{100} = 0.25\%$

18. Let the number of candidates below 15 years be x
Percentage of boys = 65%. Hence, percentage of girls =100 – 65 = 35%

$\frac{35}{100}$ x = 441

x = 441 × $\frac{100}{35}$ = 1260.

The number of candidates under 15 years = 1260
Let the total number of candidates = Y
45% of the total number of candidates are under 15 years.

$\frac{45}{100}Y = 1260$. Hence, $Y = \frac{1260 \times 100}{45} = 2800$

∴ Total number of candidates = 2800. Hence [3]

19. Percentage increase in consumption $= \left(\frac{x}{100 - x}\right) \times 100 = \frac{10}{90} \times 100 = 11\frac{1}{9}\%$

∴ Quantity of sugar that can be bought for new price= $18 + \left[\left(\frac{100/9}{100}\right) \times 18\right]$ = 18 + 2 = 20 quintals.

Hence [4]

20. Let Cost Price of 1 metre = Rs. x
CP of 33 metres = Rs. 33x
Gain = Rs. 11x
Gain Percentage = $\frac{11x}{33x} \times 100 = 33\frac{1}{3}\%$. Hence [2]

Practice Questions B (Optional)

1. A man's income before tax deduction is Rs. 5000. What is the income after tax deduction, if he pays an income tax of 5 paise in one rupee on the whole?
1] Rs. 4250 2] Rs. 4750 3] Rs. 5250 4] Rs. 5750

2. The lead ore from a certain mine yields 60% of metal, and of the metal $\frac{3}{4}\%$ is silver. The rest of the metal is lead. How much more lead is there in 8000 kg of ore compared to that of silver?
1] 6000 kg 2] 32 kg 3] 4728 kg 4] 4732 kg

3. The dimensions of a photograph are 5 inch by 7 inch. What are its dimensions if each is reduced by 30%?
1] 1.5 in. × 2.1 in. 2] 1.1 in. × 2. 5 in. 3] 3.5 in. × 4.9 in. 4] 3.9 in. × 4.5 in.

4. A certain number of eggs are bought at 4 for Rs. 5 and an equal number are bought at 9 for Rs. 10. If 15% of the eggs were broken in transportation and the remaining were sold at 2 for Rs. 3, find the profit percentage for the trader.
1] 9 % 2] 8 % 3] 8.5 % 4] 8.25 %

5. If apples are bought at 11 for Rs. 10 and sold at 10 for Rs. 11, find the profit percentage.
   1] 19 % 2] 20 % 3] 22 % 4] 21 %

6. The production of rice is doubled from 1985 to 1990. From 1990 to 1995, there was a 50% increase. What is the percentage increase in production of rice from 1985 to 1995?
   1] 150 % 2] 200 % 3] 300 % 4] 250 %

7. If x% of 350 is 21, find x.
   1] 6 2] 0.6 3] 0.06 4] 60

8. A's salary is 20% below B's salary. By how much percent is B's salary above A?
   1] $16\frac{1}{3}$ % 2] 25 % 3] 20 % 4] $16\frac{2}{3}$ %

Answer Key

1-2 2-3 3-3 4-2 5-4 6-2 7-1 8-2

CAREER AVENUES

Chapter 9 Ratios

Suggested Time : 120 minutes

The best example of a ratio is a map. The map of India drawn on a small paper actually correctly represents the distances between cities and towns. The distance between Kanyakumari and Srinagar can be found using the map, though nowhere in the map is that information directly given. What is given is a scale conversion. That dear reader, is the power of ratios.

RATIO

Comparison of two quantities of the same unit is called as ratio.
For example, let us say salary of A is Rs. 5000 and salary of B is Rs. 3000.
If we are to compare salary of A and B, we can say that

The ratio of salary of A to that of B = $\frac{\text{Salary of A}}{\text{Salary of B}} = \frac{5000}{3000} = \frac{5}{3}$

The ratio of A to B is usually written as A : B. The quantities A and B are called the terms of the ratio.
Here we have, (Salary of A) : (Salary of B) = 5 : 3

Ratio of their (A and B) salaries is NOT written as 5000 : 3000. Because, ratio is always expressed in the most reduced form. In the ratio 5000 : 3000, the terms of the ratio is divided with the common term, i.e. 1000 and written as 5 : 3.

In the map example, the "scale" given on the map is nothing but the ratio of distances on the map to distances on the ground.

Note:

1. If ratio of A to B is 5 : 3, then we CANNOT say that the value of A is 5 and the value of B is 3.
   Referring to the example above we see that salary of A is not Rs. 5 but it is Rs. 5000

2. If the ratio of A to B is 5 : 3, then we can say that the value of A is 5x and the value of B is 3x
   (Assuming that 'x' is the common factor with which the terms of the ratio is divided.)
   Referring to the above example we see that the value of 'x' is 1000.
   Salary of A = 5x = 5 × 1000 = Rs. 5000 and the salary of B = 3x = 3 × 1000 = Rs. 3000.

3. If the total salary of A and B is 8000 and the ratio of their salaries = 5 : 3, then we can find the salary of A and B, i.e. the total salary Rs. 8000 is divided into 8 parts. Out of 8 parts 5 parts is A and 3 parts is B.

Hence $\frac{5}{8}^{th}$ of 8000, i.e. $\frac{5}{8} \times 8000 = 5000$ is the salary of A and $\frac{3}{8}^{th}$ of 8000, i.e. $\frac{3}{8} \times 8000 = 3000$ is the salary of A.

	PROPERTIES OF RATIOS	**EXAMPLES**
1	The value of a ratio remains unaltered, if the first element (antecedent) and the second element (consequent) are multiplied or divided by the same quantity.	Consider the ratio a : b. (i) $\frac{a}{b} = \frac{ma}{mb}$ and (ii) $\frac{a}{b} = \frac{a/m}{b/m}$ where $m \neq 0$
2	Two or more ratios can be compared by reducing their equivalent fractions to a common denominator.	If $\frac{a}{b}$ and $\frac{x}{y}$ are the two ratios to be compared, then we have $\frac{a}{b} = \frac{ay}{by}$ and $\frac{x}{y} = \frac{bx}{by}$. Now after reducing to the common denominator we can say that

		If ay > bx, then $\frac{a}{b} > \frac{x}{y}$ and If ay < bx, then $\frac{a}{b} < \frac{x}{y}$
3	The ratio of two fractions can be expressed as a ratio of two integers.	$\frac{a}{b} : \frac{c}{d}$ then this ratio is equal to ad : bc $\frac{30}{40} : \frac{5}{7}$ = (30 × 7) : (5 × 40) = 210 : 200 = 21 : 20

PROPORTION

When two ratios are equal then the four quantities comprising them are said to be proportional. In other words, the equality of two ratios is called as proportion.

Example, if $\frac{a}{b} = \frac{c}{d}$, then a, b, c and d are in proportion.

a, b, c and d are respectively called the first, the second, the third and the fourth proportional. a and d are called the EXTREMES and b and c are called MEANS.

Note: If $\frac{a}{b} = \frac{c}{d}$, then ad = bc, i.e. when four quantities or numbers are in proportion, then the product of extremes is equal to the product of the means.

	PROPERTIES OF PROPORTION	**EXAMPLES**
1	**INVERTENDO:** If $\frac{a}{b}=\frac{c}{d}$, then $\frac{b}{a}=\frac{d}{c}$, i.e. the inverse ratio of two equal ratios are equal. This property is called INVERTENDO.	If $\frac{5}{4}=\frac{10}{8}$, then $\frac{4}{5}=\frac{8}{10}$.
2	**ALTERNENDO:** If $\frac{a}{b}=\frac{c}{d}$, then $\frac{a}{c}=\frac{b}{d}$, i.e. the ratio of antecedents and consequents of two equal ratios are equal. This property is called ALTERNENDO.	If $\frac{3}{2}=\frac{15}{10}$, then $\frac{3}{15}=\frac{2}{10}$.
3	**COMPONENDO:** If $\frac{a}{b}=\frac{c}{d}$, then $\frac{a+b}{b}=\frac{c+d}{d}$, i.e. adding 1 to both sides. This property is called COMPONENDO.	If $\frac{7}{4}=\frac{21}{12}$, then $\frac{7+4}{4}=\frac{21+12}{12}$, i.e. $\frac{11}{4}=\frac{33}{12}$
4	**DIVIDENDO:** If $\frac{a}{b}=\frac{c}{d}$, then $\frac{a-b}{b}=\frac{c-d}{d}$, i.e. subtracting 1 from both sides. This property is called DIVIDENDO.	If $\frac{5}{4}=\frac{15}{12}$, then $\frac{5-4}{4}=\frac{15-12}{12}$, i.e. $\frac{1}{4}=\frac{3}{12}$

5	**COMPONENDO – DIVIDENDO:** If $\frac{a}{b}=\frac{c}{d}$, then $\frac{a+b}{a-b}=\frac{c+d}{c-d}$, i.e. dividing the results of componendo by dividendo, we get COMPONENDO – DIVIDENDO.	If $\frac{5}{4}=\frac{15}{12}$, then $\frac{5+4}{5-4}=\frac{15+12}{15-12}$, i.e. $\frac{9}{1}=\frac{27}{3}$

VARIATION

Two quantities A and B maybe such that as one quantity changes in value, the other quantity also changes in value bearing certain relationship to the change in value of the first quantity. The three kinds of variations are (i) direct variation, (ii) inverse variation and (iii) joint variation

DIRECT VARIATION

If two quantities A and B are said to vary directly with each other, then an increase in quantity 'A' causes a proportional increase in the quantity 'B'. Also a decrease in quantity 'A' causes a proportional decrease in quantity 'B'.

This is denoted as A α B (A varies directly as B),

i.e. A = KB, where K is a constant called the constant of proportionality, i.e. $\frac{A}{B}$ = K,

Conversely," if the ratio of two quantities is a constant, then they vary directly with each other".

Usage:

1. If we say that with more hand work, chances of success increase, then we rephrase it as "Success is directly proportional to hand work"

2. If Ramesh travels for 5 hours, and increase his normal speed of driving, then he will be able to cover more than the normal distance. We can say that "Distance travelled is directly proportional to speed".

INVERSE VARIATION

If two quantities A and B are said to vary inversely with each other, then an increase in quantity 'A' causes a proportional decrease in quantity 'B'. Also a decrease in quantity 'A' causes a proportional increase in quantity 'B'.

This is denoted as A α $\frac{1}{B}$ (A varies inversely as B),

i.e. A = $\frac{K}{B}$, where K is a constant called the constant of proportionality,

i.e. AB = K, i.e. the product of two quantities is a constant.

Conversely," if the product of two quantities is a constant, then they vary inversely with each other."

Usage:

When petrol price increases, then you end up buying lesser petrol. Hence, we can say that "consumption of the petrol is inversely proportional to its price".

JOINT VARIATION

If there are three quantities A, B and C such that A varies with B when C is constant and varies with C when B is constant, then A is said to vary jointly with B and C when both B and C are varying,

i.e. A α B, when C is constant;

A α C, when B is constant $\Rightarrow$ A α BC; i.e. A= KBC, where K is the constant of proportionality

Practice Questions A (Important)

1. Arrange the following ratios in descending order of magnitude.
(6 : 7), (7 : 8), (9 : 10), (11 : 12)
1] (11 : 12), (9 : 10), (7 : 8), (6 : 7) 2] (6 : 7), (9 : 10), (7 : 8), (11 : 12)
3] (11 : 12), (7 : 8), (9 : 10), (6 : 7) 4] (11 : 12), (9 : 10), (6 : 7), (7 : 8)

2. Marks of 5 students are in the ratio 7 : 8 : 9 : 10 : 11, and their average is 90. How many of the students scored more than 75?
1] 2 2] 3 3] 4 4] 5

3. If $\frac{a}{b}=\frac{7}{8}$ then find the ratio $\left(4a-\frac{b}{2}\right):\left(4a+\frac{b}{2}\right)$.
1] 3 : 4 2] 4 : 3 3] 24 : 33 4] 13 : 32

4. If $\frac{a}{b}=\frac{7}{9}$, then find $\frac{3a+2b}{3a-2b}$.
1] 39 : 4 2] 13 : 1 3] 1 : 3 4] 32 : 3

5. The salaries of A and B are in the ratio 7 : 4 and the sum of their salaries is 121000. Find the difference in their salaries.
1] Rs. 33000 2] Rs. 77000 3] Rs. 44000 4] Rs. 32000

6. Rs. 525 is divided between A, B and C such that A gets double of what B get and B gets double of C. How much more does A get than C?
1] Rs. 300 2] Rs. 150 3] Rs. 75 4] Rs. 225

7. Rs. 1840 is divided into four parts such that a third of the first part, a fifth of the second part, an eight of the third part and a seventh of the fourth part are all equal. Find the difference between the second and the fourth part.
1] Rs. 240 2] Rs. 160 3] Rs. 140 4] Rs. 180

8. Two numbers are in the ratio 4 : 5. If 8 is added to the numbers they become in the ratio 6 : 7. Find the difference between the numbers.
1] 4 2] 16 3] 20 4] 8

9. Find the value of x in 1.7 : 5.1 = 1.9 : x.
1] 9.69 2] 5.7 3] 7.5 4] 8.67

10. It is given that x varies inversely as $y^2 - 1$. When the value of x is 24, the value of y = 10. What is the value of x when the value of y is 5?
1] 97 2] 99 3] 24 4] 198

11. A jar is filled with water and alcohol in the ratio 2 : 1, when 6 litres of the mixture is removed and replaced with 6 litres of water, the ratio of the water to alcohol becomes 4 : 1. What is capacity of the jar?
1] 10 litres 2] 15 litres 3] 18 litres 4] None of these

12. The cost of precious stone is directly proportional to the cube of its weight. A 10 gm stone cost Rs. 25000. The stone falls and breaks in to three pieces, whose weights are in the ratio 1: 1: 3. What is the difference in cost between the largest of the three pieces and the smallest?
1] Rs. 700 2] Rs. 5200 3] Rs. 650 4] Rs. 3200

13. If A : B = 3 : 5 and B : C = 4 : 7, find A : C.
1] 35 : 12 2] 12 : 35 3] 12 : 7 4] 4 : 7

14. If a : b = 4 : 5 and b : c = 6 : 7 find a : b : c.
1] 24 : 35 : 30 2] 30 : 24 : 35 3] 24 : 30 : 35 4] 4 : 30 : 7

15. If x varies directly with y, does $x^2 + y^2$ vary directly with $x^2 – y^2$?
1] Yes 2] No
3] Depends on the values of x and y 4] Cannot be determined

16. The volume of a pyramid varies jointly as its height and the area of its base. When the area of the base is 60 ft^2 and the height 14 ft, the volume is 280 ft^3. What is the area of the base of the pyramid whose volume is 390 ft^3 and whose height is 26 ft?
1] 45 sq. ft 2] 40 sq. ft 3] 36 sq. ft 4] 48 sq. ft

17. A cat pursues a rat and takes 4 leaps for every 5 leaps of the rat; but in terms of distance 3 leaps of the cat are equal to 4 leaps of the rat. Compare the distance traveled by the cat and the rat in a specified time.
1] 3 : 5 2] 16 : 15 3] 15 : 16 4] 5 : 3

18. The pressure of a certain mass of a gas varies inversely as its volume. When the pressure is 20 kg-wt/dm^2 the volume is 450 dm^3. Find the pressure (in kg-wt/dm^2) when volume is 300 dm^3 and also find the volume (in dm^3) when the pressure is 60 kg- wt/dm^2.
1] 30 & 150 2] 60 & 150 3] 20 & 150 4] 150 & 30

Answer Key

1-1	2-3	3-1	4-2	5-1	6-4	7-2	8-1	9-2	10-2
11-2	12-2	13-2	14-3	15-1	16-1	17-2	18-1		

Explanatory Answers

1. $\frac{6}{7}, \frac{7}{8}, \frac{9}{10}, \frac{11}{12}$

Applying the property of ratio of lesser inequality, we have

In the ratio $\frac{a}{b}$, if a < b, then $\frac{a+x}{b+x} > \frac{a}{b}$ (x > 0)

The given ratios could be arranged in the following way :

$\frac{6}{7}, \frac{6+1}{7+1}, \frac{6+3}{7+3}, \frac{6+5}{7+5}$

$\frac{6}{7} < \frac{7}{8} < \frac{9}{10} < \frac{11}{12}$

2. Since the ratio of the marks of the students is 7 : 8 : 9 : 10 : 11, let the marks of the students be 7x, 8x, 9x, 10x and 11x.

Average of their marks = $\frac{7x+8x+9x+10x+11x}{5} = \frac{45x}{5}$ = 9x

It is given that, 9x = 90

$\therefore$ x = 10

The marks of the students are 70, 80, 90, 100 and 110. Hence, 4 students scored more than 75 marks.

3. $\frac{a}{b}=\frac{7}{8}\Rightarrow a=\frac{7b}{8}$

$\therefore \frac{4a-\frac{b}{2}}{4a+\frac{b}{2}}=\frac{4\left(\frac{7b}{8}\right)-\frac{b}{2}}{4\left(\frac{7b}{8}\right)+\frac{b}{2}}=\frac{\frac{7b}{2}-\frac{b}{2}}{\frac{7b}{2}+\frac{b}{2}}=\frac{6b}{8b}=\frac{3}{4}$

Alternately,

Let a = 7x and b = 8x, then $\frac{\left(4a-\frac{b}{2}\right)}{\left(4a+\frac{b}{2}\right)}=\frac{4\times 7x-\frac{8x}{2}}{4\times 7x+\frac{8x}{2}}=\frac{28x-4x}{28x+4x}=24:32 \Rightarrow 3:4$

4. To get 3 and 2 as coefficients of a and b respectively, we multiply $\frac{3}{2}$ on both sides of the ratio

$\therefore \frac{a}{b}\times\frac{3}{2}=\frac{7}{9}\times\frac{3}{2}\Rightarrow\frac{3a}{2b}=\frac{21}{18}$

$\frac{3a+2b}{3a-2b}=\frac{21+18}{21-18}=\frac{39}{3}=13$ [by Componendo and Dividendo]

Alternately,

Let a = 7x and b = 9x, then $\frac{3a+2b}{3a-2b}=\frac{3\times 7x+2\times 9x}{3\times 7x-2\times 9x}=\frac{21x+18x}{21x-18x}=\frac{39x}{3x}=13.$

5. Since the ratio is 7 : 4, A's salary will be $\frac{7}{11}^{th}$ part of their total salary and B's salary will be $\frac{4}{11}^{th}$ part of their total salary.

A's salary $\frac{7}{11}\times 121000$ = Rs. 77000

B's salary $\frac{4}{11}\times 121000$ = Rs. 44000

Difference in their salary = 77000 – 44000 = 33000

Alternately,

Let salary of A = 7x and salary of B = 4x (Assuming that 'x' is the common factor with which the terms of the ratios are divided)

We have, 7x + 4x = 11x = 121000

$x=\frac{12000}{11}=11000$

Difference in their salary = 7x – 4x = 3x =3 × 11000 = 33000

6. Let C get Rs. x, B gets Rs. 2x and A gets Rs. 4x

∴ A : B : C = 4x : 2x : x

Sum of Ratios = 4x + 2x + x = 7x

7x = 525 . ∴ x = 75

A gets (4x – x = 3x), more than C.

3x = 3 × 75 = Rs. 225.

7. Let a, b, c, d be the four parts.

∴ a + b + c + d = 1840 (1)

Also $\frac{a}{3}=\frac{b}{5}=\frac{c}{8}=\frac{d}{7}=k$, i.e. a = 3k, b = 5k, c = 8k, d = 7k

Substituting the above in equation (1) we get,

3k + 5k + 8k + 7k = 1840, i.e. 23k = 1840 ⇒ k = 80

The difference between the second and the fourth part is = 7k – 5k = 2k = 2 × 80 = Rs. 160

8. Let the numbers be x and y.

Since $\frac{x}{y} = \frac{4}{5}$, let x = 4a and y = 5a

$\frac{4a+8}{5a+8} = \frac{6}{7}$

30a + 48 = 28a + 56

2a = 8

a = 4

The difference between x and y = 5a – 4a = a = 4.

9. We know that, product of extremes = product of means

i.e. $1.7 \times x = 5.1 \times 1.9$

$\therefore x = \frac{5.1 \times 1.9}{1.7} = 5.7$

10. $x \propto \frac{1}{y^2 - 1}$, and $x = \frac{K}{y^2 - 1}$, where K is the constant of proportionality.

Substituting the values of x and y in the above equation, (x = 24, y = 10), we have $24 = \frac{K}{100 - 1}$.

K = 24 × 99 and $x = \frac{24 \times 99}{y^2 - 1}$ (1)

To find the value of x, substitute y = 5 in the equation (1)

$x = \frac{24 \times 99}{(25 - 1)} = \frac{24 \times 99}{24} = 99.$

11. Since the ratio of the water to alcohol is 2 : 1, let the amount of water and alcohol in the jar be 2x and x liters.

From the mixture, 6 litres of the mixture is replaced with water.

In this 6 litres, the amount of alcohol = $\frac{1}{3} \times 6$ = 2 litres.

The amount of water = $\frac{2}{3} \times 6$ = 4 litres.

It is given that, $\frac{2x - 4 + 6}{x - 2} = \frac{4}{1}$

$\frac{2x + 2}{x - 2} = \frac{4}{1}$

2x + 2 = 4x – 8

2x = 10 → x = 5

The capacity of the jar = 2x + x = 3x = 3 × 5 = 15 litres.

12. It is given that the cost of the stone is directly proportional to cube of its weight.

Cost $\propto W^3$

Cost = $K \times W^3$, where K is the constant of proportionality.

$25000 = K \times 10^3$

K = $\frac{25000}{1000}$ = 25.

Since the weights of the stones are in the ratio, 1 : 1: 3, weight of the largest stone = $\frac{3}{5} \times 10 = 6$gm and weight of the smallest stone = $\frac{1}{5} \times 10 = 2$ gm.

The difference in cost of the stones = $K \times 6^3 - K \times 2^3 = 216K - 8K = 208K = 208 \times 25 = $ Rs. 5200

13. It is given that $\frac{A}{B} = \frac{3}{5}$ and $\frac{B}{C} = \frac{4}{7}$.

$\frac{A}{B} \times \frac{B}{C} = \frac{3}{5} \times \frac{4}{7} = \frac{12}{35}$

$\therefore$ A : C = 12 : 35

14. Here b is present in both the ratios. The numbers can be written as

a : b : c

4 : 5

6 : 7

We know that if both terms of a ratio are multiplied by the same number, the ratio remains the same. So, multiply, each ratio by a number such that b (the common term in both the ratios) acquires the same value.

	a : b : c
$4:5$ (multiplying both terms by 6)	$24:30$
$6:7$ (multiplying boht terms by 5)	$30:35$

$\therefore a:b:c = 24:30:35$

15. x is directly proportional to y, i.e. x = ky, i.e. $\frac{x}{y} = k$ (constant)

Substituting the value of x in $\frac{x^2+y^2}{x^2-y^2}$ we get $\frac{k^2y^2+y^2}{k^2y^2-y^2} = \frac{(k^2+1)y^2}{(k^2-1)y^2} = \frac{k^2+1}{k^2-1} =$ constant

$\therefore$ $x^2 + y^2$ varies directly with $x^2 - y^2$.

16. V α Ah, where V is the volume of pyramid, A is the area of base and h is the height of the pyramid.

$\therefore$ V = K $\times$ A $\times$ h, where K is the constant of proportionality

It is given that, when A = 60 ft^2, h = 14 ft, V = 280 ft^3.

Substituting the values in the above equation, we have $K \times 60 \times 14 = 280$.

$K = \frac{280}{60 \times 14} = \frac{1}{3}$. $\therefore$ The volume of pyramid, V = $\frac{Ah}{3}$.

To find the value of A, substitute V = 390 ft^3 and h = 26 ft,

$390 = \frac{A \times 26}{3} \Rightarrow A = \frac{390 \times 3}{26} = 45 ft^2$

17. Let 3 leaps of cat = 4 leaps of rat = k

One leap of cat $= \frac{k}{3}$ and one leap of rat $= \frac{k}{4}$

Cat takes 4 leaps for every 5 leaps of rat.

$\therefore \frac{\text{Distance traveled by cat}}{\text{Distance traveled by rat}} = \frac{4 \times \frac{k}{3}}{5 \times \frac{k}{4}} = \frac{\frac{4}{3}}{\frac{5}{4}} = \frac{4}{3} \times \frac{4}{5} = \frac{16}{15}$. Hence [2]

18. $P\alpha\frac{1}{V}$ where P = pressure; V = volume

$\therefore P=\frac{K}{V}$, where K = constant of proportionality. $\therefore PV = K$ (1)

When P = 20 kg wt/dm^2, V = 450 dm^3 we have

$20 \times 450 = 9000$ kg wt dm = K. $\therefore$ PV = 9000 (2)

$\text{when} V = 300\,\text{dm}^3,\ P=\frac{9000}{300}=30\ \text{kg}-\text{wt}/\text{dm}^3$

$P = 60\ \text{kg wt}/\text{dm}^2, V=\frac{9000}{60}=150\ \text{dm}^3$. Hence [1]

Practice Questions B (Optional)

1. What number must be added to each term of the ratio 37 : 29 so that it becomes 8 : 7.
   1] 8 2] 26 3] 27 4] 37

2. Two numbers are in the ratio of 8 : 5 and the difference of their squares is 156. Find the difference between the two numbers.
   1] 39 2] 3 3] 12 4] 6

3. 1250 is divided into four parts such that 4 times the first part, 3 times the second part, twice the third part are all equal to the fourth part. Find the difference between the 2nd and 3rd part.
   1] 100 2] 400 3] 150 4] 50

4. The present ages of Vishal and Varun are in the ratio 3 : 5. If the ratio of their ages 5 years ago was 8 : 15, then find the sum of their present ages.
   1] 46 2] 56 3] 48 4] 30

5. In a mixture of 60 litres, the ratio of milk to water is 5 : 1.How many litres of water should be added to the mixture to make the ratio of milk to water as 2 : 1?
   1] 12 2] 15 3] 10 4] 25

6. 10 men and 6 boys do a piece of work in 20 days while 12 men and 4 boys can do this work in 18 days. How many days will 8 men and 8 boys take to do the work?
   1] 35 2] 45 3] 22.5 4] 27.5

7. A garrison of 3300 men has provisions for 32 days when each eats 850 grams of ration per day. At the end of 7 days, reinforcement arrive and it is found that now the provisions will last 8 days less, even if each were to eat 825 grams per day. How many men comprised the reinforcement team?
   1] 1325 2] 1700 3] 1625 4] 5000

8. In class room, there are some boys and girls. The teacher brings some chocolates to distribute to the students. If each student receives a chocolate, then one boy will have no chocolate. If two boys share one chocolate and the girls get one chocolate each, then the teacher is left with three chocolates. If the ratio of the boys and girls are 4 : 1, then how many chocolates did the teacher bring?
   1] 8 2] 7 3] 6 4] 9

Answer Key

1-3 2-4 3-1 4-2 5-2 6-3 7-2 8-4

Chapter 10 Averages and Mixtures

Suggested Time : 120 minutes

Average is a representative value of a group of data. We use the word 'average' in our daily life either directly or indirectly. Though the word 'average' is used with different interpretation in daily life, in mathematics 'average' means Arithmetic Mean.

ARITHMETIC MEAN

Arithmetic Mean (AM) is the generic usage of the word average. AM is used when all values are given equal weightage and it is the mathematically central value.

For example, Average (A.M) of 1, 2, 3 and 4 is $\frac{1+2+3+4}{4} = \frac{10}{4} = 2.5$

Average of a group of entire data is defined as, $\bar{x} = \frac{\text{Sum of all data in the group}}{\text{Number of data in the group}}$

Sum of all data in the group means sum of all items in the group.

Hence, the average of 'n' quantities $X_1, X_2 \ldots X_n$, Average, $\bar{x} = \frac{X_1 + X_2 + X_3 + X_4 + \ldots + X_n}{n}$

POINTS TO REMEMBER

	POINTS	EXAMPLES
1	If the value of each item in a group of data is increased by the same value 'x', then the average of the group or items will also increase by 'x'	Average of 1, 2, 3, 4, 5 is found as Average = $\frac{1+2+3+4+5}{5} = \frac{15}{5} = 3$ Let us increase each number by 1. The new data is: 2, 3, 4, 5 and 6. The new Average = $\frac{2+3+4+5+6}{5} = \frac{20}{5} = 4$ **New Average = Old Average + 1**
2	If the value of each item in a group of data is decreased by the same value 'x', then the average of the group or items will also decrease by 'x'	Considering the previous example, Initial average = 3. Let us decrease each number by 1. The new data is: 0, 1, 2, 3 and 4. The new Average = $\frac{0+1+2+3+4}{5} = \frac{10}{5} = 2$ **New Average = Old Average – 1**
3	If the value of each item in a group of data is multiplied by the same value 'x', then the average of the group of items will also be multiplied by the same value 'x'.	Considering the previous example, Initial average = 3. Let us multiply each number by 2. The new data is: 2, 4, 6, 8 and 10. The new Average = $\frac{2+4+6+8+10}{5} = \frac{30}{5} = 6$ **New Average = Old Average $\times$ 2**

4	If the value of each item in a group of data is divided by the same value 'x', ($x \neq 0$), then the average of the group or items will also be divided by 'x'	Considering the previous example, Initial average = 3. Let us divide each number by 2. The new data is: $\frac{1}{2}$, 2, $\frac{3}{2}$, 4 and $\frac{5}{2}$. The new Average = $\frac{\frac{1}{2}+1+\frac{3}{2}+2+\frac{5}{2}}{5} = \frac{\frac{15}{2}}{5}$ $= \frac{15}{10} = \frac{3}{2}$ **New Average = $\frac{\text{Old Average}}{2}$**
5	The average of a group of items will always lie between the smallest value in the group and the largest value in the group, i.e. the average will be greater then the smallest value and less than the largest value in the group	Average of 1, 2, 3, 4, 5 is found as Average = $\frac{1+2+3+4+5}{5} = \frac{15}{5} = 3$ Smallest number in the group = 1 Largest number in the group = 5 **Average lies between 1 and 5**

An average assumes that all data points are equally important. It does not distinguish between them. However, there are cases when some data points need more significance than others. For example, there is a group of 10 students. The average salary of the boys is Rs. 3000 and that of the girls Rs. 2000. It may be wrong to conclude that the average salary of the group is Rs. 2500. Average depends on individual data points. If we do not have the information on individual data points, we cannot comment on the average of the entire group. Hence, it is necessary that we have information on the data points. For example if there is only one girl in the group, then the total salary of the group is $3000 \times 9 + 2000 \times 1$. But if there are 2 girls in the group the total salary of the group is $3000 \times 8 + 2000 \times 2$. So the total salary is dependent on the weight of the items, while the number of items in all cases is 10.

WEIGHTED AVERAGE

If $W_1, W_2, W_3 \ldots W_n$ are the weights associated with items $X_1, X_2, X_3 \ldots X_n$, then the average is given by

$$\text{Weighted average} = \frac{W_1X_1 + W_2X_2 + ... + W_nX_n}{W_1 + W_2 + ... + W_n}$$

Example

Let us consider the following data

Name of the book	Cost per book	Number of books
Mathematics	Rs. 10	20
Physics	Rs. 8	15
Chemistry	Rs. 5	15

Here there are three kinds of books and the cost per book on each kind is given. If we are asked to find the combined average cost per book, we need to take all the three kinds of book into consideration. This average is called Weighted Average.

As discussed earlier,

$$\text{Weighted Average} = \frac{\sum WX}{\sum w} = \frac{\text{Total cost of all the books}}{\text{Total number of books}}$$

$$= \frac{(20\times10)+(15\times8)+(15\times5)}{20+15+15} = \frac{395}{50} = 7.9$$

COMBINED AVERAGE

If we know the averages and the number of items of two or more related groups, the combined or composite average can be calculated with the help of the following formula:

$$\bar{x} = \frac{n_1\bar{x_1} + n_2\bar{x_2} + n_3\bar{x_3}}{n_1 + n_2 + n_3}$$

Where

$\bar{x}$ denote the combined averages of three groups.

$\bar{x_1}$, $\bar{x_2}$ and $\bar{x_3}$ denote the average of first group, second group and third group respectively.

n_1, n_2, n_3, denote the number of items in the first, second and third group respectively.

In general for 'n' groups we have, $\bar{x} = \frac{n_1\bar{x_1} + n_2\bar{x_2} + n_3\bar{x_3} + ... + n_n\bar{x_n}}{n_1 + n_2 + n_3 + ... + n_n}$

Example

There are two branches of a company employing 100 and 80 persons respectively. If the average monthly salaries for the two branches are Rs. 275 and Rs. 225 respectively, then find the average monthly salary of all the employees in the company.

The data given are

$\bar{x_1} = 275$, $\bar{x_2} = 225$, $n_1 = 100$ and $n_2 = 80$

$$\bar{x} = \frac{n_1\bar{x_1} + n_2\bar{x_2}}{n_1 + n_2} = \frac{100 \times 275 + 80 \times 225}{100 + 80} = \frac{45500}{180} = 252.78$$

GEOMETRIC MEAN

The Geometric mean (G M) of n numbers x_1, x_2, ... x_n is the nth root of their products.

$\therefore$ GM = $\sqrt[n]{x_1 \times x_2 \cdots x_n}$

Example

1. The GM of 2, 4, 16 and 32 = $\sqrt[4]{2 \times 4 \times 16 \times 32} = \sqrt[4]{2 \times 2^2 \times 2^4 \times 2^5} = \sqrt[4]{2^{12}} = 2^{\frac{12}{4}} = 2^3 = 8$

Note

1. The Geometric Mean of two numbers a and b is $\sqrt{ab}$

Example: The GM of 5 and 125 = $\sqrt{5 \times 125} = \sqrt{25^2} = 25$

2. GM is generally used to calculate the compound average growth rate.

HARMONIC MEAN

By definition, the Harmonic Mean is the reciprocal of the average of the reciprocals.

The Harmonic Mean (H M) of n numbers x_1, x_2 …. x_n is $\frac{n}{\sum_{i=1}^{n} \frac{1}{x_i}}$

Example

Find the Harmonic Mean of 1, 0.5, 10

x_i	$\frac{1}{x_i}$
1	1
0.5	2
10	0.1

$\therefore \ \sum \frac{1}{x_i} = 3.1$

Average of the three numbers = $\frac{3.1}{3}$

$\therefore$ Harmonic Mean = $\frac{3}{3.1} = \frac{30}{31}$

Note

The Harmonic Mean of two numbers 'a' and 'b' is $\frac{2ab}{a+b}$. This result is used to find the average speed when equal distances are covered at different speeds.

Example

A person travels from P to Q at 8 km/hr and then returns from Q to P at 4 km/hr, find the average speed of the entire journey.

$\therefore$ Average speed = H M = $\frac{2\times8\times4}{(8+4)}$

$= \frac{2\times8\times4}{12} = \frac{16}{3}$

$= 5\frac{1}{3}$ km/hr

Challenger Question

Anand plans to travel the distance of 25 km at three different speeds. He covers the first 10 km at 30 km/hr, another 5 km at 45 km/hr and the last 10 km at 35 km/hr. Find the average speed for the entire journey.

Average speed = weighted HM = $\frac{10+5+10}{\frac{10}{30}+\frac{5}{45}+\frac{10}{35}} = \frac{25}{\frac{1}{3}+\frac{1}{9}+\frac{2}{7}} = \frac{25}{\frac{21+7+18}{63}} = \frac{25\times63}{46} = 34.24$ km/hr

Note: This is nothing but Total distance /Total time.

MEDIAN

Median is used to define the central value in a set. The central value is identified in terms of its magnitude.

Examples

1. The set of numbers 1, 3, 5, 6, 8, 9, 10 has median as 6
2. The set of numbers 1, 3, 4, 5, 7, 9 has median $\frac{4+5}{2} = 4.5$

Note: If there are even number of items in the group, median is found by taking the average of the two middle values.

MODE

Mode is used to define the most repeated value in the group of items.

Example

1. Mode of the numbers 2, 3, 4, 4, 5, 5, 5, 6, 7, 9 is 5 since 5 is repeated most number of times
2. Mode of the numbers 2, 3, 4, 4, 5, 5, 7, 9 are 4 and 5, since 4 and 5 both repeated the same number of times.

Practice Questions A (Important)

1. Find the average of the numbers 5, 3, 8, 7, 6, 1. What is the new average if each number is increased by 3?
1] 5 and 8 2] 7 and 10 3] 6 and 9 4] 5.5 and 8.5

2. Kumar purchased two-dozen bananas at Rs. 6/dozen, another dozen at Rs. 8/dozen and 48 more bananas at Rs. 7.25/dozen. What is the average cost per dozen of the bananas that Kumar purchased?
1] Rs. 6 2] Rs. 6.50 3] Rs. 7 4] Rs. 8

3. If the average of 6, 4, 7, P and 10 is 8, find P.
1] 12 2] 11 3] 12.50 4] 13

4. The strength of a liquor is determined by the percentage of the alcohol content in it. If 20 litres of liquor A of strength 15% is mixed with 30 litres of liquor B of strength 18%, what is the strength of the resulting mixture?
1] 16% 2] 16.80% 3] 18.60% 4] 17.50%

5. The average marks secured by 36 students is determined as 52. But it is later discovered that the mark 64 was wrongly entered as 46. What is the correct average of marks of the 36 students?
1] 52.5 2] 52 3] 57 4] 51.5

6. The following table gives the distribution of marks of 22 students in a Physics test. If the average mark of the 22 students in the Physics test is 10.50, find the missing value A.

Number of students	4	5	6	7
Marks obtained in Physics	10	12	A	10

1] 5.84 2] 17 3] 10.17 4] 12.5

7. A toy plane flies along the four sides of a square at speeds of 100, 200, 300, 400 km/hr on each side respectively. Find its average speed.
1] 250 km/hr 2] 190 km/hr 3] 192 km/hr 4] 200 km/hr

8. Ramesh mixed 8 kg of sugar costing Rs. 9/kg and 4kg of sugar costing Rs. 12/kg. Find the price per kg of the mixture.
1] Rs. 9.50 2] Rs. 10.50 3] Rs. 11 4] Rs. 10

9. I buy coffee from 3 different shops at Rs. 150, 140 and 130 per kg and mix them in the proportion 8:5:3. At what price must I sell the mixture to make a profit of 20%?
1] Rs. 143.125/kg 2] Rs. 171.75/kg 3] Rs. 178.9/kg 4] Rs. 114.5/kg

10. A bag of 120 kg of sand is mixed with a heap of pure salt to make the salt 70% pure. A dealer by adding further quantity of pure salt wants to improve the quality of the heap. What quantity of pure salt should be added to the heap to improve the present mixture quality to 85%?
1] 400 kg 2] 350 kg 3] 425 kg 4] 375 kg

11. How many kilograms of sugar costing Rs. 8 per kg must be mixed with 252 kg of sugar costing Rs. 4.5 per kg so that a profit of 10% maybe made by selling the mixture at Rs. 6.6 per kg?
1] 188 kg 2] 189 kg 3] 198 kg 4] 187 kg

12. A vessel has 500 ml of pure milk. From these vessels 50 ml of the milk is removed and 50 ml of water poured into the vessel. If this operation is repeated 2 more times, what is the percentage of milk in the vessel at the end?
1] 24.30% 2] 81% 3] 72.9% 4] 75%

13. The arithmetic mean and geometric mean of two numbers are 17 and 8 respectively. Find the larger of the two numbers.

1] 32 2] 16 3] 8 4] 2

14. An amount of Rs. 143 was collected from boys and girls in a class of strength 100. If each boy contributes Rs. 1.5 and each girl contributes Rs. 1.4 find how many girls are there in the class.
1] 75 2] 25 3] 30 4] 70

15. There is a solution of alcohol and water amounting to 80 litres, with 60% of alcohol. How much of water must be added to this solution to make it a solution in which alcohol and water are equal?
1] 64 litres 2] 16 litres 3] 18 litres 4] 62 litres

16. Two vessels contain mixtures of milk and water in the ratio 7:4 in the first vessel and in the ratio 9:7 in the second vessel. In what ratio should these two mixtures be now mixed such that the resultant mixture has milk and water in the ratio 4:3?
1] 112: 77 2] 77: 112 3] 11: 80 4] 80: 11

17. What quantity of sugar costing Rs. 6.10 per kg must be mixed with 126 kg of sugar priced at Rs. 2.85 per kg so that 20% maybe gained by selling the mixture at Rs. 4.80/kg
1] 68 kg 2] 66.5 kg 3] 67.8 kg 4] 69 kg

18. A mixture of 40 kg of spirit and water contains 10% water. How much water must be added to this mixture to raise the percentage of water to 40%?
1] 20 kg 2] 10 kg 3] 22 kg 4] 18 kg

Answer Key

1-1	2-3	3-4	4-2	5-1	6-3	7-3	8-4	9-2	10-1
11-2	12-3	13-1	14-4	15-2	16-3	17-4	18-1		

Explanatory Answers

1. $\text{Average} = \frac{\text{Sum of all items}}{\text{Number of items}} = \frac{5+3+8+7+6+1}{6} = \frac{30}{6} = 5.$

If 3 is added to each number, the new mean = original mean + 3 = 5 + 3 = 8.

2. $X_1 = 2 \quad W_1 = 6$
$X_2 = 1 \quad W_2 = 8$
$X_3 = 4 \quad W_3 = 7.25$

$$\text{Average cost per dozen} = \frac{X_1W_1 + X_2W_2 + X_3W_3}{x_1 + x_2 + x_3} = \frac{(2\times6) + (1\times8) + (4\times7.25)}{2 + 1 + 4}$$

$$= \frac{12 + 8 + 29}{7} = \frac{49}{7} = \text{Rs. } 7$$

3. $$\text{Average} = \frac{\text{Sum of all items}}{\text{Number of items}} = \frac{6 + 4 + 7 + P + 10}{5}$$

$$\therefore 8 = \frac{27 + P}{5} \Rightarrow 40 = 27 + P \text{ i.e } P = 40 - 27 = 13$$

4. $n_1 = 20$ lit $\quad X_1 = 15$
$n_2 = 30$ lit $\quad X_2 = 18$

Concentration of the resulting solution, $\bar{x} = \frac{n_1\overline{x_1} + n_2\overline{x_2}}{n_1 + n_2}$

Concentration of resulting solution $= \frac{300 + 540}{50} = \frac{840}{50} = 16.80\%$

5. Total marks secured by 36 students = $52 \times 36 = 1872$
Total marks secured by 36 students (Corrected) = 1872 – 46 + 64 = 1890
$\therefore$ Correct average = $\frac{1890}{36} = 52.5$

Alternately,
Increase in the total marks when the error is corrected = 64 – 46 = 18
$\therefore$ The average of 32 students will increase by $\frac{18}{36}$, i.e. 0.5
Hence, corrected average = 52 + 0.5 = 52.5

6.

Number of students	Marks in physics	Total marks
4	10	40
5	12	60
6	A	6A
7	10	70

Total marks obtained by 22 students = 40 + 60 + 6A + 70 = 170 + 6A

$$\text{Average} = \frac{170 + 6A}{22} = 10.50$$

$\therefore$ $231 = 170 + 6A$
$6A = 61$
$\therefore$ $A = 10.17$

7. Let the side of the square be 'a'
When equal distances are covered at varying speeds, the average of the speeds is the Harmonic Mean.

$$\text{Average speed} = \frac{4a}{\frac{a}{100} + \frac{a}{200} + \frac{a}{300} + \frac{a}{400}}$$

$$= \frac{4}{\frac{12 + 6 + 4 + 3}{1200}} = \frac{4 \times 1200}{25} = 192 \text{ km/hr}$$

8. $$\text{Price per kg of the mixture} = \frac{\text{Total price}}{\text{Total quantity}}$$

$$= \frac{(8 \times 9) + (4 \times 12)}{8 + 4} = \frac{72 + 48}{12}$$

$$= \frac{120}{12} = \text{Rs. } 10$$

9. $$\text{Cost Price of the mixture} = \frac{(8 \times 150) + (5 \times 140) + (3 \times 130)}{8 + 5 + 3}$$

$$= \frac{2290}{16} = \text{Rs. } 143.125$$

$\therefore$ Selling Price = 1.2×143.125 = Rs. 171.75/kg

10. Let the amount of pure salt = x kg.
30% of the mixture = 120 kg.
i.e. $0.30 \times (120 + x) = 120$

$x = \frac{120}{0.3} - 120 = 280\,kg$

$\therefore$ In the original mixture, pure salt = 280 kg.
The new mixture contains 85% pure salt
In 100 kg of new mixture, pure salt = 85kg, sand = 15kg

Sand	Pure salt
15	85
120	y

$\therefore\ y = \frac{120 \times 85}{15} = 680kg$

$\therefore$ The required quantity of pure salt = 680 – 280 = 400 kg

11. S P = Rs. 6.6/kg.

$SP = \frac{110\ CP}{100} \Rightarrow\ CP = \frac{6.6 \times 100}{110}$ = Rs. 6 = Mean Price

$\text{Rs.}4.50/\text{kg} \searrow\ 6\ \swarrow \text{Rs.}8/\text{kg}$

$(8-6) \quad \swarrow \quad \searrow \quad (6-4.5)$

$\text{Rs.}2/\text{kg} \qquad\qquad \text{Rs.}1.50/\text{kg}$

$\therefore$ Quantity of dearer / Quantity of cheaper = $\frac{1.5}{2}$

$\therefore$ Quantity of dearer = $252 \times \frac{1.5}{2} = 126 \times 1.5 = 189$ kg

12. We know $\left(\frac{a-b}{a}\right)^n = c$

Where a = 500ml, b = 50ml, n = 3

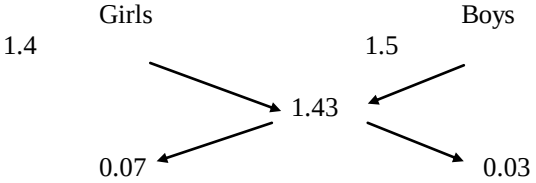
$\therefore$ Proportion of the milk = $\left(\frac{500-50}{500}\right)^3 = \left(\frac{450}{500}\right)^3 = \left(\frac{9}{10}\right)^3 = 0.729$

$\therefore$ Percentage of milk in the vessel = 100c = 72.9%

13. Let the numbers be ‘a’ and ‘b’. A.M. = $\frac{a+b}{2}$ = 17 => a + b = 34.

GM = $\sqrt{ab} = 8 \Rightarrow ab = 64$
$\therefore (34 - b)\ b = 64$, i.e. $34b - b^2 = 64$
$\therefore\ b^2 - 34b + 64 = 0$
$\therefore\ (b - 32)\ (b - 2) = 0 \Rightarrow b = 32$ or 2
$\therefore$ a = 2 or 32

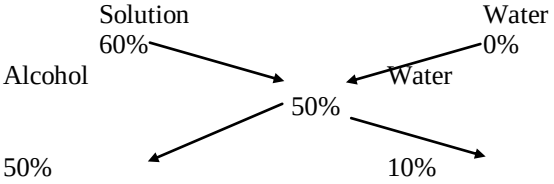
14. Mean amount = 143/100 = Rs.1.43



$\therefore \frac{\text{Boys}}{\text{girls}} = \frac{0.03}{0.07} = \frac{3}{7}$

Number of girls $= \frac{7}{10} \times 100 = 70$. Hence [4]

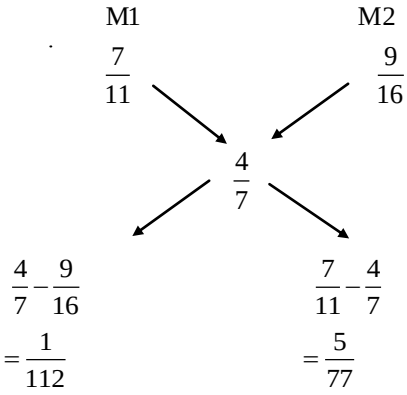
15. 60% solution of alcohol and water is mixed with pure water, (i.e. 0% concentration) to give 50% concentration solution.



The ratio of mixture to water is 50: 10 = 5: 1

Since there is 80 litres of solution initially, the amount of water to be added = $\frac{80}{5}$ = 16 litres. Hence [2]

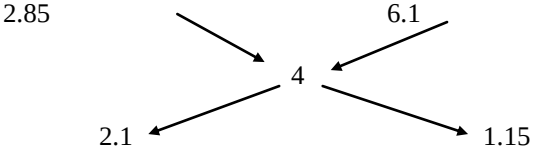
16. Concentration of first mixture M1 is $\frac{7}{11}$ and the second mixture M2 is $\frac{9}{16}$



$\therefore$ Ratio of $\frac{1^{st}\text{ mixture}}{2^{nd}\text{ mixture}} = \frac{\frac{1}{112}}{\frac{5}{77}} = \frac{1}{112} \times \frac{77}{5} = \frac{11}{80} = 11:80$. Hence [3]

17. Selling Price = Rs. 4.80 / kg

$SP = \frac{120 \times CP}{100} \Rightarrow CP = \frac{100 \times 4.8}{120} =$ Rs. 4 (Mean Price)



$$\frac{\text{Quantity of costlier}}{\text{Quantity of cheaper}} = \frac{1.15}{2.1}$$

$\therefore$ Quantity of costlier = $126 \times \frac{1.15}{2.1} =$ 69 kg. Hence [4]

18. The amount of water in the 1st mixture = $\frac{10}{100} \times 40 = 4\text{kg}$

$\therefore$ The amount of the spirit = 40 – 4 = 36 kg
In the 2nd mixture,
60 kg of spirit is contained in 100 kg of mixture

$\therefore$ 36 kg of spirit is contained in $36 \times \frac{100}{60}$ = 60 kg

$\therefore$ Water added = 60 – 40 = 20 kg. Hence [1]

Practice Questions B (Optional)

1. The average age of 30 soldiers is 25 years. The average age of first 15 of them is 24 years and the average age of last 14 of them is 26 years. What is the age of the 16th soldier?

1] 22 years 2] 25 years 3] 26 years 4] $25\frac{1}{2}$

2. Calculate the arithmetic mean for the following data:

Number of students	5	6	6	2	1
Marks	30	35	40	45	50

1] 43 2] 44 3] 37 4] 37.5

3. The average of the first 10 numbers is 16 and the average of first 25 numbers is 22. Find the average of the last 15 numbers.
1] 26 2] 22 3] 19.5 4] 28

4. The sum of seven numbers is 235. The average of first three numbers is 23 and the average of the last three numbers is 42. Find the fourth number.
1] 45 2] 43 3] 42 4] 40

5. There are 40 students in a class. Out of these students, one student whose weight is 100 kg goes away and at the same time a new student gets admitted into the class. Thus the average weight of the students increases by $\frac{1}{4}$ kg. Find the weight of the new student.
1] 105 kg 2] 115 kg 3] 107 kg 4] 110 kg

6. The average temperature of Monday, Tuesday, and Wednesday was 74° F, the average temperature of Tuesday, Wednesday and Thursday was 78° F. If the temperature on Thursday was 75° F, then find the temperature on Monday.
1] 87° F 2] 63° F 3] 64° F 4] 88° F

7. In a party, there were 8 men, when 2 of these men aged 20 and 24 left the party and 2 women entered the party, the average age of the entire group of 8 increased by 2. What was the average age of the two new women entered the party?
1] 35 2] 32 3] 30 4] 33

8. The average age of 7 trustees in a committee is the same as it was 5 year ago, after a younger man having been substituted for one of them. How much younger was he than the trustee whose place he took?
1] 32 2] 35 3] 30 4] 33

9. A motorist covers the distance of 60 km from A to B at a speed of 20 km/hr and then returns from B to A at a speed of 30 km/hr. Find his average speed for the entire journey.
1] 48 km/hr 2] 24 km/hr 3] 22 km/hr 4] 46 km/hr

10. Find the HM of two numbers whose GM and AM are 8 and 5 respectively.
1] 12.5 2] 12.3 3] 12.6 4] 12.8

Answer Key

1-3 2-3 3-1 4-4 5-4 6-2 7-3 8-2 9-2 10-4

Chapter 11 Time and Work

Suggested Time : 120 minutes

Basic Concepts of Time and Work

The basic form of the question is evaluating the work done by a set of people in comparison to another set, where the two sets may contain different number of people with varying efficiencies. Primarily the trick lies in understanding the question and separating it in form of equations. Let us begin by understanding some basic interpretations.

1. "A can complete a task in 10 hours." This implies that
   - The task is equal to 10 hours of A's effort.
   - A can do $\frac{1}{10}^{th}$ of the task in 1 hour.
   - A can do 10% of the task in 1 hour, or 50% of the task in 5 hours, or do 2 such tasks in 20 hours.

2. "B does $\frac{1}{5}^{th}$ of the task in 1 hour". This implies that
   - B does 20% of the task in 1 hour.
   - In 5 hours, B can complete the task.
   - B is twice as fast as A.
   - A takes twice the amount of time that B takes to do the task.

- If A and B were to work together, then in an hour they can do 10% + 20% of the task, i.e. 30% of the task can be completed in first hour, and on the 4th hour the task will be completed. To be more precise, the task will be completed in 3.33 hours or 3 hours 20 min.

- In other words, A and B do $\frac{1}{10}+\frac{1}{5}$ of the task in an hour. Hence, in 1 hour they can do $\frac{1}{10}+\frac{1}{5}=\frac{3}{10}^{th}$ of the task.

If $\frac{3}{10}^{th}$ of the task can be done in 1 hour, then the entire task can be done in $\frac{1}{\frac{3}{10}}=\frac{10}{3}=3\frac{1}{3}$ hours.

3. A, B and C can finish a piece of work in 2, 3 and 4 days respectively. How will the remuneration of the job be shared between them?

The remuneration for the work will be shared according to the ratio of the amount of work done by each A, B and C.

$\therefore$ Total remuneration for the work will be shared in the ratio $\frac{1}{2}:\frac{1}{3}:\frac{1}{4}$, i.e. $\frac{6}{12}:\frac{4}{12}:\frac{3}{12}$, i.e. 6 : 4 : 3.

If the total remuneration for the whole work is Rs. 2600, then

A's share = $\frac{6}{13}\times 2600$ = Rs. 1200

B's share = $\frac{4}{13}\times 2600$ = Rs. 800 and C's share = $\frac{3}{13}\times 2600$ = Rs. 600

4. For all question involving Time and Work, always find the work done in unit time (1 hour, or 1 minute or 1 day…) for each person. After reducing the work to unit time, it can be added or subtracted from the work done by another person in unit time.

5. If A is twice as good a workman as B, then A will take half of the time that B takes to do a certain piece of work.

Working simultaneously v/s working in sequence

Example 1 : A alone can do a piece of work in 40 days and B alone can do the same piece of work in 10 days. How long will they take to do the work together?

In the above example, A & B are working at the same time (simultaneously). Hence the work done in 1 day is $\frac{1}{10}+\frac{1}{40}$.

Hence $\frac{5}{40}$ of the work is done in 1 day and it would take $\frac{40}{5}$ = 8 days to do the entire work.

Example 2 : A takes 5 days to do a job and B takes 10 days to do the same job. A works for 1 day and B works for the next day, after which A works again for the third day, and the two keep alternating every day. In how much time will the job be done?

If we use the same logic and decide that in 2 days $\frac{1}{5}+\frac{1}{10}=\frac{3}{10}$ of the job is done. Hence the job would be done in 2 $\times\frac{10}{3}$ days = $\frac{20}{3}$ days, i.e. 6.66 days. This logic is incorrect.

In 6 days $3\times\frac{3}{10}=\frac{9}{10}$ th of the job is done. However, on the 6th day the job was being performed by B, and hence on the 7th day A starts the work. When A starts the work, $\frac{9}{10}$ th of the job is done and $\frac{1}{10}$ th of the job remains. Now, A takes 5 days to do the job, or does $\frac{1}{5}$ th of the work in 1 day. Hence he does $\frac{1}{10}$ th of the job in half a day. Hence the work is finished in 6.5 days and not 6.66 days.

This is a standard trap that you may find in aptitude tests!

Hence, it should be noted that once the work remaining is less than the work that can be done by any of the individuals in a day (or an hour, etc. as the case may be), then from that point onwards one should follow the manual approach and not the formulae approach.

The concept of man days / man hours, etc.

A man-day is equal to the work done by one man in one day and a man-hour is equal to the work done by one man in one hour. A man-day or man-hour is a unit of work and this concept is useful to evaluate the quantum of work done by different teams. For example, if team A comprising 8 members working 4 hours a day does a work in 6 days, and team B comprising 5 members do another task in 7 days working 8 hours a day, then we can say that task of team A was equal to $8\times4\times6$ man-hours and task of team B was equal to $5\times7\times8$ man-hours. Hence, the task of team B required more time.

In man-days or man-hours related questions, always put the work done on one side and the effort required on the other side. The effort required quantity should always be in the lowest possible units.

Negative Work

If A takes 10 days to do the job and B takes 20 days to undo the job, then we can say that B is working against A. Hence, we can say that in a day, A does $\frac{1}{10}$ th of the job and B does $\left(-\frac{1}{20}\right)$th of the job in 1 day. Hence, if it was given that both worked simultaneously, then we can say that in 1 day A and B together do $\frac{1}{10}-\frac{1}{20}$ th of the job or $\frac{1}{20}$ th of the job. So it will take 20 days for the two of them to complete the job, if they work together.

This concepts is better used in questions involving pipes, tanks and cisterns whereby a set of pipes may be attached to a tank, some of which are inlet pipes (i.e. doing the job of filling up the tank) and some of them are outlet pipes (i.e. doing the opposite job, i.e. emptying the tank).

Hence, in a specified time frame, the job done = The work done by the inlet pipe – The work done by the outlet pipe.

Short cut methods for Time and Work questions

1. **Work in percentages:** Sometimes it is faster to work in percentages than in fraction, and you can either get to the correct answer or approximate it to one of the answer options using this method. Here, instead of saying that $\frac{1}{10}$th of the work is done in 1 hour, we say that 10% of the work is done in 1 hour.
2. Let the amount of the work to be done be the equal to LCM of time required to do the work.



GURU GYAN

In questions on Time and Work, it is best to denote all activities in terms of the smallest value. For example, if a work requires a man to work for 4 weeks, working 5 days a week, and working 8 hours each day, then the smallest unit of work here is hours. Hence, measure work in terms of hours. Here, the work is equal to 4 ×5 × 8 hours = 160 hours.

However, not all work can be represented in a single unit of measurement. For example, if a work requires 12 men to work for 2 weeks, working 6 days a week, and working 5 hours each day, then the smallest unit of work here is hours, but the 12 men cannot be measured in terms of hours. So we measure the entire effort in terms of man hours. Hence, measure of work in terms of man hours is equal to 12 men × 2 × 6 × 5 hours = 720 man hours.

Sometimes, work is not measured in terms of hours, but in terms of effort by various categories of people. Again, using the above concept, try and represent work in terms of the time required by the slowest of all categories of people. For example, let 1 man can do a work in 8 days. If 4 women, in a day can do a work what 3 men can do in a day, then try to denote work in terms of woman days rather than man days. Here, as $3\text{ M} = 4\text{ W} \rightarrow \text{M} = \frac{4}{3}\text{W}$.

Now, the work requires 8 man days. Hence the work requires $8 \times \frac{4}{3}$ women days.

Practice Questions A (Important)

1. A alone can do a piece of work in 15 days and B alone can do the same piece of work in 10 days. How long will they take to do the work together?
   1] 5 days 2] 6 days 3] 12.5 days 4] 25 days

2. A and B undertook to do a piece of work for Rs. 7500. A alone could do it in 4 days and B in 6 days; with the assistance of C they finish the work in 2 days. Find the share of each (in Rs.).
   1] 3750, 2500, 1250 2] 3600, 2400, 1500
   3] 3000, 2000, 2500 4] 4500, 2500, 500

3. A can do as much work in 2 days as B can do in 3 days and B can do as much in 4 days as C can in 5 days. A piece of work takes 20 days if all work together. How long would B take to do all the work by himself?
   1] 82.5 days 2] 44 days 3] 66 days 4] 100 days

4. A and B can do a piece of work in 40 days. After working for 10 days they are assisted by C and the work is finished in 20 more days. If C does as much work in 2 days as B does in 3 days, then in how many days could A alone do the same work?

1] 65 2] 80 3] 120 4] 60

5. A tap can fill a cistern in 10 minutes, while the other can empty it in 15 minutes. The cistern being empty and if both the taps are opened together, in how many minutes will the cistern be full?

1] 20 2] 30 3] 6 4] 60

6. Three taps can fill a cistern in 30 minutes, 45 minutes and 54 minutes respectively. The cistern being empty, all the three taps are kept open and after 9 minutes the third tap is closed. In how many more minutes will the cistern be full?

1] 6.5 2] 8 3] 15 4] 6

7. A full tank is emptied in 8 hours by an outlet pipe. However, if an inlet pipe pumping in water at the rate of 5 litres per minute is opened. When the tank is full, the outlet pipe takes 12 hours to empty the tank. What is the capacity (in litres) of the tank?

1] 700 2] 4000 3] 60 4] 7200

8. A worm can climb up a burrow at 3 ft per minute but due to the slipperiness of the burrow, slips down 2 ft before it starts climbing the next minute. If the depth of the well is 30 ft, how much time will the worm take to reach the top?

1] 27 min 2] 28 min 3] 29 min 4] 30 min

9. 3 men with 5 boys can do a piece of work in 2 days and 4 men and 16 boys complete the same work in one day. How much time will it take for 1 boy together with a woman who can work twice as fast as the boy, to complete a work that is 3 times as time consuming?

1] $7\frac{1}{3}$ days 2] 8 days 3] $9\frac{1}{3}$ days 4] 28 days

10. A piece of work can be done by 10 men and 5 women together in 6 days. The same work can be done by 11 men and 8 women in 5 days. In how many days can 25 women do the same work?

1] 5 2] 6 3] 22 4] 11

11. A piece of work can be completed by A in 10 days and B in 12 days individually. If they work on the job on alternate days starting with B, how many days will the job take?

1] 11 2] $10\frac{5}{6}$ 3] 12 4] $10\ \frac{10}{11}$

12. 8 monkeys can eat 8 bananas in 8 minutes. In how many minutes will 2 monkeys eat 2 bananas?

1] 8 2] 2 3] 1 4] 4

13. There are two toy cars facing each other at a distance of 500 cm from each other. Each car moves forward by 100 cm at a speed of 50 cm/s and then moves backward by 50 cm at a speed of 25 cm/s. How long will it take for the cars to collide?

1] 12 sec 2] 14 sec 3] 16 sec 4] 13 sec

14. A alone can do a piece of work in 5 days, B can do the same piece of work in 10 days and C can do the work in 20 days. How long will they take to do the work together?

1] 3 days 2] $2\frac{2}{29}$ days 3] $2\frac{6}{7}$ days 4] $2\ \frac{5}{6}$ days

Answer Key

1-2	2-1	3-3	4-4	5-2	6-4	7-4	8-2	9-4	10-2
11-1	12-1	13-2	14-3						

Explanatory Answers

1. A's one day work $= \frac{1}{15}$ and B's one day work $= \frac{1}{10}$

$\therefore$ (A and B)'s one day work $= \frac{1}{15} + \frac{1}{10} = \frac{2+3}{30} = \frac{1}{6}$

$\therefore$ A and B together will take 6 days to do the work.

2. (A + B)'s one day work $= \frac{1}{4} + \frac{1}{6} = \frac{5}{12}$

(A + B + C)'s one day work $= \frac{1}{2}$

$\therefore$ C's one day work $= \frac{1}{2} - \frac{5}{12} = \frac{6-5}{12} = \frac{1}{12}$, i.e. C alone can complete the whole work in 12 days.

Hence, Rs. 7500 is to be divided in the proportion $\frac{1}{4} : \frac{1}{6} : \frac{1}{12}$ (or) 3 : 2 : 1

$\therefore$ A's share $= \frac{3}{6} \times 7500 =$ Rs. 3750 , B's share $= \frac{2}{6} \times 7500 =$ Rs. 2500 and

C's share $= \frac{1}{6} \times 7500 =$ Rs. 1250.

3. If A does A units of work and B does B units of work in one day, then

2A = 3B, i.e. $\frac{A}{B} = \frac{3}{2}$

If C does C units of work in one day, then $\frac{B}{C} = \frac{5}{4}$

It is given that A, B and C can complete the work in 20 days.

Ratio of the work done by A, B and C, i.e. A : B : C = 15 : 10 : 8.

Hence, B alone can complete the work in $\frac{20(15+10+8)}{10}$ days, i.e. 66 days.

4. A and B were supposed finish the work in 40 days. But they work only for 30 days.

(A + B)'s one day work $= \frac{1}{40}$, i.e. $\frac{1}{A} + \frac{1}{B} = \frac{1}{40}$

In 30 days they finish $\frac{30}{40}$, i.e. $\frac{3}{4}$ of the work.

The remaining $\frac{1}{4}^{th}$ of the work is done by C in 20 days.

C's 20 days work $= \frac{1}{4}^{th}$ of the work.

C's 2 days work $= \frac{1}{40}^{th}$ of the work = B's 3 days work.

Hence, B's 1 day work, i.e. $\frac{1}{B} = \frac{1}{120}$.

$\therefore \frac{1}{A} + \frac{1}{120} = \frac{1}{40}$

$\frac{1}{A} = \frac{1}{40} - \frac{1}{120} = \frac{3-1}{120} = \frac{1}{60}$. Hence, A alone can complete the work in 60 days.

5. In 1 minute the first tap will fill $\frac{1}{10}$ of the cistern and the other tap will empty $\frac{1}{15}$ of the cistern.

Let the cistern be filled in X minutes. $\therefore X\left(\frac{1}{10}-\frac{1}{15}\right)=1, \text{i.e.} X\left(\frac{3-2}{30}\right)=1$

X = 30 min. Hence, the cistern will be full in 30 minutes.

6. Let x be the time required to fill the cistern after the third tap is closed.

In 1 min the three taps will fill $\left(\frac{1}{30}+\frac{1}{45}+\frac{1}{54}\right)$ of cistern

$\therefore$ In 9 min they fill $9\left(\frac{1}{30}+\frac{1}{45}+\frac{1}{54}\right)=9\left(\frac{9+6+5}{270}\right)=\frac{20}{30}=\frac{2}{3}$ of cisterm

After the 3rd tap is closed, it takes x minutes to fill the cistern.

$\therefore \frac{2}{3}+x\left(\frac{1}{30}+\frac{1}{45}\right)=1$

$\frac{2}{3}+\frac{x}{18}=1$

$\frac{12+x}{18}=1$

$\therefore 12+x=18 \Rightarrow x=6$ minutes.

$\therefore$ It takes 6 minutes more to fill the cistern.

7. Work done by outlet in 1 hour = $\frac{x}{8}$, where x is capacity of the tank.

Amount of water pumped in 12 hours = 12 × 60 × 5 =3600 litres

$\therefore$ Outlet pipe removes (3600 + x) litres in 12 hours.

So in 1 hour, outlet pipe removes $\frac{3600+x}{12}$, i.e. $300+\frac{x}{12}$ liters

$\therefore 300+\frac{x}{12}=\frac{x}{8} \Rightarrow x=7200$ litres.

8. In one minute the worm climbs (3 – 2), i.e. 1 ft of the burrow

$\therefore$ In 27 min, it will climb 27 ft and in the next 1 minute it will climb 3 ft.

At the end of 28th min, as it reaches the top of the well, it will NOT slip down.

$\therefore$ It takes the worm 28 minutes to reach the top.

9. Let a man and a boy can complete the work, working separately, in M and B days respectively.

$\frac{3}{M}+\frac{5}{B}=\frac{1}{2}$ (i)

$\frac{4}{M}+\frac{16}{B}=1$ (ii)

(i) × 4 – (ii) × 3 gives $\Rightarrow \frac{1}{B}=\frac{1}{28}$

The work done by one boy and one women in a day = $\frac{1}{B}+\frac{1}{W}=\frac{1}{B}+\frac{2}{B}=\frac{3}{B}=\frac{3}{28}$.

In order to complete a job which is 3 times consuming, they will take $\frac{28}{3}\times 3$ days, i.e. 28 days.

Smart Approach

2 (3M + 5B) = 4M + 16B

$\therefore$ 6M – 4 M = 16B – 10B

$\therefore$ M = 3B

i.e. Work of 1man = Work of 3 boys
Work of 1 boy and 1 woman = 1 boy + 2 boys = Work of 3 boys
$\therefore$ Work of 4 men and 16 boys = 4M + 16B = 4(3B) + 16B = 28B
$\therefore$ 28 boys can complete the work in 1 day.
For 3 boys, it takes $3\times\left(\frac{28}{3}\right)$ days to complete the work which is 3 times as time consuming.
Hence, the number of days required = 28 days

10. Let a man and a woman can complete the work, working separately, in M and W days respectively.

$$\frac{10}{M}+\frac{5}{W}=\frac{1}{6} \quad \text{(i)}$$

$$\frac{11}{M}+\frac{8}{W}=\frac{1}{5} \quad \text{(ii)}$$

(i) $\times 8$ – (ii) $\times 5$ gives $\Rightarrow$ $\frac{1}{M}=\frac{1}{75}$ (iii)

$\therefore$ 1 man takes 75 days.

Substituting (iii) in (i) we get $\frac{10}{75}+\frac{5}{W}=\frac{1}{6}$

$\frac{5}{W}=\frac{1}{6}-\frac{2}{15}=\frac{5-4}{30}=\frac{1}{30}$ $\therefore \frac{1}{W}=\frac{1}{150}\Rightarrow$ 1 women will take 150 days.

$\therefore$ 25 women will take $\frac{150}{25}$ = 6 days.

11. In the first day B does $\frac{1}{12}$ of the job and on the second day A does $\frac{1}{10}$ of the job.

$\therefore$ In two days, A and B can complete $\frac{1}{12}+\frac{1}{10}=\frac{5+6}{60}=\frac{11}{60}$ of the job.

In 10 days (i.e. 5 sets of two days) they do $\frac{11}{60}\times 5=\frac{55}{60}=\frac{11}{12}$ of the job.

Balance $\frac{1}{12}$ of the work is done by B on the 11th day. $\therefore$ They take 11 days to do the work.

12. 8 Bananas require 8 Monkeys $\times$ 8 min, i.e.64 Monkey min. $\therefore$ 2 Bananas require 16 Monkey min. of effort
$\therefore$ 2 Monkeys have to work for 8 min to generate 16 Monkey min that is required to eat the two bananas.

13. In 4 seconds, the cars reduce the distance by 100 cm. Hence in 12 seconds, the distance is reduced to 200 cm. In the 13th and 14th second, the cars come forward by 100 cm each, and hence collide. So the gap of 500 cm is reduced in 14 seconds.

14. **Classical Approach**

Work done by A, B and C in 1 day = $\frac{1}{5}+\frac{1}{10}+\frac{1}{20}=\frac{7}{20}$ of the job.

Hence, the number of days required to complete the job = $\frac{20}{7}$ days = $2\frac{6}{7}$ days.

Alternate Approach
Work done by A, B and C in 1 day = 20% + 10% + 5% = 35% of the job.

Hence the work is completed in $\frac{100}{35}=2\frac{6}{7}$ days.

Approach by LCM method
Let work = LCM of 5, 10 and 20 units, i.e. Work = 20 units.
A takes 5 days, B takes 10 days and C takes 20 days to do the work.

If each worked independently, in one day, A does $\frac{20}{5}$ = 4 units of works, B does $\frac{20}{10}$ = 2 units of work and C does $\frac{20}{20}$ = 1 units of work.

$\therefore$ In 1 day, A, B and C can do (4 + 2 + 1), 7 units of work.

Hence, to do 20 units of work, $\frac{20}{7}$ days are required.

Practice Questions B (Optional)

1. A can do a work in 32 days and B in 40 days. They worked together for 10 days and then B leaves the work. In how many days will A alone finish the remaining work?

1] 12 2] $3\frac{1}{3}$ 3] 17.5 4] 14

2. Anish can do a piece of work in 12 days. Amit is found to be 50% more efficient than Anish. If Amit is given the piece of work, then how long will he take to do it?

1] 7 days 2] 10 days 3] 6 days 4] 8 days

3. A can do a piece of work in 9 days and B in 11 days. If they work together for 4 days, then what fraction of the work will be left unfinished?

1] $\frac{1}{19}$ 2] $\frac{29}{188}$ 3] $\frac{19}{99}$ 4] $\frac{80}{99}$

4. A can do a piece of work in 3 days and B in 5 days. How long will it take to complete the work if A and B work together?

1] 2 days 2] $1\frac{7}{8}$ days 3] 3 days 4] $1\frac{6}{7}$

5. A and B together can mow a field in 10 hours, but by himself A requires 15 hours. How long would B take to mow the same field?

1] 20 hrs 2] 22 hrs 3] 15 hrs 4] 30 hrs

6. A can do a piece of work in $10\frac{2}{3}$ days, B in $9\frac{3}{5}$ days and C in $5\frac{9}{11}$ days. In how many days can A, B and C, working together, complete the work?

1] $2\frac{15}{71}$ 2] $2\frac{2}{71}$ 3] $2\frac{50}{71}$ 4] 3

7. If a man, a women and a boy can do a piece of work in 3, 4 and 6 days respectively, what fraction of the work would remain unfinished after one man, one women, and two boys had been working at it for a day?

1] $\frac{1}{12}$ 2] $\frac{1}{6}$ 3] $\frac{1}{4}$ 4] $\frac{1}{3}$

8. A machine operator in a factory can operate either one, two or three machines simultaneously working 8 hours a day. On all three machines he contracts to produce a specified number of parts in 6 days. If at the end of 3 days one of the machines goes out of order, how many hours a day must he work for each of the last 3 days if he has to complete the order in 6 days?

1] 10 2] 9 3] 12 4] Indeterminate

9. How long will two pipes together take to fill a cistern which they can separately fill in 20 and 25 minutes respectively?

1] $11\frac{1}{3}$ min 2] 12 min 3] $11\frac{7}{9}$ min 4] $11\frac{1}{9}$ min

10. A man does task X in 1 day. In how many days would 5 men be able to do a job 5 times the size of task X?

1] 1 2] 5 3] 2.5 4] 10

Answer Key

1-4 2-4 3-3 4-2 5-4 6-3 7-1 8-3 9-4 10-1

Chapter 12 Speed and Distance

Suggested Time : 180 minutes

BASIC CONCEPTS

It is apparent that the distance you travel depends on the speed you are travelling at and the amount of time you are travelling.

$\rightarrow$ The faster you travel the further you reach.

$\rightarrow$ The longer you travel the further you reach,

i.e. distance 'd' is directly proportional to the speed, 's' and time, 't'.

$d \propto s$ and $d \propto t$

The exact relationship therefore is, Distance = Speed × Time, i.e. $d = s \times t$

From the above equation, we have

1. Speed = $\frac{\text{Distance}}{\text{Time}}$ 2. Time = $\frac{\text{Distance}}{\text{Speed}}$

Questions type- I: Basic problems on speed, time and distance

The distance between your home and the exam center is 10 km. If the exam is to start in half an hour, then how fast do you need travel to reach on time?

d = 10 km

t = 0.5 hours

The required speed is = $\frac{10}{0.5}$ = 20 km/hr

Now if you double the speed, i.e. at 40 km/hr, you will reach the center in $\frac{10}{40}$ = 0.25 hrs, or exactly half the time taken as before.

i.e. double the speed $\rightarrow$ half the time.

Half the speed $\rightarrow$ double the time.

Questions type- II : An object crossing another stationary object

There are two types of questions that can be asked under this category. They are,

A. The time taken for a train to cross a pole (or any object whose dimension is considered negligible)

The time it takes the train to cross the pole is equal to the time it takes to travel its own length.

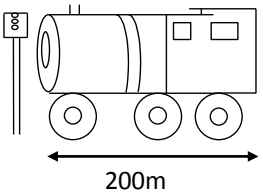
Example 1

How long will a 200 m long train travelling at 90 km/hr take to cross a signal pole?

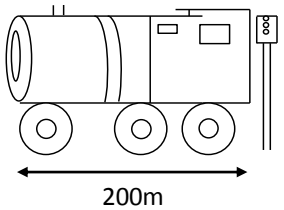
1] 8 sec 2]16 sec 3] $\frac{20}{9}$ sec 4] can't say

We first have to know the distance to be covered by the train.

Picture this



We can see from the picture above that to cross a pole a train has to indeed cover a distance equal to its own length and the picture below shows that the train has crossed the pole by travelling its own length.



Hence the above question can be solved in the following way.
d = 200 m

s = 90 km/hr = $\frac{90\times5}{18}$ m/s = 25 m/s

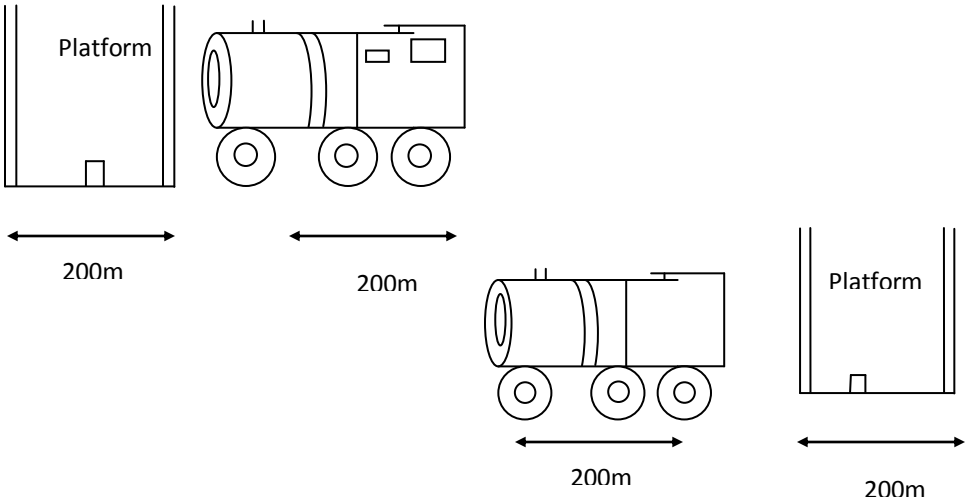
$\therefore$ Time taken = $\frac{200}{25}$ = 8 seconds

Now think.

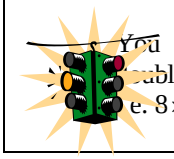
What if the pole was 1metre long? It is obvious that the train had covered this 1 m extra!!

B. The time taken for a train to cross a platform (or another stationery train)
The time it takes a train to cross a platform is the time required by the train to travel a distance equal to the length of the platform and its own length.

Example 2
How long will a 200 m long train travelling at 90 km/hr take to cross a platform 200 m long?
1] 4 sec. 2] 8 sec. 3] 16 sec. 4] cant say



So here the train has to cover 200 + 200 = 400 m and hence the time taken is = $\frac{400}{25}$ =16sec.



You can see that the speed is the same in example 1 & 2 and only the distance has doubled. So from our basics (d = s × t) we know time taken has to double too!!!, i.e. 8× 2 = 16 sec.

AVERAGE SPEED
The distance between your home and the exam center is 10 km. You have half an hour before the exam starts, how fast do you need travel to reach on time?

The required speed is = $\frac{10}{0.5}$ = 20 km/hr

In the above example we see that you had to travel at 20 km/hr, to reach the exam center on time. But does that mean you travel at 20 km/hr thought-out the distance. Won't it do if you zoom at 60 km/hr on a fly-over and crawl at 10 km/hr on a busy road with pot holes?

When we said 20 km/hr, what we meant was, on an average, over the distance you have to travel at 20 km/hr.

Hence, Average speed = $\frac{\text{Total distance}}{\text{Total time}}$

RELATIVE SPEED

The word "relative" basically means "with respect to", i.e. if two bodies are moving, what is the speed of any of them, keeping the speed of the other as zero or stationary?

Suppose, say a policeman is travelling at 50 km/hr and the thief is travelling at 40 km/hr. The initial distance between them is AB, where the thief is at point A and the police is at point B. When would the policeman catch the thief faster? When they are travelling in the same direction or opposite direction?

A. same direction

Thief ←40 km/hr Police ←50 km/hr

C……..A……B

We can see that the policeman has to travel from B to A to catch the thief. But by the time the policeman reaches B, the thief would also have moved to say point C.

B. opposite direction

Thief →40 km/hr 50 km/hr ← Policeman

A ……D……..…..……B

In this case they will meet somewhere between A and B, i.e. the policeman will have to travel more in the former case than this case.

The relative speed reduces if the bodies are moving in the same direction. In this case it is (50 – 40) = 10 km/hr. And it increases if they are moving in opposite direction, in the second case (50 + 40) = 90 km/hr

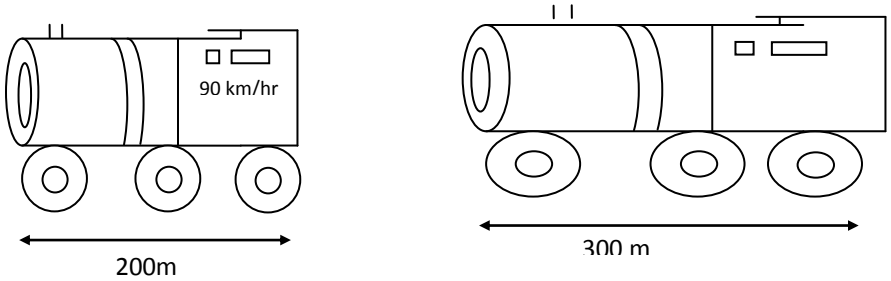
Note

1. If the 2 bodies are moving in the same direction the relative speed is the difference of the two speeds.
2. If the 2 bodies are moving in opposite direction the relative speed is the sum of the two speeds.

Questions type- III : A moving object crossing another moving object

A train 200 m long moving at 90 km/hr crosses another train 300 m long and travelling in the same direction in 25 seconds. What is the speed of the other train?

1] 72 km/hr 2] 50 km/hr 3] 18 km/hr 4] can't say



Total distance to be travelled to cross the train = 200 + 300 = 500 m

Time taken = 25 seconds

$\therefore$ Relative speed = $\frac{500}{25}$ = 20 m/s.

This means that the first train is moving at 20 m/s MORE than the second train.
Speed of second train = speed of 1st train – 20 m/s

$= 90 \times \frac{5}{18}$ m/s – 20m/s = 5 m/s . Converting 5 m/s into km/hr, we get $5 \times \frac{18}{5}$ = 18 km/hr

EFFECTIVE SPEED

It is the final speed at which a body travels if its being assisted or resisted by some other speed. For example, even if you stand still (0 m/s) on an escalator moving at 3 m/s you effective speed is 3 m/s.

Upstream/downstream speed

If is obvious that a boat would move faster downstream than upstream simply because the boat's effective speed has increased owing to the speed of the stream (i.e. water flowing).

If the speed of the boat downstream = u km/hr and the speed upstream =v km/hr
What is the speed of the boat in still water?

Speed of the boat in still water = $\frac{u+v}{2}$ km/hr

Speed of water current = $\frac{u-v}{2}$ km/hr

Practice Questions A (Important)

1. A person covers a certain distance at a certain speed. If he increases his speed by 50% then he takes 15 minutes less to cover the same distance. Find the time taken by him to cover the distance at the original speed.
   1] 45 min 2] 30 min 3] 22.5 min 4] Indeterminate

2. A car covers a certain distance going at a speed of 60 km/hr and returns back to the starting point at a speed of 40 km/hr. Find the average speed of the whole journey.
   1] 42 kmph 2] 50 kmph 3] 48 kmph 4] 52 kmph

3. A train 180 m long is running at a speed of 90 km/hr. How long will it take to pass a railway signal?
   1] 2 sec 2] 7.2 sec 3] 7 sec 4] 8 sec

4. A train whose length is 150 m passes a telegraph pole in 10 sec. Find the speed of the train in km/hr.
   1] 15 2] 4.17 3] 54 4] 45

5. A train 120 m long covers a 160 m long railway platform in 14 sec. How long will it take to pass another platform which is 150 m long?
   1] 12.5 sec 2] 12 sec 3] 13 sec 4] 13.5 sec

6. What is the time taken by a train running at 18 km/hr to cross a man standing on a platform, given that the length of the train is 120 m?
   1] 20 sec 2] 24 sec 3] 18 sec 4] 32 sec

7. How long will a train 150 m long and travelling at a speed of 45 km/hr take to cross a platform of length 250 m?
   1] 40 sec 2] 24 sec 3] 32 sec 4] 18 sec

8. Find the length of the bridge which a train 200 m long travelling at 36 km/hr can cross in 40 sec.
   1] 200 m 2] 180 m 3] 2000 m 4] 190 m

9. Walking at $\frac{4}{5}^{th}$ of the usual speed, a man reaches 15 minutes late to his office. Find his usual time.
1] 60 min 2] 45 min 3] 75 min 4] 65 min

10. An lift in a 10 storey building travels at the rate of 1 floor every 15 sec. At the ground floor and the top floor the lift stops for 1 minute. How many round trips will the lift make during a 7 hour period?
1] 75 2] 65 3] 70 4] 60

11. How many minutes will it take a police car to respond to a signal "d" km away if the police car travels at X km/hr.
1] $\frac{d}{X}$ 2] $\frac{216d}{X}$ 3] $\frac{60d}{X}$ 4] $\frac{d}{60X}$

12. A woman can finish writing a document in $2\frac{1}{2}$ hours. If she reduces her speed by 25%, how long will she take to complete the job?
1] $4\frac{1}{3}$ hrs. 2] $3\frac{1}{3}$ hrs. 3] 3.2 hrs. 4] 3 hrs.

13. Walking at a speed of 3 km/hr a man takes 8 hours to complete a journey. How much time will he require if his speed is 4 km/hr to complete the journey?
1] 6 hrs. 2] 5.5 hrs. 3] 6.5 hrs. 4] 6 hrs. 40 min.

14. Walking at 4 km/hr a man reaches his office 5 minutes late. If he walks at 5 km/hr he will be 2.5 minutes early. Find the distance of his office from his house.
1] 2.5 km 2] 2 km 3] 3 km 4] None of these

15. A girl rides her horse for 10 km at an average rate of 12 km/hr and for 12 km at an average rate of 10 km/hr. What is the average speed for the entire trip?
1] 10.80 km/hr 2] 11.80 km/hr 3] 11 km/hr 4] 12 km/hr

Answer Key

1-1	2-3	3-2	4-3	5-4	6-2	7-3	8-1	9-1	10-4
11-3	12-2	13-1	14-1	15-1					

Explanatory Answers

1. Speed is inversely proportional to time, i.e. Speed $\alpha \frac{1}{\text{Time}}$

If the speed increases to $\frac{3}{2}$ times the normal speed, then the time decreases to $\frac{2}{3}$ times the original time.

$\therefore \frac{1}{3}$ of original time = 15min.

Original time = $3 \times 15 = 45$ min
Hence [1]

2. Average speed = $\frac{2pq}{p+q} = \frac{2\times60\times40}{60+40} = \frac{2\times60\times40}{100} = 48$ km/hr

Hence [3]

3. $\text{Speed} = \dfrac{\text{Distance travelled by train}}{\text{Time}}$

$\dfrac{180}{\text{Time}} = 90 \times \dfrac{5}{18}$

Time = $\dfrac{36}{5} = 7\dfrac{1}{5}$ seconds.

Hence [2]

4. Speed = $\dfrac{\text{Length of train}}{\text{Time}} = \dfrac{150}{10} = 15$ m/s, i.e. $15 \times \dfrac{18}{5} = 54$ km/hr. Hence [3]

5. Distance travelled by train to cross a railway platform = length of train + length of platform

Speed = $\dfrac{120 + 160}{14} = \dfrac{280}{14} = 20$ m/s

$\therefore \dfrac{120 + 150}{t} = 20$

t = $\dfrac{270}{20} = 13.5$ sec.

Hence [4]

6. Speed = $\dfrac{\text{Length of train}}{\text{Time}}$

Time = $\dfrac{120}{18 \times 5} \times 18 = 24$ sec.

Hence [2]

7. Time = $\dfrac{150 + 250}{45} \times \dfrac{18}{5} = \dfrac{400 \times 18}{45 \times 5} = 32$ sec.

Hence [3]

8. Let the length of the bridge be 'x' m.

Speed = $\dfrac{\text{Length of train + Length of bridge}}{\text{Time}}$

$\therefore 36 \times \dfrac{5}{18} = \dfrac{200 + x}{40}$

$36 \times \dfrac{5}{18} \times 40 = 200 + x$

$400 = 200 + x$

$x = 200$ m

$\therefore$ Length of bridge = 200 m

Hence [1]

9. Speed is inversely proportional to time, i.e. Speed $\alpha \dfrac{1}{\text{Time}}$

When speed is $\dfrac{4}{5}$ of usual speed, time increases to $\dfrac{5}{4}$ of his usual time

Let the original time be t min.

$\therefore \dfrac{5}{4}t - t = \dfrac{1}{4}t$

$\therefore \dfrac{1}{4}$ th. of his usual time = 15 min., i.e. $\dfrac{1}{4}t = 15$

$\therefore$ Usual time = $15 \times 4 = 60$ min = 1 hr. Hence [1]

10. Time taken for one trip up = $1 \times 60 + 15 \times 10 = 60 + 150 = 210$ sec
Time taken for one trip down = $1 \times 60 + 15 \times 10 = 60 + 150 = 210$ sec
$\therefore$ Total trip up and down = 420 sec
$\therefore$ In 7 hours, number of trips = $\frac{7 \times 60 \times 60}{420} = 60$ trips. Hence [4]

11. Police car will take $\left(\frac{d}{X} \times 60\right)$ min. to respond, i.e. $\left(\frac{60d}{X}\right)$ minutes.
Hence [3]

12. She reduces the speed by 25 %, i.e. speed is reduced by $\frac{1}{4}$ th of her original speed.
The current speed = $\frac{3}{4}$ th. of the original speed.
Since the time is inversely proportional to speed, she will take $\frac{4}{3}$ of the original time.
Original time = 2.5 hrs.
Current time = $\frac{4}{3} \times 2.5 = 3\frac{1}{3}$ hrs., i.e. 3hrs. 20min. Hence [2]

13. Speed = 3 km/hr and time = 8 hrs.
Distance covered = $3 \times 8 = 24$ km.
When the speed is 4 km/hr the time taken = $\frac{24}{4} = 6$ hrs. Hence [1]

14. Let the distance be x km and the original time taken be t hrs.
When he walks at 4 km/hr, he reaches the office by 5 min. late.
Hence, $\frac{x}{4} = t + \frac{5}{60}$
When he walks at 4 km/hr, he reaches the office by 2.5 min. early.
$\frac{x}{5} = t - \frac{2.5}{60}$
$\therefore \frac{x}{4} - \frac{x}{5} = \frac{7\frac{1}{2}}{60}$
$\frac{x}{20} = \frac{15}{2 \times 60}$
$x = \frac{15}{6} = 2\frac{1}{2}$ km. Hence [1]

15. Average speed for the entire journey = Total Distance / Total Time.
$\frac{(10+12)}{\left(\frac{10}{12}+\frac{12}{10}\right)} = \frac{22}{\frac{50+72}{60}} = \frac{1320}{122} = 10.81$. Hence [1]

Practice Questions B (Optional but Important)

1. Jack has to travel from London to Manchester, a distance of 200 km. He starts at 8.00 am and travels at 40 kmph. However at 10.00 am, he realized that he needs to reach Manchester at 12.00 noon. What should be the speed at which he should travel the remaining distance?
1] 120 kmph 2] 70 kmph 3] 60 kmph 4] 48 kmph

2. Betty started from her home to school 15 minutes before her normal time. She slowed her driving speed by 20% and reached her school at normal time. What is the time taken for her to travel between her home and the school today?
1] 75 min 2] 60 min 3] 70 min 4] None of these

3. How long will a train 200 m long and travelling at 108 km/hr take to cross a man sitting in another train that is 100 m long and moving at 72 km/hr? providing the answer if the trains are travelling
a) In the same direction b) In opposite direction
1] 20 sec & 4 sec 2] 20 sec & 5 sec
3] 10 sec & 5 sec 4] 20 sec & 6 sec

4. A train crosses two people cycling in the same direction in 10 sec and 15 sec respectively. If the speeds of the two cyclists are 9 and 18 km/hr respectively, find the length of the train.
1] 105 m 2] 200 m 3] 75 m 4] None of the above

5. There are 2 bridges exactly 2 km away on a river. A fisher man rowing a boat drops his cap under the first bridge. After crossing the second bridge, he realizes his mistake and decides to row back and get the cap. He finds his cap exactly under the second bridge. How far did he travel before he realized his mistake? [Given that speed of stream = 2 km/hr and speed of boat in still water = 4 km/hr]
1] 2.5 km 2] 2.75 km 3] 3 km 4] 1 km

6. While Rohan went to another town, he covered 40 km by car at 60 km/hr, another 80 km by train, going at 100 km/hr and the remaining 20 km was covered by bus, travelling at 50 km/hr. What is the average speed for the whole journey?
1] 75 km/hr 2] 70 km/hr 3] 53.33 km/hr 4] 65 km/hr

7. A swimmer goes downstream and covers the distance between two ports in 4 hours while he covers the same distance upstream in 5 hours. If the speed of the stream is 2 km/hr, find the speed of the swimmer in still water.
1] 15 km/hr 2] 17 km/hr 3] 13.33 km/hr 4] 18 km/hr

8. Two towns P and Q are 110 km apart. A motor cycle rider starts from P towards Q at 7:00 am at a speed of 20 km/hr. Another rider starts from Q towards P at 8.00 am at a speed of 25 km/hr. Find when they will cross each other.
1] 9: 45 am 2] 9: 30 am 3] 10 am 4] None of these

Answer Key

1-3 2-1 3-1 4-3 5-3 6-1 7-4 8-3

Explanatory Answers

1. Distance travelled in 2 hours, i.e. from 8.00-10.00 am = Speed × Time = 40 × 2 = 80 km.
Distance to Manchester at 10.00 am = 200 – 80 = 120 km.
Time to Manchester = 2 hours

$\therefore$ Speed at which Jack has to travel = $\frac{120}{2}$ = 60 kmph

He needs to cover the remaining distance at 60 kmph

2. On any normal day, Betty has a fixed speed, 's' and fixed amount of time 't' to reach her school. So the distance 'd' is,

$d = s \times t$

In the given situation, Betty travelled an extra 15 minutes, but a reduced speed. So Betty covered the distance, 's' at speed $s \times \frac{80}{100}$ and in t + 15 minutes

$d = \left(s \times \frac{80}{100}\right)(t + 15)$

As the distance is the same, we can say that, $s \times t = \left(s \times \frac{80}{100}\right)(t + 15)$

Hence, $t = \frac{4}{5}t + 12$, i.e. $\frac{1}{5}t = 12$ or t = 60 minutes.

Hence, today she has taken t + 15 = 75 minutes.

3. **Same direction**

Distance to be covered = 200m (why not 200 + 200 =400? think!!)

Because the train is crossing the person whose length is insignificant.

Relative speed = 108 – 72 = 36 km/hr = 10 m/s (why do we use relative speed here & not 108 km/hr?). Because the person sitting in the train is moving at 72 km/hr. Hence, the relative speed is to be considered.

$\therefore$ Time taken $= \frac{\text{Distance}}{\text{Relative speed}} = \frac{200}{10} = 20$ sec.

Opposite direction

Distance to be covered = 200m (why not 200 + 200 =400? think!!)

Because the train is crossing the person whose length is insignificant.

Relative speed = 108 + 72 = 180 km/hr = 50 m/s (why do we use relative speed here & not 108 km/hr?). Because the person sitting in the train is also moving at 72 km/hr. Hence, the relative speed is to be considered. $\therefore$ Time taken $= \frac{\text{Distance}}{\text{Relative speed}} = \frac{200}{50} = 4$ sec.

4. 9 km/hr = $9 \times \frac{5}{18}$ = 2. 5 m/s and 18 km/hr = $2 \times 2.5 = 5$ m/s

Time taken for the train to cross the first cyclist = $\frac{\text{Length}}{\text{Relative speed}} = \frac{L}{S - 2.5} = 10$

$\therefore L = 10S - 25$ (1)

Time taken for the train to cross the second cyclist = $\frac{\text{Length}}{\text{Relative speed}} = \frac{L}{S - 5} = 15$

$\therefore L = 15S - 75$ (2)

(2) – (1) => 5S – 50 = 0

S = 10 m/s. Substituting the value of S in equation (1), we have

$L = 10 \times 10 - 25 = 75$ m.

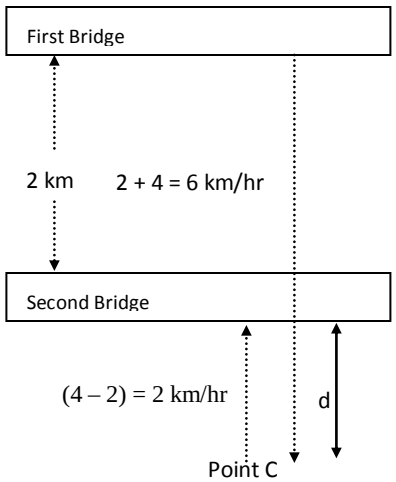
5. Let the total time from the moment he drops his cap till he collects it back is = t, in time t the cap has travelled 2 km, (i.e. the distance between the bridge) at 2 km/hr.

Hence, t = 1 hr.

Let 'C' be the point at which he realizes that he has dropped the cap, then

Time taken till point C = $\frac{2 + d}{6}$ hrs.

Time taken from C back to 2nd bridge = $\frac{d}{2}$ hrs.



$\therefore \frac{2+d}{6} + \frac{d}{2} = 1$

i.e. 2 + d + 3d = 6

4d = 4

d =1km

Hence, the fisherman has travelled 3 km downstream before he realized his mistake. Fisherman has travelled for $\frac{1}{2}$ hr on each direction.

6. Average speed = $\frac{\text{Total distance covered}}{\text{Total time}}$

Time taken in car = $\frac{40}{60} = \frac{2}{3}$ hrs.

Time taken in train = $\frac{80}{100} = \frac{4}{5}$ hrs.

Time taken in bus = $\frac{20}{50} = \frac{2}{5}$ hrs.

Average speed = $\frac{40+80+20}{\frac{2}{3}+\frac{4}{5}+\frac{2}{5}} = \frac{140}{\frac{10+12+6}{15}} = \frac{140\times 15}{28}$ = 75 km/hr

7. Let the speed of the swimmer in still water = x km/hr
Speed downstream = (x + 2) km/hr
Speed upstream = (x – 2) km/hr
Distance travelled in 4 hours while going down stream = 4 (x + 2) km
Distance travelled in 5 hours while going upstream = 5(x – 2) km
4(x + 2) = 5 (x – 2) =>4x + 8 = 5x – 10

x = 18 km/hr.

8. Let the two riders meet after x hours from the time the first rider starts.
20x + 25 (x – 1) = 110
20x + 25x – 25 = 110
45x = 135
x = $\frac{135}{45}$ = 3 hrs.

∴They meet at 10 am.

Chapter 13 Races

Suggested Time : 120 minutes

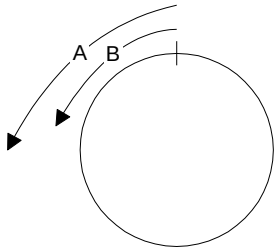
Few standard terms used in the questions

1. **Dead heat:** A race where there is no single winner, i.e. all the contestants reach the goal at the same time.
2. If B starts say 20 m ahead of A, then A is said to have given B a **head-start/start** of 20 metre or B has a 20 m advantage over A.

RACES ON CIRCULAR TRACKS

Let us consider that A and B race on a circular track with a circumference of 400 m at speeds of 50 m/s and 40 m/s respectively. They start form the same point at the same time, and travel in the same direction. Let us try solving the following questions.

I. When would they meet for the first time after they start their race?
II. Where would they meet for the 2nd and 3rd time after they start the race?
III. When would they meet for the first time at the starting point?



I. A would meet B again only when he covers the entire track and catches B from behind. This concept is similar to the scheme of 'lapping' in an Formula-1(F1) race.

A's relative speed with respect to B = 10 m/s (why do we take relative speed here?)
Distance = 400 m

Time taken for them to meet each other for the first time = $\frac{400}{10}$ = 40 sec.

So they meet for the first time in 40 seconds.

Alternate approach

Let us visualize this in another way. What if B was not moving at all. Hence, A would require to go around the circle once and he would be able to catch up with B.

Now, what if B totally moved 5 meters and stopped. Here A has to go around the circle once and come to the starting point. Then he has to travel the distance that B has traveled, i.e. 5 meters.

What if B manages to complete a full circle. Then A who was ahead of B to start of with comes back for the first time to the starting point, and not finding B (who is somewhere behind him and has not yet completed a circle) travels another round. At that time B is at the starting point. So A travels 2 circles and B travels 1 circle.

In other words, whatever distance B has traveled, A has to cover that entire distance plus one full circle to catch up with B.

Now, just think when would they meet for the second time, 3rd time etc?

II. They would meet each other every 40 seconds, i.e. for the 2nd and 3rd times they would meet each other after 80 and 120 seconds respectively.

This is because after meeting for the first time, it is like starting the race all over again. The only difference is that the starting point has now changed to where they meet for the first time. Hence in 40 seconds, A

catches up with B again. Now, a new race starts, with the starting point as the place where they meet for the 2nd time. Hence A will take another 40 seconds to catch up with B. i.e. A will catch up with B every 40 seconds uniformly.

III. To meet at the starting point, both A & B will have to be at the starting point.

A comes to the starting point every $\frac{400}{50}$ seconds and B comes to the starting point every $\frac{400}{40}$ seconds.

Hence, together they would be at starting point at LCM of $\left(\frac{400}{50}, \frac{400}{40}\right)$ seconds, i.e. 40 seconds.

IV. Therefore,

If two people A & B with speeds a & b respectively (a > b) are running around a circular track of circumference L metres starting at the same time, then

Time taken to meet for the 1st time ever = $\frac{L}{a-b}$ (If A & B are running in the same direction)

Time taken to meet for the 1st time ever = $\frac{L}{a+b}$ (If A & B are running in the opposite direction)

Time taken to meet for the 1st time at starting point is LCM of $\left(\frac{L}{a}, \frac{L}{b}\right)$

V. If three people A, B and C with speeds a, b and c respectively (a > b > c) are running around the same circular track of circumference L m and start at the same time, then

Time taken for the three to meet for the first time ever = LCM of $\left(\frac{L}{a-b}, \frac{L}{b-c}\right)$.

This is because we can say that the three meet when one of them meets the other two simultaneously, i.e. in the above formula, the point of reference is B, and the three meet when B meets with A, and B meets with C at the same time.

Obviously, we can change the point of reference to A or C.

Also time taken for the three to meet for the 1st time at the starting point = LCM of $\left(\frac{L}{a}, \frac{L}{b}, \frac{L}{c}\right)$

Practice Questions A (Optional)

1. In a 400 m dead heat race, A gives B a start of 40 m. What is the ratio of speed of A to that of B?

1] 1 : 2 2] 2 : 1 3] 10 : 9 4] 9 : 10

2. Two people A & B running around a circular track of 400 m at speeds in the ratio 10: 11. When will they meet for the first time at the starting point, if they run in (a) the same direction and (b) opposite direction?

1] (a) $\frac{400}{21}$ sec. and (b) $\frac{400}{31}$ sec. 2] (a) 20 sec. and (b) 14 sec.

3] Cannot be answered 4] None of these

3. X runs $1\frac{1}{2}$ times as fast as Y. In a race, if X gives a lead of 50 m to Y, find the distance from the starting point at which both of them will meet.

1] 100 m 2] 150 m 3] 175 m 4] 140 m

4. In a 1000 m race, Ram beats Shyam by 50 m and in the same race, Shyam beats Karan by 50 m. By what distance does Ram beat Karan?

1] 97.5 m 2] 50 m 3] 95 m 4] 92.5 m

5. In a 500 m race, the ratio of speeds of two runners P & Q is 4 : 5. If P has a start of 100 m, then what is the distance between P and Q at the finish of the race?
1] 100 m 2] Both finish the race at the same time
3] 50 m 4] 25 m

6. In a circular race of 1000 m, X and Y start from the same point and at the same time with speeds of 18 km/hr and 36 km/hr respectively. Find when they will meet for the first time on the track when they are running in the same direction and opposite direction respectively.
1] 200 sec and $66\frac{1}{3}$ sec 2] 100 sec and $66\frac{2}{3}$ sec
3] 200 sec and $66\frac{2}{3}$ sec 4] 200 sec and $66\frac{1}{6}$ sec

7. A and B can run around a circular track in 6 and 8 minutes. If they start together and run in the same direction, when will they meet each other for the first time?
1] 16 min 2] 48 min 3] 24 min 4] Indeterminate

8. A and B run a 1760 m race. A gives B a start of 55 m and still beats him by 15 seconds. If A runs at 14.08 km/hr, then find the speed of B.
1] 13. 2 km/hr 2] 14 km/hr 3] 12.8 km/hr 4] 13.6 km/hr

Answer Key

1-3 2-3 3-2 4-1 5-2 6-3 7-3 8-1

Explanatory Answers

1. Dead heat implies that both reach the goal at the same time, i.e. A cover 400 m in the same time as B cover 360 m
Ratio of the speeds = Ratio of distances covered
= 400 : 360
= 10 : 9

Shortcut
Most of the questions can be cracked by looking at the answer options. In this case, we know A has to be faster than B since A has given B a start. Hence, options [1] and [4] cannot be the answer. Option [2] cannot be the answer because A would than have to give B a start of 200 m. So the answer is option [3].

2. Since the speed of A and B are not given, we cannot find the time after which they meet each other at the starting point.

3. Since X runs $1\frac{1}{2}$ times, i.e. $\frac{3}{2}$ times as fast as Y, the time Y runs 2 m X runs 3 m, i.e. X gains 1 m for every 3 m he runs. Since he gave lead of 50 m, they will meet at (3 × 50) m, i.e. 150 m.
Hence [2]

4.

Ram	Shyam	Karan
1000	950	
	1000	950

In the time Shyam Runs 1000 m, Karan runs 950 m.

$\therefore$ In the time Shyam runs 950 m, Karan runs $\frac{950 \times 950}{100} = 902.5$ m

$\therefore$ Ram beats Karan by 1000 – 902.5 = 97.5 m

Hence [1]

5. $\frac{\text{Speed of P}}{\text{Speed of Q}} = \frac{4}{5}$

$\therefore$ In the time P runs 400 m., Q runs 500 m. But P has a start of 100 m and he has another 400 m to cover. In the time taken by P to cover this, Q can cover the entire 500 m, thus finishing the race with P.

$\therefore$ P and Q finish the race at the same time.

Hence [2]

6. Let, L = length of track = 1000 m

A's speed = $18 \times \frac{5}{18} = 5$ m/s; B's speed = $36 \times \frac{5}{18} = 10$ m/s

When they are running in the same direction,

Time taken to meet for the first time $= \frac{L}{b-a} = \frac{1000}{10-5} = \frac{1000}{10-5} = 200$ sec

When they are running in opposite direction, Time taken to meet for the first time:

$\frac{L}{b+a} = \frac{1000}{10+5} = \frac{1000}{15} = 66\frac{2}{3}$ sec

Hence [3]

7. Let the length of track = L m

Speed of A = $\frac{L}{6}$ m/min; Speed of B = $\frac{L}{8}$ m/min

Since they are running in the same direction,

Time taken to meet for first time = $\frac{L}{\frac{L}{6} - \frac{L}{8}} = \frac{L}{\frac{4L-3L}{24}} = \frac{24L}{L} = 24$ min.

Alternately,

Time taken to meet for the first time = LCM of (6 min, 8 min) = 24 min

Hence [3]

8. Time taken for A $= \frac{1760}{14.08 \times 5} \times 18 = 450$ sec.

$\therefore$ Time taken for B = 450 + 15 = 465 sec.

$\therefore$ Speed of B = $\frac{1705}{465} = 3.67$ m/s = $3.67 \times \frac{18}{5} = 13\frac{1}{5}$ km/hr.

Hence [1]

Chapter 14 Clocks

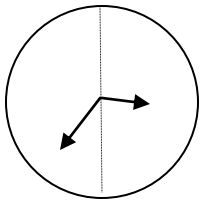
Suggested Time : 90 minutes

Question type-I Find the angle between the hour hand and minute hand at a particular time

The two cases to be considered under this question type are,

Case I : The H.H is before the M.H
Example

What is the angle between the minute hand and the hour hand when the time 3: 40?



When the time is 3:40, the H.H is before the M.H.
$\therefore$ Angle between the hands = Angle covered by M.H– Angle covered by H.H.

In 40 min, the M.H covers, 40×6^0

In 3 hrs 40 min, the H.H. covers $3 \times 30^0 + 40 \times \frac{1}{2}^0$

$\therefore$ Angle between the hands = $40 \times 6^0 - (3 \times 30^0 + 40 \times \frac{1}{2}^0) = 40 \times \left(6^0 - \frac{1}{2}^0\right) - 3 \times 30^0$

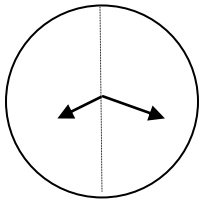
$= 40 \times \left(\frac{11}{2}^0\right) - 3 \times 30^0 = 220^0 - 90^0 = 130^0$

Therefore, when the time is H hrs M min,

If the H.H is before the M.H., then the angle between the hands = $\frac{11}{2} \times M - 30 \times H$

In the above example, H = 3 and M = 40.

Case II : The M.H is before the H.H
Example : What is the angle between the minute hand and the hour hand when the time 8: 20?



When the time is 8:20, the M.H is before the H.H.
$\therefore$ Angle between the hands = Angle covered by H.H– Angle covered by M.H.

In 20 min, the M.H covers, 20×6^0

In 8 hrs 20 min, the H.H. covers $8 \times 30^0 + 20 \times \frac{1}{2}^0$

$\therefore$ Angle between the hands = $8 \times 30^0 + 20 \times \frac{1}{2}^0 - 20 \times 6^0 = 8 \times 30^0 - 20 \times \left(6^0 - \frac{1}{2}^0\right)$

$= 8 \times 30^0 - 20 \times \left(\frac{11}{2}^0\right) = 240^0 - 110^0 = 130^0$

Therefore, when the time is H hrs M min,

If the M.H is before the H.H., then minute angle between the hands = $30 \times H - \frac{11}{2} \times M$

In the above example, H = 8 and M = 20.

In general,

At H hrs M min, the angle between the hour hand and minute hand = $\left|30 \times H - \frac{11}{2} \times M\right|$

Since we use the above derivation frequently, the reader is advised to understand and remember the derivation.
Example
What is the angle between the hour hand and minute hand when the time is 3.30?
Here H = 3 and M = 30

Hence, the angle between the hands = $\left|30\times 3-\frac{11}{2}\times 30\right| = |90-11\times 15| = 75^0$

Races and Clocks Similarity
Now, let us try a different approach to understand the question. In the races chapter, we started with asking the following question, albeit with a different set of numbers.
A and B race on a circular track with a circumference of 60 km at speeds of 60 kmph and 5 kmph respectively. They start from the same point at the same time, and travel in the same direction. When would they meet for the first time after they start their race?

In races, we answered the above question using the concept "Whatever distance B has traveled, A has to cover that entire distance plus one full circle to catch up with B".

Hence, the answer is $\frac{\text{Extra Distance that A has to travel}}{\text{A's extra speed}} = \frac{\text{Full Circle}}{\text{Relative Speed}} = \frac{60}{60-5}$.

Now, rewording the above question in the "Clocks" format,
The minute hand travels 60 dots in an hour, where as the hour hand travels only 5 dots in an hour. If the watch is set at 12.00, then would the two hands meet for the first time after they start their race?

Question type-II : **Find the time, when the angle between H.H and M.H is given**

For example,
At what time between 3 and 4 the angle between the hands of the clock is 75^0?

Let the time at which the angle between the H.H and M.H 75^0 be 3 hrs M min. The two cases that arise are,

Case I: The H.H is before M.H.

$\therefore \frac{11}{2}\times M - 30\times H = 75^0$

$\frac{11}{2}\times M - 30\times 3 = 75^0$

$\frac{11}{2}\times M = 90^0 + 75^0$

$M = \frac{165\times 2}{11} = \frac{330}{11} = 30$

i.e. at 3 hrs 30min the angle between the hands is 75^0.

Case II: The M.H is before H.H

$\therefore 30\times H - \frac{11}{2}\times M = 75^0$

$30\times 3 - \frac{11}{2}\times M = 75^0$

$\frac{11}{2}\times M = 90^0 - 75^0$

$M = \frac{15\times 2}{11} = \frac{30}{11} = 2\frac{8}{11}$

i.e. at 3 hrs $2\frac{8}{11}$ min the angle between the hands is 75^0

Practice Questions A (Optional)

1. At what time between 3 and 4 O'clock will the 2 hands of a clock be at an angle of 60° with each other?

   1] 3 hrs 27 $\frac{3}{11}$ min 2] 3 hrs $5\frac{5}{11}$ min 3] [1] or [2] 4] None of these

2. Find the time at which the hands of the clock are at right angles between 8 and 9 O'clock.

   1] 8 hrs $23\frac{7}{11}$ min 2] 8 hrs $27\frac{3}{11}$ min

   3] 8 hrs $33\frac{7}{11}$ min 4]8 hrs $26\frac{4}{11}$ min

3. At what time between 3 and 4 O'clock are the hands of a clock opposite to one another?

   1] 3 hrs $41\frac{9}{11}$ min 2] 3 hrs $49\frac{1}{11}$ min

   3] 3 hrs $53\frac{7}{11}$ min 4] 3 hrs $22\frac{8}{11}$ min

4. A clock which gains uniformly is 18 min slow at 7:25 pm on Sunday. If the clock shows the correct time at 12:00 midnight on Monday, then how much minutes did the clock gain per day?

   1] $15\frac{39}{343}$ 2] $15\frac{2}{11}$ 3] $15\frac{29}{243}$ 4] 15

5. In 24 hrs, how many times do the hands of a clock make right angles with each other?

   1] 12 2] 22 3] 44 4] 36

Answer Key

1-3 2-2 3-2 4-1 5-3

Explanatory Answers

1. Let at 3 hrs M min, the angle between the H.H and M.H. is 60^0

Case I: H.H is before M.H.

$\therefore \frac{11}{2} \times M - 30 \times H = 60^0$

$\frac{11}{2} \times M - 30 \times 3 = 60^0$

$\frac{11}{2} \times M = 90^0 + 60^0$

$M = \frac{150 \times 2}{11} = \frac{300}{11} = 27\frac{3}{11}$. i.e. at 3 hrs $27\frac{3}{11}$ min the angle between the hands is 60^0.

Case II: M.H. is before H.H.

$\therefore 30 \times H - \frac{11}{2} \times M = 60^0$

$30 \times 3 - \frac{11}{2} \times M = 60^0$

$\frac{11}{2} \times M = 90^0 - 60^0$

$M = \frac{30 \times 2}{11} = \frac{60}{11} = 5\ \frac{5}{11}$ i.e. at 3 hrs $5\ \frac{5}{11}$ min the angle between the hands is 60^0. Hence [3]

2. Let at 8 hrs M min the angle between the H.H and M.H. is 90^0

Case I: H.H. is before M.H.

$\therefore \frac{11}{2} \times M - 30 \times H = 90^0$

$\frac{11}{2} \times M - 30 \times 8 = 90^0$

$\frac{11}{2} \times M = 240^0 + 90^0$

$M = \frac{330 \times 2}{11} = \frac{660}{11} = 60$ This case is not possible because it happens at 9'O clock, not between 8 and 9.

Case II: M.H. is before H.H.

$\therefore 30 \times H - \frac{11}{2} \times M = 90^0$

$30 \times 8 - \frac{11}{2} \times M = 90^0$

$\frac{11}{2} \times M = 240^0 - 90^0$

$M = \frac{150 \times 2}{11} = \frac{300}{11} = 27\frac{3}{11}$, i.e. at 8 hrs $27\frac{3}{11}$ min the angle between the hands is 90^0. Hence [2]

3. Let at 3 hrs M min, hands of the clock be opposite to each other, i.e. angle between M.H and H.H is 180^0

Case I: H.H is before M.H.

$\therefore \frac{11}{2} \times M - 30 \times H = 180^0$

$\frac{11}{2} \times M - 30 \times 3 = 180^0$

$\frac{11}{2} \times M = 90^0 + 180^0$

$M = \frac{270 \times 2}{11} = \frac{540}{11} = 49\frac{1}{11}$

i.e. at 3 hrs $49\frac{1}{11}$ min. the angle between the hands is 180^0.

Case II: M.H is before H.H.

$\therefore 30 \times H - \frac{11}{2} \times M = 190^0$

$30 \times 3 - \frac{11}{2} \times M = 190^0$

$\frac{11}{2} \times M = 90^0 - 180^0$. This case is not possible because it does not lie between 3 and 4 O' clock.

Hence [2]

4. From 7:25 pm on Sunday to 12 midnight of Monday is a total of $(28 \times 60 + 35) = 1715$ min
During this time the clock gains 18 min

$\therefore$ In one day (24×60 min) the clock will gain $\frac{18}{1715} \times 24 \times 60 = 15\frac{39}{343}$ min. Hence [1]

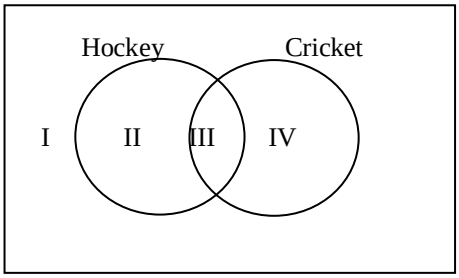
5. In 12 hours they are at right angle 22 times.

∴ In 24 hours they are at right angles $22 \times 2 = 44$ times. Hence [3]

Chapter 15 Sets

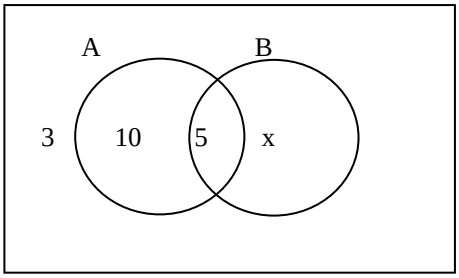
Suggested Time : 90 minutes

In a class, some students play hockey, some play cricket, some play both the sports, and some play neither. This has been represented in the Venn diagram below.



In the diagram above,
Region I represents students of the class who do not play hockey as well as cricket.
Region II represents students who play only hockey (and not cricket).
Region III represents students who play both the games.
Region IV represents students who play only cricket (and not hockey).
Region II and III represents students who play hockey.

Rule 1:
$n(A \cup B) = n(A) + n(B) - n(A \cap B)$.
Consider the diagram below. It is given that there are 15 students of type A, 16 of type B, and 5 of both types. 3 students are neither of type A nor type B.



If it is given that there are 16 people who are of type B, then please note that 16 represents the entire circle of B. Hence $x = 16 - 5 = 11$.
Hence, number of students in the class = $3 + 10 + 5 + 11 = 29$.

However, we could have also derived this value using the formula.
$n(A \cup B) = n(A) + n(B) - n(A \cap B)$.
i.e. total number of students of type A or B = student of type A + student of type B – students of both types.
= 15 + 16 – 5 = 26

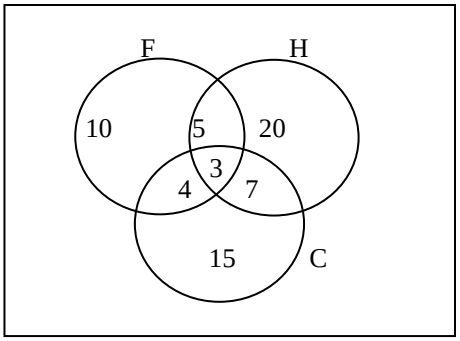
Hence, total number of students in class = type A or B + neither type A or B = 26 + 3 = 29.

Rule 2:
$n(A \cup B \cup C) = n(A) + n(B) + n(C) - n(A \cap B) - n(A \cap C) - n(B \cap C) + n(A \cap B \cap C)$.

Consider the question and the diagram given below.

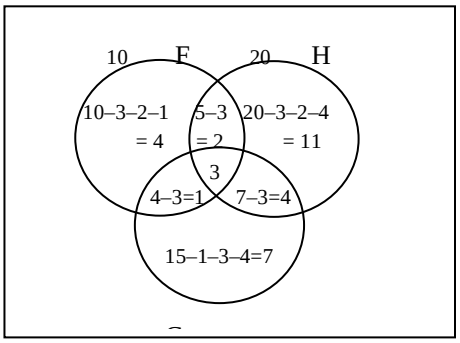
Question: 20 students play hockey, 15 play cricket, 10 play football, 3 play all the 3 games, 5 play football and hockey, 4 play football and cricket and 7 play hockey and cricket. How many students are there in the class if al students play atleast one of the three sports?

Wrong Approach



Right Approach

When it is given that 20 students play hockey, it means entire circle of hockey has 20 students. Similarly, when it is given that 5 students play football and hockey, it includes the 3 who play all three games. Hence the correct diagram is



Hence, total number of students = 4 + 2 + 11 + 1 + 3 + 4 + 7 = 32

Note: Start from the innermost part, i.e. intersection of the 3 circles. Then move out to intersection of two circles and finally calculate for regions like only H or only F or only C.

Alternately using rule 2, we have n(FUHUC) = n(F) + n(H) + n(C) – n(F∩H) – n(F∩C) – n(H∩C) + n(F∩ H ∩C).
= 10 + 20 + 15 – 5 – 7 – 4 + 3
= 32

Practice Exercise A (Important)

1. In a school 100 students study finance and 120 students study marketing. 40 students study both and 20 study neither. What is the total number of students?
1] 180 2] 200 3] 240 4] 260

2. In a school, only 3 specialisations were offered: Marketing, Finance and HR, and each of the 300 students are required to take at least one. 97 students took HR only and 73 took Finance only. How many took marketing or a combination of more than one specialisations?
1] 130 2] 154 3] 170 4] Indeterminate

3. If of an outgoing batch in a school, everyone has taken either or both of two specialisations: Finance and Marketing, and 70% have taken Finance and 60% Marketing; what percentage have taken both?
1] 20% 2] 30% 3] 40% 4] None of these

4. In Bangalore, everyone reads at least one paper and no one reads more than two. If 85% read Indian Express and 65% read the Hindu, what is the maximum possible percentage of people that read only Financial Express? Assume that there are only three publications in Bangalore?
1] 15% 2] 25% 3] 35% 4] 20%

5. In the question above, what is the maximum possible percentage of people reading Financial Express?
1] 25% 2] 35% 3] 50% 4] 65%

6. In a social gathering, 100 people wore yellow pants, 120 people green shirts, and 80 people red shoes. 30 people wore both yellow pants and green shirts, 20 people wore both green shirts and red shoes, and 25 people wore both red shoes and yellow pants. 10 people wore all the three. Now answer the following questions.

I. How many people wore yellow pants and green shirts but not red shoes?
1] 30 2] 25 3] 20 4] 15

II. How many people wore yellow pants but not green shirts or red shoes?
1] 45 2] 50 3] 55 4] 60

III. What was the total number of people in the gathering if everyone wore either red shoes, yellow pants, or green shirts?
1] 300 2] 225 3] 245 4] 235

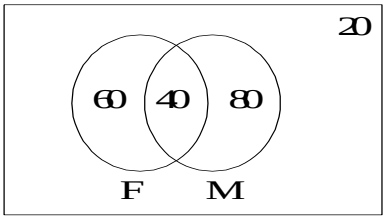
7. In a class of 70 students, one could learn table, sarod, or sitar. The enrolment for each of these classes was 4x each. Those who were taking both sitar and sarod, or sarod and tabla, or sitar and tabla were 2x in each case. x people took all three. Find the number of people who took only two classes.
1] 30 2] 40 3] 50 4] None of these

Answer Key

1-2 2-1 3-2 4-1 5-3 6(I)-3 6(II)-3 6(III)-4 7-1

Explanatory Answers

1. Number of students studying Finance only = 100 – 40 = 60.
Number of students studying Marketing only = 120 – 40 = 80.

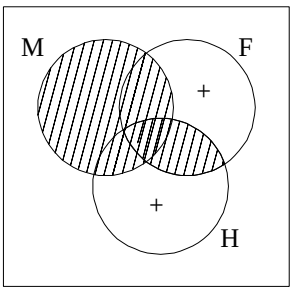


∴ Total number of students
= Finance only + Marketing only + Both + Neither
= 60 + 80 + 40 + 20 = 200
Hence, (2).

Using Set theoretic notations:
$F \cup M = F + M - F \cap M = 100 + 120 - 40 = 180.$
∴ Total number of students
= Finance or Marketing ($F \cup M$) + Neither ($\overline{F} \cap \overline{M}$)
= 180 + 20 = 200

2. Number of students studying Marketing or a combination of specialisations
= All students – Number of students taking either only Finance or only HR
= 300 – (97 + 73) = 130
Hence, (1).

Using Set notations:



$M \cup (M \cap F) \cup (F \cap H) \cup (M \cup H) = M \cup (F \cap H)$

Now, $[M \cup (F \cap H)]^{1}$

$= M^{1} \cap (F \cap H)^{1}$ = HR only + Finance only = 97 + 73 = 170

$M \cup (F \cap H) = 300 – 170 = 130$

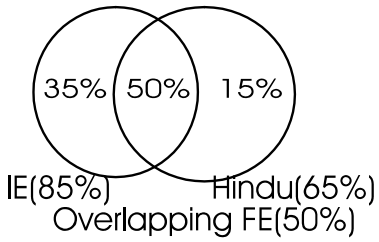
3. Percentage that has taken both
= 70% + 60% – 100% = 30%
Hence, (2).

$F \cup M = F + M – F \cap M$,

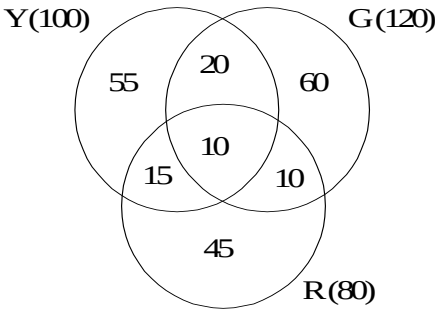
or $F \cap M = F + M – F \cup M = 70\% + 60\% – 100\% = 30\%$

4. Assume that those who read the Hindu also read IE.
So, if 85% read IE, 15% of them can read Financial Express. Hence(1).

5. Here, the trick is to minimise overlap between IE and Hindu, just the opposite of what we did above. 85% read IE and 15% do not. These 15% and another 50% read Hindu. This 50% overlaps with IE. So, FE readers include the 35% that read IE and not Hindu, and the 15% that read Hindu and not IE, i.e. 50% in total. Hence, (3).



6.



(Only green is 80)

$Y = 100$ $\quad Y \cap G = 30$

$G = 120$ $\quad G \cap R = 20$ $\quad Y \cap G \cap R = 10$

$R = 80$ $\quad R \cap Y = 25$

$\therefore \quad Y \cap G \cap \overline{R} = 30 - 10 = 20$ (only yellow and green)

$G \cap R \cap \overline{Y} = 20 - 10 = 10$ (only green and red)

$R \cap Y \cap \overline{G} = 25 - 10 = 15$ (only red and yellow)

$\therefore \quad Y \cap \overline{G} \cap \overline{R} = Y - Y \cap G - Y \cap R \cap \overline{G} = 100 - 30 - 15 = 55$ (only yellow)

Similarly, only green = 120 – 30 – 10 = 80, and
only red = 80 – 25 – 10 = 45

$\therefore$ Total
= G + yellow only + red only + yellow and red only
= 120 + 55 + 15 + 45 = 235

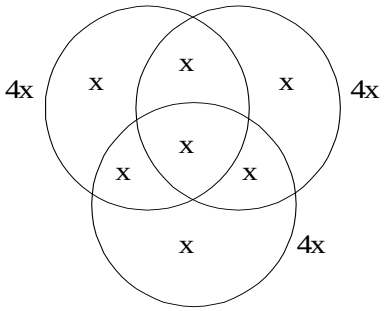
Hence, (I) -> (3), (II) -> (3), (III) -> (4).

Alternately according to (III),
Total number
$= Y \cup G \cup R$
$= Y + G + R - (Y \cap G + G \cap R + Y \cap R) + Y \cap G \cap R$
= 100 + 120 + 80 – (20 + 30 + 25) + 10 = 235
Hence, (4).

According to (I), yellow and green only
= yellow and green – all three
= 30 – 10 = 20
Hence, (3).

According to (II), yellow only
= yellow less (yellow and red) less (yellow and green only)
= 100 – 25 – 20 = 55
Hence, (3).

7. Total number
= (Table + Sitar + Sarod) – (The three taken pair wise) + (All three taken together)
or
$70 = 4x + 4x + 4x - (2x + 2x + 2x) + x \qquad \therefore \; 7x = 70$ or $x = 10$
Only two classes
= (both classes less all three) × 3
$= (2x - x) \times 3 = 3x = 30$



Hence, (1).

Practice Exercise B (Important but Optional)

1. In a class, 30 students do English and 20 students do Maths, and some students among these do both. How many more students do only English as compared to those doing only Maths?
1] 10 2] 15 3] 20 4] Indeterminate

Directions for questions 2-4:
In a class where students take either English or Maths or both, 25 boys take English and 28 girls take Maths. 20 girls take only English and 30 boys take only Maths. 3 girls take both subjects. Now answer the following questions.

2. How many students are there in the class?
1] 100 2] 103 3] 97 4] Indeterminate

3. How many students take English?
1] 48 2] 50 3] 45 4] Indeterminate

4. How many boys are there in the class?
1] 50 2] 55 3] 60 4] Indeterminate

Directions for questions 5-7:
In a certain school 21 players are on the basketball team, 26 on the hockey team, and 29 on the football team. It is known that 14 play hockey and basketball, 12 play football and basketball, 15 play hockey and football, and 8 play all the three games. For each of the following questions choose the best alternative.

5. The total number of players in the school is
1] 58 2] 45 3] 10 4] 43

6. The number of players who play football and basketball but not hockey is
1] 8 2] 4 3] 15 4] 8

7. The number of players who play only football and no other game is
1] 10 2] 8 3] 18 4] 20

DIRECTIONS for questions 8 to 10 :
A survey was conducted in a city to determine the choice of channel (DD, BBC and CNN) among viewers in viewing the news. The viewership for these three channels is 80, 22 and 15 percent respectively. Five percent of the respondents do not view the news at all.

8. What is the maximum percent of viewers who watch all the three channels?
1] 22 2] 15 3] 11 4] None of these

9. If ten percent of the respondents view DD and CNN and 12 percent view DD and BBC, then what percent of the respondents view only CNN and BBC?
1] zero 2] 2 3] 8 4] Indeterminate

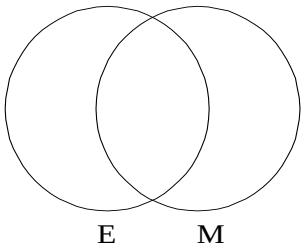
10. If 10% of the respondents view DD and CNN and 12% view DD and BBC then what percent of the respondents view all the three channels?
1] zero 2] 2 3] 8 4] Indeterminate

Answer Key

1-1 2-2 3-1 4-3 5-4 6-2 7-1 8-3 9-4 10-4

Explanatory Answers

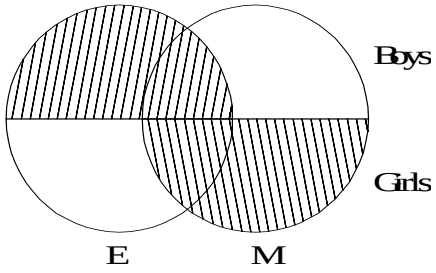
1. (Only English) – (Only Maths) = (Only English + Both) – (Only Maths + Both)
= (Total English) – (Total Math) = 30 – 20 = 10. Hence, (a).



Set theoretically, $(E \cap \overline{M}) - (M \cap \overline{E})$

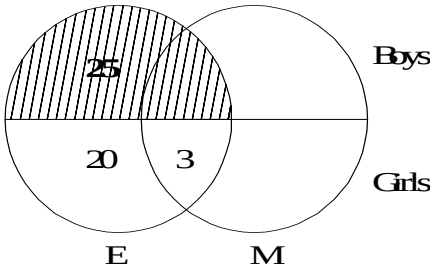
$= [(E \cap \overline{M}) + (E \cap M)] - [(M \cap \overline{E}) + ((M \cap E)] = E - M = 30 - 20 = 10$

2.



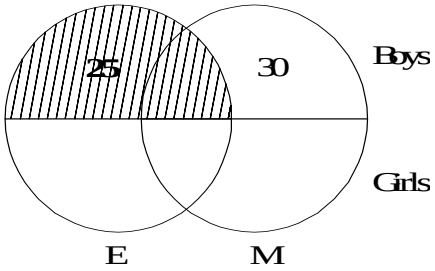
Note that the whole
= Boys taking English + Girls taking Maths + Boys taking only Maths + Girls taking only English
= 25 + 28 + 20 + 30 = 103. Hence, (2).

3.



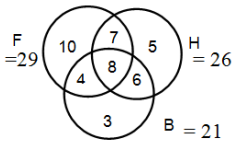
Number of students taking English
= Boys taking English + Girls taking only English + Girls taking both subjects
= 25 + 20 + 3 = 48. Hence, (1).

4.



Number of boys = Boys taking English + Boys taking only Maths
= 25 + 30 = 55. Hence, (3).

5-7. Using the given information, we can draw the Venn diagram as shown.



8. Eliminating answer choices. 22% is not possible as CNN viewership is 15%. If 15% is maximum of viewers then others will not add upto 100%, with 11% as maximum viewership who watch all the three others % can be adjusted to get total of 11%.

9. There are two unknowns and only one equation.

10. There are two unknowns and only one equation.

Chapter 16 Series Completion

Suggested Time : 60 minutes

Series completion questions are common in aptitude tests. They range from being very simple to highly complicated. There are various structures in which the questions can be asked. Here we list some of the most popular structures and also list the often used logic to create these questions. Please note that sometimes even if the question is easy and the answer is obvious, sometimes you may yet be unable to solve it. Hence it is best that you do not give more than 60-90 seconds per question.

1. Find the next term in the given series: 1, 2, 4, 7, 11, 16,?

In the series, the difference between consecutive numbers is increasing by 1. Hence the next term is 16 + 6 = 22.

2. Find the missing term in the series: 2, 6, 12,?, 30, 42, 56…

Here the series is $1^2 + 1$, $2^2 + 2$, $3^2 + 3$, …
Hence the missing term is $4^2 + 4=20$

Alternately, we could answer the question using another logic. The difference between consecutive numbers is 4, 6, x, y, 12, 14. This forms a series where the difference is increasing by 2. Hence x and y must be 8 and 10. Therefore, the missing term is 12 + 8 = 20.

3. Find the incorrect term in the series: 1, 2, 3, 5, 7, 11, 13, 17, 19…

Here all numbers other than 1 are prime numbers. Hence 1 is the incorrect term. You may think that the incorrect term is 2 as it is even and the other numbers are odd. But we are looking for a series and the remaining numbers should form a series, but some of the odd numbers such as 9 and 15 are missing. Hence, 'not prime' logic works better.

4. Find the next two terms of the series: A, B, E, D, I, G, O, J, _, _

This is a tricky one. Look closely and you will see that all vowels are coming as alternate terms. Hence there are two different series combined to form a single one. The two series are A, E, I, O, … and B, D, G, J,

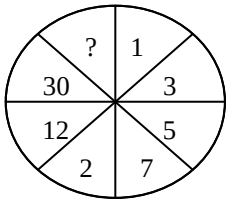
In the first series, the next term is U. In the second series, one consonant is missing after each term, and vowels do not form a part of the second series. i.e. C is missing between B and D, F is missing between D and G, H is missing between G and J. So the next term is L so that K is missing between J and L.

Hence the next two terms in the given series are U and L.

5. The first row follows a particular logic. Based on this logic, find the missing term in the second row.
Row 1: 1, 5, 25, 125, 625.
Row 2: 1, 3,?, 27, 81.

Here the series in the first row is 5^0, 5^1, 5^2, 5^3, …
Hence the missing number is the second row is $3^2 = 9$.

6. Identify the missing number in the figure.



Here, there are four odd numbers : 1, 3, 5 and 7 that form one series. Diagonally opposite numbers to each of these numbers are related. Example, $1^2 + 1 = 2$; $3^2 + 3 = 12$; $5^2 + 5 = 25$.

Hence the missing term is $7^2 + 7 = 56$

Practice Exercise A (Important)

For Questions 1-10, find the number that comes next in the sequence.

1. 54, 27, 30, 15, 18, 9, __
2. 4, 8, 11, 22, 25, 50, __
3. 2, 6, 9, 27, 30, 90, __
4. 1, 3, 3, 8, 6, 15, 10, 24, __
5. 2, 4, 7, 11, 16, 22, __
6. 5, 10, 17, 26, 37, 50, _
7. 7, 14, 23, 34, 47, 62, __
8. 2, 4, 12, 48, 240, __
9. 1, 5, 4, 10, 9, 15, 16, 20, __
10. 124, 235, 346, 457, 568, __

For questions 11 to 15, find the missing letter.

11. P R T V X _
12. U T S R _
13. O P R S V _
14. E D C B _
15. B E H _

For questions 16 to 17, given in bold figures is a series in each of the questions. The second lines follow a similar logic with one of the numbers given, the other being a, b, c, d, e. Find the number that will figure in the place of the alphabet asked in the question.

16.

	3	7	13	19	29	37
	5	a	b	c	d	e

What comes in place of 'c'?

(1) 11 (2) 17 (3) 31 (4) 23 (5) 41

17.

	4	17	66	259	1028	4101
	3	a	b	c	d	e

What comes in place of 'c'?

(1) 84 (2) 81 (3) 82 (4) 729 (5) 243

For questions 18 to 20: Find the odd one out

18. 217, 343, 154, 231, 176, 294

(1) 217 (2) 231 (3) 176 (4) 294 (5) 154

19. 1024, 256, 4096, 32, 16, 64

(1) 256 (2) 4096 (3) 64 (4) 32 (5) 1024

20. 24, 720, 120, 6, 72, 5040

(1) 24 (2) 5040 (3) 6 (4) 72 (5) 120

Answer Key

1-12	2-53	3-93	4-15	5-29	6-65	7-79	8-1440	9-25	10-679
11-Z	12-Q	13-W	14-A	15-K	16-4	17-1	18-3	19-4	20-4

Explanatory Answers

1. 54, (÷2)27, (+3)30, (÷2)15; (+3)18,(÷2)9, (+3)12

2. 4, (×2)8, (+ 3)11, (×2)22, (+ 3)25, (×2)50, (+ 3)53

3. 2, (×3)6, (+ 3)9, (×3)27, (+ 3)30, (×3)90, (+ 3)93

4. Alternate terms: 1, (+ 2)3, (+ 3)6, (+ 4)10, (+ 5)15; 3, (+ 5)8, (+ 7)15, (+ 9)24

5. 2, (+ 2)4, (+ 3)7, (+ 4)11, (+ 5)16, (+ 6)22, (+ 7)29

6. 5, (+ 5)10, (+ 7)17, (+ 9)26, (+ 11)37, (+ 13)50, (+ 15)65

7. 7, (+ 7)14, (+ 9)23, (+ 11)34, (+ 13)47, (+ 15)62, (+ 17)79

8. 2, (×2)4,(×3)12, (×4)48, (×5)240, (×6) 1440

9. Alternate terms: 1, (+ 3)4, (+ 5)9, (+ 7)16, (+ 9)25; 5,(+ 5)10, (+ 5)15, (+ 5)20

10. 124, (+ 111)235, (+ 111)346, (+ 111)457, (+ 111)568, (+ 111)679

11. P(Q)R(S)T(U)V(W)X(Y)Z

12. U T S R Q (This is Q R S T U in reverse order)

13. O P(Q) RS(TU) VW (XYZ)

14. E D C B A (This is A B C D E in reverse order)

15 B(CD)E(FG)H(IJ)K

16. The main series is formed by alternate prime numbers starting with 3.
In the same way the second series would be formed by alternate prime numbers starting with 5.
i.e, 5, 11, 17, 23… Hence (4)

17. First series:– 4, 4×4 + 1, 4×4×4 + 2…; 2nd series:– 3, 3×3 + 1, 3×3×3 + 2, …Hence (1)

18. 217, 343, 154, 231, 294 are all multiples of 7 and 176 is a multiple of 8 so the option (3)

19. 1024, 256, 4096, 16, 64 are all powers of 4, 32 is a power of 2.

20. 24, 720, 120, 6, 5040 are all factorials. So the odd one is 72.

Practice Test B (Important)

For Questions 1 to 10, find the number that comes next in the sequence.

1. 0, 7, 26, 63, __
2. 7, 3, 6, 7, 5, 11, __
3. 3, 9, 21, 45, __
4. 12, 23, 45, 56, 78, 89, 111, 122, __
5. 8, 14, 26, 50, __
6. 2, 5, 18, 41, __
7. 0, 3, 8, 15, __
8. 5, 12, 7, 9, 9, 6, __
9. 65, 35, 17, 11, __
10. 1, 2, 9, 28, __

For questions 11 to 15: Find the odd one out

11. 6561, 2187, 19683, 729, 243, 189

(1) 2187 (2) 19683 (3) 729 (4) 243 (5) 189

12. 236, 349, 464, 564, 781, 11100

(1) 464 (2) 349 (3) 781 (4) 11100 (5) 564

13. 484, 512, 343, 72, 64, 32

(1) 512 (2) 72 (3) 64 (4) 343 (5) 32

14. 81, 27, 729, 6561, 9, 59049

(1) 9 (2) 729 (3) 6561 (4) 59049 (5) 27

15. 17, 43, 59, 67, 37, 24

(1) 43 (2) 24 (3) 59 (4) 67 (5) 37

Direction for questions 16 to 20: Answer the following questions based on the following alphanumeric series

W 2 T 6 H J 9 P 8 F R 5 4 Q A 3 S Z

16. 4 of the following 5 characters are alike with respect to their position in the above sequence and form a group. Which is the one that does not belong to that group?

(1) F (2) H (3) J (4) R (5) Z

17. Which character is the 2nd to the right of the 14th character from the left hand, if the second half of the above series is reversed?

(1) A (2) R (3) 4 (4) 5 (5) Q

18. If the first half of the above series is reversed, how many such letters would you find which are immediately preceded by a letter and immediately followed by a number?

(1) None (2) One (3) Two (4) Three (5) Four

19. Which character is the 5th to the left of the 7th character from the left hand, if the first half of the above series is reversed?

(1) 8 (2) P (3) H (4) F (5) R

20. If the first half of the above series is reversed, and then all numericals taken out to form a series, then what would be 3rd number in such a series?

(1) 6 (2) 8 (3) 9 (4) 2 (5) 5

Answer Key

1-124	2-4	3-93	4-144	5-98	6-74	7-24	8-11	9-17	10-65
11-5	12-1	13-4	14-5	15-2	16–5	17–4	18–4	19–2	20–1

Explanatory Answers

1. $1^3 - 1, 2^3 - 1, 3^3 - 1, 4^3 - 1, 5^3 - 1 = 124$

2. Alternate terms: 7, 6, 5, 4 and 3, (+ 4)7, (+ 4)11

3. 3, (+ 6)9, (+ 12)21, (+ 24)45, (+ 48)93

4. 12, (+ 11)23, (+ 22)45, (+ 11)56, (+ 22)78, (+ 11)89 (+ 22)111, (+ 11)122, (+ 22)144

5. 8, (+ 6)14, (+ 12)26, (+ 24)50, (+ 48)98

6. 2, (+ 3)5, (+ 13)18, (+ 23)41, (+ 33)74

7. 0, (+ 3)3, (+ 5)8, (+ 7)15, (+ 9)24

8. Alternate series: 5, 7, 9, 11 and 12, 9, 6

9. 65, (–30)35, (–18)17, (–6)11, [–(–6)] 17 : 30, (–12)18, (–12)6, (–12)–6

10. $0^3 + 1, 1^3 + 1, 2^3 + 1, 3^3 + 1, 4^3 + 1 = 65$

11. 6561, 2187, 19683, 729, 243 are all powers of 3 so the odd man out is 189 which the option is (5)

12. 2, 3, 5, 7, 11 are all prime numbers followed by squares of 6, 7, 8, 9, 10 so the odd man out is 464. option (1)

13. 484, 512, 32, 72, 64 are all divisible by 2. 343 is divisible by 3. Hence option (4)

14. 9, 81 729, 6561, 59049 are all powers of 9. 27 is a power of 3. Hence option (5)

15. 17, 43, 59, 67, 37 are all prime numbers and 24 is not. Hence option (2)

16. Four options have one number next to it. Z does not belong to the group. Hence (5)

17. When the second part of the series is reversed, it becomes W2T6HJ9P8 ZS3AQ45RF. The 14th character is Q, and the 2nd character to its right is 5. Hence (4)

18. When the first part of the series is reversed, it becomes 8P9JH6T2WFR54QA3SZ. In this sequence H, R and A meet the requirements. Hence (4)

19. When the first part of the series is reversed, it becomes 8P9JH6T2WFR54QA3SZ. In this sequence the 7th character from the left is T and the 5th character to its left is P. Hence (2)

20. Using the above series, we can see that the 3rd numerical would be 6. Hence (1)

Chapter 17 Change of Base

Suggested Time : 60 minutes

We normally do all our mathematics in base 10 (or decimal number system). Base 10 refers to the fact that there are 10 different numericals (0, 1, 2, …9) in the number system. However we can also use any other base to perform calculations. For example, in base 2 we have only 2 digits → 0 and 1. This base is called binary number system.

Similarly in base 7, there are 7 numericals (0, 1, 2, …6) and in base 16 there are 16 numericals (0, 1, 2, …9 and A, B, …F). Please note that A here plays the role of 10 of the decimal system, B the role of 11, etc. We can have a number system of any base as long as there are unique numerical to represent.

Question Type A: Converting a base 10 number to any other base

Convert 45 of decimal number system into base 4.

$4 \mid 45$
$4 \mid 11 \rightarrow 1$
$4 \mid 2 \rightarrow 3$

Step 1: we divide 45by 4.
Step 2: we get 11 as quotient and 1 as remainder. We now divide 11 by 4.
Step 3: we get 2 as quotient and 3 as remainder. The quotient cannot be further divided by 4.
Step 4: To convert decimal number 45 into base 4, write down the last quotient and the remainders in the reverse order. i..e 2→3→1. So 45 of base 10 is 231 in base 4.

We write this as $(45)_{10} = (231)_4$

Question Type B: Converting a number of any other base into base 10

To learn this, first let us quickly understand the scientific form of writing a number.
Scientific form of base 10 or decimal number system for number 365 is $3 \times 10^2 + 6 \times 10^1 + 5 \times 10^0$

$365 = 300 + 60 + 5$
$= 3 \times 10^2 + 6 \times 10^1 + 5 \times 10^0$

Convert 321 of base 5 into decimal number system.

$3 \times 5^2 = 75$
$2 \times 5^1 = 10$
$1 \times 5^0 = 1$

Step 1: Multiply 3 with 5^2.
Step 2: Multiply 2 with 5^1.
Step 3: Multiply 1 with 5^0. Note that 5^0 is 1 (any number to the power zero is 1).
Step 4: Add the value you get. 75 + 10 + 1 = 86. Hence, 321 of base 5 is equal to 86 in base 10.

We write this as $(321)_5 = (86)_{10}$.

Converting a number of a base other than base 10 to a number in a base other than base 10.

Convert 625 of base 5 into a base 3 number.

Step 1: Convert 625 of base 5 into base 10. (Question Type B)
Step 2: Convert the base 10 number you have got into base 3. (Question Type A)

Mathematical operations on numbers in other bases
It is best that all calculations are done by converting the number into base 10 and then reconverting it into the required base. However, you could try addition, subtraction and multiplication in the given base itself if you are careful in playing with the numbers.

Before we illustrate this with an example, take a close look at the table given.

Here is a table that shows numbers in decimal number system and their conversion to other bases.

Decimal	Binary (Base 2)	Base 3	Base 4	Base 5
0	0	0	0	0
1	1	1	1	1
2	**10**	2	2	2
3	11	**10**	3	3
4	**100**	11	**10**	4
5	101	12	11	**10**
6	110	20	12	11
7	111	21	13	12
8	**1000**	22	20	13
9	1001	**100**	21	14
10	1010	101	22	20
11	1011	102	23	21

Quick question. What is 12 of the decimal system in base 2, base 3, base 4 and base 5? If you have understood the above table, then you can directly write down the values as base 2 → 1100, base 3 → 110, base 4 → 30 and base 5 → 22.

Now let us use the above table to answer a question. Add two base 4 numbers: 231 + 1212

Method 1: Convert the two numbers into base 10. Then add them up. Reconvert the answer into base 4.

Method 2: While we use the same method as in decimal system addition, the numbers that we write are in equivalent base 4 numbers. See the illustration.

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1212
+231
----
2103
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Step 1: Add the unit's digit 2 and 1. We get 3. The number 3 is written as 3 in base 4 also.
Step 2: Add the ten's digit 1 and 3. We get 4. The number 4 is written as 10. So we write 0, and carry over 1 to the hundred's place.
Step 3: Add the hundred's digits 2 and 2 and also the carry over 1. We get 5. The number 5 is written as 11 in base 4. So we write 1, and carry over 1 to the thousand's place.
Step 4: Finally add the thousand's digit and the carry over. i.e. 1 and 1. We get 2.

Note: Sometimes the word used to describe numbers of a particular base is "system". For example, instead of saying "a base 6 number", you may come across the term "in the system of 6".

Practice Test A (Optional)

1. How would you write 105 in the system of 6 in the decimal system?
   1] 41 2] 37 3] 181 4] 100

2. In a number system 12, 20, 24 are in an arithmetic progression. What is the base of the number system?
   1] 6 2] 7 3] 4 4] 5

3. Convert the number 1982 from base 10 to base 12. The result is:
   1] 1182 2] 1912 3] 1192 4] 1292

4. How would you write the base 4 number 21212 in base 10?
   1] 614 2] 258 3] 280 4] 268

5. Convert the number 365 from base 10 to base 5. The result is:
   1] 2430 2] 3240 3] 243 4] 2540

Answer Key

1-1 2-1 3-3 4-1 5-1

Explanatory Answers

1. 105 in the system of $6 = 1 \times 6^2 + 0 \times 6 + 5 = 41$
   Hence, (1) is the answer.

2. Let the base be x
   $(20)_x - (12)_x = (24)_x - (20)_x$
   $2x + 0 - [x + 2] = 2x + 4 - [2x + 0]$
   Solving, x=6

3. To convert 1982 from base 10 to 12 we have
   $12 \mid 1982$
   $12 \mid 165 - 2$
   $12 \mid 13 - 9$
   $12 \mid 1 - 1$
   $1982_{10} = 1192_{12}$
   Hence the answer is [3]

4. 21212 in the system of $4 = 2 \times 4^4 + 1 \times 4^3 + 2 \times 4^2 + 1 \times 4^1 + 2 \times 4^0 = 614$
   Hence, (1) is the answer.

5. To convert 365 from base 10 to 5 we have
   $5 \mid 365$
   $5 \mid 73 - 0$
   $5 \mid 14 - 3$
   $5 \mid 2 - 4$

   $365_{10} = 2430_5$
   Hence the answer is [1]

Chapter 18 Equations

Suggested Time : 120 minutes

SIMPLE EQUATIONS

An "equation" states that two quantities are equal, and will always contain an "unknown quantity" which we wish to find out. For example, in the equation $2x + 4 = 10$, the unknown quantity is 'x'.

To solve an equation means to find all the values of the unknown quantity. Each such a value is called a "solution" or alternatively a "root" of the equation

In the given example above,
$2x = 10 - 4$
$2x = 6$
$x = \frac{6}{2} = 3.$

The solution of the equation $2x + 4 = 10$, is $x = 2$. The value $x = 2$ said to satisfy the equation $2x + 4 = 10$.

SOLVING LINEAR EQUATIONS

A linear equation is one of the form $ax + b = 0$ where 'a' and 'b' are real numbers and the unknown quantity is 'x'. The number 'a' is called the coefficient of 'x' and 'b' is called the constant term.

Simultaneous linear equations

Sometimes equations contain more than one unknown quantity. For example, in the two equations
$3x + 7y = 27$ (1)
$5x + 2y = 16$ (2)
the unknowns are 'x' and 'y'. Such equations are called "simultaneous equations"

To solve the simultaneous equations above, we must find values of values of 'x' and 'y' which satisfy both equations at the same time.

The values of unknowns can be found in two ways.
1. By method of elimination.
2. By method of substitution

SOLUTION BY METHOD OF ELIMINATION

Solve the following simultaneous equations.
$3x + y = 28$ and $5x - y = 12$

Step 1: Eliminate one unknown and find the value of other unknown.

We first eliminate one unknown so that we can have one equation with one unknown.

In the above equations, we either can eliminate 'x' or 'y'. Note that if these two equations are added the unknown 'y' is removed or eliminated.
Adding two equations, we get
$8x = 28 + 12$
$8x = 40$
$\therefore x = \frac{40}{8} = 5$

Step 2: Finding the value of the unknown that is eliminated in the first step

To find the unknown 'y', we substitute $x = 5$ in either equation (1) or (2)
Thus from (1), we have $3 \times 5 + y = 28$
$y = 28 - 15 = 13$
The complete solution is $x = 5$ and $y = 13$.

SOLUTION BY METHOD OF SUBSTITUTION

Let us consider the following system of simultaneous equation.

$x + 2y = 4$ (1)

$2x + 3y = 7$ (2)

Instead of eliminating one unknown we can express one unknown in-terms of the other unknown to solve the system of simultaneous equations.

From (1), we have $x = 4 – 2y$

Substituting the value of x in (2), we have

$2(4 – 2y) + 3y = 7$

$8 – 4y + 3y = 7$

$y = 1$

Substituting the value of y in (1), we have

$x = 4 – 2(1) = 4 – 2 = 2.$

The complete solution is $x = 2, y = 1$.

SPECIAL CASES IN LINEAR EQUATIONS

Case 1: In the cases where there are three unknowns and two equations, it is not possible to find out the values of the unknowns uniquely. However the questions can be framed so that you may be able to find the value of one unknown uniquely.

If $a_1x + b_1y + c_1z = d_1$ and $a_2x + b_2y + c_2z = d_2$ are the two equations given and, if $\frac{b_1}{b_2} = \frac{c_1}{c_2}$, then we can find the value of 'x' uniquely.

Example: Find the value of 'x', if

$2x + 3y + 5z = 10$ (1) and

$3x + 6y + 10z = 15$ (2)

$(1) \times 2 \Rightarrow 4x + 6y + 10z = 20$ (3)

Subtracting (2) from (3), we have $x = 5$

Please note that if $\frac{b_1}{b_2} = \frac{c_1}{c_2}$, then we can easily eliminate both the unknowns 'y' and 'z'.

Note

Similarly, if $a_1x + b_1y + c_1z = d_1$ and $a_2x + b_2y + c_2z = d_2$ are the two equations given and,

if $\frac{a_1}{a_2} = \frac{c_1}{c_2}$, then we can find the value of 'y' uniquely.

if $\frac{a_1}{a_2} = \frac{b_1}{b_2}$, then we can find the value of 'z' uniquely.

The readers are advised to solve the following questions for better understanding of the above concept.

1. Find the value of 'y', if $2x + 3y + 5z = 10$ and $6x + 6y + 15z = 15$

2. Find the value of 'z', if $12x + 12y + 5z = 10$ and $6x + 6y + 15z = 15$

Case 2: In the cases where there are three unknowns and two equations, though it is not possible to find the values of the unknowns uniquely, however the questions can be framed so that you may be able to find some specific combination of unknowns.



ILLUSTRATION

Three oranges, two apples and one banana cost Rs. 56. Two oranges, three apples and four bananas cost Rs. 44. What is the cost of one orange, two apples and three bananas?
1] Rs. 24 2] Rs. 22 3] Rs. 38 4] Indeterminate

Let the cost of each orange, apple and banana be Or, A and B respectively.

$3Or + 2A + B = 56$ (1)
$2Or + 3A + 4B = 44$ (2)

$(1) + (2) \Rightarrow 5Or + 5A + 5B = 100$ dividing by 5 we get,

$Or + A + B = 20$ (3)

$4 \times (3) - (1) \Rightarrow Or + 2A + 3B = 24$

Practice Exercise A (Important)

1. Eight years ago I was three times as old as my sister, but now I am only twice as old as my sister. How old am I now?
1] 16 years 2] 24 years 3] 32 years 4] 48 years

2. Six caps, seven bats and two balls cost Rs. 28. Nine caps, eight bats and three balls cost Rs. 36. Find out the cost of each bat.
1] Rs. 4 2] Rs. 2.40 3] Rs. 1.50 4] Indeterminate

3. Among the 200 members in a party there were 40 more children than adults and 20 more adult women than adult men. How many adult men were there?
1] 30 2] 50 3] 70 4] 80

4. "How much money have you?" asked Tom's father. "Well," replied Tom, "If Dick gives me Rs. 4 he will have half as much as Harry, but if Harry gives me Rs. 4, then all of us will have the same amount." If both Dick and Harry give all they have to Tom, then how much will he have?
1] Rs. 33 2] Rs. 36 3] Rs. 42 4] Rs. 48

5. Three oranges, two apples and one banana cost Rs. 56. Two oranges, three apples and four bananas cost Rs. 44.what is the cost of one orange, two apples and three bananas?
1] Rs. 24 2] Rs. 22 3] Rs. 38 4] Indeterminate

6. In an action between two battle ships A and B, A fired 3 times as many shells as B. The total number of misses was seven times the total number of hits. The number of B's misses was 357, but B's hits exceeded A's hits by 66. What was the number of A's hits?
1] 520 2] 163 3] 105 4] 97

7. In an exhibition, Arun tries his skill at shooting. He was to receive 50 paise for hitting the "bulls eye" and had to pay 10 paise for missing the "bulls eye." He tries 40 shots but receives only Rs. 5. How many times did he hit the "bulls eye"?

1] 15 2] 30 3] 20 4] 25

8. A is 5 years older than B. If 5 years ago A was twice as old as B, then find the present age of A.

1] 20 2] 10 3] 22 4] 15

9. There is a 2-digit number, sum of whose digits is 10. Find the number among the given 4 numbers, given that it is greater than the square of twice the smaller digit but smaller than the square of the larger digit, in both cases by a difference of less than 20.

1] 19 2] 28 3] 37 4] 46

10. In a camp there are some men; each of them carries with him two horses, one with each hand. They reach a spot to fight with their enemy. Many persons died and some survived. Each survivor took a ride on a horse and ran away. If 17 men and 18 horses were found dead and 30 horses captured alive, then how many horses were there in all?

1] 48 2] 62 3] 92 4] 31

11. A couple visited five tourist spots. In each spot they took some photographs such that the number of photographs taken at any spot is one more than the number of photographs taken at the previous spot. If they took a total of 75 photographs, how many did they take at the first tourist spot?

1] 1 2] 15 3] 13 4] 14

12. Going on a 15-day tour, a person spent as many thousands of rupees each day as the number of full days left of the tour on that particular day. How much did he spend on the tour?

1] 105000 2] 107000 3] 120000 4] None of these

13. A man sold a horse for a sum of money, which was greater by Rs. 494 than half the price he paid for it, and gained thereby Rs. 150. What did he pay for the horse?

1] Rs. 419 2] Rs. 344 3] Rs. 688 4] Rs. 838

14. Two boys have 240 marbles between them. One of the boy arranges his in heaps of 6, the other boy arranges his marbles in heaps of 9. If there are 36 heaps altogether, then how many marbles has each?

1] 28 & 8 2] 168 & 72 3] 207 & 36 4] 108 & 132

15. If a sum of money is divided between three person, A, B, and C, in such a way that A and B have Rs. 42 between them, B and C have Rs. 45, and C and A have Rs. 53, then what is the total sum of money (in Rs.) is shared among A, B and C?

1] 70 2] 110 3] 87 4] 98

16. A sum of Rs. 29.30 is divided among 23 boys and girls. If a boy receives Rs. 1.30 and a girl receive Rs. 1.18, then find the number of girls.

1] 13 2] 18 3] 10 4] 5

Answer Key

1-3	2-2	3-1	4-2	5-1	6-4	7-1	8-4	9-3	10-2
11-3	12-1	13-3	14-2	15-1	16-4				

Explanatory Answers

1. Let my present age is 2x years.
$\therefore$ Sister's present age = x years.
According to the condition given in the question, $2x - 8 = 3(x - 8)$
$\therefore x = 16$
$2x = 32$

2. Let the cost of each cap, bat and ball be C, B and L respectively.
$6C + 7B + 2L = 28$ (1)
$9C + 8B + 3L = 36$ (2)
Here, $\frac{a_1}{a_2} = \frac{c_1}{c_2}$. Hence, we can find the value of B uniquely $\left[\because \frac{6}{9} = \frac{2}{3}\right]$
$\frac{3}{2} \times$ (1) – (2) $\Rightarrow 2.50\ B = 6$
$\therefore B = \frac{6}{2.50} = 2.40$

3. Let the number of adults be A.
$\therefore A + A + 40 = 200$
Number of adults, $A = \frac{200 - 40}{2} = 80.$
Let number of adult men be M. $\therefore M + M + 20 = 80$
Number of men = $\frac{80 - 20}{2} = 30.$

4. Let the amounts held by Tom, Dick, and Harry be T, D, and H respectively.
$\therefore D - 4 = \frac{1}{2} H$ (1)
$H - 4 = T + 4 = D$ (2)
As $D = H - 4$, using (1), we have $H - 8 = \frac{1}{2} H$
$\therefore H = 16.$
Substituting the value of H in (1), we get D = 12.
Substituting the value of H and D in (2), we get T = 8.
$\therefore$ When Tom gets both Dick's and Harry's amounts, he will have
$T + D + H = 16 + 12 + 8 = 36.$

5. Let the cost of each orange, apple and banana be Or, A and B respectively.
$3Or + 2A + B = 56$ (1)
$2Or + 3A + 4B = 44$ (2)
(1) + (2) $\Rightarrow 5Or + 5A + 5B = 100$ dividing by 5 we get,
$Or + A + B = 20$ (3)
$4 \times (3) - (1) \Rightarrow Or + 2A + 3B = 24$

6.

	A	**B**
Number of shells fired	x + y	x + 357 + 66 = x + 423
Number of misses	y	357
Number of hits	x	x + 66

We know that

$x + y = 3 (x + 423)$ (1)

Also, $y + 357 = 7 (x + x + 66)$ (2)

Subtracting (2) from (1), we get

$x – 357 = 3x – 14x + 3 \times 423 – 7 \times 66$

$\Rightarrow 12x = 1164$

$x = \frac{1164}{12} = 97$

7. Total number of attempts = 40

Let number of times he hits the "bulls eye" = x

$\therefore$ Number of misses = 40 – x

Hence $50x – 10 (40 – x) = 500$

$\therefore 60x = 900 \Rightarrow x = 15$

8. Let A and B be A and B years old.

$\therefore A – B = 5$ (1)

5 years ago, A was (A – 5) years & B was (B – 5) years old

$A – 5 = 2 (B – 5)$

$\therefore A – 2B = – 5$ (2)

Let us eliminate A by subtracting (2) from (1).

(1) – (2) gives B = 10 years and substituting the value of B in (1), we get,

A = 5 + 10 = 15 years.

9. It will be easy to solve these kind of questions using the answer options.

All options satisfy the conditions that sum of digits = 10.

It is given that the difference between square of the larger digit and the number is less than 20.

$9^2 – 19 > 20$, so option [1] is not true.

$8^2 – 28 > 20$, so option [2] is not true.

$46 > 6^2$, so option [4] is not true.

$\therefore$ The option [3] is the right answer.

10. Let number of men be x

$\therefore$ Number of horses = 2x

Number of persons alive = x – 17

$\therefore$ (x – 17) persons ran away with (x – 17) horses.

$\therefore$ Total Number of horses = (x – 17) + 30 + 18

$\therefore x – 17 + 30 + 18 = 2x$

$\therefore x = 31$

Hence total number of horses = 62

11. Let the number of photographs that he has taken in the first spot be x

The number of photographs that he has taken in the last spot = x + 4

$\therefore 5\left(\frac{x + x + 4}{2}\right) = 75 \Rightarrow x = 13$

Hence [3]

12. On day 1, the amount spent = 14,000.
On day 2, the amount spent = 13,000.
...
...
On day 14, the amount spent = 1000.
On day 15, the amount spent = 0.

$\therefore$ Total amount spent = $1000 \times (14 + 13 + ... + 1) = \frac{14 \times 15}{2} \times 1000 = 105000.$

Hence [1]

13. Let the man paid Rs. x for the horse

$$\left(\frac{x}{2} + 494\right) - x = 150$$

$$494 - \frac{x}{2} = 150$$

$$\frac{x}{2} = 344$$

x = 688. Hence [3]

14. Let the number of heaps be A and B

6A + 9B = 240 (1)
A + B = 36 (2)
(2) $\times 6 \Rightarrow$ 6A + 6B = 216 (3)

(1) – (3) $\Rightarrow$ 3B = 24
B = 8 and A = 28
Number of marbles = $28 \times 6 = 168$ and $8 \times 9 = 72$.
Hence [2]

15. A + B = 42 (1)
B + C = 45 (2)
C + A = 53 (3)
Taking the sum of the above three equations, we get, $2(A + B + C) = 140$
A + B + C = 70.
Hence [1]

16. Let the number of Girls be G.
$G \times 1.18 + (23 - G)\ 1.30 = 29.30$
$1.18G - 1.30G + 29.9 = 29.30$
$0.12G = 0.60$
G = 5.
Hence [4]

Practice Exercise B (Optional but Important)

1. Amit and Meghna went to New York for a 63-day holiday trip. Each day, Amit spent as many hundred dollar bills as the remaining number of days in the trips (excluding the current day), and Meghna spent as many hundred dollar bills as the number of days already spent (including the current day). How much would they have spent together on each day of the holiday?
1] $63 2] $12600 3] $6300 4] $50

2. The houses on a street are numbered 1, 2, 3, etc. up one side, then back down the other side. Opposite house No.10 is house No.23. How many houses are there on the street? [There are even number of houses.]
1] 30 2] 32 3] 34 4] Indeterminate

3. Guru and Amit are best friends. They met each other after so many years. Guru says to Amit "It's not fair that you never wished me on my birthday." Amit replies "Oh no! I forgot your birthday, By the way when is your birthday?" Guru told Amit, "12 times the date of my birth added to 31 times the month of my birth is 376." What is Guru's date of birth?
1] 21st April 2] 23rd Aug. 3] 27th May 4] None of these

4. Mohan babu bought 2 varieties of coffee beans costing Rs. 81 and Rs. 66 per kg. If he spent a total amount of Rs. 1353, and the beans were available in boxes of 1 kg, in how many possible ways could he purchase the beans?
1] 1 2] 2 3] 3 4] None of these

5. There are 15 questions in a test paper. A student is given 2 marks for a question answered correctly and he is penalized by 1 mark if he answers the question wrong. If he answers all the questions, then what is the minimum number of question he should answer correctly to get a score more than zero?
1] 5 2] 6 3] 7 4] 4

6. A pole has a fourth of its length in the mud, a third of its length in the water. If 10 m of the pole's length is above the water, then what is its length?
1] 48m 2] 60 m 3] 24 m 4] 120 m

7. Find a number which when multiplied by 11 and then diminished by 18 is equal to 15.
1] 7 2] 3 3] 33 4] 4

8. Rohan is two years younger than Mohan who is three years younger than Sohan who is four years older than Rohit who is two years older than Mohit who is three years younger than Sohit. Thus
1] Sohan is 7 years older than Mohit 2] Rohan is 3 years older than Mohit
3] Rohan is 2 years younger than Sohit 4] Rohit is 2 years older than Sohit.

9. If Rs. 66 is divided between A, B, and C so that B may have Rs. 8 more than A, and C, Rs. 14 more than B, then what is the share of C?
1] 42 2] 34 3] 20 4] 32

10. A has Rs. 12 and B has Rs. 8. After B has lost a certain sum to A, his money is only three-sevenths of A's. How much did A win from B?
1] Rs. 5 2] Rs. 2 3] Rs. 6 4] Rs. 9.20

Answer Key

1-3 2-2 3-1 4-1 5-2 6-3 7-2 8-3 9-2 10-2

Explanatory Answers

1. On the xth day, Amit spends $100 \times (63 - x)$ dollars and Meghna spends $100x$ dollars.
$\therefore$ They spend \$6300 on any day.
Hence [3]

2. It is to be noted that as there are even number of houses, opposite house number 1 is the last house, which is say house number 'n'.
Opposite house number 2 is house number $(n - 1)$.
So the sum of house numbers of opposite houses is $n + 1$.
$\therefore 10 + 23 = n + 1 \Rightarrow n = 32$.
Hence [2]

3. Let Guru's date and month of birth be D and M respectively.
Then $12D + 31M = 376$ (1)
Constraints on D and M: where $D = 1, 2, \ldots 31$ and $M = 1, 2, \ldots 12$
From (1) 'M' has to be even , since 12D is even and the Right hand side of equation is also even,
So M can be 2, 4, 6, 8, 10 or 12
$\therefore 12D = 376 - 31M$

$$D = \frac{376 - 31M}{12}$$

$$D = \frac{372}{12} + \frac{4}{12} - \frac{24M}{12} - \frac{7M}{12}$$

$$D = 31 - 2M + \frac{4 - 7M}{12}$$

$\frac{4-7M}{12}$ must be a integer (since D and (31–2M) are integers).
Substituting the values of $M = 2, 4, \ldots 12$
We find $M = 4$ and $D = 21$ only satisfies the above condition.
$\therefore$ 21 April is Guru's Birthday. Hence [1]

4. Let the number of Kgs of two coffee bean Variety be A & B

$81A + 66B = 1353$

$$B = \frac{1353 - 81A}{66}$$

$$B = 20 - A + \frac{33 - 15A}{66} \Rightarrow B = 20 - A + \frac{3(11 - 5A)}{3 \times 22}$$

$$B = 20 - A + \frac{11 - 5A}{22}.$$ (A should be an odd multiple of 11.)

When A=11, B = 20 – 11 – 2 = 7
When A = 33, B is negative
Hence only one such combination is possible. Hence [1]

5. Let the student answers x questions incorrectly. Then,
$(15 - x)\, 2 - x = 0$
We get, x = 10. When he answers 5 questions correctly, his score is zero. He has to answer 6 questions correctly, to get a score that is more than zero.
Hence [2]

6. Let the length of the post be x m

$$\frac{x}{4} + \frac{x}{3} + 10 = x \Rightarrow x - \left(\frac{x}{4} + \frac{x}{3}\right) = 10$$

$$x - \left(\frac{7x}{12}\right) = 10 \Rightarrow \frac{5x}{12} = 10$$

$$x = \frac{10 \times 12}{5} = 24 \text{ m.}$$

Hence [3]

7. Let the number be x
$11x - 18 = 15$
$11x = 33$
$x = 3$
Hence [2]

8. Since the answer choice compares the ages of people, let us assume
Rohan's age be = 10 yrs
∴ Mohan age = 12 yrs
∴ Sohan age = 15 yrs
∴ Rohit age = 11 yrs
∴ Mohit age = 9 yrs
∴ Sohit age = 12 yrs
Hence [3]

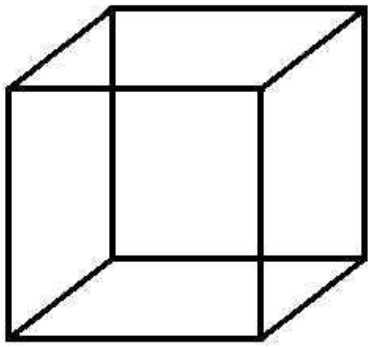
9. $B = A + 8$ (1)
$C = 14 + B$ (2)
$A + B + C = 66$ (3)
Substituting (1) in (2), we have
$C = 14 + A + 8$
$C = 22 + A$ (4)
Substituting (4) and (1) in (3), we have
$A + A + 8 + 22 + A = 66 \Rightarrow 3A = 36$
$A = 12$; $B = 20$ $C = 34$.
Hence [2]

10. Let B Lost Rs. x
$$8 - x = \left(\frac{3(12 + x)}{7}\right)$$
$56 - 7x = 36 + 3x$
$10x = 20$
$x = 2$
Hence [2]

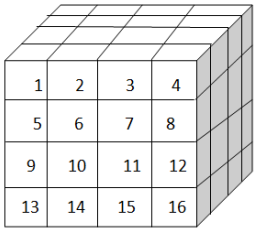
Chapter 19 Cubes and Cuboids

Suggested Time : 120 minutes

Questions regarding cutting of cubes and cuboids are fairly common in aptitude tests as they test your visualization skills. A typical question involves colouring the faces of the cube/cuboid in a particular way and then cutting the cube/cuboid in a pattern. Questions are then asked on the colours on the smaller cubes.



For example, if the cube is colored black on all faces and then 3 equidistant cuts are made from the three different sides, then how many small cubes get generated, and how many of the smaller cubes would have exactly one face coloured black. The cube below shows how the resultant cube would look. On one face, the smaller cubes have been numbered.



1. **How many cubes are created if 3 equidistant cuts are made?** 3 cuts yield 4 parts. So the number of smaller cubes would be $4 \times 4 \times 4 = 64$.

2. **Cubes which have only one of their sides painted black:** cubes numbered 6, 7, 10 and 11 on the front face of the cube. Similarly, on the other 5 faces, there would be four such cubes.

We can say that on any face, if we remove the cubes on the margins, then the inner cubes would be one side coloured black.

i.e. of a cube of dimensions $n \times n \times n$, take any face of dimension n rows and n columns. Then the number of cubes painted one side black on that face would be $(n - 2) \times (n - 2)$. For example, in the above cube, it is 2×2.

So on all 6 faces, it would be $6 \times (n - 2) \times (n - 2)$. In the above case, it would be $6 \times 2 \times 2 = 24$.

3. **Cubes which have exactly two of their sides painted black:** cubes numbered 2, 3, 8, 12, 14, 15, 9 and 5 on the front face. Similarly, on the other 5 faces, there would be eight such cubes. However some of the cubes on one face are the same as some cubes on another face. For example, the cubes numbered 2 and 3 are the same as those when counting the cubes on the top face that have two sides coloured black. Hence we need to divide the final result by 2.

Or
We can say that on any edge, if we remove the cubes on the corner of the edges, then the remaining cubes would be two side coloured black.

i.e. of a cube of dimensions n × n × n, take any edge of dimension n. Then the number of cubes painted two sides black on that edge would be (n – 2). For example, in the above cube, it is 2.

So on all 12 edges, it would be 12 × (n – 2). In the above case, it would be 12 × 2 = 24. (Alternately 6 × 8 ÷ 2).

4. **Cubes which have exactly three of their sides painted black:** cubes numbered 1, 4, 13 and 16 on the numbed face. Similarly, the other faces would also have 4 such cubes. But each cube is common to 3 faces. Hence divide by 3.

Alternately, there are 8 corners. So the answer is 8.

5. **Cubes which have none of their sides painted black:** All the numbered cubes have atleast one of their faces painted black. So the uncoloured cubes would not be visible, i.e. they are in the core of the cube above.

We can say that if we unpeel the above cube from all sides, then the core would be visible. i.e. from a cube of dimensions n × n × n, take any edge of dimension n (e.g. look at the cube vertically). Then to unpeel, we need to take away two layers – the top and the bottom. So n – 2. Would unpeel the cube vertically. We need to do this for the other two directions as well. On unpeeling, we get a smaller cube of dimensions (n – 2) × (n – 2) × (n – 2). For the above cube it would be 2 × 2 × 2 = 8.

Alternately, for better visualization sake, the unpainted cubes are below the cubes numbered 6, 7, 10 and 11. There are two such layers of cubes. Hence a total of 8 cubes are not coloured black.

Variations of these questions:
The above illustration and formulae are for cubes where all 6 faces are painted with the same colour. However, you may get questions where the faces may be painted in different colours, or instead of a n × n × n cube, you get a cuboid of dimensions x × y × z. Using the above logic and visualization, solve the questions in the practice test and you will not have a problem solving the variations.

Practice Exercise A (Important)

1. A bigger cube is painted with yellow colour and cut into 125 identical pieces. How many pieces will have one face coloured?
1] 42 2] 48 3] 54 4] None of these

2. A bigger cube is painted with Green colour and cut into 216 identical pieces. How many pieces will have two faces coloured?
1] 42 2] 48 3] 54 4] 96

3. Sixty four smaller cubes are stacked to make a bigger cube. The entire bigger cube is painted red. How many smaller cubes are unpainted?
1] 45 2] 27 3] 20 4] None of these

Directions for questions 4 – 6: Refer the data given below to answer the following questions
Thirty six smaller cubes of size 2 × 2 × 2 cm each are stacked to make a rectangular prism of size 6 × 6 × 8 cm and then the six faces of the rectangular prism is painted green.

4. How many cubes are painted on three sides?
1] 4 2] 8 3] 12 4] None of these

5. How many cubes are unpainted?
1] 4 2] 8 3] 12 4] None of these

6. How many cubes are painted on two sides?
1] 22 2] 20 3] 16 4] None of these

Directions for questions 7 – 12: Refer the data given below to answer the following questions
Two adjacent sides of a cube is painted red. The opposite sides of red coloured sides are painted green. The remaining sides are painted blue.

7. If the cube is subdivided into 27 smaller cubes, then how many smaller cubes will have its faces coloured with only green and blue?
1] 4 2] 6 3] 8 4] None of these

8. If the cube is subdivided into 64 smaller cubes, then how many smaller cubes will have its faces coloured with green and red?
1] 4 2] 6 3] 9 4] 8

9. If the cube is subdivided into 125 smaller cubes, then how many smaller cubes will have its faces coloured with only blue and red?
1] 4 2] 6 3] 12 4] None of these

10. If the cube is subdivided into 64 smaller cubes, then how many smaller cubes will have its faces coloured with red only?
1] 4 2] 6 3] 8 4] None of these

11. If the cube is subdivided into 125 smaller cubes, then how many smaller cubes will have its faces coloured with blue?
1] 24 2] 26 3] 28 4] 50

12. If the cube is subdivided into 125 smaller cubes, then how many smaller cubes will have none of its faces painted at all?
1] 25 2] 0 3] 9 4] 27

Answer Key

1-3 2-2 3-4 4-2 5-4 6-3 7-2 8-4 9-4 10-4
11-4 12-4

Explanatory Answers

1. 125 identical pieces implies a cube of dimension $5 \times 5 \times 5$. On each face, 9 cubes would be one side coloured. Hence on 6 faces, we have 54 cubes that are one side painted.

2. 216 identical pieces implies a cube of dimension $6 \times 6 \times 6$. On each edge, 4 cubes $(n - 2)$ would be two sides coloured. Hence on 12 edges, we have 48 cubes that are two sides painted.

3. 64 identical pieces implies a cube of dimension $4 \times 4 \times 4$. The unpainted cubes are in the core. Unpeeling, we get a cube of dimension $2 \times 2 \times 2$. Hence 8.

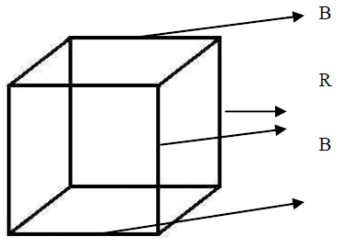
For questions 4 to 6, please note that the object is a cuboid with number of cubes = $3 \times 3 \times 4$. Using any other number is likely to lead to confusion.

4. 8 corner cubes are painted on 3 sides.

5. Unpeeling will reveal a cuboid of dimension $1 \times 1 \times 2$. i.e. 2 cubes are not painted.

6. 8 edges are of the dimension 3 cubes. Here, on each edge 1 cube would have 2 sides painted. 4 edges are of dimension 4 cubes. Here 2 cubes each have 2 sides painted. Hence total = 8 + 8 = 16.

For questions 7 to 12: refer to the cube below for better visualization. The front and one adjacent side has been marked red. The top and the bottom are labeled blue. The two unlabeled sides are green (back side and left side).



7. 27 smaller cubes implies that the dimensions are 3 × 3 × 3.
Let us take the left side which is coloured green. It shares two edges with blue. On both the edges, the middle cube would be painted green on one face and blue on another. Also the corner cube would have 2 sides painted blue and one side painted green.
Similarly, the back face also has 3 cubes which are painted only blue and green.
Hence, a total of 6 cubes.

8. 64 smaller cubes implies that the dimensions are 4 × 4 × 4.
Let us take the left side which is coulored green. It shares one edge with red. On that edge all 4 cubes would be painted green and red. Please note that one corner cube is also painted blue, but as the question is not about "only green and red", we can consider that cube too.
Similarly, the back face also has 4 cubes which are painted red and green.
Hence, a total of 8 cubes.

9. 125 smaller cubes implies that the dimensions are 5 × 5 × 5.
Let us take the top side which is coulored blue. It shares two edges with red. On one edge you have 5 cubes. 4 of these are coloured only red and blue, and one corner cube has all 3 colours. On eth other edge too, 4 of these are coloured only red and blue, and one corner cube has all 3 colours. However, in the 8 cubes that we have counted, one corner cube is common. So in all, 7 cubes with the top side as blue have the colours only blue and red.
Similarly, with the bottom side as blue, there are 7 cubes with the colours only blue and red.
Hence, a total of 14 cubes.

10. 64 smaller cubes implies that the dimensions are 4 × 4 × 4.
There are two red faces. On each red face, 4 cubes each in the center would have only red colour. Hence, a total of 8.
Also, the edge where one red face meets another also has two cubes which are only red faced. Hence a total of 10 cubes are coloured red.

11. 125 smaller cubes implies that the dimensions are 5 × 5 × 5.
The question is blue, and not 'blue only'. Hence on one blue face, all cubes would be painted blue. i.e. 25. Similarly, on the other blue face, there would be another 25 cubes painted blue. Please note that as the two blue sides are the top and bottom side, they do not have any common edges (and hence any common cube).
Hence, a total of 50 cubes.

12. 125 smaller cubes implies that the dimensions are 5 × 5 × 5.
The unpainted cubes are in the core. Unpeeling, we get a cube of dimension 3 × 3 × 3. Hence 27.

Chapter 20 Permutation and Combination

Suggested Time : 120 minutes

Important Note: This is a chapter that many students find difficult. While it is logical in nature, student's minds are sometimes closed to this chapter. If you are comfortable with logic, you would enjoy this chapter. The book explains the topic in two ways. The classical approach is used for theory and solved questions, and the logic based approach is used for practice questions. If you are not totally confident of the theory, you can still get a decent grasp of the topic by solving the practice questions. Hence, for the practice exercise, the book may not provide multiple answer options, but expect you to find out the right answer.

DIFFERENCE BETWEEN PERMUTATION AND COMBINATION

Each of the selections which can be made by taking some or all of a number of things (without regard to the order of the things in each group) is called a COMBINATION.

Example: The combination which can be made by taking the four letters p, q, r, s two at a time are six in number; namely pq, pr, ps, qr, qs, rs, each of these presenting a different selection of two letters.

Thus when r things are selected out of n, each selection is called an r – combination, and the number of ways in which such a selection can be made is called the number of combination of n things taken r at a time, and is denoted by the symbol ${}^{n}C_{r}$.

If the things in each selection are arranged in all possible orders, each of such arrangements is called a PERMUTATION.

Example: Each of the foregoing selections of the letters p, q, r, s taken two at a time, are twelve in number: namely, pq, pr, ps, qr, qs, rs, qp, rp, sp, rq, sq, sr.

Each arrangement which can be made by taking r things out of n is called an r-permutation, and the number of ways in which such an arrangement can be made is called the number of permutations of n things taken r at a time, and denoted by the symbol ${}^{n}P_{r}$.

FUNDAMENTAL PRINCIPLE: If one operation can be performed in m ways, and if corresponding to each of the m ways of performing this operation, there are n ways of performing a second operation, then the number of ways of performing the two operations together is $m \times n$.

The above principle can be extended to the case in which different operations can be performed in m, n, p,…. ways. In this case the number of ways of performing all the operations together would be $m \times n \times p \times \ldots\ldots$.

Example: A field has 4 gates, in how many ways is it possible to enter the field by one gate and come out at another?

There are 4 ways of entering the field, and with of each of these there is a choice of 3 ways of coming out (since the same gate cannot be used to come out); hence the number of ways is 4×3, or 12.

PERMUTATIONS: Each of the different arrangements which can be made by taking some or all of a number of things at a time is called a permutation.

Thus ${}^{5}P_{3}$ denotes the number of permutations or arrangements of 5 things taken 3 at a time.

$$ {}^{n}P_{r} = \frac{n!}{(n-r)!} = n(n-1)(n-2)\ldots(n-r+1) $$

Note: (i) ${}^{n}P_{n} = \frac{n!}{(n-n)!} = \frac{n!}{0!} = n!$ ($\because 0! = 1$) (ii) ${}^{n}P_{0} = \frac{n!}{(n-0)!} = \frac{n!}{n!} = 1$

RESTRICTED PERMUTATIONS:

Type I:

Example: (i) In how many permutations of 15 things taken 6 at a time will one thing (a) always occur (b) never occur?

Keeping aside the particular thing which will always occur, the number of permutations of 14 things taken 5 at a time is ${}^{14}P_5$. Now this particular thing can take up any one of the six places and so can be arranged in 6 ways. Hence the total number of permutations = ${}^{14}P_5 \times 6 = 14\times13\times12\times11\times10\times6 = 1441440$ ways.
Leaving aside the particular thing which has never to occur, the number of 14 things taken 6 at a time is ${}^{14}P_6 = 14\times13\times12\times11\times10\times9 = 2162160$ ways.
Type II: WHEN CERTAIN THINGS DO NOT OCCUR TOGETHER:

When the number of things not occurring together is two:
Find the total number of permutations when no restriction is imposed on the manner of arrangement. Then find the number of arrangements when the two things occur together. The difference of the two results gives the number of permutations in which the two things do not occur together.

Example: Find the number of ways in which 10 books can be placed on a shelf when two particular books are never together.

The two particular books can be considered as one book, and there are (10 –1), i.e. 9 books now which can be arranged in ${}^{9}P_9$ i.e. 9! ways. Now these two books can be arranged amongst themselves in 2! ways. Hence the total number of permutations in which these two books are placed together is (9! × 2!). The number of permutation 10 books without any restrictions is 10!.

Therefore, the number of arrangements in which these two books never occur together = 10! – 2! × 9! = 10 × 9! – 2 × 9! = 9! (10 – 2) = (9! × 8) ways.

(B) When the number of things not occurring together are more than two:
Example: In how many ways can 4 boys and 3 girls be arranged in a straight line so that no two girls are together?

First we arrange the position of the four boys indicated by B_1, B_2, B_3, B_4.

$\times B_1 \times B_2 \times B_3 \times B_4 \times$

This can be done in 4! ways.
Now, if the position of girls are fixed at places (including the two ends) shown by crosses, the three girls will never come together. In any of these arrangements there are 5 places for 3 girls and so the girls can sit in ${}^{5}P_3$ ways.

Hence the required number of ways in sitting of 4 boys and 3 girls is :
${}^{5}P_3 \times 4! = 5 \times 4 \times 3 \times 4 \times 3 \times 2 \times 1 = 1440$ ways.

Type III: FORMATION OF NUMBERS WITH DIGITS:

There is no restriction:
Example: Find the number of 4 digit numbers that can be made from 1,2,3,4,5
The number of 4 – digit numbers = ${}^{5}P_4 = 120$.

Number in which a particular digit occupies a particular place:
Example: Find the total number of 6 digit numbers that can be formed with the digits, 1,2,3,4,5,6 such that 5 always occupies the one's place.
In this case the one's place is fixed and the remaining five places can be filled in by the remaining 5 digits in ${}^{5}P_5$ i.e. 5! ways, i.e. 120 ways.
∴ The number of numbers in which 5 occurs in the unit's place = 120.

Numbers divisible by a particular number:
Example: Find the total number of 6 digit numbers that can be formed with the digits 2,3,4,5,6,7 which are divisible by 2.

These numbers will have 2 or 4 or 6 in the unit's place. Thus the unit's place can be filled in 3 ways. After filling up the unit's place in any one of the above ways, the remaining five places can be filled in ${}^5P_5$ ways i.e. 5! i.e. 120 ways.
$\therefore$ The total number of numbers divisible by 2 = $120 \times 3 = 360$.

(D) Numbers having particular digits in the beginning and the end:
Example: Find the total number of 6 digit numbers that can be formed with 0,1,2,3,4,5 such that the number begins with 2 and ends with 0.

Here the first and the last places are fixed and the remaining four places can be filled in 4! i.e 24 ways by the remaining four digits. Hence, the total number of numbers beginning with 2 and ending with 0 is 24.
(E) Numbers which are smaller than or greater than a particular number:
Example: Find the total number of numbers that can be formed with 3,5,7,8,9 which are greater than 50000.

In these numbers there will be 5 or a digit greater than 5 i.e 7,8 or 9 in the ten thousand's place. Thus this place can be filled in 4 ways. The remaining four places can be filled in 4! = 24 ways.
Hence the total number of numbers = $4 \times 24 = 96$.

Type IV : WORD BUILDING:

There is no restriction on the arrangement of the letters:
Example: Form words with the letters of the word 'Problem'.
The seven different letters can be arranged in ${}^7P_7 = 7! = 5040$ ways.
$\therefore$ The total number of words = 5040.

All words begin with a particular letter:
Example: form words with the letters of the word 'Publish' such that all words begin with 'b'.
All words begin with 'b'. The remaining 6 places can be filled with remaining 6 letters in 6! = 720 ways.
Hence, the total number of words = 720.

All words begin and end with particular letters:
Example: Form words with the letters of the word 'Century' such that the word begins or ends with 'u' or 't'.
The first and last position could have been filled in ${}^2P_2 = 2$ ways.
The remaining positions are filled in ${}^5P_5 = 5! = 120$ ways.
Hence the number of words = $2 \times 120 = 240$.

Vowels occur together:
Example: Form words with the letters of the word 'wheat' such that the vowels occur together.
The vowels 'e' and 'a' can be regarded as one letter. The 4 letters can be arranged in 4! = 24 ways. These two vowels can be arranged amongst themselves in 2! = 2 ways.
Hence, the total number of words = $24 \times 2 = 48$.

Consonants occur together:
Example: Form words with the letters of the word 'Chlorine' such that the consonants occur together.
The consonants 'c', 'h', 'l', 'r', 'n' can be regarded as one letter and the 4 letters can be arranged in 4! = 24 ways. These 5 consonants can be arranged amongst themselves in 5! = 120 ways.
Hence, the number of words in which consonants occur together = $24 \times 120 = 2880$.

PERMUTATIONS OF ALIKE THINGS: The number of permutations of n things taken all at a time where p of the things are alike and of one kind, q others are alike and of another kind, r others are alike and of another kind, and so on is $\frac{n!}{p!q!r!....}$

Example: Find the number of permutations of arranging the letters of the word 'INDEPENDENCE' amongst themselves? In how many arrangements will the two D's be together?

The word INDEPENDENCE has 12 letters of which 'N' is repeated 3 times, 'D' is repeated twice, 'E' is repeated 4 times, and the rest are different.

Hence the number of arrangements is given by $\frac{12!}{3!\,2!\,4!}$.

If the two D's must be together, they can be considered as one letter.
Hence there are 11 letters of which 'N' is repeated 3 times and 'E' four times.

$\therefore$ Number of arrangements = $\frac{11!}{3!\,4!}$

PERMUTATIONS OF REPEATED THINGS: The number of permutations of n different things taken r at a time, when each thing may occur any number of times is n^r.
Example: How many numbers each containing five digits can be formed?
There are in all ten digits including zero. As the first digit of the number cannot be zero, so it can be chosen in 9 ways. As repetition is permitted, the second, third, forth and fifth digits of the numbers can be any of the ten digits and so each of the remaining four places can be filled in 10 ways.

Hence the total number of 5 digit numbers $= 9 \times 10^4 = 90000$.

CIRCULAR PERMUTATIONS: Two ways of arranging five digits 1, 2, 3, 4 5, would be 12345, 54321. Now, if these arrangements are written along the circumference of a circle, the two arrangements are one and the same. Thus, circular permutations are different only when the relative order of the objects is changed; otherwise they are the same. As the number of circular permutations depend on the relative position of objects, we fix the position of one object and arrange the remaining (n – 1) objects in all possible ways. This can be done in (n – 1)! ways.

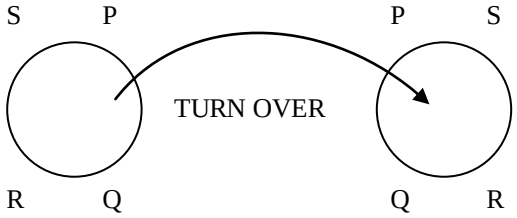
Example: (i) In how many ways can 8 people sit around a table?
Fixing the position of one of the persons, the remaining 7 people can sit around the table in 7!, i.e. 5040 ways.

(ii) In how many ways can 3 boys and 3 girls be seated at a table so that no 2 boys are adjacent?
Let the girls first take up their seats. They can sit in 2! ways. When they have been seated, there remain 3 places for the boys each between two girls. Therefore the boys can sit in 3! ways. Therefore there are 2!× 3!, i.e. 12 ways of seating.

CLOCKWISE AND COUNTER – CLOCKWISE PERMUTATIONS:
Case (i) In which counter-clockwise and clockwise are distinguishable.
While seating 4 persons P, Q, R, S around a table, the permutations are considered different.



Case (ii) Those in which counter clockwise and clockwise are not, distinguishable.
(a) While forming a garland, or stringing beads for a chain, the arrangements are not distinguishable, as the arrangements are not disturbed if we turn the garland over.



(b) The distinction between clockwise and anti-clockwise is ignored when a number of people have to be seated around a table, so as not to have the same neighbors.

Examples:
(i) Find the number of ways in which 20 beads can be arranged to form a necklace.
Fixing the position of one bead, the remaining beads can be arranged in (20 – 1)! i.e 19! ways.
As there is no distinction between the clockwise and anti-clockwise arrangements, the required number of ways = $\frac{1}{2}(19)!$

(ii) In how many ways can 5 girls sit around a table so that all shall not have the same neighbors in any two arrangements.

5 girls can sit around a table in 4! ways. As each person shall not have the same neighbors, the required arrangements = $\frac{1}{2}(4!) = \frac{24}{2} = 12$ ways

COMBINATIONS: Each of the different groups or selections which can be made by taking some or all of a number of things at a time (irrespective of order) is called as combination.
Thus ${}^5C_3$ denotes the number of combinations of 5 things taken 3 at a time.

$${}^nC_r = \frac{n!}{r!(n-r)!}, \quad \therefore {}^nC_r = \frac{{}^nP_r}{r!} \qquad \text{since } {}^nP_r = \frac{n!}{(n-r)!}$$

Note: (i) ${}^nC_r = {}^nC_{n-r}$

(ii) ${}^nC_n = {}^nC_o = 1$

(iii) If ${}^nC_x = {}^nC_y$ then either x = y or x+ y = n

(iv) ${}^nC_r + {}^nC_{r-1} = {}^{n+1}Cr$

Example: (i) If ${}^{16}Cr = {}^{16}Cr+2$, find the value of ${}^rC_4$.
Since $r \neq r + 2$, so $r + r + 2 = 16$
$\therefore 2r = 14 \Rightarrow r = 7.$
$\therefore {}^7C_4 = \frac{7!}{4!3!} = \frac{7\times6\times5}{3\times2} = 35.$

(ii) Find the number of diagonals that can be drawn by joining the angular points of a pentagon?
A pentagon has 5 angular points and 5 sides. The join of 2 angular points is either a side or a diagonal.
The number of lines joining the angular points = ${}^5C_2 = \frac{5!}{2!3!} = \frac{5\times4}{2} = 5\times2 = 10$
But the number of sides = 5
$\therefore$ The number of diagonals = 10 – 5 = 5.

(iii) A committee of 4 is to be selected from 7 boys and 5 girls. In how many ways can this be done so as to include (a) exactly 1 girl, (b) at least 1 girl.
(a) In this case we have to select 1 girl out of 5 girls and 3 boys out of 7 boys.
The number of ways of selecting 3 boys = ${}^7C_3 = \frac{7!}{4!3!} = \frac{7\times6\times5}{3\times2} = 35$
The number of ways of selecting 1 girl = ${}^5C_1 = 5$
$\therefore$ The required committee can be formed in 35 × 5 i.e 175 ways.

(b) The committee can be formed with
(i) four girls (ii) three girls and one boy (iii) two girls and two boys (iv) one girl and three boys.

$\therefore$ The committee formed in (i) ${}^5C_4$ ways, (ii) ${}^5C_3 \times {}^7C_1$ ways (ii) ${}^5C_2 \times {}^7C_2$ ways (iv) ${}^5C_1 \times {}^7C_3$ ways.

Hence the required number

$$= 5 + \left(\frac{5!}{3! \times 2!} \times 7\right) + \left[\left(\frac{5!}{3! \times 2!}\right) \times \left(\frac{7!}{2!5!}\right)\right] + \left[5 \times \frac{7!}{4!3!}\right]$$

$$= 5 + \frac{5 \times 4}{2} \times 7 + \frac{5 \times 4}{2} \times \frac{7 \times 6}{2} + \frac{5 \times 7 \times 6 \times 5}{3 \times 2} = 5 + 70 + 210 + 175 = 460 \text{ ways.}$$

TYPE 1: TO FIND THE TOTAL NUMBER OF COMBINATIONS OF N DISSIMILAR THINGS TAKING ANY NUMBER AT A TIME:

(A) When all things are different:

Each thing maybe disposed of in two ways i.e. it maybe included or rejected.

$\therefore$ The total number of ways of disposing of all the things = $2 \times 2 \times 2 \ldots$ n times = 2^n.

But this includes the case in which all things are rejected.

Hence the total number of ways in which one or more things are taken = $2^n - 1$

Example: There are 6 subjects in an examination. In how many ways can a boy pass in one or more subject?

The boy can either pass or fail in one subject.

Thus the number of ways of doing either = 2^6.

But this includes the case of failing in all subjects.

Hence, the total number of ways of passing in one or more subject = $2^6 - 1 = 64 - 1 = 63$.

(B) When all things are not different:

Suppose, out of (p + q + r ….) = x things, p are of one kind, q are of a second kind, r are of a third kind and the rest different:

Out of p things we may take, 0, 1, 2 … or p.

Hence they maybe disposed in (p + 1) ways. Similarly q alike things maybe disposed of in (q + 1) ways and r alike things in (r + 1) ways. The x different things maybe disposed in 2^x ways. This includes that case in which all are rejected.

$\therefore$ The total number of selections = $(p + 1)(q + 1)(r + 1)\,2^x - 1$

Example: Find the number of ways in which one or more letters from the phrase, 'Ten Tired Turtles' be selected.

In the phrase there are 4 t's, 3 e's, 1n, 1i, 2r, 1d, 1u, 1l, 1s

$\therefore$ The total number of selections = $(4+1)(3+1)(1+1)(1+1)(2+1)(1+1)(1+1)(1+1)(1+1) - 1$

$= 5 \times 4 \times 2 \times 2 \times 3 \times 2 \times 2 \times 2 \times 2 - 1 = 3840 - 1 = 3839$

TYPE 2: DIVISION INTO GROUPS:

To find the number of ways in which x + y different things can be divided into two groups containing x and y different things respectively.

Every time a set of x things are taken, a second set of y things are left behind. The number of combinations of (x+y) things taking x things at a time = ${}^{x+y}C_x = \frac{(x+y)!}{x!(x+y-x)} = \frac{(x+y)!}{x!\,y!}$

In general, ${}^{x+y+z}C_x \times {}^{y+z}C_y \times {}^zC_z = \frac{(x+y+z)!}{x!(y+z)!} \times \frac{(y+z)!}{y!\,z!} \times 1 = \frac{(x+y+z)!}{x!\,y!\,z!}$

Note: (i) The number of ways in which 4p things can be divided equally into 4 distinct groups in $\frac{(4p)!}{(p!)^4}$

(ii) The number of ways in which 4p things can be divided into four identical groups is $\frac{(4p)!}{4!(p!)^4}$

Examples:

(i) In how many ways can 20 things be divided into groups of 8, 7 and 5 things respectively?

The number of ways = $\frac{20!}{8!\,7!\,5!}$

(ii) In how many ways can 20 different sweets be equally divided among 4 students?

The number of ways = $\frac{(20)!}{(5!)^4}$

TYPE 3: PERMUTATIONS AND COMBINATIONS OCCURRING SIMULTANEOUSLY:

Example: How many different 5 letter words, each containing 2 vowels and 3 consonants can be formed with 5 vowels and 17 consonants?

The vowels can be selected in ${}^5C_2$ ways. Three consonants can be selected in ${}^{17}C_3$ ways.

$\therefore$ 2 vowels and 3 consonants can be selected in ${}^5C_2 \times {}^{17}C_3$ ways.

Now each group of 2 vowels and 3 consonants can be arranged in 5! ways.

$\therefore$ The total number of words = ${}^5C_2 \times {}^{17}C_3 \times 5! = 816000$.

Practice Exercise A (Important)

1. ‘a’ and ‘b’ are two digits of a two digit valid number of the form ‘ab’.

   a) How many such numbers exist so that $a \times b$ is odd?

   b) How many such numbers exists that $a \times b$ is odd and $a \neq b$?

   c) How many such numbers exist so that $a \times b$ is even, but not 0 ?

   d) How many such numbers exist so that $a \times b$ is even?

2. ‘a’ and ‘b’ are two digits of a two digit valid number of the form ‘ab’.

   a) How many such numbers exist such that $a > b$?

   b) How many such numbers exist such that $a < b$?

3. ‘abc’ is a three digit number such that $a > b > c$. How many such numbers exist?

4. ‘abc’ is a valid three digit number such that $a < b < c$. How many such numbers exist?

5. ‘abc’ is a valid three digit number such that $a < b > c$. How many such numbers exist?

6. ‘abc’ is a three digit valid number such that ‘a’, ‘b’ and ‘c’ are in an A.P. and ‘b’ is the middle term. How many such numbers exist?

7. Using the numbers 1, 2 or 3, how many distinct three digit numbers can be created?

8. Using the numbers 1, 2, and 3, how many distinct three digit numbers are created so that no two digits are the same?

9. Using the numbers 0, 1 or 2, how many distinct three digit valid numbers can be created?

10. Using the numbers 0,1 and 2, how many valid three digit numbers can be created such that no two digits are the same?

11. Using the numbers 1, 2, 3 and 4, how many three-digit numbers can be created so that the number is divisible by 4, if no digit is used more than once?

12. Using the numbers 1, 2, 3, 4 and 5 exactly once, how many four-digit numbers can be created so that

    a) they are divisible by 5

    b) they are divisible by 15

13. Using the numbers 0, 1, 2, 3, 4, and 5 exactly once, how many 5 digit numbers can be created so that they are divisible by 5?

14. An ice-cream vendor has 5 different flavors of ice-cream. Everyday I eat a double-scooper (two scoops of ice-cream). My two scoops are of different flavors, and on no two days do I eat the same combination of flavors. How many days can the ice-cream vendor serve my requirements?

15. Everybody in a room shakes hands with everyone else only once. If 190 handshakes were exchanged, how many people were in the room?

16. What is the sum of all possible three-digit numbers that can be formed using the digit 1, 2 and 3, such that the digits are not repeated ?

17. What is the sum of all four-digit numbers that can be formed using the 1, 2, 3 and 4 exactly once?

18. What is the sum of all valid four-digit numbers that can be formed using the digits 0, 1, 2, and 3 exactly once?

19. There are 3 different routes from A to B and 4 different routes from B to C. How many routes are these from A to C via B?

20. There are 4 shops in town A, 3 shops in town B and 2 godown's each in town A and B that are owned by Ghosh Babu. Shops within a town have 2 direct lines connecting one another, and shops across towns have 1 direct line connecting one another. The godowns in town A are connected to all other shops by 1 direct line. The godowns in town B is connected to all shops in town B with 2 direct lines, and with all shops in town A with 1 direct line. Finally, the godowns are connected by 2 direct lines. How many lines are there in total?

21. There are four lockers in a line numbered 1, 2, 3, and 4 respectively. In how many ways can these lockers be full if it is known that lockers that are full are consecutively numbered ?

22. I have forgotten my suitcase's three-digit password. What is the maximum number of attempts I need to make to find out the correct password?

23. Monica, Rachel and Phoebe are three friends who live in a building comprising 3 floors with each floor having 3 apartments in a line exactly one below the other. The three stay in separate apartments.

    a) What are the total number of ways in which the three can stay in the building ?

    b) What are the total number of ways in which the three friends can stay, if each stays on a separate floor ?

    c) What are the total number of ways in which the three friends can stay, if the three stay exactly one below the other ?

    d) What are the total number of ways that they can stay, if it is known that Monica and Phoebe stay on the same floor ?

`e) What are the total number of ways in which Monica stays on a floor higher than Phoebe ?

f) What are the total number of ways in which Phoebe and Rachel are neighbours (i.e. live in adjacent apartments) ?

24. My friends phone number is of seven digits. I remember that the first 3 digits are 222. I also know that of the remaining atleast one digit is 7, and no digit is either a 0 or a 2. What is the maximum number of attempts that I need to make to be certain that I get the right number?

25. There are 10 stations on the Harbour Line in Mumbai, called A to J. I have a ticket. What is the probability that it is from station C to station D ?

26. Five boys wearing red hats stand in a straight line, and 4 girls wearing blue hats stand in another straight line parallel to the boys. How many distinct triangles can be created by joining the hats of these people ?

27. There are 4 mugs adjacent to one another. They are filled with either lemon, or orange or grape juice. Adjacent mugs have different juices. In how many ways can this be done ?

28. There are 4 blue balls, 3 red balls and 11 green balls. How many balls must be drawn blindfolded, and without replacement, to ensure that

(a) you have atleast 1 blue ball.

(b) you have atleast 1 green ball.

(c) you have atleast 1 blue and 1 green ball.

(d) you have all blue balls.

(e) you have one ball each of all colours.

(f) you have 2 balls each of atleast two different colours.

(g) you have atleast 1 ball of a colour and 2 balls of another colour.

Explanatory Answers

1. a) ‘a’ can be 1, 3, 5, 7 or 9. ‘b’ also can be 1, 3, 5, 7, 9.
Hence no of ways = 5×5. i.e. with ‘a’ as 1 ‘b’ will be 1, 3, 5, 7, 9. Then with ‘a’ as 3, ‘b’ will be 1, 3, 5, 7, 9…..

b) ‘a’ can be 1, 3, 5, 7 or 9. When ‘a’ is 1, ‘b’ can be 3, 5, 7, or 9 i.e. 4 different nos. Hence total ways = $5 \times 4 = 20$.

c) As $a \times b$ has to be even, ‘a’ or ‘b’ can be either of 2, 4, 6 or 8. So if a is even, be can be eve or odd. And for a odd, b has to ve even.
Total no of ways if a is even = $4 \times 9 = 36$.
Total no of ways if a is odd = $5 \times 4 = 20$.
Total = 56

d) Of the 90 total 2-digit numbers there are 25 that are such their product of the digits is odd. Rest 65 are even.

2. a) When ‘a’ is 9, ‘b’ can be 0-8 i.e. 9 difference nos.
When ‘a’ is 8, ‘b’ can be 0-7 i.e 8 different nos.
.
.
.
Hence total number of ways = 9 + 8 + 7 + …… 1 = 45.

b) When ‘b’ is 9, ‘a’ can be 1 – 8 i.e. 8 different nos.
When ‘b’ is 8, ‘a’ can be 1 – 7 i.e. 7 different nos.
.
.
.
Hence total no of ways = 8 + 7 + 6 + …..1 = 36

3. Fix ‘a’ as 9.
When ‘b’ is 8, ‘c’ can be 7 – 0 i.e. 8 different nos.
When ‘b’ is 7, ‘c’ can be 6 – 0 i.e. 7 different nos.
.
.
.
When ‘b’ is 1, ‘c’ can be 0 i.e. 1 no only.
So with ‘a’ as 9, there are 8 + 7 + 6…1= 36 ways

Fix ‘a’ as 8.
When ‘b’ is 7, ‘c’ can be 6 – 0.
When ‘b’ is 6, ‘c’ can be 5 – 0
.
.
.
Total ways = 7 + 6 +…….1 = 28 ways.

Keep on decreasing value of ‘a’ by 1.

Hence no of ways such that a > b > c = 36 + 28 + 21 + 15 + 10 + 6 + 3 + 1 = 120

4. Fix ‘c’ as 9.
When b is 8, a can be 1 – 7 i.e. 7 different nos.
When b is 7, a can be 1 – 6 i.e. 6 different nos.
.
.
.
∴With c as 9, number of ways = 7 + 6 + ….1 = 28

Using similar logic as in question 3, keep on decreasing c by 1, and find number of ways of b and a.

Hence Ans = 28 +21 + 15 + 10 + 6 + 3 + 1 = 84

5. Fix b as 9.
a can be 1 – 8, and c can be 0 – 8. Hence no. of ways = $8 \times 9 = 72$
Fix b as 8.
A can be 1 – 7, and c can be 0 – 7. Hence no of ways = $7 \times 8 = 56$
.
.
.

Hence answer = $8 \times 9 + 7 \times 8 + 6 \times 7 + \ldots\ldots\ldots 1 \times 2 = 240$.

6. With CD (common difference) as 1, the no's can be 1, 2, 3 or 2, 3, 4 or 3, 4, 5……7, 8, 9…. i.e. 7 ways.
With CD as 2, the nos. can be 1, 3, 5 or 2, 4, 6……or 5,7,9. i.e. 5 ways
With CD as 3, the nos. can be 1,4,7 or 2,5,8 or 3, 6, 9. i.e. 3 ways
With CD as 4, the nos. can be only 1, 5 and 9.
Number of ways = 1 + 3 + 5 + 7 = 16.
However CD can be –1, –2, –3 and –4. i.e. arranged in descending order.
Hence with CD as –1, –2, –3 and –4 number of ways = 16.
However even 0 can be included for value of C. Additional numbers are (0, 1, 2), (0, 2, 4), (0, 3, 6) and (0, 4, 8).
Hence total = 16 + 16 + 4 = 36.

7. There are 3 blanks, each of which can be filled in 3 ways. Hence total no of ways = $3 \times 3 \times 3 = 27$.

8. As digit cannot be repeated, no of ways = $3 \times 2 \times 1 = 6$

9. Let the no be of the form abc. Then 'a' can be 1 or 2, while 'b' and 'c' can be either 1 or 2 or 0.
Hence total ways = $2 \times 3 \times 3 = 18$.

10. Whenever '0' is a digit, then we start from LHS.
Let the no be of the form abc.
'a' can be 1 or 2.
'b' can be 0 or one of the remaining no's, from 1 and 2.
'c' can only be the number which is leftover.
Hence total ways = $2 \times 2 \times 1 = 4$.

11. To be divisible by 4, last digit has to be even. So no must be ending with 2 or 4.
To be divisible by 4, last two digits have to be divisible by 4. Let the no. be of the form abc.
When c =2, then b = 1 or 3. a would be one of the remaining 2 numbers. Hence total ways = $2 \times 2 \times 1 = 4$ ways.
When c = 4, then b can only be 2. Hence total ways = $2 \times 1 \times 1 = 2$ ways. Total = 6 ways.

12. a) Let number be of the form a b c d.
Last digit has to be 5. c can be any of the other 4 nos. b can be any of the other 3 nos. a can be any of the other 2 nos. $\therefore$ No. of ways = $1 \times 4 \times 3 \times 2 = 24$

b) Last no. has to be 5. For tests of divisibility by 15, additionally the sum of the digits have to be multiple of 3. Only 1 set of numbers gives that. (5 + 1 + 2 + 4). Now 1, 2 and 4 can be arranged in $3 \times 2 \times 1$ ways.
So total no. of ways = 6

13. Let the no. be a b c d e. Now, e can be 0 or 5.
When e = 0, no. of ways = $5 \times 4 \times 3 \times 2 = 120$
When e = 5, no. of ways = $4 \times 4 \times 3 \times 2 = 96$
So total no. of ways = 216.

14. Let flavours be 1, 2, 3,4 and 5. So possibilities are (1, 2) (1, 3) (1, 4) (2, 3) (2, 4) (3, 4) (3, 5) and (4, 5). So total ways = 10

15. If there are n people in the noon, then no. of handshakes = Sum of numbers from 1 to n – 1. i.e. $\frac{(n-1)n}{2}$

Now, it is given that $\frac{(n-1)n}{2} = 190$

$\therefore n^2 - n - 380 = 0$ or n = 20

$\therefore$ There are 20 people in the room.

(Logic: If there were 5 people in the room, then A would shake hands with B, C, D and E. Then B would not shake hands with A again. He would shake hands with C, D and E. C would then shake hands with D and E. Finally D and E shake hands. Hence, with 5 people the answer is 4 + 3 + 2 + 1. So with 20 people, the answer would be 19 + 18 + …+ 1 = 190.

16. Let the numbers be in the units, tens and hundreds column.
Now, the total number of numbers that can be created is $3 \times 2 \times 1 = 6$ numbers.
Each of the numbers 1, 2 and 3 appear in each of the columns exactly 2 times.
So sum of all nos. in the units place = $2 \times 1 + 2 \times 2 + 2 \times 3 = 2 (1 + 2 + 3) = 12$
The sum of tens and hundreds is also = 12.
So the sum of the 6 nos. is 1332. (2 in the units place, and 1 carried over to tens place.
∴ 3 in the tens place and 1 carried over. Finally 13 is the total of the hundred's place.
Logic: For the sake of visualization, look at this table. It will help visualize other similar questions).

Hundreds	Tens	Unit
1	2	3
1	3	2
2	1	3
2	3	1
3	1	2
3	2	1

17. Using same logic as previous question.
No. of numbers = $4 \times 3 \times 2 \times 1 = 24$
So sum of the nos. in any column = $6 \times 1 + 6 \times 2 + 6 \times 3 + 6 \times 4 = 6 (1 + 2 + 3 + 4) = 60$
∴ The sum of the 24 no's = 66660. (Use same method as prev question to get to the sum).

18. A four digit number cannot start with 0.
So we can first add all four digit numbers possible with 0, 1, 2 and 3 and then subtract those that have 0 in the thousand column.
Sum of nos. with 0,1,2,3 occupying all places = $6 \times 0 + 6 \times 1 + 6 \times 2 + 6 \times 3 = 6(1 + 2 + 3) = 36$
∴ Sum of the 24 nos. = 39996.
∴ Sum of nos. that have 0 in the thousandth column = 1332 (From question 16)
∴ Answer = 39996 – 1332 = 38664.

19. There will be $3 \times 4 = 12$ routes

20. Direct lines between
Shops within town A = 2 (3 + 2 + 1) = 12
Shops within town B = 2 (2 + 1) = 6
Shops across town = $4 \times 3 = 12$
godown A with all shops = $2 \times 7 = 14$
godown in B with shops in B = $2 \times 2 \times 3 = 12$
godown in B with shops in A = $2 \times 4 = 8$
godown with each other = 2 (3 + 2 + 1) = 12
Total = 76.

21. If 1 locker is full, it can be in 4 ways.
If 2 lockers are full, it can be in 3 ways.
i.e. (1, 2) (2, 3) or (3, 4)
If 3 lockers are full, it is in 2 ways.
i.e. (1, 2, 3) or (2, 3, 4)

If 4 lockers are full, there is only 1 ways.
So total = 1 + 2 + 3 + 4 = 10 ways.

22. Total possibilities of the combination = $10 \times 10 \times 10 = 1000$
So total attempts required to know the correct sequence is 1000-1 = 999.

23. a) $9 \times 8 \times 7$
b) $9 \times 6 \times 3$
c) $9 \times 2 \times 1$ (first person has 9 choices the seconds person only 2 and the third person only 1 choice)
d) $9 \times 2 \times 7$ (Monica chooses any of 9, and phoebe chooses what remains on that floor. Rachel has the remaining 7 options open to her).
e) (i) Monica stays on top floor, then no. of ways = $3 \times 6 \times 7$
(ii) Monica stays on middle floor, then no. of ways = $3 \times 3 \times 7$
So total = $3 \times 6 \times 7 + 3 \times 3 \times 7 = 7 \times 3 \times 9$
f) (i) If phoebe chooses a corner flat (6 flats), then Rachel has to stay adjacent to her. So only 1 choice for Rachel. Hence no. of ways = $6 \times 1 \times 7$
(ii) If phoebe chooses a center flat, Rachel has 2 choices (either side) and hence no. of ways = $3 \times 2 \times 7$.
So total ways = $6 \times 7 + 6 \times 7 = 84$

24. Of the remaining 4 no's; each can be any of 8 digits (not 0 and 2).
So no. of ways = 8^4
No. of ways in which there is no 7 at all = 7^4
$\therefore$ Required answer = $8^4 - 7^4$

25. From every station, there can be 9 different tickets.
So total no of different tickets = 90. Required Probability = 1 / 90

26. No. of triangle with 2 points on line A and 1 point on line B = ${}^5C_2 \times {}^4C_1$
No. of triangles with 2 points on line B and 1 points on line A = ${}^4C_2 \times {}^5C_1$
So total ways = ${}^5C_2 \times {}^4C_1 + {}^4C_2 \times {}^5C_1$

27. No. of ways = $3 \times 2 \times 2 \times 2$
(Note that if 2nd mug is a particular juice, then the 3rd can be 2 of the other flavours. i.e. also the same as the 1st mug.)

28. a) Unless we draw 11 green and 3 red balls out, we cannot be sure. So minimum withdrawals required to ensure 1 blue ball = 11 + 3 + 1 = 15
b) 4 blue + 3 red + 1 = 8
c) 3 red + 11 green + 1 blue = 15 balls
d) 11 green + 3 red + 4 blue = 18 balls
e) 11 green + 4 blue + 1 red = 16 balls
f) 11 green + 1red + 1 blue + 1 red/blue = 14 balls
g) 11 green + 1 red/blue = 12 balls.

Chapter 21 Miscellaneous Problems

Suggested Time : 120 minutes

The syllabus of aptitude tests is never-ending. The chapters before cover all major types of questions asked in campus recruitment tests. This chapter provides you practice questions on many other areas that are less important. The questions normally are easy, and we suggest that you solve them.

THEORY & FORMULAE

Cuboid:

i) Volume of a cuboid = length $\times$ breadth $\times$ height = $\ell \times b \times h$ (cubic units)

ii) Diagonal of a cuboid = $\sqrt{\ell^2 + b^2 + h^2}$ units

The length of the longest rod that can be placed in a rectangular box or room is its diagonal.

iii) Total surface area = $2(\ell b + bh + \ell h)$ unit^2

iv) Area of four walls of a box or a room, i.e. lateral surface area = $2(\ell + b) \times h$ unit^2

v) Let the external dimensions (length, breadth, and height) of a rectangular box be ℓ, b, and h and the thickness of its sides be 't', then

a] if the box is closed, its internal dimension will be$(\ell - 2t)$,$(b-2t)$, and $(h - 2t)$.

b] if the box is open, its internal dimension will be $(\ell - 2t)$, $(b - 2t)$, and $(h - t)$.

Cube:

i) Volume of a cube = $(\text{edge})^3 = a^3$ unit^3

ii) Diagonal of a cube = $\sqrt{3}$ a unit

iii) Total surface area = $(6a^2)$ unit^2

iv) Lateral surface area = $(4a^2)$ unit^2

Cylinder

(i) Volume of cylinder = (Area of base) $\times$ height = $\pi r^2 h$ unit^3
where r is radius of base and h is the height

(ii) Lateral/Curved surface area
= Circumference of base $\times$ height = $2\pi rh$ unit^2

(iii) Total surface area = Curved surface area + Area of two ends
$= 2\pi rh + 2\pi r^2 = 2\pi r(h + r)$ unit^2

(iv) Volume of material in a hollow pipe
$= \pi (R^2 - r^2)\ \ell$, where ℓ is the length of the pipe, R is the outer radius, and r is the inner radius.

Sphere

(i) Volume of a sphere = $\frac{4}{3}\pi R^3$

(ii) Volume of hollow sphere = $\frac{4}{3}\pi R^3 - \frac{4}{3}\pi r^3 = \frac{4}{3}\pi (R^3 - r^3)$ (unit^3)

(iii) Volume of a hemi-sphere = $\frac{2}{3}\pi R^3$

Cone

(i) Volume of the cone = $\frac{1}{3}\pi r^2 \times h$

Examples:

1. Three cubes of metal whose edges are 3, 4, and 5 cm respectively, are melted down and formed into a single cube. Find its diagonal.
   1] $6\sqrt{3}$ 2] 6 3] 8 4] $8\sqrt{3}$

2. The total surface area of a cube is 726 cm^2. Find its volume.
   1] 121 cm^2 2] 144 cm^3 3] 1331 cm^3 4] 1728 cm^3

3. A road roller is cylindrical in shape. Its circular end has a diameter 0.84 m and its width is 2 m. Find the least number of complete revolutions that the roller must make in order to level a road measuring 80 m by 26.4 m.
   1] 400 2] 350 3] 300 4] 250

4. A metallic block $11 \times 7 \times 10$ cm is melted and cast into cones of radius 1.75 cm and height 8 cm. How many cones can be cast?
   1] 32 2] 30 3] 22 4] 15

5. A hollow cylinder has 792 cm^3 of material in it. If its height is 9 cm and external diameter is 16 cm, find the thickness.
   1] $\frac{1}{2}$ cm 2] 1 cm 3] $1\frac{1}{2}$ cm 4] 2 cm

6. A hemispherical hollow bowl has material of volume $\frac{436}{3}\pi$ cm^3. If its external diameter is 14 cm, find its thickness.
   1] 2 cm 2] $1\frac{1}{2}$ cm 3] 1 cm 4] $\frac{1}{2}$ cm

7. A cone, a hemisphere and a cylinder stand on equal bases and have the same height. Find the ratio of their volumes.
   1] 1 : 3 : 2 2] 1 : 2 : 3 3] 1 : 2 : 5 4] 3 : 2 : 1

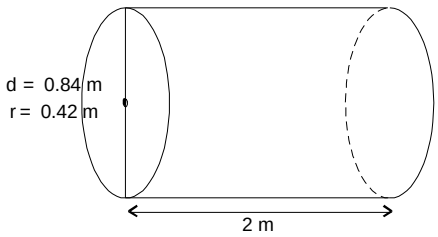
8. How many lead balls of a quarter of an inch in diameter can be cast out of the metal of a ball 3 inch in diameter, supposing no waste ?
   1] 788 2] 1528 3] 1728 4] 2546

Explanatory Answers

1. Volume of the single cube = $(3^3 + 4^3 + 5^3)$ cm^3 = 216 cm^3
   $\therefore$ Edge = $\sqrt[3]{216}$ = 6 cm
   $\therefore$ Diagonal = $6\sqrt{3}$ = 6×1.732 = 10.392 cm
   Hence [1]

2. TSA of a cube = $6(edge)^2$
   $\therefore$ $726 = 6(edge)^2$
   $\therefore$ $(edge)^2 = 726/6 = 121$
   => edge = $\sqrt{121}$ = 11 cm
   $\therefore$ Volume = $(edge)^3 = 11^3$ = 1331 cm^3. Hence [3]

3. Diameter of base = 0.84 m



$\therefore$ Radius of base = $\frac{0.84}{2}$ = 0.42 m

Width = 2m

Curved surface area

$= 2\pi r\ell = 2 \times \frac{22}{7} \times 0.42 \times 2 = 5.28\ m^2$

Area of road = $80 \times 26.4\ m^2$

$\therefore$ Number of revolutions = $\frac{80 \times 26.4}{5.28} = 400$

Hence [1]

4. Volume of block = Number of cones × Volume of cone

$\therefore 11 \times 7 \times 10 = n \times \frac{1}{3} \times \frac{22}{7} \times 1.75 \times 1.75 \times 8$

$\therefore n = \frac{11 \times 7 \times 10 \times 3 \times 7 \times 100 \times 100}{22 \times 175 \times 175 \times 8} = 30$

Hence, the number of cones cast is 30. Hence [2]

5. Volume of the hollow cylinder = 792 cm^3, height = 9 cm, external diameter = 16 cm.

$\therefore$ R = 16/2 = 8 cm.

Volume of hollow cylinder = $\pi (R^2 - r^2)h = \pi (R + r)(R - r) \times h$

$\therefore 792 = \frac{22}{7} \times (8 + r)(8 - r) \times 9$

$\Rightarrow 64 - r^2 = \frac{792 \times 7}{22 \times 9} = 28$

$\therefore r^2 = 64 - 28 = 36 \Rightarrow r = 6$ cm

$\therefore$ R – r = thickness = 8 – 6 = 2 cm

Hence, thickness = 2 cm. Hence [4]

6. Let R and r be the external and internal radii of the bowl.

R = 14/2 = 7 cm

Volume of the material = $\frac{2}{3} \pi (R^3 - r^3)$

$\therefore \frac{436}{3} \pi = \frac{2}{3} \pi (7^3 - r^3)$

$\therefore 7^3 - r^3 = 218 \Rightarrow r^3 = 343 - 218 = 125$

$\therefore r = \sqrt[3]{125} = 5$ cm

Hence, thickness = R – r = 7 – 5 = 2 cm. Hence [1]

7. Let r be the common radius. Height (h) = r.

$\therefore$ Volume of cone : Volume of hemisphere : Volume of cylinder

$= \frac{1}{3}\ \pi r^3 : \frac{2}{3}\ \pi r^3 :\ \pi r^3$

$= \frac{1}{3} : \frac{2}{3} : 1 = 1 : 2 : 3$ is the ratio of their volumes. Hence [2]

8. Volume of each 1/4 inch diameter ball

$= \frac{4}{3}\pi\left(\frac{1/4}{2}\right)^3$

$= \frac{4}{3}\pi \times \frac{1}{4^3 \times 8}$

$= \frac{\pi}{4^3 \times 3 \times 2}$

Volume of 3-inch diameter balls

$= \frac{4}{3}\pi \times \left(\frac{3}{2}\right)^3$

$= \frac{4}{3}\pi \times \frac{3^2 \times 3}{8}$

$= \frac{\pi \times 3^2}{2}\ \text{inch}^3$

$\therefore$ Required number of balls

= Volume of 3-inch dia. ball / Volume of $\frac{1}{4}$ inch dia. ball

$= \frac{\pi \times 3 \times 3 \times 4 \times 4 \times 4 \times 3 \times 2}{2 \times \pi} = 1728$ balls

Hence, the number of balls that can be cast is 1728. Hence [3]

BODMAS:

When we have to perform a series of mathematical operations, there is a rule regarding the order in which we should perform these operations. This rule is called BODMAS rule. BODMAS is an acronym which is elucidated as:
B - Bracket in the order ( ), { }, [ ]
O - of (indicating × , i.e. multiplication)
D - Division
M - Multiplication
A - Addition
S - Subtraction

Examples:

1. Using principles of BODMAS, find the value of the expression $\left(\frac{\left(8+\frac{5}{9}\div\frac{10}{36}\right)^2}{10\%\text{ of }25\times 4}\right)^2$

(1) 10 (2) 100 (3) 1 (4) 2.56 (5) none of these

2. Using BODMAS rulers, find the value of $\left[\frac{24}{7}\div\left(10\%\text{of}\frac{4\times 3}{140}\right)\right]+\left[\frac{3}{5}-\frac{1}{4}+15\%\right]$

(1) $\frac{3}{350}$ (2) 400.5 (3) 450 (4) 400 (5) none of these

3. What is the value of the following expression $\frac{(25\%\text{ of }\frac{4}{5}\text{ of }90)\div 2}{(75\%\text{ of }\frac{6}{9}\text{ of }108)\times 11}$?

1] 1/11 2] 11/54 3] 1/66 4] 1/594 4] None of these

Explanatory Answers

1. Result = $\left[\frac{(8+2)^2}{\frac{10}{100}\times 25\times 4}\right]^2 = \left[\frac{10^2}{10}\right]^2 = 10^2 = 100$. Hence (2)

2. $\left[\frac{24}{7}\div\left(\frac{10}{100}\times\frac{12}{140}\right)\right]+\left[\frac{3}{5}-\frac{1}{4}+\frac{15}{100}\right]$ = 400 + 0.5 = 400.5 Hence (2)

3. $x=\frac{\left(\frac{1}{5}\times 90\right)\div 2}{\frac{3}{4}\times 6\times 12\times 11} = \frac{9}{54\times 11}=\frac{1}{66}$

CALENDAR:

The following should be remembered:

(1) The number of days more than the complete number of weeks in a given period are called odd days.
(2) An ordinary year contains 365 days, i.e. 52 weeks 1 odd day.
(3) Every year, which is divisible by 4, e.g. (2004), is called a leap year.
(4) A leap year contains 366 days, i.e. 52 weeks and 2 odd days.
(5) Century year is a leap year if it is divisible by 400, i.e. every fourth century year is a leap year.
E.g.: 800, 1200, 1600 etc.
(6) 100 years (a century) contains 76 ordinary years + 24 leap years.
= (76 × 52) weeks + (76 × 1) odd days + (24 × 52) weeks + (24 × 2) odd days.
= [(76 × 52) + (24 × 52)] weeks + (76 + 48) odd days
= [(76 × 52) + (24 × 52)] weeks + (124) odd days
= [(76 × 52) + (24 × 52) + 17] weeks + 5 odd days,
i.e. 100 years contain 5 odd days.
200 years contain 2(5), i.e. 10 odd days and therefore (10-7), i.e. 3 odd days

Similarly 300 years contains 1 odd day, 400 years contain 4 (5) +1, i.e. 21, i.e. '0' odd days.
1st January A D was Monday ∴ we must count days from Sunday, i.e. Sunday for 0 odd days, Monday for 1 odd day and so on.

Examples:

(1) What day of the week was 3rd September 1861?
(2) It is Tuesday, March 25th. This is a leap year. What will be the day of the week on this date after 3 years?

Explanatory Answers

1. 3rd September means 1860 completed years + first 8 months of the year 1861 + 3 days in September.
1860 = (1600 + 200 + 60) years
1600 years gives no odd days.
200 years gives 3 odd days.
60 years gives (45×1+15×2 odd days) =5 odd days.
Total no of days =3+5=8, i.e. 1 odd day

From 1st Jan to 3rd September There are

January	: 3 odd days
February	: 0 odd days
March	: 3 odd days
April	: 2 odd days
May	: 3 odd days
June	: 2 odd days
July	: 3 odd days
August	: 3 odd days
September	: 3 odd days
Total	: 22 odd days, i.e. 3 weeks + 1 odd day

∴Total number of odd days = 1 + 1 = 2
∴3rd September 1861 was a Tuesday.

2. This is a leap year, so none of the next 3 years is a leap year. Each year will give one odd day so the day of the week will be 3 odd days beyond Tuesday i.e. it will be Friday.

LAWS OF INDICES

2 gets multiplied with itself 1000 times. How do you write the result?

When 2 is multiplied with itself two times, we write the result as 2×2 , i.e. 2^2

When 2 is multiplied with itself three times, we write the end result as 2^3, i.e. $2 \times 2 \times 2$

When 2 gets multiplied with itself 1000 times, we write the result as 2^{1000}. This is the way of expressing a number in Index-base form where 2 is the base and 1000 is the index.

Similarly, if 'a' is multiplied with itself 'n' times, we write the end result as a^n.

$a \times a \times a$ ..... (n times) $= a^n$,

The number a^n, is written in INDEX-BASE form.

In the expression a^n,

→ 'a' is called BASE and

→ 'n' is called POWER or INDEX or EXPONENT. The plural form of INDEX is called INDICES.

There are certain basic rules / formulae to solve the questions involving indices. These are called laws of indices. The basic rules are discussed below.

Rule 1: $a^m \times a^n = a^{m+n}$

For example, $a^4 \times a^5 = a^{4+5} = a^9$

a^4 indicates that 'a' is multiplied with itself 4 times and a^5 indicates that 'a' is multiplied with itself 5 times. Hence in ($a^4 \times a^5$), 'a' is multiplied with itself (4 + 5 = 9 times), which can be written as a^9.

Now, work out the value of $(3 \times 10^4) \times (2 \times 10^6)$.

$3 \times 10^4 \times 2 \times 10^6 = 3 \times 2 \times 10^6 \times 10^4$

$= 6 \times 10^{6+4} = 6 \times 10^{10}$

Rule: 2 $\frac{a^m}{a^n} = a^{m-n}$

For example, $\frac{a^{10}}{a^4} = a^{10-4} = a^6$

$$\frac{a^{10}}{a^4} = \frac{a \times a \times a \times a \times a \times a \times a \times a \times a \times a}{a \times a \times a \times a}$$

$$\therefore \frac{a^{10}}{a^4} = a \times a \times a \times a \times a \times a = a^6$$

Rule: 3 $\left(a^m\right)^n = a^{mn}$

For example, $\left(a^5\right)^6 = a^{30}$

a^5 indicates that 'a' is multiplied with itself 5 times and $(a^5)^6$ indicates that a^5 is multiplied with itself 6 times. Hence, totally 'a' is multiplied with itself (5 × 6 = 30 times), which can be written as a^{30}.

Rule: 4 $a^{\frac{1}{n}} = \sqrt[n]{a}$

For examples,

1. $a^{\frac{1}{2}} = \sqrt{a}$
2. $a^{\frac{3}{2}} = \left(a^3\right)^{\frac{1}{2}} = \sqrt{a^3}$
3. $a^{\frac{1}{3}} = \sqrt[3]{a}$
4. $a^{\frac{2}{3}} = \left(a^2\right)^{\frac{1}{3}} = \sqrt[3]{a^2}$

Rule: 5 $(ab)^m = a^m \times b^m$

For example, $(4 \times 2)^3 = 4^3 \times 2^3 = 64 \times 8 = 512$

Rule: 6 $\left(\frac{a}{b}\right)^m = \frac{a^m}{b^m}$

For example, $\left(\frac{3}{2}\right)^4 = \frac{3^4}{2^4} = \frac{81}{16}$

Rule: 7 $a^{m^n} \neq \left(a^m\right)^n$

Is $\left(a^3\right)^2$ equal to a^{3^2} ?

The answer is "NO"

We know that $\left(a^m\right)^n = a^{mn}$. Hence, $\left(a^3\right)^2 = a^6$.

In a^{3^2}, since 'a' is raised to the power of 3^2, i.e. 9,

$\therefore a^{3^2} = a^9$.

Hence, $\left(a^3\right)^2 \neq a^{3^2}$.

Now try and write the expression $3^{2^{3^{2^1}}}$ in the form of a^m.

To write $3^{2^{3^{2^1}}}$ in the form of a^m, we have to start working out the values of the powers from the top to bottom.

i.e. $3^{2^{3^{2^1}}} = 3^{2^{3^2}}$ $(\because 2^1 = 2)$

$3^{2^{3^2}} = 3^{2^9}$ $(\because 3^2 = 9)$

$3^{2^9} = 3^{512}$ $(\because 2^9 = 512)$

LOGARITHMS

How else can you write the equation $10^x = 100$?

The equation can also be written with the help of logarithm.

WHAT IS A LOGARITHM?
Definition: Logarithm is the exponent to which the base must be raised to produce a given number.

Let us understand the definition of the logarithm in a better way.

In the equation $10^x = 100$,
'x' is nothing but an exponent to which the base (10) must be raised to produce a given number (100). (Please go through the definition of the logarithm now)

In other words, the value of x is expressed in-terms of logarithm.
i.e. $x = \log_{10}100$
We know that 10 has to be raised to the power of 2 to produce 100.
$\therefore x = 2$
Hence, $\log_{10}100 = 2$.

How do we find the value of $\log_2 16$?
The other way of asking this question is "To what power 2 has to be raised to get 16?"
We know that $2^4 = 16$.
Hence, $\log_2 16 = 4$.

$2^4 = 16$	**..... is equivalent to ..... (means the exact same thing as)**	**$\log_2 16 = 4$**

The expression $\log_2 16 = 4$ is read as, "The logarithm of sixteen to the base 2 is 4." In the same manner, the logarithm of 1000 to the base 10 is 3.
We often just call them "logs" instead of "logarithms," simply because that's faster and easier to say, but the meaning is still the same.

Note

1. Logarithms are only defined for POSITIVE NUMBERS. Logarithms are NOT defined for NEGATIVE NUMBERS.

For example, in "$\log_a N$", 'a' and 'N' are POSITIVE NUMBERS.

2. $a^x = N$ is in index base form. $\log_a N = x$ is in logarithmic form.
3. If the base of a logarithm is not given, then we have to consider the base as 10.

Example: Find x, if $\log_x 36 = 2$

We know that $\log_x 36 = 2$ can be written as $x^2 = 36$, i.e. $\log_x 36 = 2 \Rightarrow x^2 = 36$ (or) $x = 6$

($x \neq -6$, because logarithm of negative numbers are NOT defined).

Example: Find the value of $\log_6 \sqrt[3]{216}$.

Let $\log_6 \sqrt[3]{216} = x$.

We know that $\log_6 \sqrt[3]{216} = x$ can be written as $6^x = \sqrt[3]{216}$

$6^x = \sqrt[3]{216} \Rightarrow 6^x = \left(6^3\right)^{1/3} \Rightarrow 6^x = 6^1 \Rightarrow \therefore x = 1.$

The given rules/formulae below are the concepts that we follow to solve questions involving logarithms.

Rule 1: Logarithm of addition can be written as logarithm of product. Also logarithm of product can be written as logarithm of addition.

$\log_a m + \log_a n = \log_a (m \times n)$

For example, $\log_2 3 + \log_2 7 = \log_2 (3 \times 7)$

Rule 2: Logarithm of subtraction can be written as logarithm of division. Also logarithm of division can be written as logarithm of subtraction.

$$\log_a m - \log_a n = \log_a \left(\frac{m}{n}\right)$$

For example, $\log_5 45 - \log_5 17 = \log_5 \left(\frac{45}{17}\right)$

Example: Solve x if log x = log 5 + 2 log 2

log x = log 5 + 2 log2

log 5 + 2 log2 = log 5 + log 2^2 = log 5 + log4 (or) log (5 × 4)

$\therefore$ log x = log 20 (or) x = 20

Example: Simplify log($x^2 - 9$) – log (x + 3)

log ($x^2 - 9$) – log (x + 3) = $\log \frac{x^2 - 9}{x + 3}$ = $\log \frac{(x+3)(x-3)}{(x+3)}$ =log (x – 3).

Note

1. What is the value of $\log_7 7$?

The other way of asking this question is "To what power 7 has to be raised to get 7?"
We know that $7^1 = 7$
Hence, $\log_7 7 = 1$.

Alternately,

Let x = $\log_7 7$

x = $\log_7 7$ can be written as $7^x = 7$

$\therefore x = 1$

Note

The logarithm of any quantity to the same base is unity, i.e. $\log_a a = 1$. For example, $\log_3 3 = 1$.

Here the value of 'a' cannot be zero ($a \neq 0$) because 0^0 is undefined.

2. What is the value of $\log_6 1$?

The other way of asking this question is "To what power 6 has to be raised to get 1?"
We know that $6^0 = 1$
Hence, $\log_6 1 = 0$.

Alternately,

Let x = $\log_6 1$

x = $\log_6 1$ can be written as $6^x = 1$

$\therefore x = 0$

The logarithm of 1 to any base is zero, i.e. $\log_a 1 = 0$. Here the value of 'a' cannot be zero ($a \neq 0$).

For example, $\log_4 1 = 0$ and $\log_{10} 1 = 0$

Rule 3: When you take the reciprocal of a logarithm the base and power of logarithm inter changes, i.e. $\log_b a = \frac{1}{\log_a b}$

For example, $\log_7 5 = \frac{1}{\log_5 7}$

Example: Prove that $\frac{1}{\log_x xy} + \frac{1}{\log_y xy} = 1$.

$\frac{1}{\log_x xy} = \log_{xy} x$ and $\frac{1}{\log_y xy} = \log_{xy} y$

$\therefore$ LHS = $\log_{xy}x + \log_{xy}y = \log_{xy}xy = 1$.

Rule 4: If $\log_a M = \log_b N$ and M = N, then a = b.
Here the value of M and N cannot be 0, because irrespective of the base, log 1 = 0.
Example:
If $\log_5 4 = \log_x 4$, then x = 5.

Rule 5: If $\log_a M = \log_b N$ and a = b, then M = N.
Here the value of a and b cannot be 0, because logarithm of 0 base is not defined.
Example:
If $\log_5 4 = \log_5 Y$, then Y = 4.

Example: Prove that $\log \frac{27}{16} - 2\log \frac{3}{7} + \log \frac{48}{147} = \log 3$.

LHS: $\log \frac{27}{16} - \log\left(\frac{3}{7}\right)^2 + \log \frac{48}{147}$

$= \log \frac{27}{16} - \log \frac{9}{49} + \log \frac{48}{147}$

$= \log\left(\frac{27/16}{9/49}\right) + \log \frac{48}{147}$

$= \log\left(\frac{27}{16} \times \frac{49}{9} \times \frac{48}{147}\right) = \log 3$.

SURDS

Numbers of the type $\sqrt{2}$, $\sqrt[3]{2}$, $\sqrt{3}$, $\sqrt[4]{5}$ and $\sqrt[5]{21}$ etc are called SURDS or RADICALS. Each of them represents an irrational number. $\sqrt[n]{a}$ is a surd, it is understood that
(i) 'a' is a rational number and
(ii) $\sqrt[n]{a}$ is an irrational number.
In this topic, the surds of the type $a + \sqrt{b}$ and $a + \sqrt{b} + \sqrt{c}$ are discussed from completive exams point of view.

Note
The surds of the form $a + \sqrt{b}$ is called as quadratic surd.
Example: $1 + \sqrt{2}$, $2 + \sqrt{7}$ and $8 + \sqrt{11}$

RATIONALISATION OF SURDS

Since the surd of the form $\frac{1}{a+\sqrt{b}}$ has the irrational number in the denominator, it is difficult to perform arithmetic operations on it. Hence, the denominator has to be converted into a rational number, so that the arithmetic operations can be done easily. The process of converting the denominator of the surd into a rational number without changing the value of the surd is called as rationalisation of the surd.

To convert the denominator into rational number, multiply the denominator and numerator simultaneously with the conjugate of the surd in the denominator.

Note

The conjugate of the surd, $a + \sqrt{b}$ is of the form $a - \sqrt{b}$ or $-a + \sqrt{b}$

Example: Simplify $\frac{1}{\sqrt{5}+\sqrt{3}}$

$$\frac{1}{\sqrt{5}+\sqrt{3}} = \frac{1}{\sqrt{5}+\sqrt{3}} \times \frac{\sqrt{5}-\sqrt{3}}{\sqrt{5}-\sqrt{3}}$$

$$= \frac{\sqrt{5}-\sqrt{3}}{\left(\sqrt{5}+\sqrt{3}\right)\left(\sqrt{5}-\sqrt{3}\right)}$$

$$= \frac{\sqrt{5}-\sqrt{3}}{\left(\sqrt{5}\right)^2-\left(\sqrt{3}\right)^2}$$

$$= \frac{\sqrt{5}-\sqrt{3}}{5-3}$$

$$= \frac{\sqrt{5}-\sqrt{3}}{2}$$

SQUARE ROOT OF A SURD

How do we find the square root of $a + 2\sqrt{b}$?

Method:

Let the square root of $(a + 2\sqrt{b})$ be of the form $(\sqrt{x} + \sqrt{y})$,

i.e. $\sqrt{a+2\sqrt{b}} = \sqrt{x} + \sqrt{y}$

Squaring both sides, we get $a + 2\sqrt{b} = \left[\sqrt{x} + \sqrt{y}\right]^2$

$a + 2\sqrt{b} = x + y + 2\sqrt{xy}$ (1)

Equating the rational and irrational terms on the left and right hand side of the equation (1), we get

$a = x + y$ (2)

$b = xy$ (3)

From (2) and (3) we can find the values of x and y and then find the square roof of $a + 2\sqrt{b}$ type of surd.

A shortcut for a standard question type in surds.

From equation (2) & (3) above, we notice that to find out the square root of the surd of the form $a + 2\sqrt{b}$, we need to split 'a' into two numbers 'x' and 'y' such that, $x + y = a$, and $xy = b$. Then the answer is simply $\sqrt{x} + \sqrt{y}$.

i.e. $\sqrt{a+2\sqrt{b}} = \sqrt{x} + \sqrt{y}$.

Note

If the surd is of the type $a + \sqrt{b}$, convert the surd to $a + 2\sqrt{b}$ type without changing its value and then follow the above procedure to find the square root of the surd.

Practice Questions - Set 1

1. Find the interest under simple interest for a principal of Rs. 300 at an interest of 3.75% per annum from March 5 to July 29.
   1] Rs. 28.125 2] Rs. 4.5 3] Rs. 11.25 4] Rs. 4.25

2. What principal would yield a simple interest of Rs. 25, invested at 5% for 3 months?
   1] Rs. 3125 2] Rs. 2000 3] Rs. 5000 4] Rs. 1500

3. Find the principal that would amount to Rs. 1200 at 8% per annum simple interest after 9 months.
   1] Rs. 864 2] Rs. 1272 3] Rs. 1132.08 4] Rs. 1172

4. At what rate will a sum of money amount to five times itself in 5 years, interest received being under simple interest?
   1] 100 % 2] 82.5 % 3] 68 % 4] 80 %

5. If the difference between simple interest and compound interest on a certain sum for 2 years at 10% is Rs. 50, then find the sum.
   1] Rs. 5050 2] Rs. 4700 3] Rs. 5000 4] Rs. 3750

6. A sum of money rent under compound interest amounts in 3 years to Rs. 800 and in 4 years to Rs. 840. Find the rate of interest.
   1] 5 % 2] 4.76 % 3] 2.58 % 4] 3.72 %

7. Find the compound interest on Rs. 2700 for 2 years at 4% per annum.
   1] Rs. 150.80 2] Rs. 2920.30 3] Rs. 220.30 4] Rs. 202.20

8. Arrange in descending order: $\sqrt[4]{10}, \sqrt[3]{6}, \sqrt{3}$.
   1] $\sqrt[4]{10}, \sqrt[3]{6}, \sqrt{3}$ 2] $\sqrt[4]{10}, \sqrt{3}, \sqrt[3]{6}$ 3] $\sqrt[3]{6}, \sqrt[4]{10}, \sqrt{3}$ 4] None of these

9. Arrange is ascending order: $\sqrt{5},\ \sqrt[3]{11}\ \ 2\sqrt[6]{3}$.
   1] $\sqrt[3]{11}, \sqrt{5}, 2\sqrt[6]{3}$ 2] $\sqrt{5}, \sqrt[3]{11}, 2\sqrt[6]{3}$ 3] $\sqrt{5}, 2\sqrt[6]{3}, \sqrt[3]{11}$ 4] None of these

Directions for questions 10 to 14: Refer to the data below.
Find the value of 'x' in the following questions.

10. $25^{x-3} = 125^{2x-3}$
    1] $\frac{3}{4}$ 2] 0 3] $-\frac{3}{4}$ 4] Indeterminate

11. $\log(x+1) - \log(x-1) = \log 2$
    1] 1 2] 3 3] 2 4] 0

12. $49^{x+4} = 7^2 \times (343)^{x+1}$

1] 1 2] 2 3] 3 4] 0

13. $5^{100} = x^{50}$

1] 25 2] 5 3] 125 4] Indeterminate

14. $\log_x 5 + \log_x 25 + \log_x 125 = 12$

1] $\sqrt{5}$ 2] $-\sqrt{5}$ 3] [1] or [2] 4] 5

Explanatory Answers

1. Total number of days:

March	April	May	June	July
(31–5)	30	31	30	29

Time (T) = 26 + 30 + 31 + 30 + 29 = 146 days $= \frac{146}{365}$ years $= \frac{2}{5}$ years.

$I = \frac{PRT}{100} = \frac{300}{100} \times \frac{15}{4} \times \frac{2}{5} = \frac{9}{2}$ = Rs. 4.5

2. $T = 3 \text{ months} = \frac{3}{12} \text{ year} = \frac{1}{4} \text{ year}$

$SI = \frac{PRT}{100}$; $\therefore 25 = \frac{P \times 5}{100} \times \frac{1}{4}$

$P = \frac{25 \times 100 \times 4}{5}$ = Rs. 2000

3. $A = P + SI$ $T = \frac{9}{12} = \frac{3}{4}$ year

$A = P + \frac{P\,R\,T}{100}$

$A = P\left(1 + \frac{RT}{100}\right)$

$1200 = P\left[1 + \left(\frac{8}{100} \times \frac{3}{4}\right)\right]$

$A = P\left(\frac{53}{50}\right) \Rightarrow P = \frac{1200 \times 50}{53}$

P = Rs. 1132.08

4. Let the principal be Rs. x and the amount after 5 years = Rs. 5x

$A = P\left[1 + \frac{RT}{100}\right]$

$5x = x\left[1 + \frac{R \times 5}{100}\right]$

$5 = \left[\frac{100 + 5R}{100}\right]$ $\Rightarrow 500 - 100 = 5R$

i.e. 5R = 400

$R = \frac{400}{5} = 80\%$

Alternately,

Let the principal be Rs. 100.

The amount after 5 years = Rs. 500

The interest earned for 5years = Rs. 400

The interest earned for one year = $\frac{400}{5}$ = Rs. 80

The rate of interest = $\frac{80}{100} \times 100 = 80\%$

5. $(CI - SI)_{\text{for 2 years}} = P\left(\frac{R}{100}\right)^2$

$50 = P\left(\frac{10}{100}\right)^2 = \frac{P}{100}$

$\therefore$ P = Rs. 5000

Alternately,

Let the principal be Rs. 100

When the rate of interest is 10%, the amount at the end of 2 years under SI and CI are Rs. 120 and Rs. 121 respectively.

In this case the difference between SI and CI = Rs. 1

When the difference is Rs. 1, the principal = Rs. 100

Hence, when the difference is Rs. 50, the principal = $\frac{100 \times 50}{1}$ = Rs. 5000

6. Amount in 3 years, A_3 = Rs. 800

Amount in 4 years, A_4 = Rs. 840

$\therefore$ Interest for 1 year = 840 – 800 = Rs. 40

$\text{Rate} = \frac{I \times 100}{P\ T} = \frac{40 \times 100}{800 \times 1} = 5\%$

(Note A_3 is principle of 4th year)

7. $A = P\left(1 + \frac{R}{100}\right)^n$

$= 2700\left(1 + \frac{4}{100}\right)^2$

$= 2700\ (1.04)^2 = 2920.32$

$\therefore\ CI = A - P = 2920.32 - 2700 = \text{Rs. } 220.32$

8. $\sqrt[4]{10},\ \sqrt[3]{6},\ \sqrt{3}$

LCM of 4, 3, 2 is 12

$\therefore\ \sqrt[4]{10} = 10^{\frac{1}{4}} = 10^{\frac{3}{12}} = \left(10^3\right)^{\frac{1}{12}} = \sqrt[12]{1000}$

$\sqrt[3]{6} = 16^{\frac{1}{3}} = 6^{\frac{4}{12}} = \left(6^4\right)^{\frac{1}{12}} = \sqrt[12]{1296}$

$\sqrt{3} = 3^{\frac{1}{2}} = 3^{\frac{6}{12}} = \left(3^6\right)^{\frac{1}{12}} = \sqrt[12]{729}$

$\sqrt[12]{1296} > \sqrt[12]{1000} > \sqrt[12]{729}$

$\therefore\ \sqrt[3]{6} > \sqrt[4]{10} > \sqrt{3}$

Hence [3]

9. $\sqrt{5},\ \sqrt[3]{11},\ 2\sqrt[6]{3}$

LCM of 2, 3, 6 is 6.

$\sqrt{5} = 5^{\frac{1}{2}} = 5^{\frac{3}{6}} = \left(5^3\right)^{\frac{1}{6}} = \sqrt[6]{5^3} = \sqrt[6]{125}$

$\sqrt[3]{11} = 11^{\frac{1}{3}} = 11^{\frac{2}{6}} = \left(11^2\right)^{\frac{1}{6}} = \sqrt[6]{11^2} = \sqrt[6]{121}$

$2\sqrt[6]{3} = \sqrt[6]{2^6 \times 3} = \sqrt[6]{64 \times 3} = \sqrt[6]{192}$

$\therefore\ \sqrt[6]{121} < \sqrt[6]{125} < \sqrt[6]{192}$

$\therefore\ \sqrt[3]{11} < \sqrt{5} < 2\sqrt[6]{3}$. Hence [1]

10. $(25)^{x-3} = (125)^{2x-3}$

$\therefore\ \left(5^2\right)^{x-3} = \left(5^3\right)^{2x-3}$

$\therefore\ 5^{2x-6} = 5^{6x-9}$

Since bases are equal, powers can be equated

$\therefore\ 2x - 6 = 6x - 9$

$3 = 4x$

$x = \frac{3}{4}$. Hence [1]

11. $\log(x + 1) - \log(x - 1) = \log 2$

$\therefore \log\left(\frac{x+1}{x-1}\right) = \log 2$. We know that if $\log_{10}M = \log_{10}N$, then M = N.

$\therefore\ \frac{x+1}{x-1} = 2 \Rightarrow\ x + 1 = 2x - 2$

$\therefore x = 3.$ Hence [2]

Note When the base of the logarithm is not given, consider the base as 10.

12. $49^{x+4} = 7^2 \times (343)^{x+1}$

$\left[7^2\right]^{x+4} = 7^2 \times (7^3)^{x+1}$

$\therefore\ 7^{2x+8} = 7^{2+3x+3}$

Since bases are equal, powers can be equated

$\therefore 2x + 8 = 5 + 3x$ => $x = 3$. Hence [3]

13. $5^{100} = x^{50}$

$\therefore\ x = 5^{\frac{100}{50}} = 5^2 = 25$

$\therefore\ x = 25$

And also $(-25)^{50} = 5^{100}$. Hence, x can be either 25 or – 25. Hence [4]

14. $\log_x 5 + \log_x 25 + \log_x 125 = 12$

$\therefore \log_x 5 + 2\log_x 5 + 3\log_x 5 = 12$

$\therefore \log_x 5 (1+2+3) = 12$

$\therefore \log_x 5 = \frac{12}{6} = 2$

$\log_x 5 = 2$ can be written as $x^2 = 5$

$\therefore x^2 = 5 \Rightarrow x = \pm\sqrt{5}$

Logarithms is not defined for the negative base. Hence, $x = \sqrt{5}$. Hence [1]

Practice Questions - Set 2

1. Arrange in descending order of magnitude: $\sqrt[4]{3}; \sqrt[6]{10}; \sqrt[12]{25}$

1] $\sqrt[6]{10}, \sqrt[12]{25}, \sqrt[4]{3}$ 2] $\sqrt[4]{3}, \sqrt[12]{25}, \sqrt[6]{10}$ 3] $\sqrt[4]{3}, \sqrt[6]{10}, \sqrt[12]{25}$ 4] $\sqrt[6]{10}, \sqrt[4]{3}, \sqrt[12]{25}$

Directions for questions 2 and 3: Refer the data below.

Simplify the following:

2. $\sqrt{49} + \sqrt{0.49} + \sqrt{0.0049}$

1] 7.077 2] 7.77 3] 7.07 4] 7.007

3. $\frac{\sqrt{63} \times \sqrt{7}}{\sqrt[3]{27}}$

1] 7 2] $\frac{\sqrt{7}}{3}$ 3] $\frac{\sqrt{7}}{\sqrt{27}}$ 4] 9

4. Simplify: $\frac{(16.4 + 20.5)^2 + (20.5 - 16.4)^2}{(20.5)^2 + (16.4)^2}$

1] 2.28 2] 2.04 3] 1.684 4] 2

5. Which of the following numbers is bigger?

(i) $(10^3)^3$ (ii) $10^{(3^3)}$ (iii) $10(3)^3$ (iv) $(10^3)^{2^2}$

1] i 2] iv 3] iii and i 4] ii

6. Find the value of x if $81^{2x+3} = 243^{5-x}$

1] 0 2] 1 3] $\frac{1}{2}$ 4] –1

7. $\frac{45.55 \times 45.55 + 44.45 \times 44.45 + 2 \times 45.55 \times 44.45}{35.55^2 + 185.55^2 - 2 \times 35.55 \times 185.55} =$

1]5 2] $\frac{9}{25}$ 3] $\frac{3}{5}$ 4] $-\frac{3}{5}$

8. $\dfrac{4.6^3 + 3.4^3}{4.6^2 - 4.6 \times 3.4 + 3.4^2} =$

1] 8 2] 7.68 3] 7.71 4]s 4.6

9. Arrange the following in ascending order.

$2^{2^{2^2}}$, $2^{2^{2^2}}$, 2^{222}, 222^2

1] $222^2, 2^{2^{2^2}}, 22^{22}, 2^{222}$ 2] $2^{2^{2^2}}, 222^2, 22^{22}, 2^{222}$

3] $2^{2^{2^2}}, 22^{22}, 2^{222}, 222^2$ 4] None of these

10. If $a^x = b$, $b^y = c$ and $c^z = a$, then xyz is equal to

1] $\dfrac{1}{abc}$ 2] abc 3]1 4]0

Explanatory Answers

1. Arrange in descending order of magnitude: $\sqrt[4]{3}; \sqrt[6]{10}; \sqrt[12]{25}$

The LCM of 4, 6, 12 is 12.

$\sqrt[4]{3} = (3)^{\frac{1}{4}} = 3^{\frac{3}{12}} = \sqrt[12]{3^3} = \sqrt[12]{27}$

$\therefore \sqrt[6]{10} = (10)^{\frac{1}{6}} = 10^{\frac{2}{12}} = \sqrt[12]{10^2} = \sqrt[12]{100}$

$\sqrt[12]{25} = \sqrt[12]{25}$

Now arranging in descending order: $\sqrt[12]{100} > \sqrt[12]{27} > \sqrt[12]{25}$, i.e. $\sqrt[6]{10} > \sqrt[4]{3} > \sqrt[12]{25}$.

Hence [4]

2. $\sqrt{49} + \sqrt{0.49} + \sqrt{0.0049} = 7 + 0.7 + 0.07 = 7.77$

Hence [2]

3. $\dfrac{\sqrt{63} \times \sqrt{7}}{\sqrt[3]{27}} = \dfrac{\sqrt{7 \times 9} \times \sqrt{7}}{3} = \dfrac{7 \times 3}{3} = 7$

Hence [1]

4. $\dfrac{(20.5+16.4)^2 + (20.5-16.4)^2}{(20.5)^2 + (16.4)^2} = \dfrac{2\left(20.5^2 + 16.4^2\right)}{\left(20.5^2 + 16.4^2\right)} = 2$ $\left[\because (a+b)^2 (a-b)^2 = 2\left(a^2 + b^2\right)\right]$

Hence [4]

5. $(10^3)^3 = 10^9$

$10^{3^3} = 10^{27}$

$10(3)^3 = 10(27) = 270$

$(10^3)^{2^2} = (10^3)^4 = 10^{12}$

Clearly 10^{27} is the greatest of the four numbers

Hence [4]

6. $81^{2x+3} = 243^{5-x}$

$\left[(3)^4\right]^{2x+3} = \left[[3]^5\right]^{5-x}$

$\therefore 3^{8x+12} = 3^{25-5x}$

Since bases are equal, exponents can be equated

$\therefore 8x + 12 = 25 - 5x \Rightarrow 13x = 13$

$\therefore x = 1$

Hence [2]

7. $$\frac{45.55\times45.55+44.45\times44.45+2\times45.55\times44.45}{35.55^2+185.55^2-2\times35.55\times185.55}$$

$$=\frac{(45.55+44.45)^2}{(35.55-185.55)^2}=\frac{(90)^2}{(150)^2}=\frac{90\times90}{150\times150}=\frac{3\times3}{5\times5}=\frac{9}{25}$$

Hence [2]

8. $$\frac{4.6^3+3.4^3}{4.6^2-4.6\times3.4+3.4^2}=\frac{(4.6+3.4)(4.6^2-4.6\times3.4+3.4^2)}{4.6^2-4.6\times3.4+3.4^2}=8$$

Hence [1]

9. $2^{2^{2^2}}$, 22^{22}, 2^{222}, 222^2

$2^{2^{2^2}} = 2^{2^4} = 2^{16}$

$22^{22} = (2\times11)^{22} = 2^{22}\times11^{22}$

$(222)^2 = (2\times111)^2 = 2^2\times111^2$

Comparing $2^2 \times 111^2$ and 2^{16}, i.e. $\frac{2^{16}}{2^2\times111^2}=\frac{2^{14}}{111^2}=\frac{2^{7\times2}}{111^2}=\frac{\left(2^7\right)^2}{111^2}=\frac{128^2}{111^2}$

$\frac{128^2}{111^2} > 1$ (or) $128^2 > 111^2$. Hence $2^{16} > 222^2$

$\therefore 222^2 < 2^{16} < 22^{22} < 2^{222}$

Hence [1]

10. $a^x = b \Rightarrow \log_a b = x$

$b^y = c \Rightarrow \log_b c = y$

$c^z = a \Rightarrow \log_c a = z$

$\therefore x\,y\,z = \log_a b \,.\, \log_b c \,.\, \log_c a$

$= \log_a b \log_b a \quad (\because \log_a m \times \log_n a = \log_n m)$

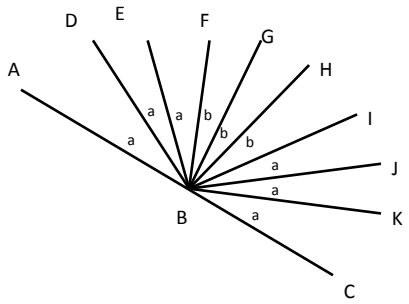
$= \log_a a = 1 \qquad (\because \log_a a = 1)$

Hence [3]

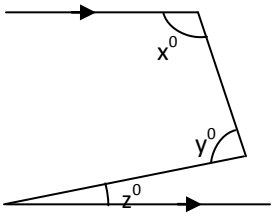
Practice Questions - Set 3

1. In the figure, AC is a line segment. Line segments are drawn from B to D, E, G, H, I, J, K respectively. Which of the following angles has a degree measure that can be found from the figure?

1] $\angle$ FBG 2] $\angle$ DBG 3] $\angle$ EBG 4] $\angle$ GBI

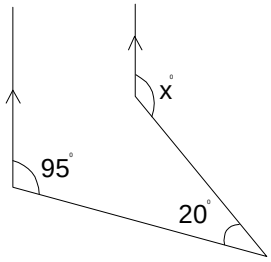


2. The value of $(x^0 + y^0 - z^0)$ from the diagram given below is



1] 120^0 2] 150^0 3] 180^0 4] 210^0

3. The value of x from the diagram given below

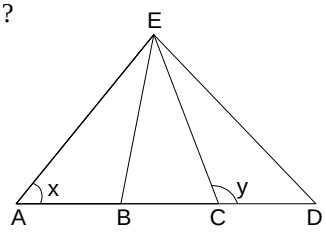


1] 115^0 2] 105^0 3] 95^0 4] Cannot be determined

4. The base angle of an isosceles triangle is 30 less than its vertical angle. Then its vertical angle is _____.

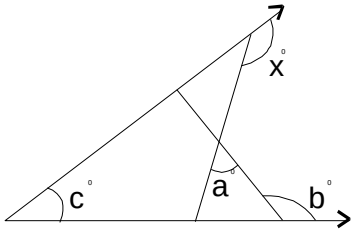
1] 90^0 2] 80^0 3] 60^0 4] 50^0

5. In the given figure (not drawn to scale), AB = BE and BC = CE. ABCD is a straight line. Then which of the following relate y to x ?



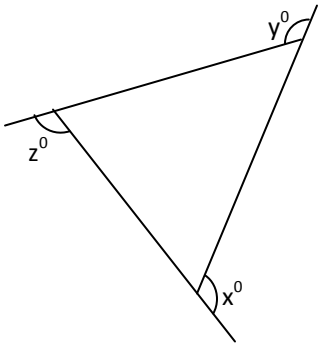
1] y = x 2] y = 2x 3] y = 3x 4] y = 4x

6. Which of the following relate x in terms of a, b and c?



1] 180 – a + b – c 2] 180 + a – b + c
3] 180 – a + b + c 4] 180 – a – b – c

7. What is the value of $(x^0 + y^0 + z^0)$ from the diagram given below ?

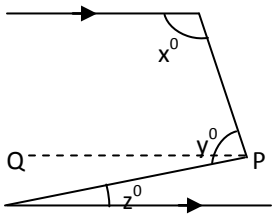


1] 1 right angle 2] 2 right angles 3] 3 right angles 4] 4 right angles.

8. The diameter of the wheel of a car is 63 cm. Calculate the number of times the wheel will rotate in travelling through a distance of 3.762 km.
1] 1900 2] 1200 3] 1000 4] 800

Explanatory Answers

1. A, B, C lie on a straight line
$\therefore 6a + 3b = 180 \Rightarrow 2a + b = 180$
$\therefore \angle$ DBG has a degree measure that can be found .
Hence [2]

2. 

Draw PQ ‖ to the given lines. Using co-interior angles are supplementary and alternate angles are equal we have,

$y = 180 – x + z$

$\Rightarrow x + y – z = 180^0$

Hence [3]

3. Construction: Extend CD to touch BF at E.

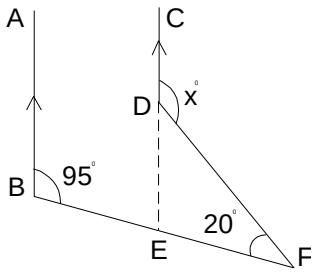
CE ‖ AB

$\therefore \angle DEF = 95$ (corresponding with $\angle ABE$)

$\angle CDF = \angle DEF + \angle EFD$ (exterior angle theorem)

$\therefore x = 95^0 + 20^0 = 115^0$

Hence [1].



4. Let the two base angles of the triangle be x.

$\therefore$ The vertical angle = (x + 30).

$\therefore 2x + (x + 30) = 180^0$

$\therefore 3x = 180 – 30 = 150^0$

$\therefore x = 150/3 = 50^0$

the angles of the triangle are 50^0, 50^0, 80^0.

$\therefore$ the vertical angle is 80^0

Hence [2]

5. Δ ABE is isosceles, since AB = BE.

$\therefore \angle EAB = \angle AEB = x$

$\therefore \angle EBC = 2x$ (exterior angle theorem)

ΔBCE is isosceles, since BC = CE.

$\therefore \angle EBC = \angle BEC = 2x$

$\therefore \angle ECD = 2x + 2x$ (exterior angle theorem)

$\therefore y = 4x$

Hence [4].

6. $\angle DCE = 180 – b$ (linear pair with b)

$\therefore \angle AEC = 180 – (a + 180 – b) = b – a$

$\therefore AEB = 180 – (b – a) = 180 – b + a$

$\therefore x = c + 180 – b + a$ (exterior angle theorem)

$\therefore x = 180 + a – b + c$

Hence [2].

7. Sum of exterior angles = $x^0 + y^0 + z^0$. We know that sum of interior angles of a triangle = 180^0

i.e. (180 - x) + (180 - y) + (180 - z) = 180

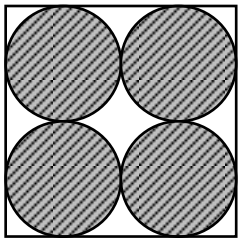
$\Rightarrow$ x + y + z = 360 i.e. 4 right angles

Hence [4]

8. Total distance travelled = 3.762 km = 3.762×10^5 cm.
Diameter of the wheel = 63 cm.
$\therefore$ Number of rotations = $\frac{3.762 \times 10^5}{\frac{22}{7} \times 63}$ = 1900 times.
Hence [1]

Practice Questions - Set 4

1. Four equal circles are drawn as shown in the figure inside a square of side 12 cms. Find the area of the unshaded region .



1] 144 - 36π 2] 144 + 72π 3] 144 - 72π 4] 144 - 16π

2. Two cylindrical buckets have their diameters in the ratio 3:1 and their heights are as 1:3. Their volumes are in the ratio of :
1] 1:2 2] 2:3 3] 3:1 4] 3:4

3. On a certain sum, the simple interest at the end of $6\frac{1}{4}$ years becomes $\frac{3}{8}$ of the sum. What is the rate percent?
1] 7% 2] 6% 3] 5% 4] $5\frac{1}{2}$%

4. You have reached Utopia where you find two kinds of precious stones, rubies and emeralds. The worth of a ruby and that of an emerald is Rs. 4 crores and Rs. 5 crores respectively while their weights are 0.3 kg and 0.4 kg respectively. You have a bag that can carry a maximum of 12 kgs. How many rubies and emeralds would you carry such that their total value is maximised?
1] 20 rubies and 15 emeralds 2] 8 rubies and 24 emeralds
3] 0 rubies and 30 emeralds 4] None of the above

5. If in $(BE)^2$ = MPB, the letters B, E, P and M stand for distinct digits, then M equals
1] 1 2] 6 3] 3 4] 9

6. An airplane leaves an airport at 3.00 p.m. and flies due north at 400 kilometres per hour. At 5.00 p.m. another plane leaves the same airport and flies due east at 500 kilometres per hour. What will be the shortest distance in kilometres between the two planes at 8.00 p.m. assuming the earth is flat?
1] 3500 2] 2500 3] $100\sqrt{75}$ 4] 4000

7. An ice-cream company makes a popular brand of ice-cream in a rectangular shaped bar 6 cm long, 5 cm wide and 2 cm thick. To cut costs, the company had decided to reduce the volume of the bar by 20%. The thickness will remain the same, but the length and width will be decreased by the same amount. The new length L will be
1] 5.5 2] 5 3] 4.5 4] 4

8. $(2^2 + 4^2 + 6^2 + ... 20^2) - (1^2 + 3^2 + 5^2 + ... 19^2)$ is equal to
   1] 100 2] 420 3] 210 4] 55

9. A square is circumscribed by a circle of diameter 6 cms. What is the length in cms. of each side of the square?
   1] 6 2] $3\sqrt{2}$ 3] 2 4] 3

10. An aircraft is flying at a height of 3000 m from the ground at an average speed of 400 kms/hr while descending. It has to travel 5 kms before it can touch down on the runway and start operating its brake system. How far is it from the touch-down point if it had been on the ground?
    1] 5 kms 2] 4 kms 3] 3 kms 4] None of these

Explanatory Answers

1. Clearly the diameter of each of the circles is 12 / 2 = 6 cms.
   ∴ the radius of each of the circles = 6 / 2 = 3 cms.
   ∴ the area of the shaded region = $4\{\pi(3)^2\}$ = 36π sq cms.
   Also the area of the square is $12^2 = 144$ sq cms.
   Hence the area of the unshaded region is $144 - 36\pi$ sq cms. Hence (1)

2. Let the diameters be 3r and r and heights be h and 3h respectively.
   ∴ the ratio of their volumes = $\dfrac{\pi\left(\frac{3r}{2}\right)^2 h}{\pi\left(\frac{r}{2}\right)^2 (3h)}$ = 3 : 1. Hence (3)

3. Let the sum be x. Then S.I = 3x / 8 ∴ Rate = $\dfrac{100\left(\frac{3x}{8}\right)}{x\left(\frac{25}{4}\right)} = \dfrac{300x}{8} \times \dfrac{4}{25x} = 6\%$ Hence (2)

4. Clearly buying a ruby weighing 0.3 kg and worth Rs. 4 crores is a better deal than buying an emerald stone weighing 0.4 kg worth Rs. 5 crores.
   Hence the best deal would be to buy and carry only 40 rubies which is worth 160 crores. Hence (4)

5. $(BE)^2 = MPB \approx 19^2 = 361$
   ∴ M corresponds to 3. (Note: since a square of 2-digit number is a three digit number, BE <32).
   Hence (3)

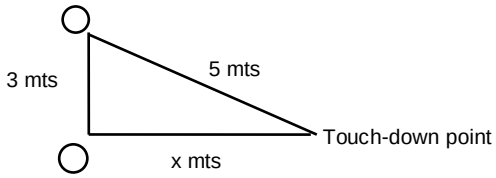
6. Distance of the second plane from the starting point at 8.00 pm = 500 x 3 = 1500 Km.
   Distance of first plane from the starting point at 8.00 pm = 400 x 5 = 2000 Km.
   Shortest Distance = $\sqrt{(2000)^2 + (1500)^2} = 100\sqrt{20^2 + 15^2} = 100\sqrt{625}$ = 2500 Km. Hence (2)
   Note : use Pythagoras triplet : 3×500, 4×500, 5×500

7. Volume of ice-cream = 6 X 5 X 2 = 60 cc
   New volume = 80% of 60 = 48 cc.
   Let length and width decrease by x cms each. New volume = (6-x) (5-x)2 = 48
   (6-x)(5-x) = 24
   On solving we get x ≅ 0.57
   ∴ new length = (6 - 0.57) = 5.42 ≈ 5.5. Hence (1)

8. $(2^2 + 4^2 + ... +20^2) - (1^2 + 3^2 + ... 19^2) = (2^2 - 1^2) + (4^2 - 3^2) + (6^2 - 5^2) + (8^2 - 7^2) + ... + (20^2 - 19^2)$
$= (2 + 1)(2 - 1) + (4 + 3)(4 - 3) + .. (20 + 19)(20 - 19) \quad = 3 + 7 + 11 + ... + 39$
This is an AP with a = 3, d = 4, n = 10
$S_n = \frac{n}{2}$ (first term + last term) = $\frac{10}{2}$ (3 + 39) = 210. Hence (3)

9. Diagonal of the square = Diameter of circle = 6 cms
or $a\sqrt{2} = 6$ cms.
Side of square = $a = \frac{6}{\sqrt{2}} = 3\sqrt{2}$ cms. Hence (2)

10.



The above shows the current position of the aircraft.
Using Pythagoras' theorem we get x = 4 mts. Hence (2)

Chapter 22 Coordinate Geometry

Suggested Time : 180 minutes

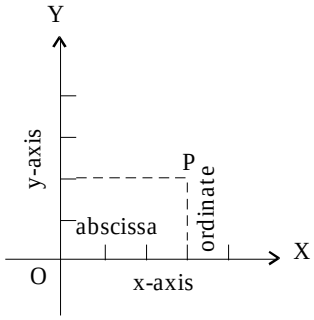
Analytical geometry is that branch of mathematics which treats geometric figures algebraically, i.e. in which geometric figures are studied by means of equations.

Coordinate geometry is that branch of geometry in which two numbers, called coordinates, are used to indicate the position of a point in a plane and which makes use of algebraic methods in the study of geometric figures.

Directed line: A directed line is a straight line on which (i) a fixed point called the origin, (ii) a unit of length, and (iii) a positive direction have been assumed. The origin is denoted by the letter 'O' and positive direction is indicated by placing an arrow head on the direction of the line.

Number scale: Real numbers which include positive numbers, negative numbers, and zero are used to make the number scale. We make the origin (point 'O') to correspond to the number zero. Any number to the right of zero is positive and the one on the left of zero is negative for a horizontal number scale and any number to the top of zero is positive and that to the bottom of zero is negative for a vertical number scale.

Cartesian coordinates: These are used to give the position of a point by placing it in relation to the axes of reference which are formed by combining the two number scales at right angles to each other such that their zero points coincide. The horizontal number scale is called the x-axis and the vertical number scale the y-axis.



The point where the two scales cross each other is called the **Origin**. The two together are called **Rectangular Axes** or **Coordinate Axes**. The distance of a point from the y-axis is called the **x-coordinate** or the **Abscissa** and the distance of the point from the x-axis is called the **y-coordinate** or the **Ordinate**.

NOTE:

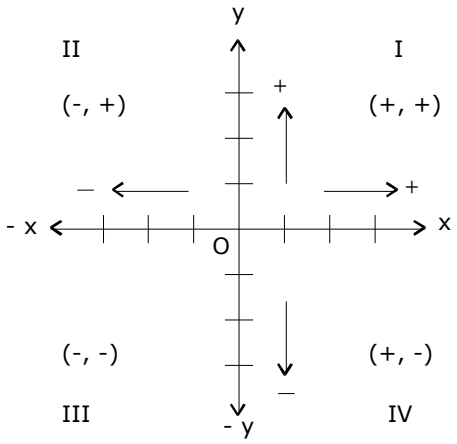
(i) The coordinates of the origin are (0, 0).

(ii) The coordinates of any point on the x-axis is (x, 0), i.e. y-coordinate is zero.

(iii) The coordinates of any point on the y-axis is (0, y), i.e. x-coordinate is zero.

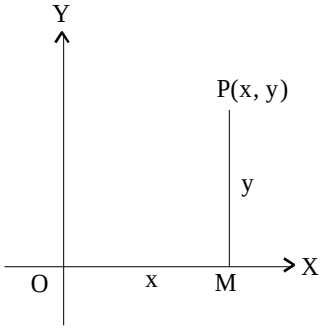
QUADRANTS

The coordinate axes separate the plane into four regions called **Quadrants**. The quadrants are numbered I, II, III, IV in the counter clockwise direction. A point of the plane may lie in any quadrant.

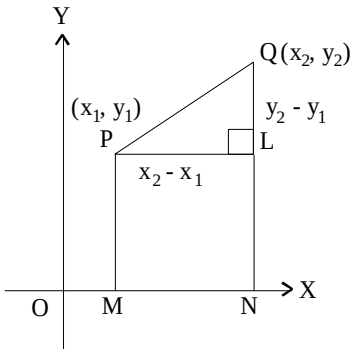


Location of point	x-coordinate	y-coordinate
Quadrant I	Positive	Positive
Quadrant II	Negative	Positive
Quadrant III	Negative	Negative
Quadrant IV	Positive	Negative

Plotting a point: A point can be plotted by measuring its proper distances from the axes. Thus any point P(x, y) can be plotted by measuring OM equal to x on the x-axis and MP perpendicular to OM and equal to y.



Distance formula: Let $P(x_1, y_1)$ and $Q(x_2, y_2)$ be two points. From P to Q draw PM and QN perpendicular on the x-axis and PL perpendicular on QN.



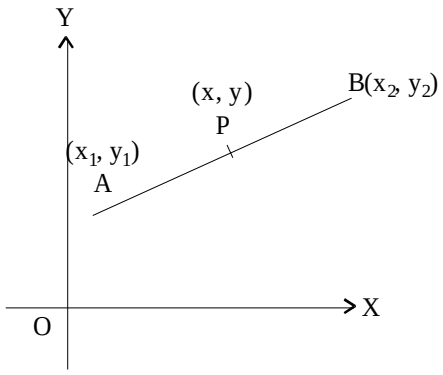
Then $PL = x_2 - x_1$ and $QL = y_2 - y_1$.
In right-angled PLQ,
$PQ^2 = PL^2 + LQ^2$ (Pythagoras theorem)
$\therefore PQ^2 = (x_2 - x_1)^2 + (y_2 - y_1)^2$
$\Rightarrow PQ = \sqrt{(x_2 - x_1)^2 + (y_2 - y_1)^2}$

NOTE:

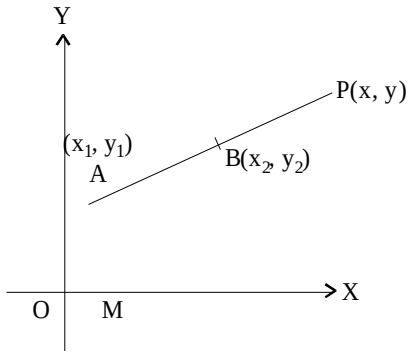
The distance from the origin O(0,0) to a point P(x, y) is given by $OP = \sqrt{x^2 + y^2}$.

DIVISION OR SECTION FORMULAE

A. Internal division: Let point P(x, y) divide line AB internally in the ratio m : n. Then the coordinates of P are, $x = \frac{mx_2 + nx_1}{m+n}$ and $y = \frac{my_2 + ny_1}{m+n}$.



B. External division: Let point P(x, y) divide line AB externally in the ratio m : n such that $\frac{AP}{PB} = \frac{m}{n}$. Then the coordinates of P are $x = \frac{mx_2 - nx_1}{m - n}$ and $y = \frac{my_2 - ny_1}{m - n}$.



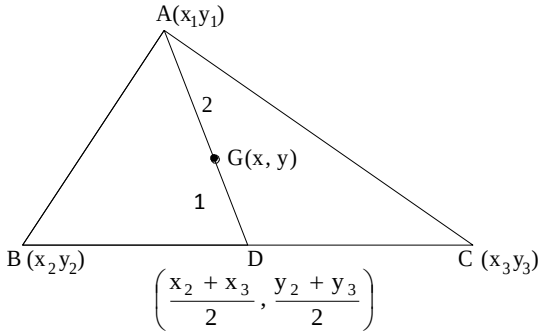
C. Mid-point formula: If the point P(x, y) divides the line AB in the same ratio, i.e. m = n, then the coordinates of P are,

$$x = \frac{mx_2 + nx_1}{m + n} = \frac{mx_2 + mx_1}{m + m} = \frac{m(x_2 + x_1)}{2m} = \frac{x_2 + x_1}{2}$$

$$y = \frac{my_2 + ny_1}{m + n} = \frac{my_2 + my_1}{m + m} = \frac{m(y_2 + y_1)}{2m} = \frac{y_2 + y_1}{2}.$$

CENTROID OF A TRIANGLE

The centroid of a triangle is the point of intersection of its medians. The centroid G divides the median internally in the ratio 2 : 1.

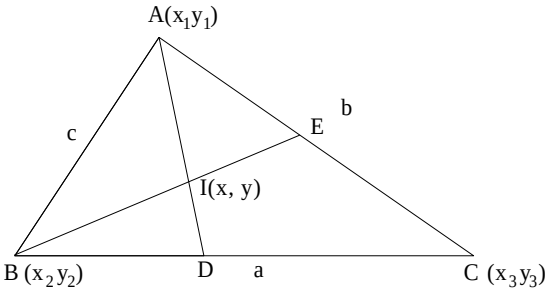


Thus, the coordinates of the centroid G are $x = \frac{x_1 + x_2 + x_3}{3}$ and $y = \frac{y_1 + y_2 + y_3}{3}$.

INCENTRE OF A TRIANGLE

The point of concurrence of the internal bisectors of the angles of a triangle is called the incentre of the triangle and is generally denoted by I.

Let A(x_1, y_1), B(x_2, y_2), and C(x_3, y_3) be the vertices of a triangle. Let AD and BE be the bisectors of $\angle A$ and $\angle B$ intersecting at the incetre I(x, y). Let 'a' be the length of side opposite to $\angle A$, and 'b' the length of side to $\angle B$, and 'c' the length of the side opposite to $\angle C$.



Thus the coordinates of I are,

$x = \frac{ax_1 + bx_2 + cx_3}{a+b+c}$, $y = \frac{ay_1 + by_2 + cy_3}{a+b+c}$.

□

SOLVED EXAMPLES-1

E-1 Find the radius of the circle that has its centre at (0, –4) and passes through ($\sqrt{11}$, 1).

Sol: The radius of the circle

$= \sqrt{(\sqrt{11}-0)^2 + [1-(-4)]^2}$

$= \sqrt{11+(5)^2} = \sqrt{11+25} = \sqrt{36} = 6$ units

E-2 Find the coordinates of the point which divides internally the join of points A(8, 9) and B(–7, 4) internally in the ratio 2 : 3.

Sol: Let P(x, y) divide the points A(8, 9) and B(–7, 4) internally in the ratio 2 : 3.

$$\therefore x = \frac{mx_2 + nx_1}{m+n} = \frac{2(-7)+3(8)}{2+3} = \frac{24-14}{5} = 2$$

$$y = \frac{my_2 + ny_1}{m+n} = \frac{2(4)+3(9)}{2+3} = \frac{8+27}{5} = 7$$

Hence, the coordinates of P are (2, 7).

E-3 Find the third vertex of a triangle if two of its vertices are (–1, 4) and (5, 2) and its centroid is at the point (0, 3).

Sol: Let A(–1, 4), B(5, 2), and C(x, y) be the vertices of ΔABC with centroid G(0, 3).
Centroid divides the median in the ratio 2 : 1.

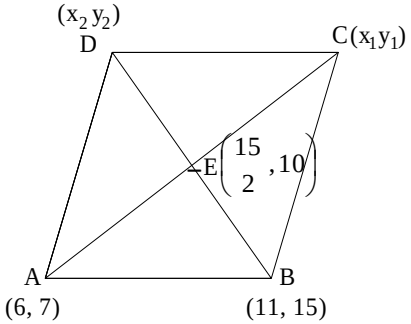
$\therefore \left(\frac{-1+5+x}{3}, \frac{4+2+y}{3}\right) = (0, 3)$

$\Rightarrow \frac{-1+5+x}{3} = 0, \frac{4+2+y}{3} = 3$

$\therefore 4 + x = 0, 6 + y = 9 \qquad \therefore x = -4, y = 3$

$\therefore C(x, y) = (-4, 3)$

E-4 Two consecutive vertices of a parallelogram are (6, 7) and (11, 15). If the diagonals intersect at $\left(\frac{15}{2}, 10\right)$, find the other vertices.



Sol: Let A(6, 7), B(11, 15), C(x_1, y_1), and D(x_2, y_2) be the vertices of a parallelogram. AC and BD have the same midpoint E$\left(\frac{15}{2}, 10\right)$.

Midpoint of AC = $\left(\frac{15}{2}, 10\right)$.

$\therefore \left(\frac{x_1+6}{2}, \frac{y_1+7}{2}\right) = \left(\frac{15}{2}, 10\right)$

$\therefore \frac{x_1+6}{2} = \frac{15}{2}, \frac{y_1+7}{2} = 10 \ \therefore \ x_1 = 15 - 6, y_1 = 20 - 7$

$\therefore x_1 = 9, y_1 = 13$

Midpoint of BD = $\left(\frac{15}{2}, 10\right)$

$\therefore \left(\frac{x_2+11}{2}, \frac{y_2+15}{2}\right) = \left(\frac{15}{2}, 10\right) \ \therefore \frac{x_2+11}{2} = \frac{15}{2} \ \frac{y_2+15}{2} = 10,$

$\therefore x_2 = 15 - 11, y_2 = 20 - 15 \qquad \therefore x_2 = 4, y_2 = 5$

$\therefore C(x_1, y_1) = (9, 13)$ and $D(x_2, y_2) = (4, 5)$

E-5 Prove that the points (–1, –1), (1, 1), and $(-\sqrt{3}, \sqrt{3})$ form an equilateral triangle.

Sol: Let A(–1, –1), B(1, 1), and C$(-\sqrt{3}, \sqrt{3})$ be the given points.

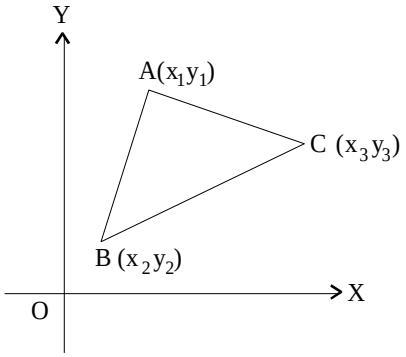
$AB = \sqrt{[1-(-1)]^2+[1-(-1)]^2} = \sqrt{(1+1)^2+(1+1)^2} = \sqrt{2^2+2^2} = \sqrt{8}$,

$BC = \sqrt{(-\sqrt{3}-1)^2+(\sqrt{3}-1)^2} = \sqrt{3+2\sqrt{3}+1+3-2\sqrt{3}+1} = \sqrt{8}$,

$CA = \sqrt{(-1+\sqrt{3})^2+(-1-\sqrt{3})^2} = \sqrt{1-2\sqrt{3}+3+1+2\sqrt{3}+3} = \sqrt{8}$

Since AB = BC = CA = $\sqrt{8}$, the three points form an equilateral triangle.

AREA OF TRIANGLE



Let A(x_1, y_1), B(x_2, y_2), and C(x_3, y_3) be the vertices of a triangle. Then area of ΔABC

$= \frac{1}{2}\ [(x_1y_2 - x_2y_1) + (x_2y_3 - x_3y_2) + (x_3y_1 - x_1y_3)]$

$= \frac{1}{2}\ [x_1(y_2 - y_3) + x_2(y_3 - y_1) + x_3(y_1 - y_2)]$

NOTE:

(i) If one vertex of the triangle is the origin and the remaining vertices are the points (x_1, y_1) and (x_2, y_2), its area would be ($x_1y_2 - x_2y_1$).

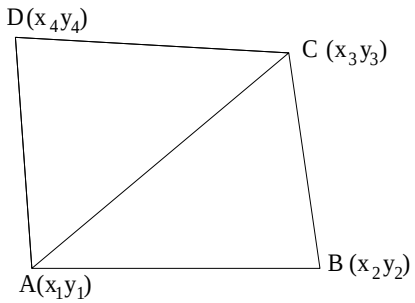
(ii) Aid to memory: Write the abscissae and the ordinates of the vertices in two columns with the abscissa and ordinate of the first column occurring twice. Multiply diagonally attaching a –ve sign to the multiplication from right to left and +ve sign to the multiplication from left to right. Divide the whole result by 2.

$$\begin{matrix} x_1 & y_1 \\ x_2 & y_2 \\ x_3 & y_3 \\ x_1 & y_1 \end{matrix} \quad \begin{matrix} - & + \end{matrix} \qquad \triangle = \frac{1}{2} \begin{vmatrix} x_1 & y_1 \\ x_2 & y_2 \\ x_3 & y_3 \\ x_1 & y_1 \end{vmatrix}$$

AREA OF A QUADRILATERAL

The area of any quadrilateral can be found by splitting up the quadrilateral into two triangles and adding up their areas.

Thus, area of a quadrilateral whose vertices are $A(x_1, y_1)$, $B(x_2, y_2)$, $C(x_3, y_3)$, and $D(x_4, y_4)$ is as given below:



Area of quad. ABCD = $\Delta ABC + \Delta ACD$

$$= \frac{1}{2}\ [(x_1y_2 - x_2y_1) + (x_2y_3 - x_3y_2) + (x_3y_1 - x_1y_3)]$$

$$+ \frac{1}{2}\ [(x_1y_3 - x_3y_1) + (x_3y_4 - x_4y_3) + (x_4y_1 - x_1y_4)]$$

$$= \frac{1}{2}\ [(x_1y_2 - x_2y_1) + (x_2y_3 - x_3y_2) + (x_3y_4 - x_4y_3) + (x_4y_1 - x_1y_4)]$$

COLLINEAR POINTS AND CONDITION OF COLLINEARITY OF THREE POINTS

Collinear points are points which lie on the same straight line. If three points (x_1, y_1), (x_2, y_2), and (x_3, y_3) are collinear, then the area of the triangle formed by them is zero, i.e. $\frac{1}{2}(x_1y_2 - x_2y_1 + x_2y_3 - x_3y_2 + x_3y_1 - x_1y_3) = 0$

SOLVED EXAMPLES-2

E-1 If the area of the triangle formed by the points (1, 2), (2, 3), and (a, 4) is 5 unit^2; find a.

Sol: The given three points are (1, 2), (2, 3), and (a, 4).

Area of the triangle

$$= \frac{1}{2}\ [x_1(y_2 - y_3) + x_2(y_3 - y_1) + x_3(y_1 - y_2)]$$

$$= \frac{1}{2}\ [1(3 - 4) + 2(4 - 2) + a(2 - 3)]$$

$$= \frac{1}{2}\ [1(-1) + 2(2) + a(-1)] = \frac{1}{2}\ (3 - a)$$

But area = 5 unit^2

$\therefore \frac{1}{2}\ (3 - a) = 5 \Rightarrow 3 - a = 10 \qquad \therefore a = -7$

E-2 Find the area of the quadrilateral formed by the points (0, –1), (–2, –3), (–4, 2), and (0, 4).

Sol: Let A(0, –1), B(–2, –3), C(–4, 2), and D(0, 4) be the vertices of the quadrilateral.

$$\text{Area of (ABCD)} = \frac{1}{2}\,[x_1y_2 - x_2y_1 + x_2y_3 - x_3y_2 + x_3y_4 - x_4y_3 + x_4y_1 - x_1y_4]$$

$$= \left|\frac{1}{2}\,[0(-3) - (-2)(-1) + (-2)(2) - (-4)(-3) + (-4)(4) - 0(2) + (0)(-1) - 0(4)]\right|$$

$$= \left|\frac{1}{2}\,[0 - 2 - 4 - 12 - 16 - 0 + 0 - 0]\right|$$

$$= \left|\frac{1}{2}\,[-34]\right| = 17 \text{ unit}^2$$

E-3 Show that the points (5, 7), (1, 3), and (–1, 1) are collinear.

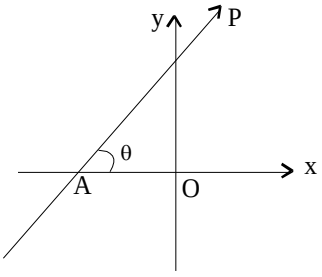
Sol: Let the points A(5, 7), B(1, 3), and C(–1, 1) be the vertices of a triangle.

$$\text{Area} = \frac{1}{2}\,[x_1(y_2 - y_3) + x_2(y_3 - y_1) + x_3(y_1 - y_2)]$$

$$= \frac{1}{2}\,[5(3 - 1) + 1(1 - 7) + (-1)(7 - 3)] = \frac{1}{2}\,[5(2) + 1(-6) - 1(4)] = \frac{1}{2}\,[0] = 0$$

Since area = 0, the points A, B, and C are collinear.

Slope or gradient of a straight line: The slope (or gradient) of a line is the tangent of the angle which the part of the line above the x-axis makes with the positive direction of the x-axis. The slope of a line is indicated by the letter 'm'.



If AP be the portion of a line above the x-axis and $\angle XAP = \theta$, then slope of AP = m = tan θ .

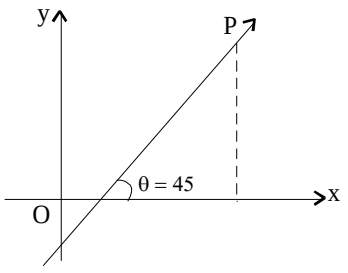
NOTE:

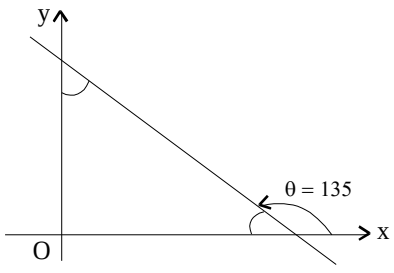
(i) θ *is measured positively from* 0^0 *to* 180^0.

(ii) To calculate slope, the angle is measured from the positive direction of the x-axis in the anticlockwise direction to the part of the line above the x-axis.

(iii) When $\theta = 45^0$, $m = \tan 45^0 = 1$.

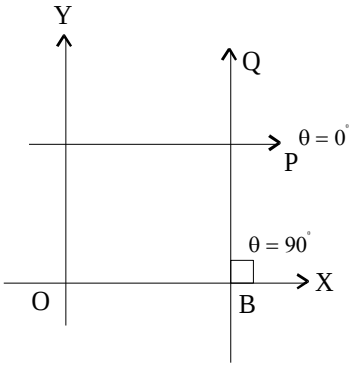
(iv) When $\theta = 135^0$, $m = \tan 135^0 = \tan (180 - 45) = -\tan 45^0 = -1$.





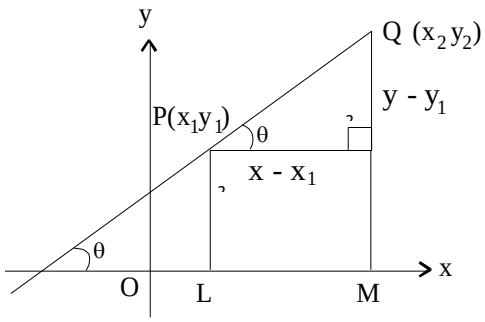
(v) If a line is parallel to the x-axis, its inclination is 0^0 *and so its slope is* $\tan 0^0$, *i.e. 0.*

(vi) If a line is perpendicular to the x-axis, its inclination is 90^0, and so its slope is tan 90^0, i.e. .



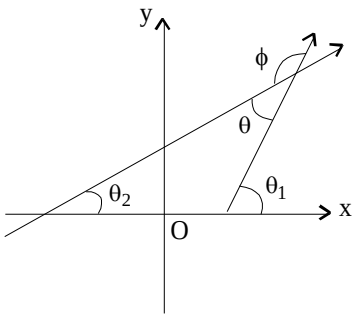
Slope of a line joining two points: Let $P(x_1, y_1)$ and $Q(x_2, y_2)$ be two points. Let θ be the inclination of the line with the x-axis.

Then, $m = \tan\theta = \frac{QM}{PM} = \frac{y_2 - y_1}{x_2 - x_1}$.



ANGLE BETWEEN TWO LINES

Consider two lines whose inclinations to the x-axis are θ_1 and θ_2.



Let θ be the interior angle between the given lines, then

$\theta_1 = \theta + \theta_2$ (exterior angle theorem)

$\therefore \theta = \theta_1 - \theta_2$

$\therefore \tan\theta = \tan(\theta_1 - \theta_2) = \frac{\tan\theta_1 - \tan\theta_2}{1 + \tan\theta_1 \tan\theta_2}$

If $m_1 = \tan\theta_1$ and $m_2 = \tan\theta_2$, we have $\tan\theta = \frac{m_1 - m_2}{1 + m_1 m_2}$.

NOTE:

(i) *If the lines are parallel, then* $\theta = 0^0$.

$\therefore \tan\theta = \tan 0 = 0$, *i.e.* $\frac{m_1 - m_2}{1 + m_1 m_2} = 0$ $\quad \therefore \quad m_1 - m_2 = 0 \Rightarrow m_1 = m_2$

(ii) *If the lines are perpendicular, then* $\theta = 90^0$, *i.e.* $= \frac{m_1 - m_2}{1 + m_1 m_2} = \infty$.

$\therefore$ *The denominator* $1 + m_1 m_2$ *should be equal to zero,*

i.e. $1 + m_1 m_2 = 0 \Rightarrow m_1 m_2 = -1$ *or* $m_2 = \frac{-1}{m_1}$.

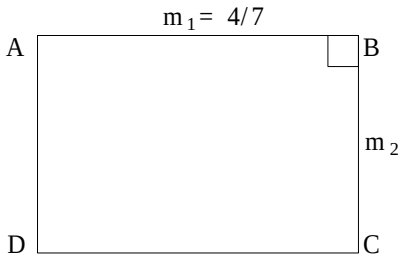
SOLVED EXAMPLES-3

E-1 Find the slope of a line parallel to the line which passes through (–a, 0) and (0, b).

Sol: Slope of the line through the points A(–a, 0) and B(0, b) is $m = \frac{b - 0}{0 - (-a)} = \frac{b}{a}$.

$\therefore$ Slope of the line parallel to AB is b/a.

E-2 In a rectangle ABCD, slope of AB = $\frac{4}{7}$. Find the slope of BC.



Sol: Slope of AB = $m_1 = \frac{4}{7}$.

Since BC $\perp$ AB, $m_2 = -\frac{1}{m_1} = -\frac{1}{4/7} = -\frac{7}{4}$.

Hence, slope of BC = $-\frac{7}{4}$.

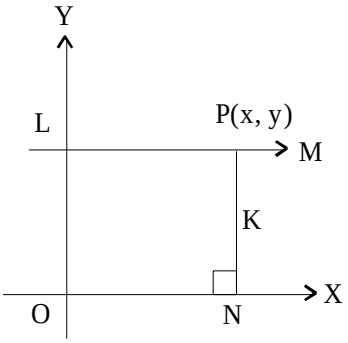
LOCUS

Locus of a point is the path which it describes when moving under a certain given condition. The equation of the locus is a relation between x and y which is satisfied by the coordinates of all points of the locus and by no others.

EQUATION OF A LINE

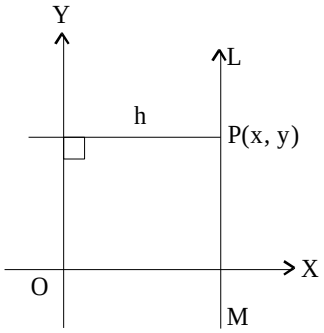
(i) A straight line parallel to x-axis at a given distance from it – The equation of line LM parallel to x-axis and at a distance 'k' from it is given by y = k (i.e. constant).

NOTE: Equation of the x-axis is y = 0.

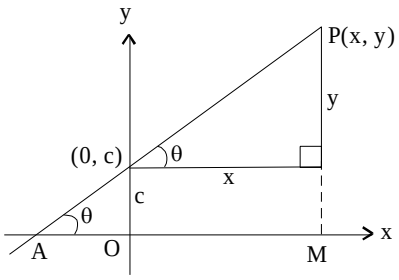


(ii) A straight line parallel to y-axis at a given distance from it – The equation of line LM parallel to y-axis and at a distance h from it is given by x = h (i.e. constant).

NOTE: Equation of the y-axis is x = 0.



(iii) **Slope-intercept form:** The equation of the straight line whose slope (i.e. gradient) is m and whose intercept on the y-axis is c is given by y = mx + c.

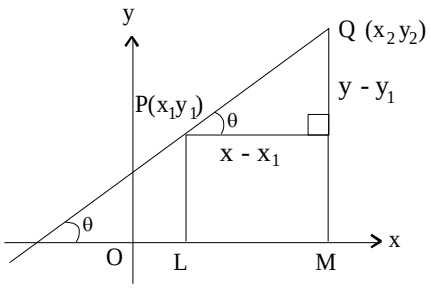


NOTE: If the straight line passes through the origin, then its equation is given by, y = mx (as y-intercept c = 0).

(iv) **Point-slope form:** The equation of the straight line passing through the point $P(x_1, y_1)$, and having a gradient m is given by $y - y_1 = m(x - x_1)$.

(v) **Two point form:** The equation of the straight line passing through two given points $P(x_1, y_1)$ and $Q(x_2, y_2)$ is given by

$$y - y_1 = \left(\frac{y_2 - y_1}{x_2 - x_1}\right)(x - x_1).$$



(vi) **Intercept form:** The equation of a straight line which makes intercepts 'a' and 'b' on the x-axis and y-axis respectively is given by

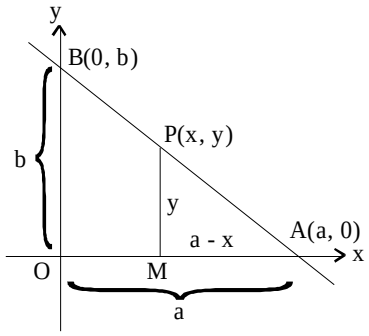
$$\frac{x}{x-\text{intercept}} + \frac{y}{y-\text{intercept}} = 1, \text{ i.e. } \frac{x}{a} + \frac{y}{b} = 1.$$

(vii) **Perpendicular form or normal form:** The equation of a straight line in terms of the length of the perpendicular 'P' from the origin to the line and the angle α which the perpendicular makes with the x-axis is given by $x \cos + y \sin = p$.

NOTE:

(i) p is always positive.

(ii) α the angle between the normal and the positive x-axis is such that $0 \leq x \leq 2$.



(viii) **General equation of the straight line:** All the different forms of equations of the straight line are all first degree equations in both x and y. Every equation of the form ax + by + c = 0, is the equation of a straight line provided 'a' and 'b' are not zero simultaneously.

NOTE:

(i) If b = 0, then a must not be equal to zero and the above equation becomes x = – c/a, which is the equation of a straight line parallel to the y-axis and – c/a units from it.

(ii) *If $b \neq 0$, then dividing by b*

we get, $\frac{ax}{b} + y + \frac{c}{b} = 0$, *i.e.* $y = -\frac{ax}{b} - \frac{c}{b}$, *i.e. it is of the form* $y = mx + c$, *where slope* $= m = -\frac{a}{b}$ *and y-intercept* $= c = -\frac{c}{b}$.

(iii) *Gradient of a line = slope* $= m = -\frac{a}{b}$, *i.e.* $\frac{\text{coefficient of x}}{\text{coefficient of y}}$.

(iv) *Let the lines be* $a_1x + b_1y + c_1 = 0$ *and* $a_2x + b_2y + c_2 = 0$ *and* θ *be the angle between the lines then*

$$\therefore \tan\theta = \frac{m_1 - m_2}{1 + m_1m_2} = -\frac{a_1}{b_1} - \left(-\frac{a_2}{b_2}\right) \div 1 + \frac{a_1a_2}{b_1b_2}$$

$$= \frac{-a_1b_2 + a_2b_1}{b_1b_2 + a_1a_2}$$

□

SOLVED EXAMPLES-3

E-1 Find the equation of a line passing through the point (5, –3) and parallel to x-axis.

Sol: The equation of a line parallel to x-axis is y = k.
Since the line passes through (5, –3), this point should satisfy y = k.
$\therefore$ y = –3 is the required equation of the line.

E-2 Find the equation of a line parallel to the segment with end points (6, 8), (3, 4) and having intercept –2 on the y-axis.

Sol: Slope $m = \frac{y_2 - y_1}{x_2 - x_1} = \frac{4-8}{3-6} = \frac{-4}{-3} = \frac{4}{3}$

y-intercept = c = –2

$\therefore$ Equation of the line is $y = mx + c$, i.e. $y = \frac{4}{3}x - 2$, i.e. $3y = 4x - 6$,

$\therefore 4x - 3y - 6 = 0$

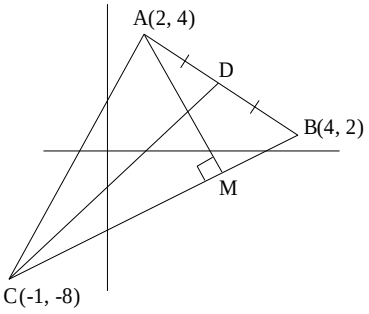
E-3 If A(2, 4), B(4, 2), and C(–1, –8) are the vertices of ΔABC. Find the equation of (i) side BC, (ii) altitude AM, and (iii) median CD.

Sol: (i) Equation of side BC is given by

$$y - y_1 = \left(\frac{y_2 - y_1}{x_2 - x_1}\right)(x - x_1),$$

where $B(x_1, y_1) = (4, 2)$ and

$C(x_2, y_2) = (-1, -8)$



$$\therefore y - 2 = \frac{-8-2}{-1-4}(x - 4)$$

$$\therefore y - 2 = 2(x - 4)$$

$$\therefore y - 2 = 2x - 8 \Rightarrow 2x - y - 6 = 0$$

(ii) AM $\perp$ BC $\quad \therefore$ Slope of AM $= -\dfrac{1}{\text{Slope of BC}} = -\dfrac{1}{2}$

Since AM passes through A(2, 4) we have, equation of AM is,

$$y - 4 = -\frac{1}{2}(x - 2)$$

$$\therefore 2y - 8 = -x + 2 \qquad \therefore \quad x + 2y - 10 = 0$$

(iii) D is the midpoint of AB.

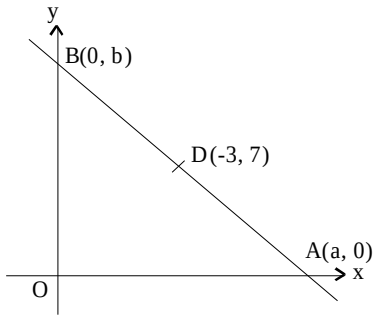
$$\therefore D \equiv \left(\frac{4+2}{2}, \frac{2+4}{2}\right) = \left(\frac{6}{2}, \frac{6}{2}\right) = (3, 3).$$

$$\therefore \text{Equation of CD is } y - (-8) = \frac{3-(-8)}{3-(-1)}[x - (-1)], \text{ i.e. } y + 8 = \frac{11}{4}(x + 1),$$

$$\therefore 4y + 32 = 11x + 11 \qquad \therefore 11x - 4y - 21 = 0$$

E-4 Point (–3, 7) is the midpoint of the portion intercepted by the line between the coordinate axes. Find the equation of the line.

Sol: Let the line passing through the point D(–3, 7) cut the x-axis at A(a, 0) and B(0, b).
D is the midpoint of AB.



$\therefore D(-3, 7) = \left(\frac{a+0}{2}, \frac{0+b}{2}\right)$

$\therefore \frac{a}{2} = -3 \Rightarrow a = -6$ and $\frac{b}{2} = 7 \Rightarrow b = 14$

$\therefore$ The equation of the line is, $\frac{x}{a} + \frac{y}{b} = 1$.

$\therefore \frac{x}{-6} + \frac{y}{14} = 1 \ \therefore \frac{-7x+3y}{42} = 1 \ \therefore -7x + 3y = 42 \ \therefore 7x - 3y + 42 = 0$

E-5 Find the equation of a line passing through the point (3, –2) and parallel to the line $3x + 4y - 5 = 0$.

Sol: Slope of the line $= -\frac{a}{b} = -\frac{3}{4}$

$\therefore$ Slope of a parallel line $= -\frac{3}{4}$

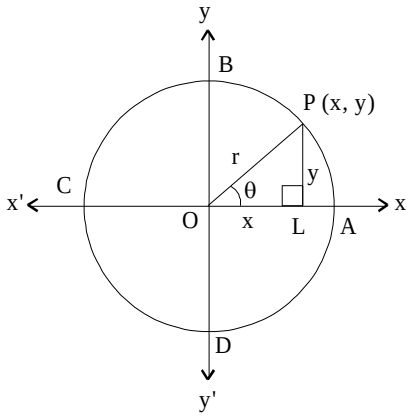
$\therefore$ Equation of the required line is $y - (-2) = -\frac{3}{4}(x - 3)$.

$\therefore 4y + 8 = -3x + 9 \Rightarrow 3x + 4y - 1 = 0$

Chapter 23 Trigonometry

Suggested Time : 120 minutes

Trigonometric ratios: Trigonometric ratios express the relationship which exists between the size of one angle and the lengths of two edges in a right angled triangle.



Consider a circle with radius 'r' and centre as the origin. Let a moving point P(x, y) lie on the circumference of the circle as shown in the figure.

Let $\angle AOP = \theta$. Draw $PL \perp OA$.

$\therefore$ OL = x, PL = y, and OP = r

In right-angled ΔOLP, for acute angle θ :

(i) sine (sin) is defined as the ratio between the lengths of perpendicular (i.e. opposite side) and hypotenuse.

$\therefore \sin\theta$ = Perpendicular or opp. side / Hypotenuse

$$= \frac{PL}{OP} = \frac{y}{r}$$

(ii) cosine (cos) is defined as the ratio between the lengths of base (or adjacent side) and hypotenuse.

$\therefore \cos\theta$ = Base or adjacent side / Hypotenuse = $\frac{OL}{OP} = \frac{x}{r}$

(iii) tangent (tan) is defined as the ratio between the lengths of perpendicular (or opposite side) and base (or adjacent side).

$\therefore \tan\theta$ = Perpendicular or opposite side / Base or adjacent side = $\frac{PL}{OL} = \frac{y}{x}$

(iv) cotangent (cot) is defined as the ratio between the lengths of base (or adjacent side) and perpendicular (or opposite side).

$\therefore \cot\theta$ = Base or adjacent side / Perpendicular or opposite side = $\frac{OL}{PL} = \frac{x}{y}$

(v) secant (sec) is defined as the ratio between the lengths of hypotenuse and base (or adjacent side).

$\therefore \sec\theta$ = Hypotenuse / Base or adjacent side = $\frac{OP}{OL} = \frac{r}{x}$

(vi) cosecant (cosec) is defined as the ratio between the lengths of the hypotenuse and perpendicular (or opposite side).

$\therefore \text{cosec}\,\theta$ = Hypotenuse / Perpendicular or opposite side = $\frac{OP}{PL} = \frac{r}{y}$

NOTE:

(i) *Each trignometrical ratio is a real number and has no unit.*

(ii) $\cot\theta = \frac{1}{\tan\theta}$ *or* $\tan\theta = \frac{1}{\cot\theta}$ *or* $\cot\theta \ \tan\theta = 1$

(iii) $\sec\theta = \frac{1}{\cos\theta}$ *or* $\cos\theta = \frac{1}{\sec\theta}$ *or* $\cos\theta \ \sec\theta = 1$

(iv) $\operatorname{cosec}\theta = \frac{1}{\sin\theta}$ *or* $\sin\theta = \frac{1}{\operatorname{cosec}\theta}$ *or* $\sin\theta \ \operatorname{cosec}\theta = 1$

For standard angles we have,

Angle (In degrees and radians)	0^0	30^0	45^0	60^0	90^0
Sin	0	$\frac{1}{2}$	$\frac{1}{\sqrt{2}}$	$\frac{\sqrt{3}}{2}$	1
Cos	1	$\frac{\sqrt{3}}{2}$	$\frac{1}{\sqrt{2}}$	$\frac{1}{2}$	0
Tan	0	$\frac{1}{\sqrt{3}}$	1	$\sqrt{3}$	∞
Cot	∞	$\sqrt{3}$	1	$\frac{1}{\sqrt{3}}$	0
Sec	1	$\frac{2}{\sqrt{3}}$	$\sqrt{2}$	2	∞
Cosec	∞	2	$\sqrt{2}$	$\frac{2}{\sqrt{3}}$	1

FUNDAMENTAL IDENTITIES:

(i) $\sin^2\theta + \cos^2\theta = 1 \Rightarrow \sin^2\theta = 1 - \cos^2\theta$ and $\cos^2\theta = 1 - \sin^2\theta$

(ii) $1 + \tan^2\theta = \sec^2\theta \Rightarrow \sec^2\theta - \tan^2\theta = 1$, i.e. $(\sec\theta - \tan\theta)(\sec\theta + \tan\theta) = 1$

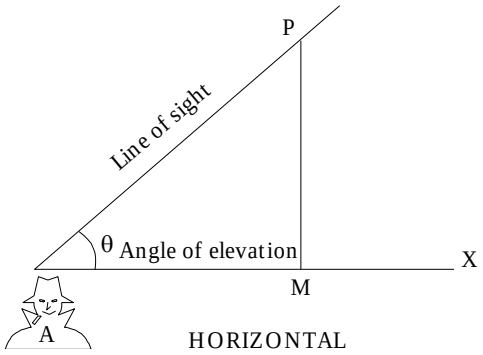
(iii) $1 + \cot^2\theta = \operatorname{cosec}^2\theta - \cot^2\theta = 1$, i.e. $(\operatorname{cosec}\theta - \cot\theta)(\operatorname{cosec}\theta + \cot\theta) = 1$

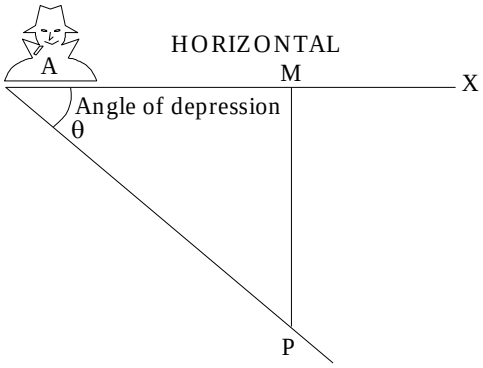
ANGLES OF ELEVATION AND DEPRESSION

Suppose a person A is looking towards an object P which is at a higher level than A. Let AX be the horizontal line drawn through the Observer. Then $\angle$PAX is called the ***Angle of Elevation*** of P from A.

Thus, the angle of elevation is the angle between the line of sight and a particular horizontal line through the observer.

Conversely, if the observer is at a higher level than the object, then the angle between the line of sight and a particular horizontal line through the observer is called the ***Angle of Depression*** of the object P from A.





NOTE:

(i) In both cases the horizontal line is the starting line and so the angle is to be measured from the horizontal line through the observer.

(ii) Numerically, angle of elevation is equal to the angle of depression.

(iii) $\sqrt{2}$ *= 1.414 and* $\sqrt{3}$ *= 1.732.*

Chapter 24 Data Interpretation

Suggested Time : 120 minutes

Cracking the Data Interpretation (DI) section can be the easiest or the most challenging parts of any aptitude test. We list the basic definitions to answer questions in data interpretation. Once the definitions are known, you are then prepared to work out examples and practice questions.

LIST OF DEFINITIONS AND FORMULAE

The following is a list of terminology used in data interpretation with examples and calculation shortcuts. Use the following data to understand the meaning of each of the terms.

The following table gives the ODI averages of Indian batsman between 2002 and 2005.

Year	Tendulkar	Sehwag	Dravid	Yuvraj	Kaif
2002	55	70	42	28	21
2003	48	41	38	37	33
2004	42	28	51	43	39
2005	40	54	67	82	35

1. **Increase** or 'absolute increase' or 'actual increase': This is the simple difference between the numbers.
   E.g. What has been the increase in Dravid's batting average from 2002 to 2005 ?
   Ans : 67 – 42 = 25
   E.g. Whose batting average has increased the most between 2002 and 2005?
   Ans : Yuvraj Singh → from 28 to 82.

2. **Decrease** or absolute decrease: Simple difference in numbers or magnitude, when difference is negative.
   E.g. What has been the decrease in Tendulkar's batting average from 2002 to 2005?
   Ans : 55 – 40 = 15

The following table gives the sales and profit of a company for 4 years.

Year	Sales (Rs. crores)	Profit (Rs. crores)
1997	250	40
1998	350	60
1999	400	20
2000	450	75

3. **Percentage Increase** or Growth: Simple increase from base value as a percentage of the base value.
   E.g. What has been the percentage increase in sales from 1997 to 2000?

Ans : $\frac{450-250}{250}\times 100=\frac{200}{250}\%=80\%$

Formulae: $\frac{\text{New Amount}-\text{Previous Amount}}{\text{Previous Amount}}\times 100=\frac{\text{Final}-\text{Initial}}{\text{Initial}}\times 100=\frac{\text{Difference}}{\text{Initial}}\times 100$

Short cuts : For the above calculation, we can also use the following methods:

1] $\frac{200}{250}=\frac{4}{5}=80\%$

2] $\left(\frac{450}{250}\times 100\right)-100=180\%-100\%=80\%$

3] $\left(\frac{450}{250}\right)-1=1.8-1=0.8\text{ or }80\%$

E.g. When has the growth in profit been the highest ?

Ans : 1998 = $\frac{60-40}{40}\times 100=\frac{20}{40}\times 100=50\%$;

1999 = No growth (it is a negative growth);

2000 = $\frac{75-20}{20}\times 100=\frac{55}{20}\times 100=275\%$.

$\therefore$ Highest growth has been in 2000

4. **Percentage Decrease** or Decline: The formula is same as percentage increase, i.e. Simple decrease from previous figure as a percentage of the previous figure.
E.g. What has been the percentage decrease in profits from 1998 to 1999?

Formulae : $\frac{\text{New Amount}-\text{Previous Amount}}{\text{Previous Amount}}\times 100=\frac{\text{Final}-\text{Initial}}{\text{Initial}}\times 100=\frac{\text{Difference}}{\text{Initial}}\times 100$

Ans : $\frac{20-60}{60}\times 100=\frac{-40}{60}\%=-66.66\%$

Note: The " – " sign signifies that there has been a decrease. $\therefore$ The decrease is of 66.66 %.

Important: Percentage decrease can be over 100 % only if final value turns negative.

5. **Change and percentage change:** Change is similar to increase / decrease and percentage change is similar to percentage increase / decrease. However, change is calculated without taking the signs into consideration.
In the earlier example, what was the change in profits for the three years.
Change in 1997-98 is from 40 to 60, i.e. 20.
Change in 1998-99 is from 60 to 20, i.e. 40. (Note that it is not written as " – 40").
Change in 1999-2000 is from 20 to 75, i.e. 55.

Practice Questions A (Optional)

Directions for questions 1 to 5: Four companies A, B, C and D constitute the ITES sector in Toronto. The following table shows the percentage share of sales, profits and employees of the 4 companies.

Company	Sales	Profits	Employees
A	40%	25%	50%
B	20%	30%	20%
C	10%	20%	10%
D	30%	25%	20%

1. What is the ratio of profits to sales for company A?
   1] 0.625 2] 0.25 3] 0.56 4] Indeterminate

2. What is the ratio of sales of company A to that of company B?
   1] 2 : 1 2] 5 : 2 3] 1 : 2 4] Indeterminate

3. Which company has the highest sales per employee ratio?
   1] A 2] B 3] C 4] D

4. If profits of C are 10 crore and account for 20% of it's sales, which company has the best profit margin (profit as percentage of sales)?
   1] A 2] B 3] C 4] D

5. By what percentage should A reduce its employee force if it wishes to equal the profit per employee ratio of C?
   1] 50% 2] 25% 3] 75% 4] 22.5%

Directions for questions 6 to 11: Ghosh Babu recently has acquired four companies: ACT; FACT; NEXT; JEST. He noticed that the sales of JEST is half that of ACT whereas the profits of JEST are twice as much as profits of ACT. The expenses of NEXT are Rs.3 crore less than that of JEST whereas the profit of FACT is Rs.1 crore less than that of NEXT. The expenses of ACT is 3 times that of JEST. It is known that the sales of NEXT is Rs.15 crore or one – fourth of that of FACT. All figures are of 1998 – 99.

Ghosh Babu is also informed that the sales of JEST is Rs.10 crore more than that of NEXT and the expenses of ACT are 90% of its own sales. *Profit = Sales – Expenses*

6. The total sales of all the four companies is (Rs. crore) _____
   1] 200 2] 150 3] 125 4] 160

7. By how much (in Rs. crore) do the expenses of ACT exceed that of NEXT?
   1] 34 2] 43 3] 33 4] 47

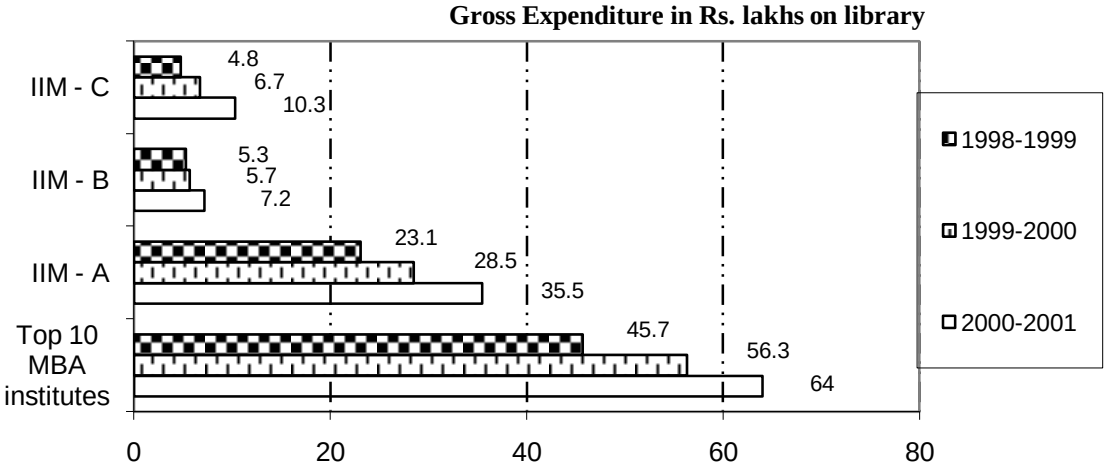
8. Which company had the maximum profits?
   1] ACT 2] FACT 3] NEXT 4] JEST

9. By how much (in Rs. crore) do the expenses of FACT exceed the profits of ACT?
   1] 51 2] 52 3] 53 4] 56

10. What are the total profits (in Rs. Crore) of the four companies?
   1] 15 2] 20 3] 25 4] 40

11. The profits of FACT form what percentage of the expenses of NEXT?
   1] 16.67% 2] 20% 3] 15% 4] 33.33%

Directions for questions 12 to 16: Refer to the graph below



Note: (Top 10 MBA institutes include IIM A, B and C).

12. Of the three IIMs which institute showed a minimum actual increase in library spend between 1998-2001?
   1] IIM-A 2] IIM-B 3] IIM-C 4] IIM-A & IIM-B

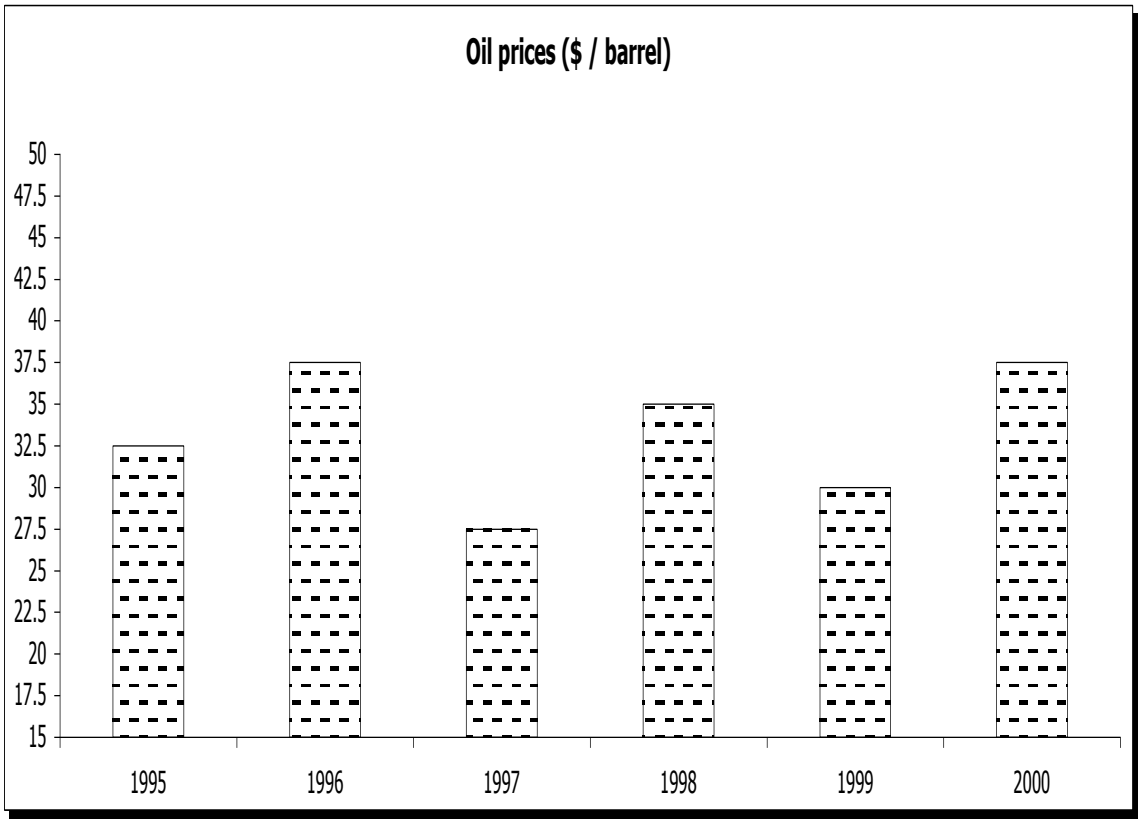
13. Of IIMs – A, B and C, which institute recorded a maximum percentage increase in library spend during the period 1998-2001?
   1] IIM-A 2] IIM-B 3] IIM-C 4] IIM-B & C

14. Approximately what was the percentage of library spending of IIM-B as compared to IIM-A in 2000-01?
   1] 10% 2] 15% 3] 20% 4] 25%

15. In 1999-2000, the combined gross spend on library, of IIMs A, B and C comprised approximately what percentage of gross spend on library for the top 10 B-schools?
   1] 40% 2] 50% 3] 60% 4] 70%

16. Which institute among the top 10 showed the least increase in library spend from 1998-99 to 1999-2000?
   1] IIM – A 2] IIM – B 3] IIM – C 4] Indeterminate

DIRECTIONS for questions 17 to 21: Refer to the graph below.



17. Oil prices in 1996 are what percent of the average oil prices from 1995 – 2000?
    1] 112.5 2] 125 3] 100 4] 88.88

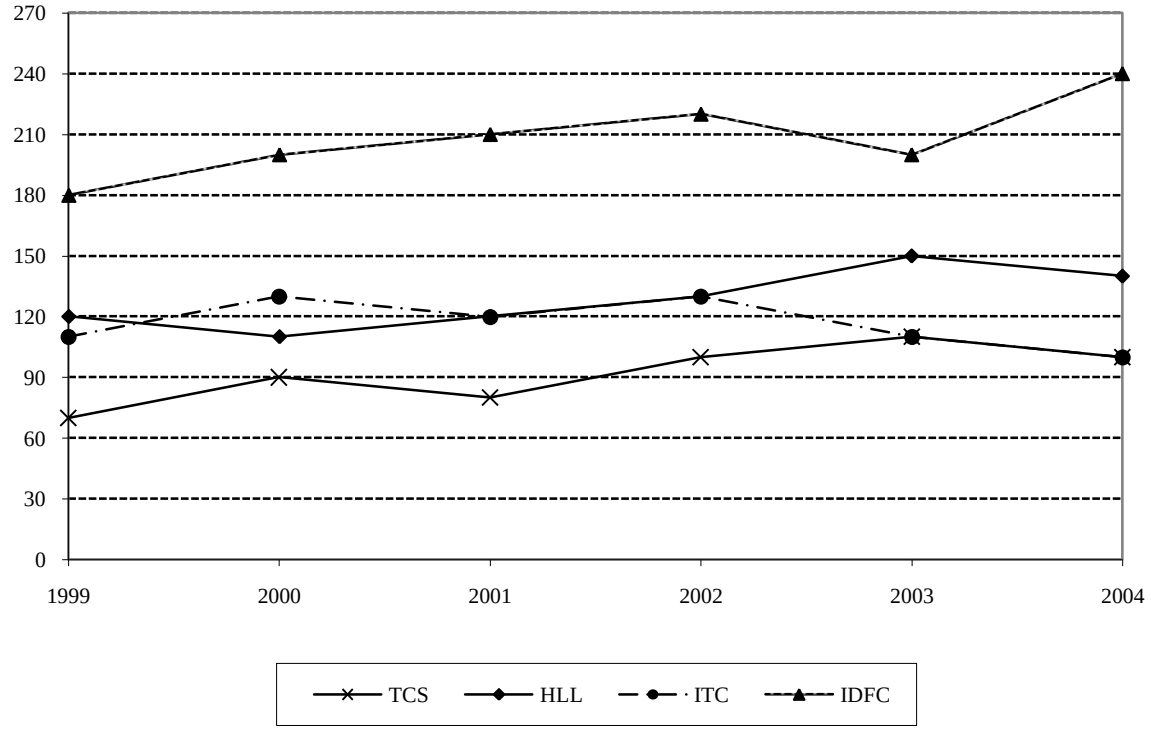
18. For how many years has the oil price been lesser than the average oil prices for the period 1995-2000?
    1] 2 2] 3 3] 4 4] Indeterminate

19. In which year has the percentage change in oil prices been the highest over that of its previous year?
    1] 1998 2] 1997 3] 2000 4] 1999

20. By what percentage does the increase in oil prices from 1997 to 1998 exceed the increase in oil prices from 1995-1996?
    1] 27.5% 2] 30% 3] 35% 4] 50%

21. The average annual growth rate in oil prices from 1995 to 2000 is _____
    1] 3% 2] 5% 3] 2.5% 4] 1%

Directions for questions 22 to 25: Refer to the graph given below that shows the end-of-year market values of four shares for the period 1999-2004.



22. Which share showed the greatest percentage increase in market value in any year during the entire period?
1] ITC 2] HLL 3] TCS 4] IDFC

23. In which year was the greatest absolute change in market value for any share recorded?
1] 1999 2] 2001 3] 2003 4] 2004

24. Mr. Natwarlal sold 1 share of TCS and 1 share of ITC to buy 1 share of IDFC in 1999. In which later year can he sell IDFC to buy 1 share each of TCS and HLL without any gain/loss?
1] 2000 2] 2001 3] 2003 4] 2004

25. Hitenbhai sold 1 share of TCS and 1 share of HLL to buy 1 share of IDFC in 2000. In which later year would Hitenbhai's notional loss from this decision due to share value changes have been the most?
1] 2001 2] 2002 3] 2003 4] 2004

Explanatory Answers

1. [4]. The value cannot be found as the exact values of sales and expenditure is not known. It should be noted that only the values of percentage break-ups are indicated.

For better understanding, let total sales be s and total profit be p.

Can we find the value of $\frac{25\%\text{ of p}}{40\%\text{ of s}} = \frac{0.25p}{0.40s} = \frac{5}{8} \times \frac{p}{s}$ without knowing the values of p and s?

2. [1]. The percentage break-ups for the same base value are indicated.
Thus, ratio = 0.4 / 0.2 = 2 : 1

3. [4]. Sales per employee = Total sales / Number of employees.
As the absolute values of the parameters involved are not given, it is not possible to zero in on the numerical values.
However, comparisons in terms of ratios involved can be made. Thus, company D has highest sales / employee ratio.
Ratio = 0.3 / 0.2 = 1.5

4. [3]. One should note that the values given in the question are redundant and answer can be found as ratio of profits and sales percentages.

5. [3]. Profit per Employee of A = 0.25 / 0.5 = 0.5
Profit per Employee of C = 0.2 / 0.1 = 2
Thus, for A to have profit per employee ratio of 2, its number of employees (x) should be calculated thus.
0.25/x = 2, and hence x = 0.125, i.e. reduced from 0.5 to 0.125 or 75%.

Explanatory Answers for questions 6 to 10

Create a table as follows:

Company	Sales	Expenses	Profits
ACT	50	45	5
FACT	60	58	2
NEXT	15	12	3
JEST	25	15	10

6-2 7-3 8-4 9-3 10-2 11-1

12. IIM – B [1.9]. Hence, [2].

13. IIM – C [114.5%] ; Smaller the base higher the percentage change. Hence, [3].

14. 20%; Approximate to $\left[\frac{7}{35}\right]$ = 20%. Hence, [3].

15. 70%; $\left[\frac{40.9}{56}\right] = \left[\frac{5}{7}\right] = 5 \times 0.14 = 0.7$. Hence, [4].

16. Indeterminate as details on any of the other 7 colleges in the top 10 are not individually known.

17. Average oil price = $\frac{32.5+37.5+27.5+35+30+37.5}{6} = 33.33$.

Price in 1999 as percentage of average = $\frac{37.5}{33.33} \times 100 = 112.5$. Hence, [1].

18. Oil prices have been less than the average in 1995, 1997, and 1999. Hence, [3].

19. Two major changes are decrease of 10 in 1997 and increase of 7.5 in 1998.

Percentage change in 1997 = $\frac{27.50-37.5}{37.5} \times 100 = -26.66\%$.

Percentage change in 1998 = $\frac{35-27.5}{27.5} \times 100 = 42.85\%$. Hence, [1].

20. Increase in price from 1997 to 1998 = 7.5.
Increase in price from 1995 to 1996 = 5.
Increase in 1997 – 98 as percentage of 1995 – 96 = 50%. Hence, [4].

21. Prices in 1995 = 32.5. Prices in 2000 = 37.5

Growth = $\frac{37.5-32.5}{32.5} \times 100 = 15.38\%$.

This growth has happened across 5 periods.

Hence, average growth rate = $\frac{15.38}{5} = 3.07\%$. Hence, [1].

22. For the same absolute change, percentage increase is always higher for a smaller base. In this question for 1999, TCS has the lowest absolute values for its share price. Therefore, the first step should be to find the highest percentage increase for TCS between any two consecutive years and then proceed in increasing magnitude of share price , which in this case would be TCS followed by HLL, ITC and then IDFC.

The highest absolute change for TCS is between 1999 and 2000 when it goes from 70 to 90 which is a percentage increase of almost 30%. For any other company, the absolute change has to be more than 20 to show a higher percentage increase. The only time this happens is between 2003 and 2004 for IDFC when it goes from 200 to 240 which is a 20% increase and hence lower than TCS. Hence [3]

23. The trick is to just find the line segment between any two consecutive years with the greatest slope. IDFC share prices increases form 200 to 240 in 2004. Hence, [4].

24. IDFC = 240, HLL = 140, TCS = 100. Hence, [4].

25. In 2003, combined value of TCS and HLL = 260 while IDFC = 200, Loss = Rs. 60. Hence, [3].

Chapter 25 Logical Reasoning

Suggested Time : 60 minutes

The first principle of logic is an understanding of logical constructs. There are various type of constructs used in logical reasoning; the interpretation of the most often used among them are described below.

a. Conditional logic: (Either … Or …)
In the either …or construct, only one of the conditions holds true. For e.g. In "either X or Y" type of a statement, if X is true then Y is false. However if Y is true then X is false. Similarly, one can also reverse the order of arrangement of the two conclusions. i.e. if X is false, then we know that Y is true, and if Y is false, then we derive that X must be true. i.e. $X_T \rightarrow Y_F$ or $X_F \rightarrow Y_T$ or $Y_T \rightarrow X_F$ or $Y_T \rightarrow X_F$.

In an either …or construct, a sentence is given, followed by a set of statements. From the set, you have to choose a pair which can be deduced or inferred from the opening sentence.

Example: Either I am happy or I am smiling.

a. I am happy
b. I am smiling
c. I am not happy
d. I am not smiling

1. ac only 2. bc only 3. ad & bc 4. ab only

In this case, the answer is simply the one where one and only one of the actions is happening. Hence the answer is either ad or bc or da or cb. It is clear in the main statements that at any given time, one of the actions (i.e. smiling or being happy) has to happen.

Jack is either a dancer or not a singer.

Here, consider the statement again in the format Either X Or Y. Here X = is a dancer, and Y = not a singer. We know that in an either…or construct $X_T \rightarrow Y_F$ or $X_F \rightarrow Y_T$ or $Y_T \rightarrow X_F$ or $Y_T \rightarrow X_F$.

Note that the negative of a negative is a positive. Hence Not Y would be Not of "Not a singer" which means "Is a Singer".

There are two possible combinations.
1. Jack is a singer; Jack is a dancer or
2. Jack is not a dancer; Jack is not a singer.

In this case, if one statement of the premise is negative and the other is positive, the answer is either both statements are negative or both are positive.

Two more conclusions can be reached if we reverse the order of the two conclusions reached.
1. Jack is a dancer; Jack is a singer or
2. Jack is not a singer; Jack is not a dancer.

b. Cause & Effect type

In this kind of questions, two statements are given with a causal relationship between them.

Example: If he cleans his room it will rain in town. Thus statement 'A' (if he cleans the room) leads to statement 'B' (rains in town). But not necessarily vice versa. i.e. every time it rains in town does not necessarily mean he has cleaned his room.

In this type of questions check for nature of relationships between 'A' & 'B'. If 'A' is the cause is it a necessary and sufficient condition for 'B' to happen?

If 'B' happens does 'A' necessarily happen? That is, does a reverse causal relationship hold true?

Example: If I drop a coin(A), then my wish will come true (B).

a. I dropped a coin
b. My wish came true
c. My wish did not come true
d. I did not drop a coin

1. ab only 2. cd and ab 3. bd only 4. ca, cd & ba

Here, the construct is of the type "if X happens then Y happens". In a construct of this type, the only two conclusions that are possible are
1) X has happened, hence Y will happen.
2) Y as not happened, hence X could not have happened.

In the above example, if A happens, then B will happen. Hence answer is (2)

Example: Only if I drop a coin (A), does my wish come true (B).

a. I dropped a coin
b. My wish came true
c. My wish did not come true
d. I did not drop a coin

1. ab only 2. cd and ba 3. bd only 4. ab, ba, cd & dc.

Here, the construct is of the type "if and only if X happens, then Y happens". This is similar to X = Y. In a construct of this type, the conclusions that are possible are
1) X has happened, hence Y will happen.
2) X has not happened, hence Y will not happen.
3) Y has happened, so X must have happened.
4) Y as not happened, hence X could not have happened.

In the above example, B happens only if A happens. Hence answer is (4)

Practice Exercise A (Important)

Directions for questions: Each question has a main statement followed by four statements labeled A, B, C and D. Choose the ordered pair of statements where the first statement implies the second, and the two statements are logically consistent with the main statement.

1. Either the monkey is not angry, or he wants a banana.
   A. The monkey wants a banana.
   B. The monkey is not angry.
   C. The monkey does not want a banana.
   D. The monkey in angry.
   1] CB only 2] DA only 3] AB only 4] CB and DA

2. Either Shyam is a villain, or he is a hero.
   A. Shyam is a hero.
   B. Shyam is a villain.
   C. Shyam is not a villain.
   D. Shyam is not a hero.
   1] CD only 2] BA only 3] CD & BA 4] DB & CA

3. Whenever Rajeev uses the hat, he dreams about magicians.
   A. Rajeev did not dream about magicians.
   B. Rajeev used the hat.
   C. Rajeev dreamt about magicians.
   D. Rajeev did not use the hat.
   1] AD 2] DC 3] CB 4] DA

4. If I talk to my professors, then I do not need to take a pill for headache.
   A. I talked to my professors.
   B. I did not need to take a pill for headache.
   C. I needed to take a pill for headache.
   D. I did not talk to my professors.
   1] AB only 2] DC only 3] CD only 4] AB and CD

Answer Key

1-4 2-4 3-1 4-4

Explanatory Answers

1. In the Either…Or case, if X is false, then we know that Y is true, and if Y is false, then we derive that X must be true. i.e. $X_T \rightarrow Y_F$ or $X_F \rightarrow Y_T$ or $Y_T \rightarrow X_F$ or $Y_T \rightarrow X_F$. Hence, the possible options are BC, CB, AD and DA. From amongst the answer option the best answer is [4].

2. In the Either…Or case, if X is false, then we know that Y is true, and if Y is false, then we derive that X must be true. i.e. $X_T \rightarrow Y_F$ or $X_F \rightarrow Y_T$ or $Y_T \rightarrow X_F$ or $Y_T \rightarrow X_F$. Hence, the possible options are BD, DB, AC and CA. From amongst the answer option the best answer is [4].

3. In an If…Then statement, the two conclusions possible are 1) X has happened, hence Y will happen.
   2) Y as not happened, hence X could not have happened. Hence the conclusions are BC or AD. Hence [1]

4. In an If…Then statement, the two conclusions possible are 1) X has happened, hence Y will happen.
   2) Y as not happened, hence X could not have happened. Hence the conclusions are AB or CD. Hence [4]

Chapter 26 Deductive Logic

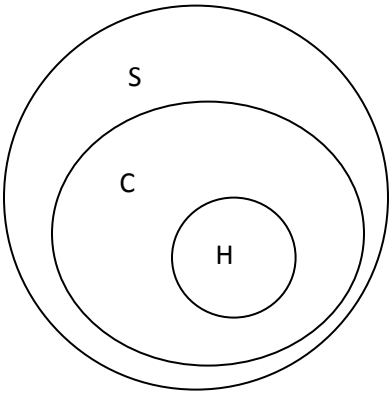
Suggested Time : 60 minutes

This type of problems involves arriving at a conclusion based on a few given premises.

Example:

All cooks make soup. ⟶ Premise I
All housewives are cooks. ⟶ Premise II
∴ All housewives make soup. ⟶ Conclusion

This logic can be represented as a Venn diagram.



S- Soup
C- Cook
H – Housewife

There can two types of deductive logic questions. In Type 1, you are expected to choose three sentences from a set of options that would make a logical connection. Hence any two statements can be premises, and the third would be a conclusion. This type of question is very easy to solve.

Example: Choose the three sentences that make a logical combination.

A. Some dogs are deaf.
B. Deaf dogs are intelligent.
C. Jimmy is a dog.
D. Jimmy is deaf dog.
E. Jimmy is intelligent.

1. ABE 2. BDE 3. CDB 4. EBD

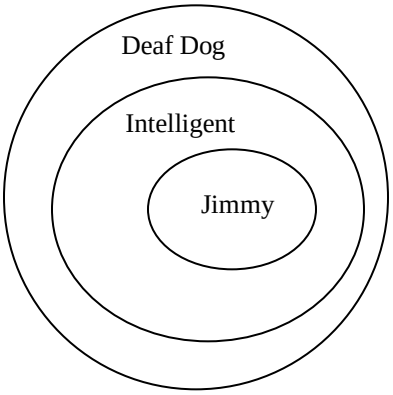
In Type 2, you are expected to choose the option in which the last sentence can be deduced from the first two.

Example: Choose the options in which the third part of the sentence is logically derived from the previous two.
A. Some fish are pink; Pink fish live in the sea; Gold fish do not live in the sea.
B. Sheela is Ashish's sister; Jagan is Sheela's husband; Jagan & Ashish are related.
C. All scorpions sting; This is a scorpion; This will sting.
D. All flowers smell good; All roses smell good; All roses are flowers.
1. only B 2. only C 3. B and C 4. only A

Some strategies to tackle deductive logic questions

1. Work backward from the choices.
In the first example, only choices ending with E can be considered right. Of these, only choice 2 can be logically inferred from the sentences given.

2. Whenever possible represent relationships as Venn diagrams as shown for option 2.



3. Check logic of all sets in Type 2. More than one set may be correct.
In example for Type 2 given above, the right choice is 3, i.e., 'B' & 'C' are correct.

4. Eliminate choices where a statement negates or just repeats a previous statement.
Example: Bachchan acts well; Bachchan sings well; Bachchan only acts.

5. Always assume the statements at face value, even if it does not conform to your common knowledge and only check for logical connects between the statements.
In example for Type 2, 'D' is not a correct option. Though common knowledge says that Roses are flowers, it does not logically follow from the first two sentences.

Practice Exercise A (Important)

Directions for questions: Each question consists of five statements followed by options consisting of three statements put together in a specific order. Choose the option which indicates a valid argument, that is, where the third statement is a conclusion drawn from the preceding two statements.

1.
A. All research companies employ engineers.
B. SCT employs engineers.
C. SCT is a research company.
D. Some research companies employ engineers.
E. SCT employs only engineers.

1] ABC 2] ACB 3] CDB 4] ACE

2.
A. Air conditioners increase green house gases in the environment
B. Increase in green house gases is hazardous to health.
C. Air conditioners are hazardous to health.
D. Some air conditioners do not cause increased green house gases.
E. Some air conditioners is not hazardous to health.

1] CBA 2] BDE 3] CDE 4] BAC

3.

A. Toffees are not sweets.
B. Some toffees are sweet.
C. All sweets are tasty.
D. Some toffees are not tasty.
E. No toffee is tasty.

1] CEA 2] BDC 3] CBD 4] EAC

4.

A. Some towns are polluted.
B. All polluted towns should be destroyed.
C. Town Bennai should be destroyed.
D. Town Bennai is polluted.
E. Some towns should be destroyed.

1] BDE 2] BAE 3] ADE 4] CDB

5.

A. No police is a criminal
B. Shyam is not a criminal.
C. Shyam is a police.
D. Ram is not a police.
E. Ram is a criminal

1] ACB 2] ABC 3] ADE 4] ABE

6.

A. Shooters like guns.
B. Boys are shooters.
C. Balaram is a shooter.
D. Balaram likes guns.
E. Balaram may use guns.

1] BCA 2] ADC 3] ABE 4] ACD

7.

A. All actors are handsome.
B. Some actors are popular.
C. SRK is handsome.
D. SRK is popular actor.
E. Some popular actors are handsome.

1] ACD 2] ABE 3] DCA 4] EDC

8.

A. Modern industry is a technology driven
B. BTI is a modern industry.
C. BTI is technology driven.
D. BTI may be technology driven.
E. Technology driven industry is modern.

1] ABC 2] ABD 3] BCA 4] EBC

9.
A. All Software professionals are blue eyed people.
B. Some smart people are not blue eyed people.
C. Some babies are blue eyed.
D. Some babies are smart.
E. Some smart people are not Software professionals.

1] BCD 2] ABE 3] CBD 4] None of the above

10.
A. Finance professionals are in great demand.
B. Ram and Sita are in great demand.
C. Ram is in great demand.
D. Sita is in great demand.
E. Ram and Sita are finance professionals.

1] ABE 2] ECD 3] AEB 4] EBA

Answer Key

1-2 2-4 3-1 4-2 5-1 6-4 7-2 8-1 9-2 10-3

Explanatory Answers

1. In option 2, RC < E and SCT < RC. Hence, SCT < E.

2. In option 4, GHG < HTH and AC < GHG. Hence, AC < HTH.

3. In option 1, S = Ta and To ≠ Ta. Hence, S ≠ To.

4. In option 2, PT = D and ST = PT. Hence, ST = D.

5. In option 1, P ≠ C and S = P. Hence, C ≠ S

6. In option 4, S = G and B = S. Hence, B = G.

7. In option 2, A < H and A = P. Hence, P < H.

8. In option 1, MI = TD and BTI = MI. Hence, TD = BTI.

9. In option 2, SoPro = BEP and SmPe ≠ BEP. Hence SmPe ≠ SpPro.

10. In option 3, FP = GD and R&S = FP. Hence, R&S = GD.

Chapter 27 Analytical Reasoning

Suggested Time : 120 minutes

Analytical Reasoning is based on the basic constructs used in Logical Reasoning. The questions require you to organize and create a structure to solve the question. Almost all aptitude tests have a large number of questions on Analytical Reasoning.

Here is a sample:

Questions are based on the following:
In a group there are five persons A, B, C, D, E. There is exactly a tap dancer, a preacher and a swimmer in the group, and a person cannot have more than one profession. E is the husband of a member of the group and the two are the only married couple in the group. B is the brother of C. B is neither a preacher nor a swimmer. None of the women is either a tap dancer or a swimmer. A and D are unmarried and neither of them is a tap dancer or a preacher or a swimmer.

1. Which of the following people is E's wife?
   1] A 2] B 3] C 4] D

2. Who is the swimmer?
   1] A 2] E 3] B 4] C

3. Who is a tap dancer?
   1] B 2] A 3] C 4] D

Let us follow the steps required to answer the question.

Step 1: Core Question and Deciding Structure: There are 5 people and some professions, with no person following 2 professions. This can be visually represented in a table. Visual representation would help as reading the data over and over again can then be avoided, and information search is fast.

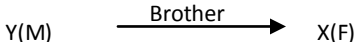
Step 2: Creating the table structure: Here is a sample table structure that could be constructed.

	Tap Dancer	Preacher	Swimmer
A			
B			
C			
D			
E			

Step 3: Read each line of the case, and map the data on the table. You may have to modify the table structure as you go. For e.g. when you read that you also have to include the gender and the relation between people, you may decide to write it next to the names of the 5 persons.

Step 4: Create your own short-forms to represent the data or relationships. Some examples are
Using M and F for males and females
Using marked arrows to describe relationships.

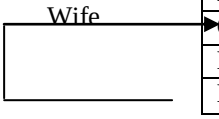
For E.g., to represent Y is the brother of X who is a female, we can construct the following diagram



Step 5 : You may have to read the data more than once to be able to map out the entire information on the table. Your table may look as follows.

	Tap Dancer	Preacher	Swimmer
A(Unmarried)	N	N	N
B(M)		N	N
C			
D(Unmarried)	N	N	N
E(M)			

This leaves you with only one conclusion, that C is a female and is married to E. Your final table may look as follows.



	Tap Dancer	Preacher	Swimmer
A(Unmarried)	N	N	N
B(M)	Y	N	N
C(F)	N	Y	N
D(Unmarried)	N	N	N
E(M)			

Hence, E has to be the swimmer.

While there may be many styles of answering the question, structuring the data is very useful, and it helps in answering the questions faster, and with higher accuracy. You should create your own structures, short-forms and shortcuts to answer the questions.

Not all questions require a tabular structure, and you have to decide what would be the best structure be

Questions are based on the following:
Exactly seven persons – P, Q, R, S, T U and V – participate in and finish all of a series of swimming races. There are no ties for any position at the finish of any of the races.
V always finishes somewhere ahead of P.
P always finishes somewhere ahead of Q.
Either R finishes first and T finishes last, or S finishes first and U or Q finishes last.

1. If in a race V finishes fifth, which of the following must be true?
   1] S finishes first. 2] R finishes second.
   3] T finishes third 4] Q finishes fourth.

2. If in a race R finishes first, V can finish no lower than
   1] second 2] third 3] fourth 4] fifth

3. If in a race S finishes second, which of the following can be true?
   1] P finishes before R. 2] V finishes before S.
   3] P finishes before V 4] U finishes before V.

4. If in a race S finishes sixth and Q finishes fifth, which of the following can be true?
   1] V finishes first or fourth. 2] R finishes second or third.
   3] P finishes second or fifth. 4] U finishes third or fourth.

Here, each question requires a change in structure. It is evident from the question, as each question is clearly stating a new case (If this happens, then what will happen).

The steps to solve this question set would be marginally different.

Step 1: Understand and represent the common data for easy reading.
For e.g.
V always finishes somewhere ahead of P. (V>P)
P always finishes somewhere ahead of Q. (P>Q)
Therefore, V>P>Q

Either R finishes first and T finishes last, or S finishes first and U or Q finishes last.
This is an either… or construct.
Hence, if (R finishes first and T finishes last) is true, then (S finishes first and U or Q finishes last) is false.
i.e. either R = 1 and T = 7 or S = 1 and U / Q = 7
i.e. only R or S could be in the first position and T, U or Q can be in the last position.

Step 2: Read the information given in the question carefully and understand the question clearly. Let us look at question 1

1. If in a race V finishes fifth, which of the following must be true?
   1] S finishes first. 2] R finishes second.
   3] T finishes third 4] Q finishes fourth.

The words "Must be true" means have to be true in all cases.

Step 3: Create a structure to map the data in the question

1	2	3	4	5	6	7
				V		

Step 4: Now solve the question using the common data

Using the conclusion V > P > Q, we have

1	2	3	4	5	6	7
				V	P	Q

If Q is 7th, then R cannot be 1st . Hence S is 1st .

1	2	3	4	5	6	7
S				V	P	Q

Step 5: Do not try and fit all the positions, as there could be multiple situations possible. What you have now got is sufficient to solve the question.

The answer is [1]. i.e. S is in the first position…confirmed.

Let us look at question 4

4. If in a race S finishes sixth and Q finishes fifth, which of the following can be true?
   1] V finishes first or fourth. 2] R finishes second or third.
   3] P finishes second or fifth. 4] U finishes third or fourth.

"Can be true" means may be true or maybe false. i.e. it should not be necessarily true or necessarily false.

1	2	3	4	5	6	7
				Q	S	

Now, if S finishes 6th, then R has to be 1st and T has to be 7th. (since either R or S have to be 1st). So we have

1	2	3	4	5	6	7
R				Q	S	T

While this is normally sufficient to answer the question, we can also go ahead and see what are the possible cases for the other alphabets. We know that V > P. Hence there are only 3 possible cases. U in the 2nd place, or in the 3rd place, or in the 4th place, and the other two places filled by V and P.

Case 1

1	2	3	4	5	6	7
R	U	V	P	Q	S	T

Case 2

1	2	3	4	5	6	7
R	V	U	P	Q	S	T

Case 3

1	2	3	4	5	6	7
R	V	P	U	Q	S	T

Now the question asks, "which of the following can be true".

1] V finishes first or fourth.
2] R finishes second or third.
3] P finishes second or fifth.
4] U finishes third or fourth.

From among the answer options, only 4th option is correct.
i.e. U finishes third or fourth can be true, or can be false (as it can also finish 2nd).

You can now answer the other two questions. The answers are 2-3, and 3-4

Practice Exercise A (Important)

DIRECTIONS for questions 1 to 5: Refer to the case let given below.
Six people – A, B, C, D, E, and F compete in a race. About the result of the race, the following is known:
1] D finished third.
2] A did not finish last.
3] B finished either just before E or just after E.
4] There were no ties.

1. Which of the following is a possible order of the competitors at the finish, from first to last?
   1] A, D, E, B, F, C
   2] E, B, D, C, F, A
   3] B, E, D, F, A, C
   4] A, B, D, E, C, F

2. If F finishes before A but after D, which of the following must be true?
   1] C must finish either fourth or sixth.
   2] E must finish either first or second.
   3] B must finish either first or third.
   4] F must finish either second or fourth.

3. If D finishes before E, which of the following must be false?
   1] A finishes fifth.
   2] C finishes second.
   3] B finishes last.
   4] F finishes second.

4. Which of the following is a complete list of people who could finish second?
   1] B, F
   2] B, F, A
   3] B, F, A, E
   4] B, F, A, E, C

5. If B finishes first, which of the following could be true?
   1] C finishes before E and D. 2] A finishes before F.
   3] F and C finish before A. 4] E finishes before B and D.

DIRECTIONS for questions 6 to 10: Refer to the case let below.
B and S do not play guitar. M does not play sitar and guitar. Tabla is not played by S and A. A and R do not play the sitar. M and R do not play tabla or violin. Each person plays only one instrument. The 5th instrument is a flute.

6. R plays the…
   1] Tabla 2] Flute 3] Guitar 4] Sitar

7. Which instrument is played by A?
   1] Violin 2] Flute 3] Guitar 4] Tabla

8. Who plays the Flute?
   1] M 2] S 3] B 4] R

9. The Sitar is played by…
   1] B 2] A 3] S 4] M

10. The instrument played by B is…
    1] Violin 2] Flute 3] Sitar 4] Tabla

DIRECTIONS for questions 11 - 15, refer to the data below.
The roads to travel to different cities from starting point A are described below. I, J, K can be reached only from E. You cannot reach E from D or C. A leads to B and C. From C one can reach G. B leads to E. D can be reached from C. D leads to F. G also leads to F. L can be reached only from G via F. M can be reached only from D via F. All distances are the same (1 km). All the roads are one-way only.

11. If one starts from A, one can reach all cities using different routes except
    1] L 2] K 3] M 4] None

12. If one reached D, one can go to
    1] J & G 2] F & L 3] F & M only 4] F only

13. If one reaches M, he must have travelled through
    1] B 2] B & K 3] C & D 4] G & F

14. If one has travelled to K, one must have travelled
    1] 1 km. 2] 3 km. 3] 2 km. 4] 4 km.

15. To reach M, one must travel
    1] 1 km. 2] 3 km. 3] 4 km. 4] 6 km.

DIRECTIONS for questions 16 to 20, refer to the data below.
On rolling 6 dice, it is found that
1] Three of the dice show the same number. The rest show different numbers.
2] Only one die shows 6.
3] Not more than 3 dice show 4 or more.

16. What is the minimum possible total of numbers on the faces if the three dice having same number show 2?
    1] 16 2] 14 3] 12 4] 9

17. What is the maximum total if 4 of the dice show less than 4?
    1] 29 2] 32 3] 17 4] 22

18. What would be the maximum total if 3 dice are faulty and have numbers only on 5 faces?
    1] 31 2] 24 3] 34 4] Indeterminate

19. If only 1 die shows 1, what is the maximum number of dice with numbers greater than 4?
    1] 3 2] 1 3] 2 4] Indeterminate

20. What is the maximum number that can be on the face of the 3 dice which show the same number?
    1] 2 2] 4 3] 3 4] 5

Answer Key

1-3	2-2	3-1	4-4	5-2	6-3	7-1	8-1	9-3	10-4
11-4	12-3	13-3	14-2	15-3	16-1	17-4	18-4	19-3	20-3

Explanatory Answers

For questions 1 to 5:

1. [3] is the answer – [1] violates condition 1, [2] violates condition 2, and [4] violates condition 3.

2. D finishes 3rd, A does not finish last – given that F finishes after D but before A => D is 3rd, F 4th, A 5th, and C 6th; also B and E have to be 1st and 2nd – condition 3. Hence, the answer is [2].

3. If D finishes before E, it means E and B between them have to be in places 4 and 5 or 5 and 6 (condition 3), which means nobody else can be at position 5. Answer is [1].

4. Everybody except D can finish second – since there are no other conditions mentioned. Hence, the answer is [4].

5. If B is first, E has to be 2nd and D 3rd – conditions [1] and [2]. Hence, [1] obviously cannot be true, [3] cannot be true as it means A finishes last, which is against condition [2], E cannot finish before B as B is first. Hence, your answer is [2].

For questions 6 to 10:

	B	S	A	M	R
G	×	×		×	
S			×	×	×
T		×	×	×	×
V				×	×
F					

Consider the following matrix: The given conditions can eliminate various options as represented by crosses above. The remaining is easy.

M must play the flute hence tick there and cross flute for others. This implies that R has to play the Guitar.

Hence A has to play the violin. Which makes S play the Sitar and finally B gets the Tabla.

6. Answer is [3].

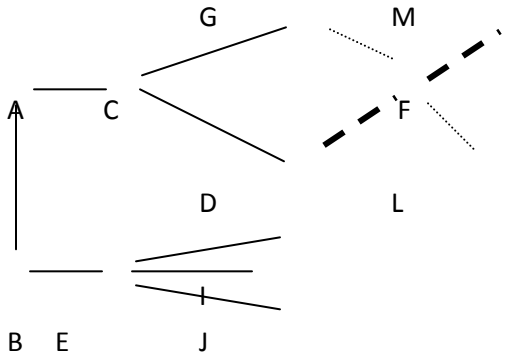
7. Answer is [1].

8. Answer is [1].

9. Answer is [3].

10. Answer is [4].

For Answers 11-15 :
The route map is as shown below :



Now, all the questions can be answered easily.

For Answers 16-20 :

16. Min. possible total = 2 + 2 + 2 + 1 + 3 + 6 = 16. Answer is (1)

17. Max. possible total = 3 + 3 + 3 + 2 + 5 + 6 = 22. Answer is (4)

18. Since the numbers missing from the 3 faulty dice are not known, the answer is (4)

19. Directly from condition (3), the answer is (1)

20. The maximum number on the 3 dice that show the same can be 3, since, any number greater than 3 would violate at least one of the conditions. Answer is (3)

Chapter 28 Synonyms & Antonyms

Suggested Time : 45 minutes

Synonyms & Antonyms
Synonyms are words that are similar in meaning, while antonyms are just the opposite. For example: 'elucidate' and 'explain' are synonyms, while 'ebb' and 'flow' are antonyms. To answer questions on synonyms & antonyms, arriving at the correct meaning of the words is important.

A. Put The Word In Context and Test Words For Positive And Negative Connotations.
In our daily speech, we combine words into phrases and sentences; rarely do we use a word by itself. This can cause words that we have little trouble understanding in sentences to suddenly appear unfamiliar when we view them in isolation. For example, take the word "prejudice". Most people don't recognize it in isolation. Yet most people recognize it from the popular book "Pride and prejudice". From this, you may know that the word is likely to have a negative connotation, and is related to pride.

Now, let us look at a few options.
(A) bias (B) disorder (C) skill (D) chivalry

Chivalry is not correct as it is a positive trait. Skill seems out of context. Disorder is not as well related to Pride as Bias is. Hence (A)

B. Watch Out For Eye-Catchers.
On medium and hard problems some answer-choices will catch your eye by reminding you of some part of the original word or some common meaning of the word. Be wary of these choices; they are eye-catchers.
Example:

BANAL:
(A) something that is banned
(B) what cannot be banned
(C) ordinary
(D) a harsh tone

BANAL may be a tough word for some. Now, two of the options seem to be around the word BAN. These are eye-catchers and are not the correct answers. Another option is suggesting that that the word is linked to NASAL, which means "of the nose", and in the Indian context, someone who sings from the nose (nasal tone). This is also an eye-catcher, and the correct answer is prosaic or ordinary.

C. Be Alert To Secondary (Often Rare) Meanings Of Words.
On problems of average difficulty, the test setters often use common words but with their uncommon meanings. An example of antonym will illustrate this.

Example:
AIR:
(A) suppress
(B) speak
(C) expose
(D) aerate

The common meaning of AIR is concerned with atmosphere, i.e. to ventilate. Options like aerate are synonyms. When you use the word AIR in other contexts, like "AIR your views", you conclude that speak or expose are synonymous. The opposite of freedom to AIR your view is to suppress a fact. Hence (a).

D. GRAMMAR

The parts of speech in a synonym/antonym problem are consistent throughout the problem. Hence, if the given word is a verb, then every answer-choice will be a verb as well. This fact often helps you determine whether a word is being used in a secondary sense because words often have different meanings depending on their use as nouns, verbs, or adjectives.

Example:

AIR:
(A) release
(B) differ
(C) expose
(D) enshroud

AIR is commonly used as a noun--indicating that which we breathe. But every answer-choice is a verb. Hence, AIR in this case must also be a verb. A secondary meaning for AIR is to discuss publicly. The opposite is to ENSHROUD, to hide, to conceal. Hence, the answer is (D).

Note: Never spend more than a few seconds on an synonym / antonym problem! If you don't know the given word, use the above techniques to eliminate as many answer-choices as possible; guess from the remaining ones; then move on.

Practice Exercise A (Optional)

DIRECTIONS for questions: For the word given at the top of each table, match the dictionary definitions on the left (A, B, C, D) with their corresponding usage on the right (E, F, G, H). Out of the four possibilities given in the boxes below the table, select the one that has all the definitions and their usages correctly matched.

1. Account

	Dictionary Definition		**Usage**
A	Consider	E	You are accountable for the consequences of your mischief
B	To be held responsible	F	You are account for every penny spent.
C	The act of calculating something	G	I gave an account of the accident to the police
D	A narrative	H	A man is accounted to be innocent until he is proven guilty.

1	
A	**H**
B	**F**
C	**E**
D	**G**

2	
A	**H**
B	**E**
C	**F**
D	**G**

3	
A	**G**
B	**F**
C	**E**
D	**H**

4	
A	**F**
B	**G**
C	**H**
D	**E**

2**. Accord**

	Dictionary Definition		**Usage**
A	Give or Grant	E	His behavior does not accord with his principle
B	Agree	F	Out of her own accord she is a spinster
C	United, uniform	G	The foreign visitor was accorded a warm welcome.
D	Without intervention	H	The voice of all the singers was in one accord.

1	
A	**H**
B	**F**
C	**E**
D	**G**

2	
A	**H**
B	**E**
C	**F**
D	**G**

3	
A	**G**
B	**E**
C	**H**
D	**F**

4	
A	**F**
B	**G**
C	**H**
D	**E**

3. **Accelerate**

	Dictionary Definition		**Usage**
A	More quickly in velocity	E	Increase in population accelerated the growth of poverty.
B	Increase or extend	F	When they reached the highway the driver accelerated the car.
C	Increase in rate	G	We want the government to accelerate the measure for social development
D	Increase in amount	H	The follies of the king accelerated the downfall of his kingdom.

1	
A	**H**
B	**F**
C	**E**
D	**G**

2	
A	**H**
B	**E**
C	**F**
D	**G**

3	
A	**G**
B	**E**
C	**H**
D	**F**

4	
A	**F**
B	**G**
C	**H**
D	**E**

4. **Run**

	Dictionary Definition		**Usage**
A	Move Swiftly	E	As soon as we appeared she ran off.
B	Flee for escape	F	He will run into debt if he goes on spending without care.
C	Become or get	G	Sealing wax will run out if you heat it.
D	Extend spread	H	She came running towards me.

1	
A	**H**
B	**F**
C	**E**
D	**G**

2	
A	**H**
B	**E**
C	**F**
D	**G**

3	
A	**G**
B	**E**
C	**H**
D	**F**

4	
A	**F**
B	**G**
C	**H**
D	**E**

5. Reverse

	Dictionary Definition		**Usage**
A	Turn the other way round	E	If I ask my brother to do something he will do just the reverse.
B	Work in a contrary direction	F	After his father's death his attitude towards his family completely changed.
C	Opposite in nature or order	G	The new government has reversed all the policies of the previous government.
D	Complete change of action	H	Let me see the reverse side of the cloth.

1	
A	**H**
B	**F**
C	**E**
D	**G**

2	
A	**H**
B	**E**
C	**F**
D	**G**

3	
A	**G**
B	**E**
C	**H**
D	**F**

4	
A	**F**
B	**G**
C	**H**
D	**E**

6. **Simple**

	Dictionary Definition		**Usage**
A	Not difficult	E	He's so simple that he will believe anything you tell him.
B	Direct	F	Even a child can solve this simple puzzle
C	Humble	G	He writes in a simple language.
D	Easily deceived	H	A simple life will attract others.

1	
A	**H**
B	**F**
C	**E**
D	**G**

2	
A	**H**
B	**E**
C	**F**
D	**G**

3	
A	**G**
B	**E**
C	**H**
D	**F**

4	
A	**F**
B	**G**
C	**H**
D	**E**

7. **Short**

	Dictionary Definition		**Usage**
A	Measuring little from end to end	E	Let's go for a short walk.
B	Brief	F	She is vertically challenged. i.e. short.
C	Not tall	G	They ran short of water in the party.
D	Deficiency	H	Short hair is cooler in summer.

1	
A	**H**
B	**F**
C	**E**
D	**G**

2	
A	**H**
B	**E**
C	**F**
D	**G**

3	
A	**G**
B	**E**
C	**H**
D	**F**

4	
A	**F**
B	**G**
C	**H**
D	**E**

8. **Presume**

	Dictionary Definition		**Usage**
A	To believe that something is true without proof	E	I wouldn't presume to advice someone as clever as you.
B	To be bold enough/knowledge to do so	F	She married again on the presumption that her first husband was dead.
C	Take unfair advantage of	G	We can't draw any conclusion based on presumption.
D	Supposing	H	They presume on her generosity too much.

1	
A	**H**
B	**F**
C	**E**
D	**G**

2	
A	**H**
B	**E**
C	**F**
D	**G**

3	
A	**G**
B	**E**
C	**H**
D	**F**

4	
A	**F**
B	**G**
C	**H**
D	**E**

9. **Damn**

	Dictionary Definition		**Usage**
A	Condemn	E	His soul is damned
B	Curse	F	Her contribution towards the project was made damned.
C	Unimportant	G	Work pressures have damned peoples social life.
D	No value	H	Critics damned the book.

1	
A	**H**
B	**F**
C	**E**
D	**G**

2	
A	**H**
B	**E**
C	**F**
D	**G**

3	
A	**G**
B	**E**
C	**H**
D	**F**

4	
A	**F**
B	**G**
C	**H**
D	**E**

10. Rate

	Dictionary Definition		**Usage**
A	Frequency in which something moves	E	Your pay will be based on your performance.
B	Fixed or charged	F	The book is only second – rate
C	Class or standard	G	I will do nothing until I hear from you at any rate.
D	Whatever happens	H	If you run your business at this rate probably you have to close them one day.

1	
A	**H**
B	**F**
C	**E**
D	**G**

2	
A	**H**
B	**E**
C	**F**
D	**G**

3	
A	**G**
B	**E**
C	**H**
D	**F**

4	
A	**F**
B	**G**
C	**H**
D	**E**

Directions for questions 11-15 : Select the word nearly opposite in meaning to the capitalised word.

11. SHIRK
    1] accept 2] protest 3] deny 4] defy

12. JETTISON
    1] sail 2] throw 3] preserve 4] ruin

13. INCOMPATIBLE
    1] efficient 2] wise 3] harmonious 4] steady

14. NONCHALANT
    1] concerned 2] casual 3] noisy 4] polished

15. PRESCIENCE
    1] early science 2] ignorance 3] knowledge 4] routine

Directions for questions 16-20 : Each question has four items. Select the item that does not belong to this group.

16. 1] Abstract 2] Conceptual 3] Material 4] Ideational

17. 1] Hive 2] Web 3] Nest 4] Hub

18. 1] Apt 2] Relevant 3] Appropriate 4] Adept

19. 1] Equestrian 2] Neigh 3] Derby 4] Bark

20. 1] Scapegoat 2] Refugee 3] Pariah 4] Outcast

Answer Key

1-2	2-3	3-4	4-2	5-3	6-4	7-2	8-3	9-2	10-2
11 -1	12 -3	13 -3	14 -1	15 -2	16 -3	17 -4	18 -4	19 -4	20 -1

Explanatory Answers

Not required for 1 to 15.

16. Abstract, Conceptual and Ideational are intangible. ---------- material is the odd one
17. Hive, Web and Nest are dwellings ---------- hub is the odd one
18. Apt, Relevant and Appropriate are related terms with respect to their meanings -------- Adept is skillful and therefore the odd one
19. Neigh, Derby and Equestrian are related to horses ---------- bark is not.
20. Refugee, Pariah and Outcast are terms related to exile ---------- Scapegoat is the odd one

Chapter 29 Analogies

Suggested Time : 45 minutes

Analogies are pairs of words which share a relationship. For instance, an Igloo is to an Eskimo, what a Kennel is to a Dog.

In analogy questions, the relationship between the words is more important than the meanings of the words themselves. The following are examples of analogical relationships between words with strategies to crack the analogies questions.

A. Before You Look at The Answer-Choices, Think of a Short Sentence That Expresses The Relationship Between The Two Words.

Example:
FISH : SCHOOL ::
How are FISH and SCHOOL related? Well, a group of fish is called a school.

Example:
JOURNALIST : TYPEWRITER ::
Paraphrase: A journalist uses a typewriter as a tool of his trade.

Example:
DOUBTS : DISPEL : :

1] fear : overcome	2] clouds : clear
3] blood : kill	4] curtain : push

Paraphrase: "You dispel a doubt". Similarly, you overcome fear.

B. If More Than One Answer-Choice Fits Your Paraphrase, Make Your Paraphrase More Specific.

Example:
SHIP : HULL : :

1] gun : trigger	2] helicopter : runway
3] car : bonnet	4] scooter : wheels

Paraphrase: "Part of the ship is hull". However, this paraphrase eliminates only choice (2). But when we define hull as the frame or body of the ship, then we can infer the closest answer to be (3).

Note: The parts of speech are consistent throughout an analogy problem. Hence, if the given pair is an adjective and a noun, then each answer-pair will be an adjective and a noun, in that order. This helps you determine the intended meaning when one (or both) of the given words has more than one part of speech.

C. Eliminate Answer-Choices That Do Not Have A Clear And Reasonably Necessary Relationship.

Educated guessing is a very useful technique. If you can eliminate one or more answer-choices, you will probably increase your score by guessing.

Example:
CORROSION : IRON ::

1] lazy : energy	2] disease : vision
3] atrophy : muscle	4] computer : mouse

In choices (1) and (3) there are clear and reasonably necessary relationships between the words of each pair: a LAZY person lacks ENERGY, and ATROPHY means "the wasting away of MUSCLE." But in choice (2) there is no necessary relationship between the words; most DISEASES have no effect on VISION. Hence, eliminate choice

(2). Corrosion is a verb, but computer is not. Eliminate (4). The correct answer is (3) since CORROSION is the wasting away of IRON, just as ATROPHY is the wasting away of MUSCLE.

Note: Be careful when eliminating answer-choices to hard analogy problems because the relationship may not be strong, or it may actually be between esoteric (rare) meanings of the words. This is often what makes a hard analogy problem hard.

CLASSIFICATION OF ANALOGIES

In the last section, we analyzed the structure of an analogy problem; in this section, we will analyze the various types of analogies. The simplest analogies are synonyms & antonyms, which we have already discussed.

A. MEMBER AND CLASS

Member and class analogies are usually easy problems. In this type of analogy, the first word may be an element of the class that the second word describes, or vice versa. This category can also be classified as "Type Of." E.g. SNAKE : INVERTEBRATE

B. DEGREE OF INTENSITY

The test setters consider these problems to be hard. However, once you get used to them, they can become routine. In a degree-of-intensity analogy, the two words express a similar concept, but one word is stronger, harsher, or more intense than the other. INTEREST : OBSESSION

C. PART TO WHOLE

In this type of analogy, the first word is part of the second word. The order can also be whole to part.
E.g. ACTORS : TROUPE ::

D. DEFINITION

This is probably the most common type of analogy problem. In a definitional analogy one of the words can be used to define the other. E.g. SANCTUARY : REFUGE

E. LACK OF

This sub-category of the definition type of analogy is important and common enough to warrant a name. In this type of analogy one word describes the absence of the other word. E.g. DISHEARTENED : HOPE.

F. MANNER

This type of analogy describes the manner, way, or style by which an action is accomplished. E.g.
RATTLE : SPEAK

G. FUNCTION

This type of analogy describes the purpose or function of something. E.g. MNEMONIC : MEMORY

H. ACTION & SIGNIFICANCE

In this type of analogy one word describes an action and the other word indicates the significance of the action. E.g. COURTSY : REVERENCE ::

Practice Exercise A (Important)

Directions for questions: Each question has a pair of CAPITALISED words followed by four pairs of words. Choose the pair of words which best expresses the relationship similar to that expressed in the capitalised pair.

1. WE : OUR
   1. It : It's
   2. Her : She
   3. Who : Whose
   4. They : Them

2. FOOD : GOURMET
   1. Book : Critic
   2. Art : Connoisseur
   3. Sports : Fan
   4. Craft : Skill

3. COMPARE : LIKEN
   1. Hyperbole : Overstatement
   2. Overwrite : Scribble
   3. Prescription : Medicine
   4. Primal : Lateral

4. BRAND : PRODUCT
   1. Dalda : Rath
   2. Aircraft : Flying Machine
   3. Ram : Boy
   4. Sports car : Automobile

5. DROUGHT : FAMINE
   1. Poverty : Plenty
   2. Drip : Fluid
   3. Camera : Film
   4. Criticism : Resentment

6. PREVARICATE : EQUIVOCATE
   1. Sleepy : Exhausted
   2. Speculate : Verify
   3. Convey : Intimate
   4. Intrude : Incite

7. DIVIDE : UNITE
   1. Split : Apart
   2. Marriage : Divorce
   3. Fission : Fusion
   4. Chasm : Gap

8. CHEQUERED : SQUARE
   1. Cone : Triangle
   2. Braided : Strand
   3. Flexible : Plastic
   4. Blackboard : Slate

9. EAGER : EXCITED
   1. Perform : Activate
   2. Excel : Perform
   3. Angry : Irate
   4. Praise : Exalt

10. JUST : ARBITRARY
    1. Order : Chaos
    2. Fair : Whimsical
    3. Hope : Surprise
    4. Proper : Improper

Answer Key

1-3 2-2 3-1 4-3 5-4 6-3 7-3 8-2 9-4 10-2

Explanatory Answers

1. We : Our ----- the relationship is that of possessive form.
2. Food : Gourmet ----------Gourmet is an expert of food and drinks.
   Similarly, a connoisseur is an expert in art. ---------- Option (2)
3. Compare : Liken ---- synonymous relationship
4. Brand : Product ---------- a brand differentiates the product from other similar products.
   Similarly, Ram (name) differentiates Ram from rest of the boys.
5. Drought : Famine ----- cause and effect relationship.
6. Prevaricate : Equivocate ----- relationship is synonymous
7. Divide : Unite ---- relationship of antonyms. Note that (2) could have been the answer if it was given as divorce : marriage.
8. Chequered : Square -------- A property of square is that it could be chequered. The word chequered is an adjective that describes square better.
   Braided : Strand ---------- strand is part of the braid.
9. Eager : Excited ----- Cause and effect relationship
10. Just : Arbitrary ---- The words are not total opposites, they just describe methods in a process.

CAREER AVENUES

Chapter 30 Grammar based questions

Suggested Time : 120 minutes

Sentence completion and sentence correction are both dependant on the grammar skills. The field of grammar is huge and complex--tomes have been written on the subject. This complexity should be no surprise since grammar deals with the process of communication. Aptitude tests grammar tests only a small part of standard written English. Grammar can be divided into two parts: Mechanics and Usage.

Mechanics concerns punctuation, capitalization, etc. It is not tested on the aptitude tests nearly as often as is usage. So don't spend too much time worrying whether the comma is in the right place or whether a particular word should be capitalized.

Usage concerns how we choose our words and how we express our thoughts: in other words, are the connections between the words in a sentence logically sound, and are they expressed in a way that conforms to standard idiom? This is the part of grammar that the aptitude tests concentrate on.

Seven major categories of usage are tested:

A. PRONOUN ERRORS
A pronoun is a word that stands for a noun, known as the antecedent of the pronoun. The key point for the use of pronouns is this: pronouns must agree with their antecedents in both number (singular or plural) and person (first, second, or third).

Example:
Sanjay has yet to receive his degree.
Here, the pronoun *his* refers to the noun *Sanjay*.
Following is a list of the most common pronouns:

PRONOUNS
I, me, we, us any, she, her, they, none, he, him, them, all, it, these, most, anyone, those, more, either, some, who, each, that, which, many a, both, what, nothing, ourselves, you, one, another, any, everything, few, mine, several, his, hers, others, this, that. Most popular rules that govern usage of pronouns are

i. A pronoun should be plural when it refers to two nouns joined by *and*.
Example:
Jaya and Kamala believe *they* passed the final exam.
The plural pronoun *they* refers to the compound subject *Jaya and Kamala*.

ii A pronoun should be singular when it refers to two nouns joined by *or* or *nor*.
Example:
Neither Jaya *nor* Kamala believes *they* passed the final. (Wrong)
Correct: Neither Jaya *nor* Kamala believes *she* passed the final.

iii. A pronoun should refer to one and only one noun or compound noun.
If a pronoun follows two nouns, it is often unclear which of the nouns the pronoun refers to.
Example:
The breakup of the Soviet Union has left *nuclear weapons* in the hands of unstable, nascent *countries*. It is imperative to world security that *they* be destroyed.

Although one is unlikely to take the sentence to mean that the countries must be destroyed, that interpretation is possible from the structure of the sentence. It is easily corrected:

The breakup of the Soviet Union has left *nuclear weapons* in the hands of unstable, nascent *countries*. It is imperative to world security that ***these*** **weapons** be destroyed.

B. SUBJECT-VERB AGREEMENT
Within a sentence there are certain requirements for the relationship between the subject and the verb.

i. The subject and verb must agree both in number and person.
Example:
We have surpassed our sales goal of one million dollars.
Here, the first person plural verb *have* agrees with its first person plural subject *we*.

ii. When the subject and verb are reversed, they still must agree in both number and person.
Example:
Attached are copies of the contract.
Here, the plural verb *are attached* agrees with its plural subject *copies*. The sentence could be rewritten as
Copies of the contract *are attached*.

C. MISPLACED MODIFIERS
i. As a general rule, a modifier should be placed as close as possible to what it modifies.
Example:
Following are some useful tips for protecting your person and property from the police.
As written, the sentence implies that the police is a threat to your person and property. To correct the sentence put the modifier *from the police* next to the word it modifies, *tips*:
Following are some useful tips from the police for protecting your person and property.

ii. When a phrase begins a sentence, make sure that it modifies the subject of the sentence.
Example:
Coming around the corner, a few moments passed before I could recognize my old home.
As worded, the sentence implies that the moments were coming around the corner. The sentence can be corrected as follows:
As I came around the corner, a few moments passed before I could recognize my old home. OR
Coming around the corner, I paused a few moments before I could recognize my old home.

D. FAULTY PARALLELISM

i. For a sentence to be parallel, similar elements must be expressed in similar form.
ii. When two adjectives modify the same noun, they should have similar forms.
Example:
The topology course was both *rigorous* and *a challenge*. Since both *rigorous* and *a challenge* are modifying *course*, they should have the same form:

The topology course was both *rigorous* and *challenging*.

iii. When a series of clauses is listed, the verbs in each clause must have the same form.
Example:
During his trip to Uzbekistan, the PM will *discuss* ways to stimulate trade, *offer* economic aid, and *trying* to forge a new coalition with moderate forces in Afghanistan.

In this example, the first two verbs, *discuss* and *offer*, are active. But the third verb in the series, *trying*, is passive. The form of the verb should be active:

During his trip to Uzbekistan, the PM will *discuss* ways to stimulate trade, *offer* economic aid, and *try* to forge a new coalition with moderate forces in Afghanistan.

iv. When the first half of a sentence has a certain structure, the second half should preserve that structure.
Example:
To acknowledge that one is an alcoholic is *taking* the first and hardest step to recovery.

The first half of the above sentence has an infinitive structure, *to acknowledge*, so the second half must have a similar structure:

To acknowledge that one is an alcoholic is *to take* the first and hardest step to recovery.

E. FAULTY VERB TENSE

A verb has four principal parts:

1. Present Tense

a. Used to express present tense.
He studies hard.
b. Used to express general truths.
During a recession, people are cautious about taking on more debt.
c. Used with *will* or *shall* to express future time.
He will take the GRE next year.

2. Past Tense

a. Used to express past tense.
He took the GRE last year.

3. Past Participle

a. Used to form the *present perfect tense*, which indicates that an action was started in the past and its effects are continuing in the present. It is formed using *have* or *has* and the past participle of the verb.
He has prepared thoroughly for the GRE.
b. Used to form the *past perfect tense*, which indicates that an action was completed before another past action. It is formed using *had* and the past participle of the verb.
He had prepared thoroughly before taking the GRE.
c. Used to form the *future perfect tense*, which indicates that an action will be completed before another future action. It is formed using *will have* or *shall have* and the past participle of the verb.
He will have prepared thoroughly before taking the GRE.

4. Present Participle (*-ing* form of the verb)

a. Used to form the *present progressive tense*, which indicates that an action is ongoing. It is formed using *is*, *am*, or *are* and the present participle of the verb.
He is preparing thoroughly for the GRE.
b. Used to form the *past progressive tense*, which indicates that an action was in progress in the past. It is formed using *was* or *were* and the present participle of the verb.
He was preparing for the GRE.
c. Used to form the *future progressive tense*, which indicates that an action will be in progress in the future. It is formed using *will be* or *shall be* and the present participle of the verb.
He will be preparing thoroughly for the GRE.

F. PASSIVE VOICE

The passive voice removes the subject from the sentence. It is formed with the verb *to be* and the past participle of the main verb.
Passive: The bill was resubmitted.
Active: The MP has resubmitted the bill.
Unless you want to de-emphasize the doer of an action, you should favor the active voice.

Example:

In the past few years and to this day, many teachers of math and science had chosen to return to the private sector.
(A) had chosen to return to the private sector.
(B) having chosen to return to the private sector.
(C) chose to return to the private sector.
(D) have chosen to return to the private sector.
(E) have chosen returning to the private sector.

Choice (A) is incorrect because it uses the past perfect *had chosen*, which describes an event that has been completed before another event. But the sentence implies that teachers have and are continuing to return to the private sector. Hence, the present perfect tense should be used.

Choice (B) is incorrect because it uses the present progressive tense *having chosen*, which describes an ongoing event. Although this is the case, it does not capture the fact that the event began in the past.

Choice (C) is incorrect because it uses the simple past *chose*, which describes a past event. But again, the sentence implies that the teachers are continuing to opt for the private sector.

Choice (D) is the correct answer because it uses the present perfect *have chosen* to describe an event that occurred in the past and is continuing into the present.

Choice (E) is incorrect because it leaves the thought in the sentence uncompleted.

G. IDIOM & USAGE
There are hundreds of idioms that test makers often test. The most popular are listed.

i. Accept/Except:
Accept means "to agree to" or "to receive." *Except* means "to object to" or "to leave out."
We will *accept* (receive) your manuscript for review.
No parking is allowed, *except* (leave out) on holidays.

ii. Adapted to/for/from
Adapted to means "naturally suited for." *Adapted for* means "created to be suited for." *Adapted from* means "changed to be suited for."
The polar bear is *adapted to* the subzero temperatures.
For any "New Order" to be successful, it must be *adapted for* the continually changing world power structure.
Lucas' latest release is *adapted from* the 1950 B-movie "Attack of the Amazons."

iii. Affect/Effect:
Effect is a noun meaning "a result."
Increased fighting will be the *effect* of the failed peace conference.
Affect is a verb meaning "to influence."
The rain *affected* their plans for a picnic.

iv. Among/Between:
Between should be used when referring to two things, and *among* should be used when referring to more than two things.
The young lady must choose *between* two suitors.
The fault is spread evenly *among* the three defendants.

v. Being that vs. Since:
Being that is nonstandard and should be replaced by *since*.
(Faulty) *Being that* darkness was fast approaching, we had to abandon the search.
(Better) *Since* darkness was fast approaching, we had to abandon the search.

vi. Conform to (not *with*):
Stewart's writing does not *conform to* standard literary conventions.

vii. Double negatives:
(Faulty) *Scarcely nothing* was learned during the seminar.
(Better) *Scarcely anything* was learned during the seminar.

viii. Fewer/Less:
Use *fewer* when referring to a number of items. Use *less* when referring to a continuous quantity.
In the past, we had *fewer* options.
The impact was *less* than what was expected.

ix. Identical with (not *to*):
This bid is *identical with* the one submitted by you.

x. In contrast to (not *of*):
In *contrast to* the conservative attitudes of her time, Mae West was quite provocative.

xi. Independent of (not *from*):
The judiciary is *independent of* the other branches of government.

xii. Not only . . . but also:
In this construction, *but* cannot be replaced with *and.*
(Faulty) Patel is *not only* the top salesman in the department *and also* the most proficient.
(Correct) Patel is *not only* the top salesman in the department *but also* the most proficient.

xiii. Regard vs. Regards:
Unless you are giving best wishes to someone, you should use *regard*.
(Faulty) In *regards* to your letter, we would be interested in distributing your product.
(Correct) In *regard* to your letter, we would be interested in distributing your product.

xiv. Speak to/with:
To *speak to* someone is to tell them something:
We *spoke to* the Inspector about the alleged embezzlement.
To *speak with* someone is to discuss something with them:
Sanjay *spoke with* his friend Dara for hours yesterday.

xv. Whether vs. If
Whether introduces a choice; *if* introduces a condition. A common mistake is to use *if* to present a choice.
(Faulty) He inquired *if* we had decided to keep the gift.
(Correct) He inquired *whether* we had decided to keep the gift.

Practice Exercise A (Important)

DIRECTIONS for questions 1 to 10: The sentences below are divided into four parts (a), (b), (c), (d). Select the part which has an error.

1. (a) A few
(b) invitees
(c) made short speeches
(d) appropriate for the occasion

2. (a) The audience
(b) was perceptively thinner
(c) after the interval
(d) on the first night

3. (a) We prefer
(b) going by car
(c) to travel
(d) by train

4. (a) Preoccupied by
(b) personal problems
(c) as he was, he paid little
(d) attention to his business

5. (a) I wish
(b) you would have
(c) seen him
(d) in that condition

6. (a) Eve teasing is
(b) not a phenomena
(c) of
(d) recent origin

7. (a) The police
(b) are now
(c) in the possession
(d) of the documents

8. (a) She was so forbidding
(b) that
(c) none of we men
(d) dared to oppose her

9. (a) The reason for
(b) the delay is because
(c) we have had difficulty
(d) in obtaining materials

10. (a) A grant of
(b) Rs.700,000 was made
(c) towards the rehabilitation
(d) of the temple at Halebidu

DIRECTIONS for questions 11 to 17: Read the following sentences and identify errors of grammar, usage and spelling.
Mark (a) if the sentence has ONE error.
Mark (b) if the sentence has TWO errors.
Mark (c) if the sentence has THREE errors.
Mark (d) if the sentence has FOUR OR MORE errors.

11. Nights after nights he used to pour over his books with patient determination of a scholar.

12. These types of errors occur less seldom in good writing.

13. I think myself that we ought to reconsider the matter again.

14. Pessimism is the tendency to see the dark side of life and pessimist is someone who has this overall outlooks.

15. Many Indian states have no dearth of talent, such as Punjab.

16. With its recent renovation this most unique of Bombay's restaurants can claim to be even more unique.

17. He resigned from the dictatorship and took up teaching with the view ultimately to become a consultant.

DIRECTIONS for questions 18 to 27: From the options given below choose the grammatically correct sentence.

18. (a) Albert Einstein was thought to have so little promise at graduation that no school or university bothered to offer him a job.
(b) Albert Einstein was thought to have had so little promise at graduation that no school or university bothered to offer him a job.
(c) At graduation Albert Einstein was thought to have little promise, so that no school or university bothered to offer him a job.
(d) At graduation, Albert Einstein was thought to have a little promise, so that no school, nor any university bothered to offer him a job.

19. (a) Many a men get praised for their reserve and so-called shyness when he is simply too proud to risk making a fool of himself.
(b) Many a men gets praised for their reserve and so-called shyness when he is simply too proud to risk making a fool of himself.
(c) Many a men get praised for their reserve and so-called shyness when he is simply too proud to risk making a fool of himself.
(d) Many a men gets praised for his reserve and so-called shyness when he is simply too proud to risk making a fool of himself.

20. (a) My brother who resigned from his job two months before is presently unemployed.
(b) My brother who resigned his job since two months is presently unemployed.
(c) My brother who resigned his job two months ago is at present unemployed.
(d) My brother who resigned his job two months is currently unemployed.

21. (a) Playing superb strokes, Tracy Austin was knocked off the court by Ms. Navratilova.
(b) Playing superb strokes, Tracy Austin was knocked of the court by Ms. Navratilova.
(c) Tracy Austin was knocked off the court by Ms. Navratilova playing super strokes.
(d) Ms. Navratilova, playing superb strokes, knocked Tracy Austin off the court.

22. (a) He was a man who liked honesty and would show magnitude, even comparison to an worthy opponent.
(b) He was a man who liked honesty and would show magnanimity, even compassion, to a worthy opponent.
(c) He was a man who liked honesty and would show magnanimousness and also compassion to a worthy opponent.
(d) He was a man who liked honesty and would show magnitude and compassion to a worthy opponent.

23. (a) Most of children in India are learning English these days.
(b) Many more of children in India are learning English these days.
(c) More children in India are learning English.
(d) Many more of children than earlier are learning English these days.

24. (a) A policeman and criminal will obviously interpret things differently.
(b) A policeman and a criminal will obviously interpret things differently.
(c) The interpretation of a criminal and policeman will obviously differ.
(d) The interpretation of criminal and a policeman will obviously differ.

25. (a) No more inviting a smile had he ever seen.
(b) Not more an inviting smile he had ever seen.
(c) No smile more inviting had he ever seen.
(d) He had never seen no more an inviting smile.

26. (a) His speech conjures up an utopia of freedom, and equality and the justice.
(b) His speech conjures up a utopia of the freedom and equality and justice.
(c) His speech conjures up an utopia of freedom, equality and justice.
(d) His speech conjures up a utopia of freedom, equality, and justice.

27. (a) I suppose every writer feels there is something they would like to say -- that's on the tip of one's tongue.
(b) I suppose every writer feels there is something they would like to say -- that's on the tip of their tongue.
(c) I suppose every writer feels there is something he should like to say -- that's on the tip of their tongue.
(d) I suppose every writer feels there is something he would like to say -- that's on the tip of his tongue.

DIRECTIONS for questions 28 to 35: Each of the following questions below has sentences with a blank. From the choices given select the appropriate word to fill in the blank.

28. She lost her temper; I wasn't surprised that she _____.
(a) lost (b) had (c) would (d) did

29. My father was in the hospital _____ 6 weeks.
(a) during (b) since (c) for (d) throughout

30. I don't enjoy tennis, and I don't much like cricket _____.
(a) either (b) neither (c) also (d) too

31. The principal got very angry; _____ out of his room.
(a) he even told me to get (b) he told me even to get
(c) he told even me to get (d) he told me to get even

32. No sooner did she agree to marry him than she _____ to have serious doubts.
(a) begin (b) began (c) begun (d) had begun

33. _____ there before, I know what to expect.
(a) Being (b) As being (c) Having been (d) In being

34. If I _____ enough money next year, I will go to Japan.
(a) will have (b) shall have (c) should have (d) have

35. It was really astonishing that she _____ to talk to you.
(a) refused (b) refuses (c) should refuse (d) has refused

Answer Key

1-d	2-b	3-c	4-a	5-b	6-b	7-c	8-c	9-b	10- c
11-b	12-a	13-b	14-a	15-a	16-b	17-c	18-b	19-d	20-c
21-d	22-b	23-c	24-b	25-c	26-d	27-d	28-d	29-c	30-a
31-a	32-b	33-c	34-d	35-a					

Explanatory Answers

1. appropriate to
2. perceptibly
3. to traveling
4. Preoccupied with
5. you had seen
6. not a phenomenon
7. in possession
8. none of us men
9 . the delay is that
10. towards the restitution ...
11. Night after night ... pore ...
12. occur seldom ...
13. ... think we ought to reconsider the matter
14. ... outlook
15. Many Indian states such as Punjab ...
16. 'unique' cannot be 'more' or 'most'
17. He resigned the directorship and took up
18. 'thought to have had' is the only correct alternative.
19. Many a man gets praised ... for his reserve. Alternative (b) is bad because it has 'their reserve' which is wrong.
20. 'Resign a job' and 'two months ago' are correct. If you have marked an alternative with 'since' do refer to the rule in your note.
21. Only the last alternative correctly tells who played superb strokes and knocked off whom.
22. 'magnitude' is a matter of size or extent -- magnanimity is the correct choice.
23. 'More of' is wrong.
24. Whereas 'a criminal and a policeman' are two different people 'a criminal and policeman' could give the impression of being a single person.
25. 'No smile more inviting' is the only correct alternative.
26. Alternative (b) with 'the freedom etc.' makes it awkward. (d) is the only correct alternative.
27. 'Every writer' is singular.

28-35. Not required

Practice Exercise B (Optional but Important)

Directions for questions 1 to 5: Mark the part that has an error in it. Mark 5, if there is no error.

1. The phenomenon of (1) suicide bombers have become (2) a grim (3) new reality in the region. (4) No error. (5)

2. The central government (1) not only funds (2) the space program (3) and also funds (4) nuclear research. No error (5)

3. An overly complex (1) IT system can make even (2) the simplest (3) tasks difficult. (4) No error (5)

4. As winter closed in, (1) hopes fade (2) that a devastating earthquake will lead to (3) a new era of peace. (4) No error. (5)

5. Online MBAs (1) is one of the (2) fastest growing segments (3) in business education. (4) No error (5)

Directions for questions 6-10: Choose from among the 5 options, the one which best explains the underlined idiom in the following statements.

6. The city was back to square one in their effort to build a new bridge.
   (1) back to the beginning
   (2) totally confused
   (3) going around in circles
   (4) fighting amongst each other
   (5) going very slow

7. There was much bad blood between the three brothers.
   (1) corruption
   (2) unpleasant feelings between people
   (3) petty fights
   (4) doubts and suspicion
   (5) bias by the parents

8. The woman is a ball of fire and is always busy doing something
   (1) an active and energetic person
   (2) a hot tempered person
   (3) constantly shouting and screaming orders
   (4) a pushy person
   (5) none of these

9. There were several no-shows at the concert last night.
   (1) poorly made shows
   (2) persons who makes a reservation for something and then neither comes nor cancels it
   (3) performers booed away
   (4) audience not allowed entry even with valid tickets
   (5) none of these

10. She is nobody`s fool. You will not have to worry about her at all when she goes to New York.
    (1) a smart person
    (2) homely person
    (3) a person who cannot take care of himself
    (4) over-smart and aggressive
    (5) educated and confident

Directions for questions 11-15: In each of the following sentences, some part of the sentence is underlined. Beneath each sentence you will find five ways of phrasing the underlined part. Select the best answer keeping in mind correctness, effectiveness of expression, grammar, choice of words and sentence construction.

11. Your friends displayed <u>a very unique</u> bravery
(1) most very unique (2) a very unique (3) very unique
(4) a unique (5) none of these

12. Neither North Korea nor south Korea <u>are willing to wage</u> a war at this stage.
(1) is willing to wage (2) are willing to wage (3) both are willing to wage
(4) willing to wage (5) none of these

13. The officer returned with the news for which I was not in the least <u>preparing</u>
(1) preparation (2) prepared (3) preparing
(4) prepare (5) none of these

14. The man took his hands <u>out his pockets</u>
(1) out him pockets (2) on his pockets (3) out of his pockets
(4) out on his pockets (5) none of these

15. How I wish it has been <u>me</u>
(1) I (2) myself (3) me (4) mine (5) none of these

Answer Key

1-2	2-4	3-5	4-1	5-2	6-1	7-2	8-1	9-2	10-1
11-4	12–1	13-2	14-3	15-1					

Explanatory Answers

1. "have" refers to "the phenomenon", which is singular. So the correct usage is "has become", Hence (2)
2. "Not only" should be linked with "but also". Hence (4)
3. There is no error. Hence (5)
4. The words "As winter closed in "is in past tense, the rest of the sentence is in present continuous tense. Hence (1)
5. "Online MBAs" is plural. Hence it should be "are one of the". Hence (2)
6. The phrase means 'back to the drawing board, or planning stage, or basics." Hence (1)
7. The phrase means 'a feeling of animosity and distrust'. Hence (2)
8. The phrase means 'a live-wire of lively person'. Hence (1)
9. The phrase means 'people who do not turn up, as promised'. Hence (2)
10. The phrase means 'not one who will be easily fooled' or not 'naïve'. Hence (1)
11. Very and unique are adjectives. Hence (4)
12. It points out to either North Korea or south Korea, i.e, Only one country at a time. Hence singular verb "is willing to wage …" should come. Hence (1)
13. The word "Returned" is past tense, hence the end of the statement should be "prepared". Hence (2)
14. Out is an adverb, Hence a preposition of is required to connect the noun. Hence (3)
15. me cannot stand for it which is nominative case. Hence (1)

CAREER AVENUES

Chapter 31 Jumbled Sentences

Suggested Time : 60 minutes

Jumbled sentences are questions that require you to construct a whole coherent paragraph out of sentences mixed up in order. These questions are tests on logic, common sense and speed. It interesting to know that this question doesn't examine any previously learnt concept, neither is it based on any particular logic. Extensive practice is the key.

However, we have a few strategies for you. Study them and use them in the exercise that follow and you'll be comfortable with them.

Paragraph jumbles questions are presented in different forms.

a) Four/five sentences either numbered 1, 2, 3, 4, 5 or lettered a, b, c, d, e are given in a jumbled form. The sentences are followed by choices of four possible combinations of arrangements of these sentences. You have to choose the most logical order of arrangement of these sentences to form a meaningful, coherent paragraph.

E.g.

A. The managers are themselves parts of various hierarchies of managerial influence, coupled to one another by explicit and implicit codes of behaviour and information network.
B. Management science and MIS have introduced new elements of complexity into the manager's world.
C. Management deals with complex social systems joined by physical and informational mechanisms to other social systems that are the responsibility of other managers.
D. If this seems to make the social system appear complex, it is so intended.

1. DCAB 2. ADCB 3. DBAC 4. CADB

b) A set of six sentences is given, in which the first and the sixth sentences are fixed at their respective positions. The four sentences that falls between the first and the sixth sentences are presented in a jumbled form. You are asked to choose the logical order of sentences from the choices.

E.g.

1. If the phenomenon of down gradation is understood and accepted, brands have to continuously seek options that offer better value to consumers.

A. Therefore, given an option, the consumer was willing to be satisfied by greater value from trusted brands rather than downgrade.
B. Tea in polypacks doubled their volume share between 1989 and 1991 in Bombay.
C. Biscuits in family packs witnessed phenomenal growth of such brands even during periods of inflation.
D. The onset of the down gradation phenomenon seems to be a good time to launch new products, or better still, create new sub-categories.

6. For instance, the more expensive gel toothpaste category upped its share in Bombay by six percent at the expense of popular and economy brands.

1. BCAD 2. CDAB 3. ADBC 4. DCBA

Strategies to solve jumbled sentences

The strategies for tackling jumbled sentences questions are given below.

1. Check out the options given. If they all start (or end) with a different sentence, choose the option that is right. Thus, you need not spend valuable time in reading all sentences too closely.
Example:

A. But the fiery youth can work wonders only if guided by wisdom and sagacity of elders.
B. This is a responsibility for the youth.
C. The great task of national reconstruction demands the sweat and blood of its citizens.
D. Because they have zeal and vigour to take up challenges.

1. ACDB 2.CBDA 3. BDAC 4.DABC

Of the choices, the one starting with 'C' is obviously right.

2. If choices are too closely related, identify logical pairs of sentences and pick out the best option.
Example:

A. Freedom is man's birthright.
B. Freedom implies that we should not encroach upon the freedom of others.
C. But, often, social life is so interdependent that one's freedom could mean another's restrictions.
D. But un-chartered freedom endangers social life.

1. BACD 2. ADBC 3. DBAC 4. DCBA

In this 'A' & 'D' are a sequence as are 'B' & 'C'. So, in the choices given the option is 2.

3. Sentences beginning with pronouns, conjunctions etc. will give an idea of the sequence.
Example:

A. He was bone-weary and soul-weary and found himself muttering, "Either I can't manage this place, or it's unmanageable."
B. To his horror, he realized that he had become the victim of a an amorphous, unwitting, unconscious conspiracy to immerse him in routine work that had no significance.
C. It was one of those nights in the office when the office clock was moving towards four in the morning and Bennis was still not through with the incredible mass of paper stacked before him.
D. He reached for his calendar and ran his eyes down each hour, half-hour and quarter-hour to see where his time had gone that day, the day before, the month before.

1. ABCD 2. CADB 3. BDCA 4. DCBA

The first sentence should be C. Hence the answer is 3.

4. Try working backward also, in the six sentence variety, if there is no obvious clue in the first sentence.
Example:

1. Picture a termite colony, occupying a tall mud hump on an African plain.
A. Hungry predators often invade the colony and unsettle the balance
B. The colony flourishes only if the proportion of soldiers to workers remains roughly the same, so that the queen and workers can be protected by the soldiers and the queen and soldiers can be serviced by the workers.
C. But its fortunes are presently restored, because the immobile queen, walled in well below ground level, lays eggs not only in large enough numbers but also in varying proportions required.
D. The hump is alive with worker termites and soldier termites going about their distinct kinds of business.
6. How can we account for her mysterious ability to respond like this to events on the distant surface?

1. BADC 2. DBAC 3.ADBC 4.BDCA

In this the last sentence has to be 'C', because the questions asked in '6'is in response to something the queen does. An explicit mention of the queen is made only in 'C'. The first sentence has to be D, since it is the logical continuation about the hump. Hence the choice is '2'.

Practice Exercise A (Optional)

Directions for questions 1-5: Arrange the sentences A, B, C and D to form a logical sequence between sentence 1 and 6.

1. 1. Why are horses the same?
A. It may be old and lame, and in time it will die.
B. A particular horse 'flows' naturally.
C. But there is something all horses have in common.
D. You probably don't think they are at all.
6. But the 'form' of the horse is eternal and immutable.
1. DCAB 2. CABD 3. CBDA 4. DCBA

2. 1. Buddhism is a way to salvation.
A. But Buddhism is more severely analytical.
B. In the Christian tradition there is also a concern for the fate of human society conceived as a whole, rather than merely as a sum or network of individuals.
C. Salvation is a property, or achievement, of individuals.
D. Not only does it dissolve society into individuals; the individual in turn is dissolved into component parts and instants, a stream of events.
6. In modern terminology, Buddhist doctrine is reductionist.
1. BCAD 2. ADBC 3. CBAD 4. CDAB

3. 1. Matrilineal systems of land inheritance advantaged women in many respects, especially in granting them economic and social security.
A. Women, in particular, were profoundly affected by these changes.
B. The large joint family estates came to be partitioned; there was an increasing penetration of market forces and patriarchal ideologies spread in influence.
C. These systems, however, did not remain fixed over time.
D. Interventions by the colonial and post-colonial states, and the processes of social change which these set in motion, eroded customary practices.
6. At the same time, their customary exclusion from major authority in public bodies mean that they were unlikely to be the ones directing the changes.
1. BDCA 2. CDBA 3. CDAB 4. CADB

4. 1. It is a nostalgia trip.
A. A 130-year-old steam locomotive, out of work for almost a century, is getting back on the rails.
B. The railway ministry conceived the stunt, the tourism ministry saw the bucks.
C. So the Fairy Queen is out of New Delhi's railway museum into a workshop.
D. The plan is to hitch the Queen with two modern coaches and run it on a round-trip between Delhi and Agra.
6. The trip will only be an occasional novelty, the old world charm costing 20,000/- per person.
1. DCBA 2. ACBD 3. ABCD 4. BADC

5. 1. Whenever a major company announces a work force reduction, you can read it in the papers the next day.
A. Do they head to the post office and start mailing resumes, hoping to land a job just like their old one?
B. But, curiously, there's little follow-up ; what happens to those thousands of ex-employees?
C. Lay-off stories have become a sobering staple of business journalism over the past decade.
D. Do they ever make as much money as they used to?
6. Or are they consigned to forever chase behind their former salaries?
1. DABC 2. ABCD 3. CBAD 4. BCAD

Directions for questions 6-15 : Choose the most logical order of sentences from among the four given choices to construct a coherent paragraph.

6. A. However, the real challenge today is unlearning, which is much harder.
   B. But the new world of business behaves differently from the world in which we grew up.
   C. Learning is important for both people and organisations.
   D. Each of us has a "mental model" that we've used over the years to make sense.
   1. DBCA 2. CADB 3. DACB 4. CBDA

7. A. A large number of intellectuals believe that the North is using its military and economic powers to force unequal contracts on the South.
   B. The make-believe ethical issue of the sanctity of law camouflages the unethicality of the entire transaction, which is a travesty of the ethical concept of the greatest good for the greatest number.
   C. Once these contracts are made, the North uses the facade of legality and ethics to pin-down the South.
   D. Thus it suffers from the flaw that the law one of the useful means to implement ethics has fouled the ethicality of the ends.
   1. DACB 2. CBDA 3. ACBD 4. BDAC

8. A. The fact that he could find absolutely nothing to substantiate their wild claims made no difference.
   B. We always gave the poor man a cup of tea, and he grew quite fond of some of the animals.
   C. The neighbours, now thoroughly indignant, kept bombarding the local health authorities.
   D. On an average, twice a week, the poor inspector was forced to come up to the house.
   1. DBAC 2. CDAB 3. ADBC 4. CADB

9. A. How times change.
   B. Lakme once bestrode the Indian cosmetic scene.
   C. Today, international cosmetic giants are giving Indian women wider choice.
   D. Time was when you found the world's best cosmetics only in the suitcases of flights to India.
   1. BCAD 2. CBAD 3. DABC 4. DBCA

10. A. 'Surrender at Dhaka' aims to fill this gap.
    B. Despite the importance of the event, there is no serious book about the conflict.
    C. It also majorly altered the geographic equations in South Asia.
    D. The 1971 war changed the political geography of the sub-continent.
    1. BDCA 2. DCBA 3. ADBC 4. CBAD

11. A. There was a twist - the notes were all as tattered as notes could get.
    B. What came out was a large garland made of currency notes.
    C. When the RBI governor came to inaugurate the new printing press, the local politicians gave him a gift-wrapped box.
    D. The unsuspecting governor opened the box in full view of the gathering.
    1. CDBA 2. BDCA 3. ABDC 4. BCAD

12. A. Most computer users develop disorders because they ignore warnings like a sore shoulder.
    B. Thus begins a search for relief : ice, yoga, herbs.
    C. They keep panting and dragging until scar tissue forms.
    D. But cures are elusive.
    1. DBCA 2. DABC 3. BDCA 4. ACBD

13. A. It was a hot afternoon and the accountant was fanning himself breathlessly.
    B. 'All right' he said 'give me two of them'.
    C. A salesman became very enthusiastic about the capabilities of a new micro-computer.
    D. This led him to claim to a public accountant that the machine would do half his work for him.
    1. BDCA 2. BCAD 3. CDAB 4. CBAD

14. A. This is a fair question to ask people selling financial advice.
    B. "If you're so smart, why aren't you rich"?
    C. The question would not stymie Jay Leno, who has achieved success both as an anchor and as a business person.
    D. It is a fair question to put to the economists too.
    1. BADC 2. DCAB 3. BDAC 4. CBAD

15. A. What if anything awaits beyond the grave?
    B. To hope for life in the hereafter is a part of human nature.
    C. Easter at its essence is about a belief in the victory of life over death.
    D. It is one faith's response to the mystery that has haunted mankind since our first ancestor gazed into the abyss of death.
    1. BDAC 2. CDAB 3. BCAD 4. CABD

Answer Key

1-4	2-3	3-2	4-3	5-3	6-2	7-3	8-2	9-3	10-2
11-1	12-4	13-3	14-1	15-2					

Explanatory Answers

1. Option (4) – Statement 1 has to be followed by D (It answers the question asked in 1)
   Statements DC are in sequence
   Statement A has to precede statement 6

2. Option (3) – Statement C elaborates on the 1st sentence, therefore C follows 1 (i.e. 1C)
   Statements AD are in sequence

3. Option (2) – Statement C logically follows 1(Option 1 is ruled out)
   Sentence D follows C as it tells us what happened over time. (i.e. CD)
   Sentence B elaborates D (i.e. 1CDB)

4. Option (3) – A has to follow 1 and D immediately follows C.

5. Option (3) – Sentence C logically follows 1
   Statements BA are in sequence and AD are in sequence

6. Option (2) – It has to start with C and A only fits in after C. DB explains A

7. Option (3) – The para cannot begin with C or D, therefore option 1 & 2 are ruled out
   Statements AC are in sequence & BD are in sequence, therefore option (3)

8. Option (2) – The para cannot begin with A, hence option 3 is ruled out. A has to come somewhere after C.

9. Option (3) – Sentence D is the introductory sentence. Sentence A cannot be the last sentence and sentence BC are in sequence

10. Option (2) – The para cannot start with B, A or C.

11. Option (1) – The para cannot start with A or B. Sentence C is the introductory sentence.

12. Option (4) – The para cannot start with B or D

13. Option (3) – Statement CD are in sequence. A has to come somewhere after D.

14. Option (1) – The para cannot start with D or C. Statements AD are in sequence (D cannot precede A)

15. Option (4) – Sentence ABD are in sequence

Chapter 32 Critical Reasoning

Suggested Time : 120 minutes

A Critical Reasoning question is presented as a short paragraph, followed by one or two questions. The short paragraph is an argument. An argument consists of three parts:

i) the premise – the evidence
ii) the conclusion – the main point of the argument and
iii) an assumption – the unstated or missing premises without which the argument would be untenable.

In certain cases the conclusion may not be given and the question may require that you supply the conclusion.

The premises are pieces of evidence that the author uses to arrive at the conclusion. The conclusion is valid depending on the strength of the premises or evidences.

Let's look at an example to understand these terms better.

Revenue from this magazine is at an all time low. Newsstand sales are dropping and there are no new subscribers. Therefore sweeping editorial changes are necessary if the magazine is to survive.

Conclusion: Sweeping editorial changes are necessary for the magazine to survive.

Premises: Revenue down
Newsstand sales low
No new subscribers

Assumption: Only old editorial policy was responsible.

Structural clues that help you locate a conclusion in an argument.
Certain key words can help you isolate the conclusion and the evidence in a stimulus; clues that signal evidence includes - since, because, due to etc.
Clues that signal conclusion include – so, therefore, thus, consequently, hence, as a result etc.

Basic Approach
Always look at the question first.

The basic question types are:

- Find conclusion
- Supply conclusion
- Identify assumption
- Strengthen argument
- Weaken argument
- Find an inference
- Mimic argument

Find the conclusion
In this question type the conclusion is already given. You have to choose from the choices the statement that is already given in the passage as the conclusion of the argument. Look out for key words like hence, so, thus, therefore etc.

At the same time conclusion may also begin without these key words.

Illustration
When young students first look at modernist abstract painting, their eyes are assailed by a seemingly meaningless mass of squiggles. It is only after a study of the history of art and the forces which led up to abstraction that it is

possible to appreciate the intellectual sophistication of modern art. Thus, a high-school study of modern art should always begin with a study of the history of art.

Which of the following is the main point of the passage above?
A. To understand the history of art, it is necessary to study modern art.
B. Young students are unable to appreciate fully the complexities of modern art.
C. An understanding of the history of art is essential to an understanding of modern art.
D. To understand abstract art, students must first study the history of art.

Analysis:
Premise #1 : Students don't understand modern art.
Premise #2 : To understand modern art it is necessary to study art history
Conclusion: Students of modern art must first study art history.

Answers: A. Reverses P # 2 B. P # 1C. P # 2 D. correct answer

Supply the conclusion
In this question type the argument consists of merely a set of evidences. You have to supply the conclusion. This implies that the answer is not a sentence given in the passage but must include and rest on all the evidences.

Illustration
Fewer elected officials are supporting environmental legislation this year than at any time in the last decade. In a study of thirty elected officials, only five were actively campaigning for new environmental legislation. This comes at a time when the public's concern for the environment is growing by leaps and bounds.

Which of the following conclusions is best supported by the passage above?

A. More elected officials are needed to support environmental legislation.
B. Elected officials have lost touch with the concerns of the public.
C. The five elected officials who actively campaigned for new environmental legislation should be congratulated.
D. If the environment is to be saved, elected officials must support environmental legislation.

Note: 1. Answer not a sentence from passage.
2. The answer must include and rest on all premises.

Analysis
Premise #1 : Fewer elected officials are supporting environmental legislation.
Premise #2 : Public concern for the environment is growing.

Answers:
A. Ignores premise #2
B. Ignores environment
C. Ignores public concern
D. Correct answer.

Find the assumption
To answer this question, you have to identify the conclusion that is already given in the paragraph. An assumption is a statement that is not given as the premise but is required if the conclusion has to be valid. Without the assumption we cannot arrive at the conclusion. As the assumption is an unstated premise, the answer is not a statement given in the paragraph. Choose the choice that would connect to the conclusion as choice, therefore conclusion.

i.e. *Assumption = choice, therefore conclusion*

Illustration
Many people believe that gold and platinum are the most valuable commodities. To the true entrepreneur, however gold and platinum are less valuable than opportunities that can enable him to further enrich himself. Therefore, in the world of high-finance, information is the most valuable commodity.

The author of the passage above makes which of the following assumptions?

A. Gold and Platinum are not the most valuable commodities.
B. Entrepreneurs are not like most people.
C. The value of information is incalculably high.
D. Information about business opportunities is accurate and will lead to increased wealth.

Approach:
i. There could be several assumptions.
ii. You are required to find the assumption on which the conclusion depends.

Analysis
Conclusion: In the world of high finance information is the most valuable commodity.

Answers
A. Irrelevant B. Irrelevant C. Same as conclusion D. Correct answer.

In strengthen - weaken the argument, the type of premises given should be examined. The premises could be presented in the form of

statistical data – numbers, percentages, ratios etc.
causal data - that is the data may be in the form of cause- effect relationship, or
analogy – arriving at a conclusion through likening it to a similar situation.

Strengthen argument
Here the conclusion is already given in the argument. One of the choices if true further supports or reinforces the conclusion. This means that the conclusion is further strengthened if a choice is true.
The question is phrased in one of the following ways:

Which of the following, if true, strengthens the argument?
Which of the following, if true, least weakens the argument?
Which of the following, if true, adds credence to the argument?
Which of the following, if true, reinforces the argument?

Illustration
It has recently been proposed that we adopt an all volunteer army. This policy was tried on a limited basis several years ago, and was a miserable failure. The level of education of the volunteers was unacceptably low, while levels of drug use and crime soared among army personnel. Can we trust our national defense to a volunteer army? The answer is clearly, "No".

Which of the following statements, if true, most strengthens the author's claim that an all volunteer army should not be implemented?
A. The general level of education has risen since the first time an all volunteer army was tried.
B. The proposal was made by an organization called Citizens for Peace.
C. The first attempt to create a volunteer army was carried out according to the same plan now under proposal and under the same conditions as those that exist today.
D. A volunteer army would be less expensive than an army that relies on the draft.

Approach:
Look out for 3 types of logic.

1. Statistical 2. Analogy 3. Causal

Analysis: In the above argument the logic is analogy. “the idea did not work then, it won’t work now”.

Answers:
A. Weakens B. Irrelevant C. Correct answer D. Irrelevant

Weaken the argument

In this question type the conclusion is already in the argument but with a questionable assumption. One of the choices if true, weakens the conclusion.

The question is phrased in one of the following ways.
Which of the following, if true, weakens the conclusion?
Which of the following, if true, least strengthen the argument?
Which of the following, if true, jeopardizes the argument?
Which of the following, if true, questions the validity of the argument?
Which of the following, if true, damages the argument?

Illustration
The recent turnaround of the LEX corporation is a splendid example of how an astute chief executive officer can re-channel a company’s assets toward profitability. With the new CEO at the helm, LEX has gone, in only three business quarters, from a 10 million dollar operating loss to a 22 million dollar operating gain.

A major flaw in the reasoning of the passage above is that
A. The passage assumes that the new CEO was the only factor that affected the corporation’s recent success.
B. The recent success of the corporation may be only temporary.
C. The chief executive officer may be drawing a salary and bonus that will set a damaging precedent for this and other corporations.
D. the author does not define ‘profitability’.

Approach : Look out for the type of logic
1. Statistical 2. Analogy 3. Causal

Analysis
The logic is causal.
Cause : CEO, Effect : Turnaround

Answers
A. Correct answer
B. Irrelevant
C. Irrelevant
D. Irrelevant

Find the inference
In this question type the conclusion is already given. If the statements in the argument are true, one of the choices also must be true. If there are two premises it is possible for us to make two inferences.
The questions appear as follows:
If the statements in the argument are true, which of the following must also be true?
Which of the following can be inferred from the passage?
Which of the following is implied in the passage?

Illustration
In film and videotape, it is possible to induce viewers to project their feelings onto characters on the screen. In one study when a camera shot of a woman’s face was preceded by a shot of a baby in a crib, the audience thought the woman’s face was registering happiness. When the same shot of the woman’s face was preceded by a shot of a lion running toward the camera, the audience thought that the woman’s face was registering fear. Television news teams must be careful to avoid such manipulation of their viewers.

Which of the following can be inferred from the passage?

A. Television news teams have abused their position of trust in the past.
B. The expression on the woman's face was, in actuality, blank.
C. A camera shot of a baby in a crib provoked feelings of happiness in the audience.
D. Audiences should strive to be less gullible.

Approach:
1. Inferences often have very little to do with conclusion.
2. For each premise you can have an inference.

Answers:
A. Outside scope B. Outside scope C. Correct answer D. Outside scope
Another possible inference is
A camera shot of a lion provoked feeling of fear in the audience.

Mimic the argument: Here one of the choices is similar / parallel to the given argument in logic and structure.

Illustration
World class marathon runners do not run more than six miles a day when they are training. Therefore, if you run more than six miles a day, you are not world-class.

Which of the following statements supports its conclusion in the same manner as the argument above?

A. Sprinters always run in the morning. If it is morning, and you see someone running, it will not be a sprinter.
B. Paint never dries in less than three hours. If it dries in less than three hours it is not paint.
C. Little league games are more fun for the parents than for the children who actually play. Therefore, the parents should be made to play.
D. If a car starts in the morning, chances are it will start again that evening. Our car always starts in the morning, and it always starts in the evening as well.

Approach:
Simplify terms. The sentence is of the format "If A → not B, therefore if B → not A"
Analysis
A. If A, B. If B not A
B. If A not B. If B not A
C. If A, B not C.
D. If A, B. A therefore B

Practice Exercise A (Optional)

Directions for questions: *Read each of the short passages given below and answer the questions that follow.*

1. At present people in Bangalore spend around two thousand rupees a year on entertainment, eight hundred rupees more than in 1990. This increase is because people between the age group 16 and 40 have accounted for an increase in population. It can be inferred from the passage that

1. More than forty percent of Bangalore population is between the ages 16 and 40.
2. The population has increased since 1990.
3. Before 1990 children below the age 16 were more in number than those above 16.
4. People between the ages 16 and 40 spent more on entertainment than do other age groups.

2. It is believed that people who live in the coastal regions of India live longer due to the inclusion of fish in their daily diet. But recent studies show that people in certain other regions who do not consume fish at all also live longer. Which of the following can be inferred from the passage?

   1. People in non-coastal areas may be taking a substitute for fish.
   2. It is difficult to compare dietary habits of people.
   3. Intake of fish may not be attributed to longevity.
   4. Objectivity is almost impossible in comparative studies.

3. Infosys goes beyond resumes and conducts group discussions in its search for personnel for top jobs. The group discussion technique extends the validity of the selection process. It can be inferred from the passage that

   1. Group discussion has emerged as a key technique in selection processes.
   2. Resumes are not sufficient enough to judge a person's worth.
   3. Comparison between interviews is possible only in group discussions.
   4. Qualitative variations are not evident in resumes.

4. Mindless criticism in literary circles is not a new phenomenon. Objective analysis is a far dream unless a literary critic is an author himself. It can be inferred from the passage that

   1. Harsh criticism is a phenomenon only in literary fields.
   2. Objective analysis of literary works is possible only when critics are not prejudiced.
   3. A critic who is an author himself can give more objective criticism.
   4. Attitude toward literary works vary from critic to critic.

5. Students say they are tired of essay tests. They believe that the grading is entirely based on the teacher's perception and opinion. It can be inferred from the passage that

   1. Essay tests are considered as a primitive form of evaluation.
   2. Essay tests are criticized for the subjectivity involved.
   3. These tests are characterized by objectivity.
   4. It is difficult to assess student achievement through these tests.

6. Twenty years ago university students opted mainly for sciences and arts. Now the trend is engineering and management courses, as these are lucrative professions. Students will again opt for sciences and arts if these disciplines are as remunerative as the others. The argument above is based on

   1. an attempt to refute generalisation.
   2. an appeal to set values and beliefs.
   3. an analogy between past and the future.
   4. a hypothesis that is least verifiable.

7. If John is elected as the union president of the college, he must be a final year student. It can be inferred from the passage that

   1. only John can be elected as union president.
   2. only final year students can be elected as union president.
   3. some final year students must be selected to the union.
   4. John did not refuse to serve the union.

8. Analysing quantified data qualitatively has begun to generate major breakthroughs in social science research, which brings us to Prof. Rao's fascinating doctoral thesis on joint family system. Which of the following can be properly inferred from the passage?

1. Quantified data are the most important source in social science research.
2. Prof. Rao is the first researcher to carry out qualitative analysis.
3. Social science researchers are contributing immensely to quantitative analyses.
4. Prof. Rao's thesis offers new insights into its subject, based on qualitative analysis of quantified data.

9. Statistics reveal that during the past two years, an increasing number of youth between seventeen and twenty five years of age have been injured or killed in Bangalore road accidents. Ninety percent of these men were riding 100 cc motor bikes. It can be inferred from the passage that people would most likely favour legislation that prohibited

1. young men belonging to the given age group riding 100 cc bikes.
2. young men under the age of seventeen riding two wheelers.
3. young men riding two wheelers on highways.
4. young men riding powerful bikes at peak hours.

10. In the near future, blood banks will start screening donors for HIV virus. The screening test disqualifies nearly 8 percent of donors. It can be inferred from the statements that
1. The incidence of HIV positive cases will go up to eight percent.
2. The demand for blood from banks will fluctuate.
3. Donations are likely to become less frequent.
4. The blood supply in blood banks is likely to go down.

Answer Key

1 - 4 2 - 3 3 - 2 4 - 3 5 - 2 6 - 3 7 - 2 8 - 4 9 - 1 10 - 4

Chapter 33 Fill-in-the-blanks

Suggested Time : 90 minutes

The sentence completions form the most straightforward part of any aptitude test, and most students do well on them. A sentence completion question consists of an incomplete sentence. The sentence may have one or two blanks. From the five answer choices given, you are required to choose an appropriate word or words that can serve as the missing word/ words that will best complete the meaning of the given sentence.

DIRECTIONS: Each sentence below has one or two blanks; each blank indicates that some thing has been omitted. Beneath the sentence are five lettered words or set of words. Choose the word or set of words for each blank that best fits the meaning of the sentence as a whole.

Q. A thinking man resists the vulgar prosperity that _____ ever to _____ , by preserving and communicating heroic sentiments.
A. disintegrates ..... banality
B. transforms ..... neutrality
C. decays ..... luxury
D. defers ..... poignancy
E. retrogrades ..... barbarism

The theory underlying a sentence completion stimulus is that the meaning and the tone of a sentence can be gathered without the missing word/words. If so, it is possible for you to supply word(s) to complete the sentence.

Invariably in every sentence completion question there will be a clue or two that could be used as leads toward identifying the correct choice word(s).

These clues could be structural (grammatical) or semantic (meaning) clues. The word(s) that you choose should agree and merge with the sentence structurally and functionally or semantically.

Every sentence completion question can be analyzed for the negative or positive nature of words to be filled in.

WHAT IS MEASURED

I. Skill of vocabulary

Sentence completion question is a test of your word power and diction. It is only if you know the meaning of word(s) that you will be able to choose confidently, the appropriate word to fill the incomplete sentence. Moreover the question sentence also may have difficult words in them, knowing the meaning of which is essential for understanding the sentence in totality and to choose the appropriate word(s).

Look at the following question:

Q. The department can use information received from dissatisfied employees and _____ to launch an investigation against the company.
A. pettifoggers
B. buccaneers
C. ambassadors
D. whistleblowers

The knowledge of the meaning of the words in the choices is essential for you to select the answer to this question with confidence. Let's analyze the meaning of these words.

A pettifogger is one who practices chicanery or unlawful business. Can this word fit the sentence? No.

A buccaneer is a pirate; again not a word that would go with dissatisfied employees to supply information about a company.
A whistleblower is one who informs about corruption or wrongdoing. Yes, this fits the sentence.
An ambassador is not one to do anything with a company. In fact this is the first choice you could eliminate.

II. Skill of semantics and syntax

A sentence has a semantic dimension (meaning) and a syntactical dimension (word arrangement). Knowledge of these two is a mandatory condition for you to be able to choose the appropriate word. You must understand the meaning of a sentence in order to identify the appropriate choice word that fills the blank and completes the sentence coherently.

You should have a fair understanding of the functions of structural words in a sentence such as unless, rather, yet, while, unfortunately, in contrast, despite.

Look at the following question:

Q. The meaning of words is destroyed if you change them, where as the meaning of a thought _____ whatever word you think proper to use.
A. alters
B. transforms
C. remains
D. settles
E. belies

When you read the sentence you must recognize the function of whereas in the sentence. It signals a shift in the meaning of the sentence.

One thing is destroyed whereas the other is not; the answer thus is remains.

The semantic feature of a sentence refers to its meaning, and how meaning of the given words in a sentence help us to identify the word to complete the sentence.

Study the following sentence:

Q. A cult brand _____ the imagination of a small group who spread the word, make converts, help turn a fringe product into a _____ name.
A. ceases ..... redoubtable
B. elevates ..... dynamic
C. fires ..... dream
D. seizes ..... mainstream
E. pools ..... envious

The first clue is the word imagination; hence pools, ceases and seizes can easily be eliminated. You have to work with fires and elevates. Fire one's imagination is a standard idiomatic usage. Moreover the meaning of the sentence implies turning a fringe (peripheral) product into (opposite of it) a dream product.

III. Skill of identifying clues

A sentence completion question is not merely a test of your vocabulary. It also tests your ability to analyze the given sentence and to identify the structural and semantic clues that decide on the meaning and nature of the word(s) to be chosen.

You will find one or two clues in a sentence completion question. Common key words such as and or but are all the more valuable for us in sentence completion. These are structural clues. (These are dealt with in detail later)

SENTENCE COMPLETION STRATEGIES

In order to answer sentence completion questions correctly, you should first read the entire sentence as you see it, without trying to fill in the word or words. Desist from working on each blank separately. The second part of the sentence may have a bearing on the first part of the sentence. And both words must fit into the meaning of the sentence; therefore read your choice into the sentence supplying and evaluating both words.

Each sentence contains a few crucial clues that determine the answers, and finding these clues will guide you to the correct answer. Look for what is directly stated. In sentence completion questions, you may also find poetic and literary language, which tests your understanding of figurative usage. The correct answer is the one that is directly implied in the sentence.

- Determine where the sentence is heading, using structural clues.
- Anticipate answers in your words.
- Look for answers that are similar to yours.
- Fix words in the sentence to see if they gel with the whole sentence.

Sentence completion strategy in a nutshell.

Strategy 1
Read the entire sentence.
In both one and two blank sentences read the entire sentence before you inspect the choices. An overall understanding of the sentence is essential.

Strategy 2
Identify structural clues to determine the nature of the word(s) required.
You can determine the positive or negative tone of the expected word using structural clues. And identify the words whose synonym, antonym, parallel word or cause/ effect that you have to supply.

Strategy 3
Substitute your own words.
Place your own words in the blanks assuming that answer choices are not given, keeping in mind the clues you have identified.

Strategy 4
Identify words similar to the ones you substituted.
If the word you have substituted goes with the requirement of the sentence, you are sure to find a word(s) similar in meaning and tone to the ones you have substituted.

Strategy 5
Read the sentence back with word(s) filled in.
Now read the entire sentence with the words filled in to see if the word(s) fit literally and figuratively.
To sum up with examples, some strategies for answering these questions are:
A. Before You Look At The Answer-Choices, Think Of A Word That "Fits" The Sentence.
Example:
……..at having done poorly on the CAT, Laxmi began to question her abilities. Her self-confidence was .....

(A) Elated, appeased
(B) Crestfallen, destroyed
(C) Levitated, elevated
(D Surprised, sustained

If somebody does poorly, she is likely to be crestfallen (despairing) and has begun to question herself, then her self-confidence would be destroyed. Hence, the answer is (B).

B. Be Alert To Transitional Words.
Transitional words tell you what is coming up. They indicate that the author is now going to draw a contrast with something stated previously, or support something stated previously.

i. *Contrast Indicators*
To contrast two things is to point out how they differ. In this type of sentence completion problem, we look for a word that has the opposite meaning (an antonym) of some key word or phrase in the sentence. Following are some of the most common contrast indicators:

But, Yet, Despite, Although, However, Nevertheless

Example:
Although the warring parties had …… a number of disputes, past experience made them ..... to express optimism that the talks would be a success.
(A) reduced, rash
(B) included, ambivalent
(C) discussed, overjoyed
(D) settled, reticent

"Although" sets up a contrast between what has occurred--success on some issues—"settled" and what can be expected to occur--success for the whole talks. Hence, the parties are reluctant to express optimism. The common word "reluctant" is not offered as an answer-choice, but a synonym—"reticent"--is. The answer is (D).

ii. *Support Indicators*
Supporting words support or further explain what has already been said. These words often introduce synonyms for words elsewhere in the sentence. Following are some common supporting words:

And, Also, Furthermore, Likewise, In Addition, For

Example:
Govinda is an ……. and ..... speaker, equally caustic toward friend or foe--a true curmudgeon.
(A) earthy, lofty
(B) opprobrious, vituperative
(C) verbose, unstinting
(D) loud, retiring

"And" in the sentence indicates that the missing adjectives are similar in meaning, "opprobrious," which is very negative and vituperative--the only negative word—means "abusive." Hence, the answer is (B).

iii. *Cause And Effect Indicators*
These words indicate that one thing causes another to occur. Some of the most common cause and effect indicators are

Because, For, Thus, Hence, Therefore, If , Then .

Example:
Because the House has the votes to …….. a presidential veto, the President has no choice but to .....
(A) approve, object
(B) accept, abdicate
(C) return, capitulate
(D) override, compromise

Since the House has the votes to pass the bill or motion, the President would be wise to compromise and make the best of the situation. The answer is (D).

Practice Exercise A (Important)

DIRECTIONS: Each of the questions below has sentences with one or more blanks. From the choices given select the appropriate word to fill in the blanks.

1. If you want to slim, you should avoid eating _____ foods such as bread and potatoes.
(a) fatty (b) starchy (c) greasy (d) sugary

2. Morphia is sometimes used _____ severe pain.
(a) to stifle (b) to suffocate (c) to smother (d) to deaden

3. I have it on very good _____ that Mr. Jones will be retiring in September.
(a) reference (b) authority (c) information (d) inference

4. If you want this mango tree to bear good fruit next year, you will have _____ it.
(a) to pare (b) to nip (c) to whittle (d) to prune

5. He lost his temper and _____ his teeth in rage.
(a) ground (b) curled (c) rasped (d) rattled

6. I wish I could get out of the habit of _____ when I feel embarrassed.
(a) blooming (b) firing (c) flaming (d) blushing

7. After searching for half an hour, the children found their ball in the _____ by the lilac bushes.
(a) underground (b) underworld (c) undergrowth (d) underneath

8. I don't _____ to know much about art, but I know what I like.
(a) believe (b) pretend (c) imagine (d) represent

9. My boss must be the most _____ man in the world. He is so obstinate.
(a) pigheaded (b) featherheaded (c) lighthead (d) hardheaded

10. When the ship docked at Bombay, they found a _____ in the hold.
(a) stowaway (b) trespasser (c) interloper (d) squatter

11. If you don't want your bicycle to be stolen, you should put _____ on it.
(a) a handcuff (b) a fetter (c) a latch (d) a padlock

12. If you paint the walls cream, I think it will _____ the colour of the curtains better.
(a) take out (b) throw out (c) lift out (d) bring out

13. Before you decided to buy that house, you ought to have it _____.
(a) overlooked (b) observed (c) scanned (d) surveyed

14. The cat jumped upon the table and began _____ the milk from the jug/
(a) to lap (b) to suck (c) to spoon (d) to strain

Answer Key

1-a 2-c 3-c 4-c 5-b 6-d 7-d 8-b 9-a 10-d
11-c 12-a 13-a 14-b

Practice Exercise B (Optional)

Directions for questions: In each of the following sentences, part/parts of the sentence is/are left blank. Beneath each sentence four different ways of completing the sentence are indicated. Choose the best alternative from among the four.

1. In pursuance of their decision to resist what they saw as anti-labour policies, the company employees' union launched agitation to_____
   1. show their virility
   2. reaffirm their commitment to the company
   3. bring down the government
   4. demonstrate their strength

2. The safest general characterization of the European philosophical tradition as it has developed up to now, with all its diverse proponents, is that is consists of a_____ Plato.
   1. series of footnotes to
   2. set of prologues to
   3. collection of chapters on
   4. string of commentaries to

3. The interest generated by the soccer World Cup is _____compared to the way cricket_____ the nation.
   1. milder, fascinates
   2. lukewarm, electrifies.
   3. tepid, inspires
   4. unusual, grips

4. No doubt, it was our own government but it was being run on borrowed ideas, using _____ solutions.
   1. worn out
   2. second hand
   3. impractical
   4. appropriate

5. The telephone symbolises that awkward _____ in all communication technologies; while it _____ to bring us together, it keeps us apart.
   1. paradox, needs
   2. irony, intends
   3. paradox, tries
   4. irony, wishes

6. A positive emotion like love is said to be not only a great stress - buster _____ .
   1. it also drains you emotionally.
   2. but also a contributor in improving our immune system.
   3. but also decreases our life span.
   4. it leads to cardio-vascular dysfunctioning.

7. Running and workouts may not affect non - asthmatics but _____ .
   1. may aggravate the condition among those suffering from the disease.
   2. may alleviate the condition among asthmatic patients.
   3. may neutralise the condition among asthmatics.
   4. may have no impact on people who suffer from the disease.

8. Dominating husbands are egoistic and consider themselves _____ .
   1. less intelligent than their spouses.
   2. on par with and equal to their spouses.
   3. overwhelmed by their emotions.
   4. superior to and more intelligent than their spouses.

9. World energy consumption will increase by more than one third by 2015, even if _____.
   1. oil prices decrease.
   2. oil prices rise and people conserve more energy.
   3. people do not conserve energy.
   4. oil price is checked.

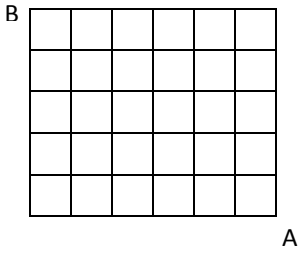
10. Bank employees, who were opposed to automation till recently, appear _____ .
    1. to have finally rejected the proposal.
    2. no say in the matter whatsoever.
    3. to have reconciled to the needs of computerization.
    4. to have turned down the idea for computerization.

Answer Key

1-4 2-1 3-2 4-2 5-3 6-2 7-1 8-4 9-2 10-3

Practice Questions

1. A is lost in a city which has roads in the form of a 7 × 6 grid as shown in the figure. He has to reach B who is at the diagonally opposite end of the city. All routes from one node to another are equidistant. In how many ways can he travel to meet B given that he takes the shortest path?



1] 42 2] 462 3] 56 4] 1716

2. A box contains 20 balls, of which 4 are white, 7 are black and 9 are red. At least how many balls must one pick up to get 9 balls of the same colour?
1] 17 2] 18 3] 19 4] 20

3. In a number system 12, 20, 24 are in an arithmetic progression. What is the base of the number system?
1] 6 2] 7 3] 4 4] 5

4. A company is setting up its internal telephone network, and providing extension numbers to each telephone. All numbers lie between 001 and 999. If a department has to have 8 consecutive extension numbers, all starting with 9, how many are the possible options?
1] 92 2] 93 3] 100 4] None of these

5. Question: What is the average weight of the class?
Statements:
A. The ratio of the average weight of all the boys and girls is 3:2
B. The number of boys and girls in the class are 40 and 60 respectively

1] Statement A alone is sufficient to answer the question.
2] Statement B alone is sufficient to answer the question.
3] Statement A and B are together required to answer the question.
4] Both statements A and B together are also not sufficient to answer the question.

6. Question: If x, y, z are positive integers, then is 5x + 6y + 7z divisible by 5?
Statements:
A. 5x + 3z is divisible by 5
B. y + 2z is divisible by 10.

1] Statement A alone is sufficient to answer the question.
2] Statement B alone is sufficient to answer the question.
3] Statement A and B are together required to answer the question.
4] Both statements A and B together are also not sufficient to answer the question.

DIRECTIONS for questions 7 and 8: In the questions below there are four statements which express the same idea. Choose the alternative that is most clear and concise.

7. 1] A marketer would do well to remember that a brand's equity lies with the consumer.
2] A marketer would do well to remember that the equity of a brand is lying with the consumer.
3] That a brand's equity lies with the consumer is something a marketer would do well to remember.
4] A marketer would do well in case he remembered that a brand's equity lies with the consumer.

8. 1] Microsoft has come a long way after 1980, when it was started by Software engineer Bill Gates.
2] In 1980, Microsoft was started by Software engineer Bill Gates and has come a long way since then.
3] Started in 1980 by Software engineer Bill Gates, Microsoft has come a long way.
4] Software engineer Bill Gates in 1980 started Microsoft which has come a long way.

Directions for questions 9 -10 : In each of the following sentences, part/parts of the sentence is/are left blank. Beneath each sentence four different ways of completing the sentence are indicated. Choose the word / phrase that best completes the sentence.

9. The life-history of some of the past Indian freedom leaders amply shows how a mix of resilience and _____ helped them _____ their obstacles before achieving the pinnacle of glory.
1] fortitude …. win over
2] will-power …. procrastinate
3] fortitude …surmount
4] determination …. rectify

10. The novel comes as a _____ to all women dumped by_____ males who will feel provoked at the title of the book, or at the sight of the picture on the cover.
1] surprise ….eclectic
2] comfort ….narcissistic
3] solace ….non-conformist
4] prelude ….chauvinistic

11. Question: Is the price of a pen an odd number?
Statements:
A. The price of three pens, four pencils and six sharpeners is an even number.
B. The price of a sharpener is same as the price of a pencil.

1] Statement A alone is sufficient to answer the question.
2] Statement B alone is sufficient to answer the question.
3] Statement A and B are together required to answer the question.
4] Both statements A and B together are also not sufficient to answer the question.

12. Question: If $a > x$ what is the value of x?
Statements:
A. $a^x = x^a$
B. $a \neq x$

1] Statement A alone is sufficient to answer the question.
2] Statement B alone is sufficient to answer the question.
3] Statement A and B are together required to answer the question.
4] Both statements A and B together are also not sufficient to answer the question.

Directions for questions 13 to 15 : The questions are independent of one another.

13. There needs to be a significant change in the evaluation policy of many post graduation colleges when recruiting from graduation colleges. Most post-grad colleges look at academic performance, and as a result many deserving candidates are not even evaluated by these colleges. Aptitude tests, on the other hand, are better indicators of future academic performance, and deserve a look in.

Which of the following is an assumption of the passage?

1] Aptitude tests have proven themselves as cost-effective tools to weed out poor performers.
2] Academic standards in graduation colleges is not upto mark.
3] Colleges currently are not using correct parameters for evaluation.
4] Aptitude tests are not yet popular amongst post-grad colleges as a tool for prospective student assessment.

14. You are a guest in a friend's house and he has two children, a boy, Jackie, and a girl, Sujata. You know that all statements of one of them is always true and of the other are always false. You ask both of them the way to the neighborhood mall. You get the following replies.

Jackie : I do not know the way to the mall. Sujata does not know the way to the mall.
Sujata : I do not know the way to the mall and Jackie knows the way to the mall.

Based on the above, I can infer that
1] Jackie speaks the truth
2] Sujata speaks the truth
3] Indeterminate
4] Data Inconsistent

15. Modern warfare does not revolve around the usage of standard weapons and armaments only. Recently China held India hostage, when an artificial lake was formed in a region in Tibet under the control of the Chinese, and the overflow of the lake or a leakage in the dam could overflow the Sutlej and hence flood the entire Sutlej basin in India. China refused permission to Indian specialists to inspect the lake, and held the country ransom to a "water bomb". Again, the US is arm-twisting India to support US on major issues like Iraq and terrorism, in return for supporting India's demand for a UNSC seat. These are symptoms of war, albeit of a different kind.

Which of the following would qualify as war, according to the definition of war described above?

1] The refusal of the Taliban government in Afghanistan to permit Indian forces to free hostages by use of force, of an Indian plane hijacked in India and taken to Afghanistan.
2] The refusal of a groom to marry just before the marriage ritual if the bride's family did not provide him a Maruti 800 car in dowry.
3] The US slapping high duties on Chinese exports to USA, in retaliation to rising trade deficit of USA with China, and the Chinese refusal to devalue its currency to set into order the imbalance in trade.
4] A strike called by farmers in the rest of Maharashtra to protest against the decision of the government to cut power in the rest of the state to provide uninterrupted power to the capital Mumbai, when the power was actually generated outside Mumbai.

16. The weather during Ghosh Babu's vacation was strange. It rained on 15 different days, but it never rained for a whole day. Rainy mornings were followed by clear afternoons. Rainy afternoons were preceded by clear mornings. There were 12 clear mornings and 13 clear afternoons in all. How long was the vacation?
1] 15 days 2] 20 days 3] 25 days 4] 40 days

17. Organising a party requires a lot of effort. In a buffet salad party, the host had 4 different vegetables for salads. How many different salads can a guest make from the four vegetables - lettuce, cucumber, carrot, and mushrooms, if even a single vegetable is a salad?
1] 12 2] 16 3] 15 4] none of these

18. Two teams participating in a competition had to take a test in a given time. Team B chose a test with 300 questions, and team A the test with 10% less questions. Team A completed the test 3 hours before schedule

while team B completed it 6 hours before schedule. If team B answered 7 questions more than team A per hour, how many questions did team A answer per hour?
1] 15 2] 18 3] 21 4] 24

19. Mr.Ghosh wants to fence his square shaped field with each side measuring 120 ft. If the posts have to put at a distance of 12 ft each, how many posts are required?
1] 38 2] 39 3] 40 4] 36

20. Two successive discounts of x per cent and y per cent are given. When is the net discount highest given x + y = k (k = constant)?
1] k > x > y 2] x < y < k
3] x = y 4] x = k, y = 0 or y = k, x = 0

21. Surekha works for a book marketing company and gets a stipend of Rs.3800. She also gets commission on books sold at the rate of Rs.8 per copy for first 25 copies, Rs.10 per copy for the next 25 copies, Rs.12 per copy for the next 50, and for every copy sold thereafter, she gets Rs.15 per copy. If in a month she makes Rs.4430, how many books did she sell?
1] 58 2] 60 3] 65 4] None of these

22. For a real number x, [x] denotes the integral part of x. That is [x] is the largest integer less than or equal to x. For example [1.33] = 1 and [–1.33] = –2. What is the value of $\left[\frac{1}{2}\right]+\left[\frac{1}{2}+\frac{1}{100}\right]+....\left[\frac{1}{2}+\frac{99}{100}\right]$?
1] 49 2] 50 3] 48 4] 51

23. In a charity show an NGO invites donors to raise money for earthquake relief. The NGO manages to get 40% of the targeted funds from 60% of the donors. 10% of the donors are unable to make any payment. If the NGO still manages to raise the targeted amount, then how much more do the remaining donors pay over the average paid by all donors who attended the charity show?
1] 50% 2] 100% 3] 200% 4] 150%

24. In a test of 12 questions, a candidate gets 1 mark for every right answer, 0 for unanswered questions, and a negative 0.25 for every wrong answer. What are the total number of ways in which a particular candidate may score a positive integral value of net score?
1] 13 2] 17 3] 21 4] 23

25. In a wholesale shop, 3 tonnes of rice and 2 tonnes of wheat cost Rs. 800, one tonne of wheat and two and a half tonnes of barley cost Rs. 850 and 3 tonnes of maize and half tonne of barley cost Rs. 3150. If I plan to buy 5 tonnes each of the 4 grains, what is my total outflow?
1] 8000 2] 9000 3] 9160 4] Indeterminate

26. Three students go on a picnic. A does not carry water and snacks. B does not carry the camera or the umbrella. C does not carry the football and the cap. If each of the six items was definitely carried by atleast one of the members, and each member carried 2 articles only, than who carried the snacks ?
1] A 2] B 3] C 4] Indeterminate

27. In a football match, a total of 30 goals were scored. Team A has 4 forwards and team B has 3 forwards. In the match, all goals were scored by forwards, and each forward scored atleast one goal. If no player from the two teams scored the same number of goals, who won the match?
1] A 2] B 3] Draw 4] Indeterminate

28. 4 books were distributed among 3 friends, such that each received atleast 1 book. X received Ayn Rand's Fountainhead. Y did not receive any Ayn Rand book. Z did not receive Stephen Covey's 7 habits or Shiv Khera's Born to win. If the fourth book was Ayn Rand's Night of January 16th, who received the 7 habits?
1] X 2] Y 3] Z 4] Indeterminate

29. There are 100 statements written on the board. Statement 1 is "only 1 statement among the 100 is false". Statement 2 is " only 2 statements among the 100 are false". Hence each nth statement reads "Only n statements among the 100 are false". How many statements on the board are true if the last statement is "Only 100 statements among the 100 are false"?
1] 0 2] 1 3] 50 4] Indeterminate

30. From a list of four movies, four friends discuss their favourite movies. Atleast 2 friends vote for Matrix, not more than 3 vote for Mummy, 1 votes for Alien and 2 vote for Jerry McGuire. If two friends have exactly voted for 2 different movies, and 2 friends for exactly 3 different movies, then how many votes did Matrix get?
1] 2 2] 3 3] 4 4] Can't say

31. A particular shop sells goods at cost. A person buys goods worth Rs.60 from the shop and gives the shopkeeper a Rs.100 note. As the shopkeeper does not have change to give his customer, he exchanges this Rs.100 note for ten Rs.10 notes from the bank to settle accounts with the customer. The next day, the banker returns the Rs.100 note claiming it to be a counterfeit and takes two Rs.50 notes from the shopkeeper. If the note was actually a counterfeit, what is the total loss to the shopkeeper?
1] Rs.140 2] Rs.100 3] Rs.160 4] Rs.200

32. In a strike called by a union, a part of the 1500 employees of the company reported for work while the others were on strike. The next day, 4% of the striking workers reported for work and 6% of the working employees joined the strike. If number of striking workers on both days were the same, how many workers were striking?
1] 500 2] 600 3] 900 4]Indeterminate

Instructions for next 2 questions: Each question below contains six statements followed by four sets of combinations of three. Choose the set in which statements are logically related. i.e. where one of them is logically derived from the other two.

33. A. All executives are managers.
B. All managers are not executives.
C. All managers are doers.

D. All executives are paid well.
E. All managers are paid well.
F. Executives are paid well.
1] ADE 2] BED 3] ABE 4] ACF

34. A. Some machines are energy efficient.
B. Energy efficient machines are expensive.
C. The NX-C10 is a machine.
D. NX-C10 is energy efficient.
E. NX-C10 is expensive.
F. All machines are expensive.
1] CDB 2] ABF 3] CDF 4] FCE

35. When Sona and Tina went for shopping initially Sona had twice the money than Tina. They together bought things amounting to Rs.250. Out of which Tina's share was 60%. At the end Sona was left with thrice the amount that Tina had. What was the amount with Tina at the beginning?
1] Rs.325 2] Rs.350 3] Rs.375 4] Rs.400

36. What is the number of digits in $48^4 \times 5^{12}$?
1] 13 2] 12 3] 14 4] 16

37. Using the digits 0, 1, 2, 3, 4, 5 and 6, how many 4 digit numbers can be formed such that no digit is repeated and the four digit number is divisible by 5?
1] 220 2] 240 3] 120 4] 100

38. When a truck travels at 60 mph, it uses 30% more diesel to travel any distance than it does when it travels at 50 mph. The truck can travel 20 miles on a gallon of diesel if it is travelling at 50 mph. The truck has only 10 gallons of diesel and is 160 miles from its destination. It takes 20 minutes for the truck to refuel. How long will it take the truck to reach its final destination, if the truck is driven at 60 mph?
1] 160 minutes 2] 180 minutes 3] 192 minutes 4] 190 minutes

39. If a and b are the roots of the equation $x^2 - 10x + 16 = 0$, the value of $(1 - a)(1 - b)$ is
1] –7 2] 7 3] 16 4] –16

40. From a bag containing 100 balls, one ball weighs 9 grams and all the other weigh 10 grams each. Using a simple balance where balls can be kept on either pan, what is the minimum weighs required to identify the defective ball?
1] 3 2] 4 3] 5 4] 7

41. There are 10 ants A_1 to A_{10} who live in 10 houses H_1 to H_{10} respectively. A_1 and A_{10} exchange homes, then A_2 and A_9 exchange homes. A_3 and A_8, and then A_4 and A_7 also exchange homes. A_5 then swaps his home with each of the other ants, starting first with the resident of H_1, then swapping his new home with resident of H_2, then his new home with resident of H_3 and so on, until he finally rests in H_{10}. Who is now in H_9?
1] A_1 2] A_9 3] A_{10} 4] None of these

42. There are 4 switches S_1, S_2, S_3, and S_4 in one room connected to four bulbs B_1, B_2, B_3 and B_4 in another room. Ram has forgotten which switch is for which bulb. He can at any time switch ON exactly 2 switches and walk to the other room to check which bulbs have are glowing. Every time he switches ON 2 switches, he scores 2 points. What is the minimum number of points he scores before he is correctly able to identify the switches and their corresponding bulbs?
1] 2 2] 4 3] 6 4] 8

43. Blue ties go only with yellow or brown jackets. Red ties go only with brown or blue jackets. White shoes go only with blue ties or brown jackets. Which of the following is a valid dressing combination?
1] White shoes, Brown jackets and Blue ties.
2] Brown jacket, Red tie and White shoes.
3] White shoes, Blue jacket and Red ties.
4] All of the above.

44. Let A = x and B = y and C = z
Step 1: Swap contents of A and B
Step 2: Swap contents of B and C
Step 3: Swap contents of A and C
Perform the above 3 steps 3 times. Now what is the content of A, B and C respectively?
1] x, y, z 2] x, z, y 3] z, x, y 4] y, z, x

45. 36 identical chairs must be arranged in rows with the same number of chairs in each row. Each row must contain at least 3 chairs and there must be at least 3 rows. A row is parallel to the front of the room. How many different arrangements are possible ?
1] 2 2] 4 3] 5 4] 6

46. If x, y, z are chosen from the 3 numbers $-3, \frac{1}{2}$, 2 without repetition. What is the largest possible value of $\frac{x^3}{y}\ z^2$?
1] 214 2] 144 3] 432 4] 168

47. The pressure of a given volume of gas varies directly as the temperature. When the temperature is constant, the product of the pressure and volume is constant. When the volume is 200 and pressure is 250, temperature is 100. What will the temperature be when pressure is 300 and volume is 400 ?
1] 120 2] 240 3] 360 4] 480

48. The incomes of two people are in the ratio 4 : 5 and expenditures in the ratio 7 : 8 respectively. The total savings between the two of them is Rs.3000 and that is split in the ratio 1 : 2. Find their total income.
1] Rs.15,000 2] Rs.18,000 3] Rs.22,500 4] Rs.27,000

49. 8 gallons of milk are taken out of a cask filled with milk and replaced by water. Later 6 gallons of the mixture are taken out of the cask and replaced with water, with the result that the cask now contains equal proportions of milk and water. What is capacity of the cask ?
1] 28 gallons 2] 16 gallons 3] 24 gallons 4] 36 gallons

50. The cost of production in a factory has a fixed component and an amount that varies directly with the number of units produced. The units are sold at 20% profit. When 250 units are produced, the price is Rs.120, and the price drops to Rs.96 when the production is doubled. What will the price be when 1000 units are produced?

1] Rs.70 2] Rs.80 3] Rs.86.8 4] Rs.84



Explanatory Answers

1. A needs to cover 11 segments. Each time he has a choice of either going north or west. He can only have exactly 5 North's. So in his route when he chooses North determines his route. In the 11 segments he can choose 5 North's in $^{11}c_5$ ways.
i.e. 462.
Alternately, he has to choose exactly 6 West's. This can be done in $^{11}c_6$ ways = $^{11}c_5$ = 462.

2. $r = 9 = k + 1 \Rightarrow r = 8$
Here n = 1 as there is only 1 colour with 9 balls.
$\therefore$ 4 + 7 + (kn + 1)
= 4 + 7 + [8(1) + 1]
= 20 balls.
Hence [4]
Alternately
9 balls of the same colour is only possible for red balls. In the worst case, you may draw 19 balls and still not get all red. So you have to draw 20 balls to ensure that you have all red colour balls.

3. Let the base be x
$(20)_x - (12)_x = (24)_x - (20)_x$
$2x + 0 - [x + 2] = 2x + 4 - [2x + 0]$
Solving, x=6

4. The numbers can start from any number between 900 and 992 (both included); so there are 93 choices. Hence, [2].

5. Both the statements together & either statement alone is not sufficient to give a unique solution

6. A) is not sufficient
B) 5x + 6y + 7z = 5x + 5y + 5z + (y + 2z)
y + 2z is divisible by 5, because y + 2z is divisible by 10.
Hence 5x + 6y + 7z is divisible by 5.

7. Option 1 is correct.

8. Option 3 is correct.

9. Option 3 is correct.

10. Option 2 is correct.

11. Statement A alone is sufficient.

12. Combination of the two statement will also not give the answer
A) $a^x = x^a$ Four cases arise.
1) a = x 2) a = 2; x = 4 3) a = 4 x = 2 4) a = -2 and x = -4
But in two cases a > x
B) $a \neq x$. does not help.

13. Best answer is 4. Cannot be 2. Nowhere it is hinted that quality of education provided is bad. Only what has been hinted is that evaluation process is not upto the mark.

14. An important question. Please remember that when two statements are linked by an and, we have to consider that as one statement. Now False of A and B leads to 3 cases

1] only A False 2] only B false, and c] both false.
So Jackie speaks the truth. i.e. Both do not know the way to the mall. And Sujata has to speak false. So from the two parts of her statement, even if one is false, that is sufficient. The second part of her statement is false.

15. The kind of 'war' being spoken here is blackmail, arm-twisting etc. between nations. That is the underlying current in option 3, as when US failed to convince China to devalue its currency, it starts seeking other options, in trying to force China to do so.

16. 5 totally clear days and 15 half and half. In the 15 half and half days: 7 <= 12 of the clear mornings were followed by rain, 8 <= 13 of the clear afternoons were preceded by rain. 20 days of the vacation in all for Ghosh Babu.

17. You can choose or reject each vegetable.
$\therefore$ Two choices for each
-> Total number of choices = 2^4.
This includes one case where no vegetable is selected.
$\therefore$ Answer = $2^4 - 1 = 15$.
Hence, [3].

18. Number of questions in the test taken by team A = 270.
Let team A answer x number of questions for y hours.
Then xy = 270.
Team B answers x + 7 questions for y – 3 hours.
Hence, (x + 7)(y – 3) = 300.
$\therefore$ x = 18 and y = 15 or team A answers 18 questions per hour.

19. If we take two opposite sides and start from one corner, then 11 posts each can be put. On the remaining two sides, as the corners already have posts, only 9 posts are required.
Total = $11 \times 2 + 9 \times 2 = 40$.

20. The discount is highest when the product $\frac{100 - x}{100} \times \frac{100 - y}{100}$ is lowest.
Above product is highest when x = y and lowest when x or y is at the limits.
=> x = k; y = 0 or y = k, x = 0.

21. $3800 + (8 \times 25) + (10 \times 25) + (12 \times 15) = 4430$.
Hence, books sold = 25 + 25 + 15 = 65 books.

22. For $n < 50 \left[\frac{1}{2} + \frac{n}{100}\right] = 0$ and for $n \geq 50$ and $n \leq 99$, $\left[\frac{1}{2} + \frac{n}{100}\right] = 1$.
$\therefore$ The value of the given sum = 50.
Hence, [2].

23. 40% funds come from 60 + 10 = 70% donors. $\therefore$ the rest 30% pay 60% or double of the average. i.e. 100 % more than the average. Hence [2]

24. Using no incorrect answers, there are 12 ways, with scores of 1 to 12. Using 4 incorrect answers, there are 7 ways, i.e. 2 to 8 correct. Note that 1 correct is not possible as the net score would be 0. Similarly, with 8 incorrect, 2 ways are possible i.e. 3 and 4 correct. Hence total number of ways = 12 + 7 + 2 = 21 ways. Hence [3]

25. $6t + 4w = 1600$; $2w + 5b = 1700$; $6m + b = 6300$.
Adding, we get $6t + 6w + 6b + 6m = 9600$.
$\therefore 5t + 5w + 5b + 5m = 8000$.
Hence [1]

26. Option [4]

27. As information given is that each forward scored ATLEAST 1 goal, we cannot infer who won the match. Hence [4]

28. X and Z got the two Ayn Rand's books, but who received the other two cannot be inferred. Hence [4]

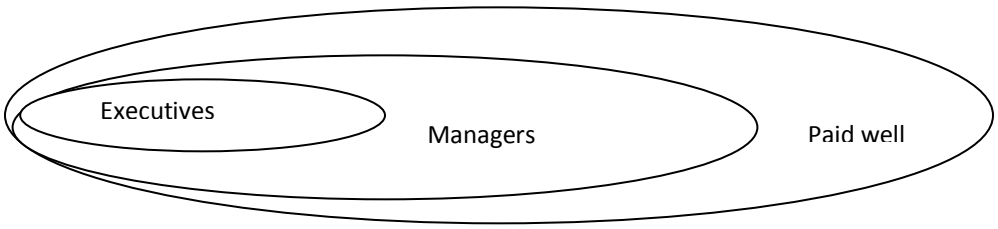
29. Only statement 99 is true, as all statements except 99 are false. Hence [2]

30. There are a total of 10 votes, of which exactly 3 go to Alien and Jerry McGuire together. The remaining 7 are distributed between Matrix and Mummy. Maximum votes that Matrix can get is 4 and maximum votes that Mummy can get are 3. Hence [3]

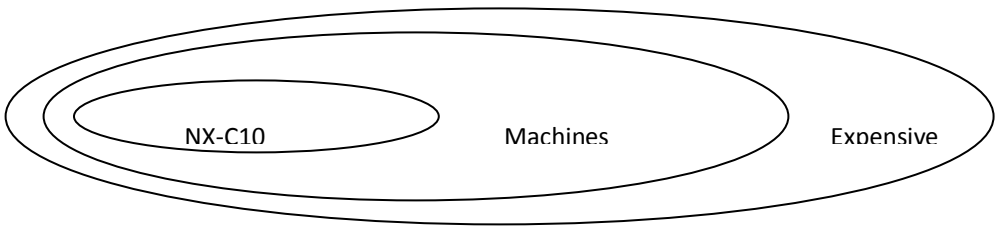
31. The shopkeeper has lost Rs.100 only. The entire transaction can also be viewed as the shopkeeper getting a counterfeit Rs.100 note against which he has given goods worth Rs.60 and change worth Rs.40.

32. Let striking workers : working employees = x : y.
0.04x move to work and 0.06y join the strike. As there will be no difference in value of x and y after this exchange.
$0.4x = 0.06y$ or $x : y = 6 : 4$ or $3 : 2$.
$\therefore$ Number of striking workers = $\frac{3}{5} \times 1500 = 900$.

33. Option (1)



34. Option (4)



35. Initial money with Sona = s and Tina = t.
Purchases made by them 250%.
Tina's purchases $\frac{60}{100} \times 250 = 150\%$.
Sona's purchases = 100%.
Money with them at the end (s – 100) and (t = 150).
But s = 2t .... (i) and (s – 100) = 3(t – 150) .... (ii).
$\therefore$ 2t – 100 = 3t – 450 $\therefore$ 350 = 3t – 2t $\therefore$ 350 = t

36. $48^4 \times 5^{12} = 3^4 \times 2^{16} \times 5^{12}$
$= 81 \times 2^4 \times 10^{12}$
$= 81 \times 16 \times 10^{12}$
$= 1296 \times 10^{12}$ i.e. it has 16 digits. Hence, [4].

37. Units place can be filled by either 0 or 5.
When 0 is in the units place, each of the other places can be filled by $6 \times 5 \times 4 = 120$ ways.
When 5 is in the units place, 0 cannot be in the thousandth place. Hence, each of the three places can be filled by $5 \times 5 \times 4$ ways = 100 ways.
Total ways = 220.
(Note: be careful and alert, there are 7 digits and not 6)

38. Driving time = $\frac{160}{60} = 2\frac{2}{3}$ hours = 160 minutes.
You need to decide whether or not the truck must stop for petrol. At a speed of 60 mph, the truck will use 30% more fuel, so it will need 1.3 gallons to travel 20 miles. Thus, x (amount of oil needed to travel 160 miles) must satisfy $\frac{160}{20} = \frac{x}{1.3}$ or x = 10.4 gallons.
So, if the truck is driven at 60 mph, it will have to stop for petrol, since it has only 10 gallons. $\therefore$ The total time needed = 160 + 20 = 180 minutes.

39. The equation is $x^2 - 10x + 16 = 0$
$\therefore$ a + b = – (–10) = 10 ; ab = 16 $\Rightarrow$ b = $\frac{16}{a}$

$\therefore$ (1 – a) (1 – b) = 1 – (a + b) + ab = 1 – 10 + 16 = 7
Hence the answer is [2]
Alternately
As a and b are roots of $x^2 - 10x + 16 = 0$, $(x - a)(x - b) = x^2 - 10x + 16$
For x = 1, $(1 - a)(1 - b) = 1^2 - 10 + 16 = 7$

40. 3^5=243. Upto 243 balls can be checked within 5 weighings. Hence [3]

41. The ant in H_{10} will go to H_9. That is A_1. Hence [1]

42. Let the four switches be A, B, C, D. Let the two switches that are switched on first be A and B. In the second round, if he switches on say B and C, then he can identify which light stands for B, A, C and D. Hence he scores 4 points. Hence [2]

43. Option [2]. Option 1 is not correct, as there is a 'or' case.

44. At the end of one round of operations, it is found that only contents of B and X are exchanged. Similarly at the end of 2 round, contents of B and C will be exchanged and again at the end of the third round, B has z and C has y. Hence [2]

45. Let c = number of chairs in a row, r = number of rows.
cr = 36.
We need to know how many ways we can write 36 as a product of 2 integers each greater than or equal to 3.
36 = 1× 36, 2 × 18, 3 × 12, 4 × 9, 6 × 6, 9 × 4, 12 × 3, 18 × 2, and 36 × 1.
Of these, 5 i.e. (3 × 12, 4 × 9, 6 × 6, 9 × 4, 12 × 3) satisfy the requirements.

46. Clearly to maximise $\frac{x^3z^2}{y}$, x should be 2, z should be –3, and y should be $\frac{1}{2}$. Max $\frac{x^3z^2}{y}$ = 144.

47. $\frac{P_1V_1}{T_1} = \frac{P_2V_2}{T_2}$

$\therefore \frac{200\times250}{100} = \frac{300\times400}{T} \rightarrow T = \frac{300\times400\times100}{200\times250} = 240$

48. Income = 4x and 5x.
Expenses = 7y and 8y.
Savings = 1000 and 2000.
$\therefore$ 4x – 7y = 1000 and 5x – 8y = 2000
Solving, x = 2000 and 9x = 18,000.

49. After first round, cask holds 8 gallons of water and rest milk.
Suppose its volume is x gallons in total.

$\therefore$ Now when 6 gallons are taken out, amount of water taken out is $6\times\frac{8}{x}$.

$\therefore$ Final volume of water = $8 - \frac{48}{x} + 6$.

$\therefore 14 - \frac{48}{x} = \frac{x}{2} \rightarrow x = 24$

50. The cost (C) = F + qv, where F is the fixed component, q is the number of units produced and v the variable cost per component.

$\therefore$ Sales price = $1.2\times\frac{C}{q} = 1.2\left(\frac{F}{q}\right) + 1.2v$

$120 = 1.2\left(\frac{F}{250}\right) + 1.2v$ or $100 = \left(\frac{F}{250}\right) + v \rightarrow 25{,}000 = F + 250v$ .... (i)

and $96 = 1.2\frac{F}{500} + 1.2v$ or $80 = \frac{F}{500} + v \rightarrow 40{,}000 = F + 500v$ .... (ii)

By (ii) – (i),
$\therefore$ 250v = 15,000 $\rightarrow$ 500v = 300,000
$\therefore$ F + 1000v = 40,000 + 30,000 = 70,000
$\therefore$ Sales price of 1000 units = $1.2\left(\frac{F}{1000}\right) + 1.2v = 1.2\times70 = 84$